Fired Heaters for General Refinery Services

API STANDARD 560
SECOND EDITION, SEPTEMBER 1995

American Petroleum Institute
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Fired Heaters for General Refinery Services

Manufacturing, Distribution and Marketing Department

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FOREWORD

This standard is based on the accumulated knowledge and experience of petroleum refiners, fired heater manufacturers, and engineering contractors. The objective of this publication is to provide a purchase specification to facilitate the procurement and manufacture of fired heaters for general refinery services.

The primary purpose of API standards on heat transfer equipment is to establish minimum mechanical requirements. This limitation in scope is one of charter as opposed to interest and concern. Energy conservation is of concern and has become increasingly important in all aspects of equipment design, application, and operation. Thus, innovative, energy-conserving approaches should be aggressively pursued by the manufacturer and the user during these steps. Alternative approaches that may result in improved energy utilization should be thoroughly investigated and brought forth. This is especially true of new equipment proposals, since the evaluation of purchase options will be based increasingly on total life costs as opposed to acquisition cost alone. Equipment manufacturers, in particular, are encouraged to suggest alternatives to those specified when such approaches achieve improved energy effectiveness and reduced total life costs without sacrifice of safety or reliability.

This standard requires the purchaser to specify certain details and features. Although the purchaser may desire to modify, delete, or amplify sections of this standard, it is strongly recommended that all modifications, deletions, and amplifications be made by supplementing this standard, rather than by rewriting or by incorporating sections of this standard into another complete standard.

This standard has incorporated into its appendix two previous API publications. API Recommended Practice 532, *Measurement of the Thermal Efficiency of Fired Process Heaters* is Appendix G; and API Recommended Practice 533, *Air Preheat Systems for Process Heaters* is Appendix F.

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Certain serious adverse health effects are associated with asbestos, among them the serious and often fatal diseases of lung cancer, asbestosis, and mesothelioma (a cancer of the chest and abdominal linings). The degree of exposure to asbestos varies with the product and the work practices involved.

Consult the most recent edition of the Occupational Safety and Health Administration (OSHA), U.S. Department of Labor, Occupational Safety and Health Standard for Asbestos, Tremolite, Anthophyllite, and Actinolite, 29 *Code of Federal Regulations* Section 1910.1001; the U.S. Environmental Protection Agency, National Emission Standard for Asbestos, 40 *Code of Federal Regulations* Sections 61.140 through 61.156; and the U.S. Environmental Protection Agency (EPA) rule on labeling requirements and phased banning of asbestos products, published at 54 *Federal Register* 29460 (July 12, 1989).

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CONTENTS

Page

SECTION 1—GENERAL
1.1 Scope 1
1.2 Alternative Designs 1
1.3 Conflicting Requirements 1
1.4 Definition of Terms 1
1.5 Nomenclature 3
1.6 Referenced Publications 4
1.7 Proposals 9
1.7.1 Purchaser's Responsibilities 9
1.7.2 Vendor's Responsibilities 9
1.8 Documentation 9
1.8.1 Drawings for Purchaser's Review 9
1.8.2 Foundation Loading Diagrams 10
1.8.3 Documents for Purchaser's Review 10
1.8.4 Certified Drawings and Diagrams 10
1.9 Final Records 10

SECTION 2—DESIGN CONSIDERATIONS
2.1 Process 10
2.2 Combustion 10
2.3 Mechanical 11

SECTION 3—TUBES
3.1 General 11
3.2 Extended Surface 12
3.3 Materials 12

SECTION 4—HEADERS
4.1 General 13
4.2 Plug Headers 13
4.3 Returns Bends 13
4.4 Materials 14

SECTION 5—PIPING, TERMINALS, AND MANIFOLDS
5.1 General 14
5.2 Allowable Movement and Loads 14
5.3 Materials 14

SECTION 6—TUBE SUPPORTS
6.1 General 15
6.2 Loads and Allowable Stress 16
6.3 Materials 17

SECTION 7—REFRACTORIES AND INSULATION
7.1 General 17
7.2 Brick and Tile Construction 18
7.3 Castable Construction 18
7.4 Ceramic Fiber Construction 19
7.5 Multicomponent Lining Construction 20
7.6 Materials 20

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SECTION 8—STRUCTURES AND APPURTENANCES

8.1 General 20
8.2 Structures 21
8.3 Header Boxes, Doors, and Ports 21
8.3.1 Header Boxes 21
8.3.2 Doors and Ports 21
8.4 Ladders, Platforms, and Stairways 22
8.5 Materials 22

SECTION 9—STACKS, DUCTS, AND BREECHING

9.1 General 22
9.2 Design Considerations 23
9.3 Allowable Stresses 23
9.4 Static Design 24
9.5 Wind-Induced Vibration Design 24
9.6 Materials 26

SECTION 10—BURNERS AND AUXILIARY EQUIPMENT

10.1 Burners 26
10.2 Sootblowers 27
10.3 Fans and Drivers 28
10.4 Damper Controls 28

SECTION 11—INSTRUMENT AND AUXILIARY CONNECTIONS

11.1 Flue Gas and Air 29
11.1.1 Flue Gas and Combustion Air Temperature 29
11.1.2 Flue Gas and Combustion Air Pressure 29
11.1.3 Flue Gas Sampling 29
11.2 Process Fluid Temperature 30
11.3 Auxiliary Connections 30
11.3.1 Purge Steam Connections 30
11.3.2 Vent and Drain Connections 30
11.4 Tube-Skin Thermocouples 30
11.5 Access to Connections 30

SECTION 12—SHOP FABRICATION AND FIELD ERECTION

12.1 General 31
12.2 Steel Fabrication 31
12.3 Coil Fabrication 32
12.4 Painting and Galvanizing 32
12.5 Refractories and Insulation 33
12.6 Field Erection 33

SECTION 13—INSPECTION, EXAMINATION, AND TESTING

13.1 General 34
13.2 Weld Inspection and Examination 34
13.3 Castings Examination 34
13.4 Examination of Other Components 35
13.5 Testing 36
13.5.1 Pressure Testing 36
13.5.2 Refractory Testing 36
13.5.3 Studded Tube Testing 36
13.5.4 Positive Materials Identification 36

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December 27, 1999 12:18:25

APPENDIX A—EQUIPMENT DATA SHEETS 37

APPENDIX B—PURCHASER'S CHECKLIST 67

APPENDIX C—PROPOSED SHOP ASSEMBLY CONDITIONS 73

APPENDIX D—STRESS CURVES FOR USE TO DESIGN TUBE SUPPORT ELEMENTS 77
D.1 Scope 77
D.2 Referenced Publications 77
D.3 Casting Factor 77
D.4 Minimum Cross Sections 77
D.5 Maximum Design Temperatures 77
D.6 Corrosion Resistance 78
D.7 Proprietary Alloys 78
D.8 Stress Curves 78
D.9 Data Sources 78

APPENDIX E—CENTRIFUGAL FANS FOR FIRED HEATER SYSTEMS 93
E.1 General 93
E.2 Design 94
E.3 Accessories 98
E.4 Inspection, Testing, and Preparation for Shipment 101
E.5 Vendor's Data 102

APPENDIX F—AIR PREHEAT SYSTEMS FOR FIRED PROCESS HEATERS 105
F.1 General 105
F.2 General Considerations in Selecting an Air Preheat System 105
F.3 Interaction of the Air Preheat System with the Fired Heater 106
F.4 Selecting an Air Preheat System 107
F.5 Safety, Operability, and Maintenance 112
F.6 Selecting System Temperatures 113
F.7 Combustion Air and Flue Gas Quantities 116
F.8 Duct Design and Damper Selection 117
F.9 Guidelines for Specifying Equipment 129
F.10 Environmental Impact of Air Preheat Systems 135
F.11 Preparing an Inquiry 136
APPENDIX F.A—FLUE GAS DEW POINT DICUSSION AND REFERENCES 137

APPENDIX G—MEASUREMENT OF THE THERMAL EFFICIENCY OF FIRED PROCESS HEATERS 141
G.1 General 141
G.2 Testing 144
G.3 Determination of Thermal Efficiency 144
APPENDIX G.A—MODEL FORMAT FOR DATA SHEETS 151
APPENDIX G.B—MODEL FORMAT FOR WORK SHEETS 159
APPENDIX G.C—SAMPLE WORK SHEETS FOR AN OIL-FIRED HEATER WITH NATURAL DRAFT (G.3.2.1) 171
APPENDIX G.D—SAMPLE WORK SHEETS FOR A GAS-FIRED HEATER WITH PREHEATED COMBUSTION AIR FROM AN INTERNAL HEAT SOURCE (G.3.2.2) 181

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December 27, 1999 12:18:25

APPENDIX G.E—SAMPLE WORK SHEETS FOR A GAS-FIRED HEATER WITH PREHEATED COMBUSTION AIR FROM AN EXTERNAL HEAT SOURCE (G.3.2.3) 189
APPENDIX G.F—ESTIMATING THERMAL EFFICIENCY FOR OFF-DESIGN OPERATING CONDITIONS 195

Figures
1—Typical Heater Types 5
2—Typical Burner Arrangements (Elevation View) 6
3—Heater Components 7
4—Air Preheat System Using Regenerative, Recuperative, or Heat Pipe Unit 8
5—System Using Indirect Closed System Air Preheater with Mechanical Circulation 8
6—External Heat Source for Air Preheating 9
D-1—Carbon Steel Castings: ASTM A 216, Grade WCB 80
D-2—Carbon Steel Plate: ASTM A 283, Grade C 81
D-3—$2^1/_4$Cr-1Mo Castings: ASTM A 217, Grade WC9 82
D-4—$2^1/_4$Cr-1Mo Plate: ASTM A 387, Grade 22, Class 1 83
D-5—5Cr-$^1/_2$Mo Castings: ASTM A 217, Grade C5 84
D-6—5Cr-$^1/_2$Mo Plate: ASTM A 387, Grade 5, Class 1 85
D-7—19Cr-9Ni Castings: ASTM A 297, Grade HF 86
D-8—Type 304H Plate: ASTM A 240, Type 304H 87
D-9—25Cr-12Ni Castings: ASTM A 447, Grade HH, Type II 88
D-10—Type 309H Plate: ASTM A 240, Type 309H 89
D-11—25Cr-20Ni Castings: ASTM A 297, Grade HK 90
D-12—Type 310H Plate: ASTM A 240, Type 310H 91
D-13—50Cr-50Ni-Cb Castings: ASTM A 560, Grade 50Cr-50Ni-Cb 92
F-1—Recommended Minimum Metal Temperature 115
F-2—Sample Flow Sheet for Duct Design and Damper Selection 118
F-3—Moody's Friction Factor 119
F-4—Pressure Drop for Flue Gases in Ducts 121
F-5—Equipment Lengths (*LD*) for Multiple Piece Miter Elbows of Round Cross Section 125
F-6—Branch Loss Coefficients 126
F-7—Duct Zones 127
F.A-1—Dew Point of Flue Gas Versus Sulfur in Fuel Oil (Test Data from Industrial Boilers) 138
F.A-2—Dew Point of Flue Gas Versus Sulfur in Fuel Oil (Test Data from Industrial Boilers and Data of Attig and Sedor) 138
F.A-3—Dew Point of Flue Gas Versus Sulfur in Fuel Oil (Test Data from Industrial Boilers and Data of Energy Technology, Inc.) 139
G-1—Instrument and Measurement Locations 142
G-2—Typical Aspirating (High-Velocity) Thermocouple 143
G-3—Typical Heater Arrangement with Nonpreheated Air 145
G-4—Typical Heater Arrangement with Preheated Air from an Internal Heat Source 146
G-5—Typical Heater Arrangement with Preheated Air from an External Heat Source 147
G.B-1—Enthalpy of H_2O, CO, CO_2, and SO_2 168
G.B-2—Enthalpy of Air, O_2, and N_2 169

Tables
1—Tubeseat Dimensions 12
2—Extended Surface Materials 13
3—Extended Surface Dimensions 13

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December 27, 1999 12:18:25

4—Tube Materials and ASTM Specifications 14
5—Tube Center-to-Center Dimensions 14
6—Plug Header and Return Bend Materials 15
7—Maximum Allowable Forces, Moments, and Movements 16
8—Maximum Design Temperatures for Tube Support Materials 17
9—Maximum Temperatures for Anchor Tips 18
10—Minimum Yield Strength and Modulus of Elasticity for Structural Steel 24
11—Minimum Shape Factors and Effective Diameters for Wind Loads 25
12—Minimum Clearance for Natural Draft Operation 26
13—Materials of Construction 28
14—Service Factors 29
15—Welding Filler Materials 31
D-1—Sources of Data Presented in Figures D-1 Through D-13 79
E-1—Maximum Shaft Runout Indicator Readings 97
E-2—Maximum Severity of Defects in Castings 101
F-1—Fittings 122
F-2—Recommended Damper Types 129
F.A-1—Flue Gas Dew Point Data from Oil-Fired Industrial Boilers 139
G-1—Limits on Variability of Data 144

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Fired Heaters for General Refinery Services

SECTION 1—GENERAL

1.1 Scope

1.1.1 This standard covers the minimum requirements for the design, materials, fabrication, inspection, testing, preparation for shipment, and erection of fired heaters, air preheaters, fans, and burners for general refinery service.

1.1.2 In a fired heater, heat liberated by the combustion of fuels is transferred to fluids contained in tubular coils within an internally insulated enclosure.

Note: A bullet (●) at the beginning of a paragraph indicates that a decision by the purchaser is required. These decisions should be indicated on the data sheets (see Appendix A) or stated in the inquiry or purchase order. Decisions should be indicated on the check list (see Appendix B).

1.2 Alternative Designs

The vendor may offer alternative designs in addition to the base design when permitted by the inquiry. Any variance with this standard or the purchaser's specification shall be clearly indicated in the proposal.

1.3 Conflicting Requirements

● **1.3.1** In case of conflict between this standard and the inquiry or the purchase order, the purchaser shall specify the order of precedence.

1.3.2 In the absence of a specified order of precedence, the vendor shall obtain written approval from the purchaser before proceeding with the work affected by a conflict between the proposal and this standard or the purchase order.

1.4 Definition of Terms

1.4.1 *Air heater or air preheater* is a heat transfer apparatus through which combustion air is passed and heated by a medium of higher temperature, such as the products of combustion, steam, or other fluid.

1.4.1.1 *Direct air preheater* is an exchanger which transfers heat directly between the flue gas and the combustion air. A regenerative air preheater uses heated rotating elements and a recuperative design uses stationary tubes, plates, or cast iron elements to separate the two heating media.

1.4.1.2 *Indirect air preheater* is a fluid-to-air heat transfer device. The heat transfer can be accomplished by using a heat transfer fluid, process stream, or utility stream which has been heated by the flue gas or other means. A heat pipe air preheater uses a vaporizing/condensing fluid to transfer heat between the flue gas and air.

1.4.2 *Arch* is a flat or sloped portion of the heater radiant section opposite the floor.

1.4.3 *Atomizer* is a device used to reduce a liquid fuel oil to fine mist. Atomization media are steam, air, or mechanical.

1.4.4 *Anchor or tieback* is a metallic or refractory device that holds the refractory or insulation in place.

1.4.5 *Backup layer* is any refractory layer behind the hot face layer.

1.4.6 *Balanced draft heater* uses a fan to supply combustion air and to remove flue gases.

1.4.7 *Breeching* is the heater section where flue gases are collected after the last convection coil for transmission to the stack or the outlet duct work.

1.4.8 *Bridgewall, division, or gravity wall* separates two adjacent heater zones.

1.4.9 *Bridgewall temperature* is the flue gas temperature leaving the radiant section.

1.4.10 *Burner* introduces fuel and air into a heater at the desired velocities, turbulence, and concentration to establish and maintain proper ignition and combustion. Burners are classified by the types of fuel fired, such as: oil, gas, or combination of gas and oil. A secondary consideration in classifying burners is the means by which combustion air is mixed with the fuel.

1.4.11 *Casing* is the metal plate used to enclose the fired heater.

1.4.12 *Castable* is an insulating concrete poured or gunned in place to form a rigid refractory shape or structure.

1.4.13 *Ceramic fiber* is a fibrous refractory insulation composed primarily of silica and alumina. Applicable forms include blanket, board, module, rigidized blanket, and vacuum-formed shapes.

1.4.14 *Coil pressure drop* is the difference between the coil inlet pressure and the coil outlet pressure between terminals, excluding the effect of static head.

1.4.15 *Convection section* is the portion of the heater in which the heat is transferred to the tubes primarily by convection.

1.4.16 *Corbel* is a projection from the refractory surface generally used to prevent flue gas bypassing the convection section tubes.

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1.4.17 *Corrosion allowance* is the additional material thickness added to allow for material loss during the design life of the component. It is the corrosion rate times tube design life, expressed in inches (millimeters).

1.4.18 *Corrosion rate* is the reduction in the material thickness due to the chemical attack from the process fluid or flue gas or both, expressed in mils per year (millimeters per year).

1.4.19 *Crossover* is the interconnecting piping between any two heater coil sections.

1.4.20 *Damper* is a device for introducing a variable resistance for regulating volumetric flow of gas or air.

1.4.20.1 *Butterfly damper* is a single-blade damper pivoted about its center.

1.4.20.2 *Louver damper* is a damper consisting of several blades each pivoted about its center and linked together for simultaneous operation.

1.4.21 *Draft* is the negative pressure (vacuum) of the flue gas measured at any point in the heater, expressed in inches of water column (pascals).

1.4.22 *Draft loss* is the pressure drop through duct conduits or across tubes and equipment in air and flue gas systems.

1.4.23 *Duct* is a conduit for air or flue gas flow.

1.4.24 *Efficiency, fuel* refers to the heat absorbed divided by the net heat of combustion of the fuel as heat input, expressed as a percentage.

1.4.25 *Efficiency, thermal* refers to the total heat absorbed divided by the total heat input, expressed as a percentage.

1.4.26 *Erosion* is a reduction in material thickness due to mechanical attack from a fluid, expressed in inches (millimeters).

1.4.27 *Excess air* is the amount of air above the stoichiometric requirement for complete combustion, expressed as a percentage.

1.4.28 *Extended surface* refers to the heat transfer surface in the form of fins or studs attached to the heat absorbing surface.

1.4.29 *Extension ratio* is the ratio of total outside exposed surface to the outside surface of the bare tube.

1.4.30 *Flue gas* is the gaseous product of combustion including the excess air.

1.4.31 *Forced draft heater* supplies combustion air by a fan or other mechanical means.

1.4.32 *Fouling allowance* is a factor to allow for a layer of residue that increases pressure drop, usually a build up of coke or scale on the inner surface of a coil, expressed as inches (millimeters). This value shall be used in calculating the fouled pressure drop.

1.4.33 *Fouling resistance* is a factor used to calculate the overall heat transfer coefficient. The inside fouling resistance shall be used to calculate the maximum metal temperature for design. The external fouling resistance is used to compensate the loss of performance due to deposits on the external surface of the tubes or extended surface.

1.4.34 *Guillotine blind* is a single blade device that is used to isolate equipment or heaters.

1.4.35 *Header or return bend* is the common term for a 180-degree cast or wrought fitting that connects two or more tubes.

1.4.36 *Header box* is the internally insulated structural compartment, separated from the flue gas stream, which is used to enclose a number of headers or manifolds. Access is afforded by means of hinged doors or removable panels.

1.4.37 *Heat absorption* is the total heat absorbed by the coils excluding any combustion air preheat, expressed in British thermal units per hour (megawatts).

1.4.38 *Heat flux density, average* is the heat absorbed divided by the exposed heating surface of the coil section. Average flux density for an extended surface tube shall be indicated on a bare surface basis with extension ratio noted, expressed in British thermal units per hour per square foot (kilowatts per square meter).

1.4.39 *Heat flux density, maximum* is the maximum local heat transfer rate in the coil section, expressed in British thermal units per hour per square foot (kilowatts per square meter).

1.4.40 *Heat release* is the total heat liberated from the specified fuel, using the lower heating value of the fuel, expressed in British thermal units per hour (megawatts).

1.4.41 *Heating value, higher (HHV),* is the total heat obtained from the combustion of a specified fuel at 60°F, expressed in British thermal units per pound or per cubic foot (kilojoules per kilogram or per cubic meter).

1.4.42 *Heating value, lower (LHV),* is the higher heating value minus the latent heat of vaporization of the water formed by combustion of hydrogen in the fuel, also called the net heating value, expressed in British thermal units per pound or per cubic foot (kilojoules per kilogram or per cubic meter).

1.4.43 *Hot face layer* is the refractory layer exposed to the highest temperatures in a multilayer or multicomponent lining.

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1.4.44 *Hot face temperature* is the temperature of the refractory surface in contact with the flue gas or heated combustion air. The hot face temperature is used to determine refractory or insulation thickness and heat transmitted. The design temperature is used to specify the service temperature limit of the refractory materials.

1.4.45 *Induced draft heater* uses a fan to remove flue gases and maintain a negative pressure in the heater to induce combustion air without a forced draft fan.

1.4.46 *Manifold* is a chamber for the collection and distribution of fluid to or from multiple parallel flow paths.

1.4.47 *Metal fiber reinforcement* is stainless steel needles added to castable for improved toughness and durability.

1.4.48 *Monolithic lining* is a single-component lining system.

1.4.49 *Mortar* is a refractory material preparation used for laying and bonding refractory bricks.

1.4.50 *Multicomponent lining* is a refractory system consisting of two or more layers of different refractory types; for example, castable and ceramic fiber.

1.4.51 *Multilayer lining* is a refractory system consisting of two or more layers of the same refractory type.

1.4.52 *Natural draft heater* is a unit in which a stack effect induces the combustion air and removes the flue gases.

1.4.53 *Normal heat release* is the design heat absorption of the heater divided by the calculated fuel efficiency expressed in British thermal units per hour (megawatt).

1.4.54 *Pass or stream* is a flow circuit consisting of one or more tubes in series.

1.4.55 *Pilot* is a smaller burner that provides ignition energy to light the main burner.

1.4.56 *Plenum or windbox* is a chamber surrounding the burners that is used to distribute air to the burners or reduce combustion noise.

1.4.57 *Plug header* is a cast return bend provided with one or more openings for the purpose of inspection, mechanical tube cleaning, or draining.

1.4.58 *Primary air* is that portion of the total combustion air that first mixes with the fuel.

1.4.59 *Protective coating* is a corrosion-resistant material applied to a metal surface; for example, on casing plates behind porous refractory materials to protect against sulfur in the flue gases.

1.4.60 *Radiant section* is that portion of the heater in which heat is transferred to the tubes, primarily by radiation.

1.4.61 *Radiation or setting loss* is the heat lost to the surroundings from the casing of the heater and the ducts and auxiliary equipment (when heat recovery systems are used), expressed as percent of heat release.

1.4.62 *Secondary air* is the air supplied to the fuel to supplement primary air.

1.4.63 *Setting* is the heater casing, brickwork, refractory, and insulation, including the tiebacks or anchors.

1.4.64 *Shield section* contains those tubes that shield the remaining convection section tubes from direct radiation.

1.4.65 *Sootblower* is a mechanical device for discharging steam or air to clean heat-absorbing surfaces.

1.4.66 *Stack* is a vertical conduit used to discharge flue gas to the atmosphere.

1.4.67 *Strakes or spoilers* are metal stack attachments that prevent wind-induced vibration.

1.4.68 *Target wall* is a vertical refractory firebrick wall which is exposed to direct flame impingement on one or both sides.

1.4.69 *Temperature allowance* is the number of degrees Fahrenheit (Celsius) to be added to the process fluid temperature to account for flow maldistribution and operating unknowns. The temperature allowance is added to the calculated maximum tube metal temperature or the equivalent tube metal temperature to obtain the design metal temperature.

1.4.70 *Terminal* is a flanged or welded projection from the coil providing for inlet and outlet of fluids.

1.4.71 *Tube guide* is used with vertical tubes to restrict horizontal movement while allowing the tube to expand axially.

1.4.72 *Tube retainer* is used to restrain horizontal radiant tubes from lifting off the intermediate tube supports during operation.

1.4.73 *Tube support or tube sheet* is any device used to support tubes.

1.4.74 *Vapor barrier* is a metallic foil placed between layers of refractory as a barrier to flue gas flow.

1.4.75 *Volumetric heat release* is the heat released divided by the net volume of the radiant section, excluding the coils and refractory dividing walls, expressed in British thermal units per hour per cubic foot (kilowatts per cubic meter).

1.5 Nomenclature

The type of heater is normally described by the structural configuration, radiant tube coil configuration, and burner arrangement. Some examples of structural configurations

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are cylindrical, box, cabin, and multicell box. Examples of radiant tube coil configurations include vertical, horizontal, helical, and arbor. Examples of burner arrangements include upfired, downfired, and wallfired. The wallfired arrangement can be further classified as sidewall, endwall, and multilevel.

Figure 1 illustrates some typical heater types.

Figure 2 illustrates typical burner arrangements.

Various combinations of Figures 1 and 2 can be used. For example, Figure 1C can employ burner arrangements 2A, 2B, or 2C. Similarly, Figure 1D can employ burner arrangements 2A or 2D.

Figure 3 shows typical components.

Figures 4, 5 and 6 show typical combustion air preheat systems.

1.6 Referenced Publications

● **1.6.1** The editions of the following standards, codes, or specifications that are in effect at the time of publication of this standard shall, to the extent specified herein, form part of this standard. The applicability of changes that occur after the time of the inquiry shall be mutually agreed upon by the purchaser and the vendor.

AISC[1]

Specification for Design, Fabrication, and Erection of Structural Steel for Buildings

API

RP 530	*Calculation of Heater Tube Thickness in Petroleum Refineries*
Std 611	*General-Purpose Steam Turbines for Refinery Service*
EA 7301	*Guidelines on Noise*

ASCE[2]

7-88	*Minimum Design Loads for Buildings and Other Structures*

ASME[3]

B16.5	*Steel Pipe Flanges and Flanged Fittings*
B16.9	*Wrought Steel Buttwelding Fittings*
B16.11	*Forged Steel Fittings, Socket-Welding and Threaded*
B16.28	*Wrought Steel Buttwelding Short Radius Elbows and Return Bends*
B31.3	*Chemical Plant and Petroleum Refinery Piping*
B36.10	*Welded and Seamless Wrought Steel Pipe*

Boiler and Pressure Vessel Code, Section IX, "Welding and Brazing Qualifications"

AWS[4]

D1.1	*Structural Welding Code*
D14.6	*Specification for Welding of Rotating Elements of Equipment*

ICBO[5]

Uniform Building Code

MSS[6]

SP-53	*1995 Quality Standard for Steel Castings and Forgings for Valves, Flanges and Fittings and Other Piping Components Magnetic Particle Examination Method*
SP-55	*1983 Quality Standard for Steel Castings for Valves, Flanges and Fittings and Other Piping Components Visual Method (R-1990)*
SP-93	*1987 Quality Standard for Steel Castings and Forgings for Valves, Flanges and Fittings and Other Piping Components Liquid Penetrant Examination Method (R-1992)*

SSPC[7]

SP-3	*Power Tool Cleaning*
SP-5	*White Metal Blast Cleaning*
SP-6	*1991 Surface Preparation Specification No. 6 Commercial Blast Cleaning*
SP-10	*Near-White Blast Cleaning*

1.6.2 The following standards, codes, or specifications are also referenced in this standard:

ABMA[8]

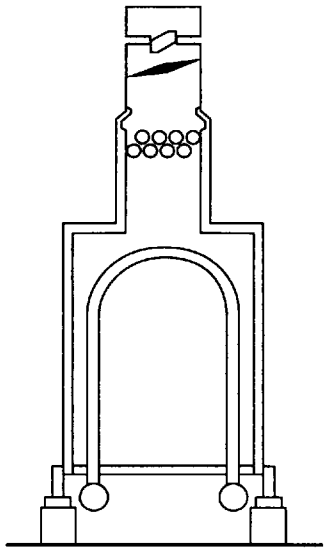
Standard 9	*Load Ratings and Fatigue Life for Ball-Bearings*

AMCA[9]

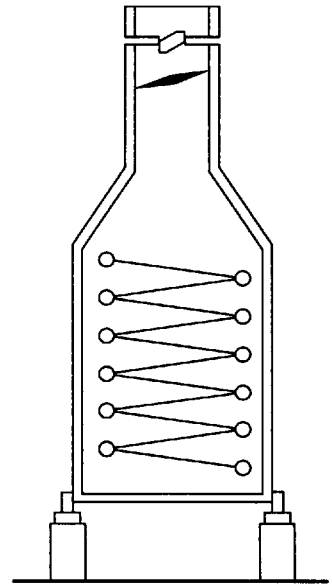
99-86	*Standards Handbook*
99-2404-78	*Drive Arrangements for Centrifugal Fans*
201	*Fans and Systems*
203	*Field Performance Measurement of Fan Systems*
210	*Laboratory Methods of Testing Fans for Rating*
801	*Power Plant Fans—Specification Guidelines*

[1]American Institute for Steel Construction, 400 North Michigan Avenue, Chicago, Illinois, 60611.
[2]American Consulting Engineers, 1015 15th Street, N.W., Washington, D.C. 20005.
[3]American Society of Mechanical Engineers, 345 East 47th Street, New York, New York 10017.
[4]American Welding Society, 550 N.W. LeJeune Rd., Miami, Florida 33135.
[5]International Conference of Building Officials, 5360 Workman Mill Road, Whittier, California 90601-2298.
[6]Manufacturers Standardization Society, 127 Park Street, N.E., Vienna, Virginia 22180.
[7]Steel Structures Painting Council, 4400 Fifth Avenue, Pittsburgh, Pennsylvania 15213-2683.
[8]American Bearing Manufacturing Association, 1200 19th Street, N.W., Suite 300, Washington, D.C. 20036.
[9]Air Movement & Control Association, 30 West University Drive, Arlington Heights, Illinois 60004.

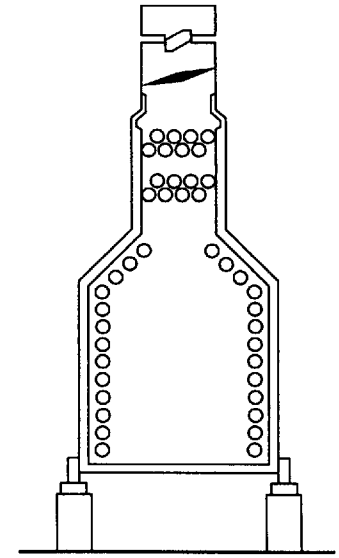
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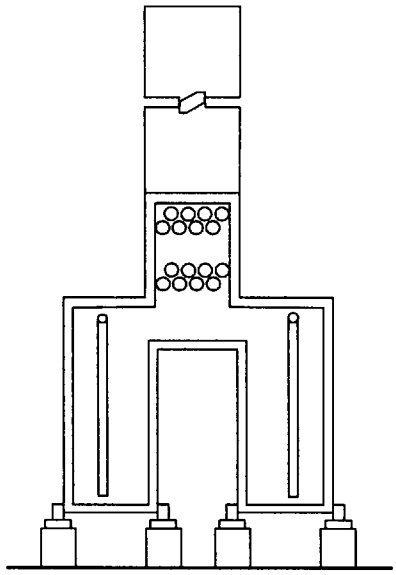
TYPE A—BOX HEATER WITH ARBOR COIL



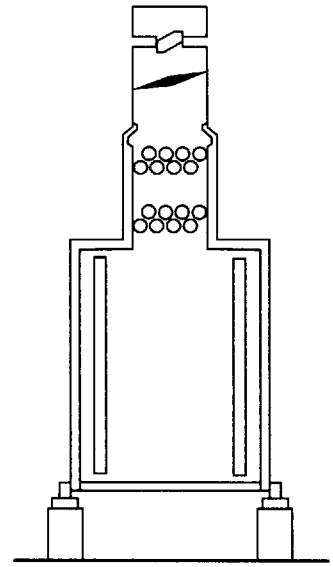
TYPE B—CYLINDRICAL HEATER WITH HELICAL COIL



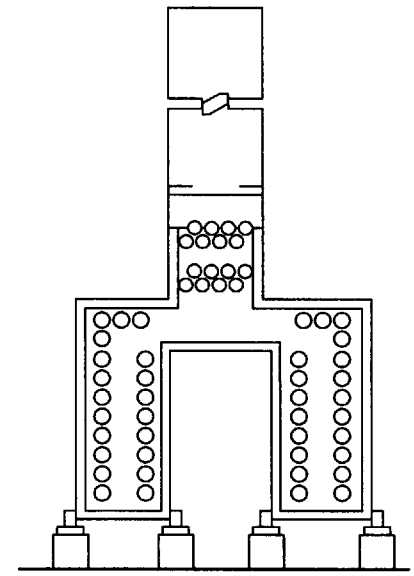
TYPE C—CABIN HEATER WITH HORIZONTAL TUBE COIL



TYPE D—BOX HEATER WITH VERTICAL TUBE COIL



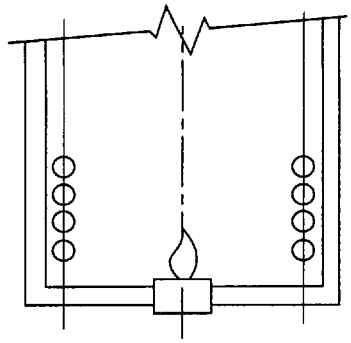
TYPE E—CYLINDRICAL HEATER WITH VERTICAL COIL



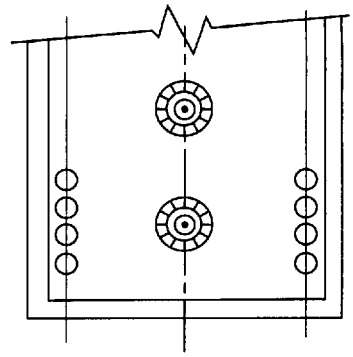
TYPE F—BOX HEATER WITH HORIZONTAL TUBE COIL

Figure 1—Typical Heater Types

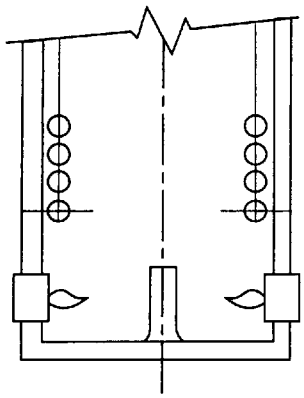
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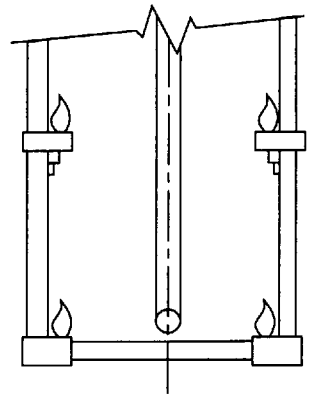
TYPE A—UPFIRED



TYPE B—ENDWALL FIRED



TYPE C—SIDEWALL FIRED



TYPE D—SIDEWALL FIRED MULTI-LEVEL

Figure 2—Typical Burner Arrangements (Elevation View)

802 *Power Plant Fans—Establishing Performance using Laboratory Models*
803 *Power Plant and Industrial Fans—Site Performance Test Standard*

API

RP 500A *Classification of Locations for Electrical Installations in Petroleum Refineries, 4th edition 1982, Reaffirmed 1987*
RP 554 *Process Instrumentation and Control*
RP 555 *Process Analyzers*
Std 610 *Centrifugal Pumps for General Refinery Services*
Std 673 *Special Purpose Centrifugal Fans for General Refinery Services*

ASM[10]

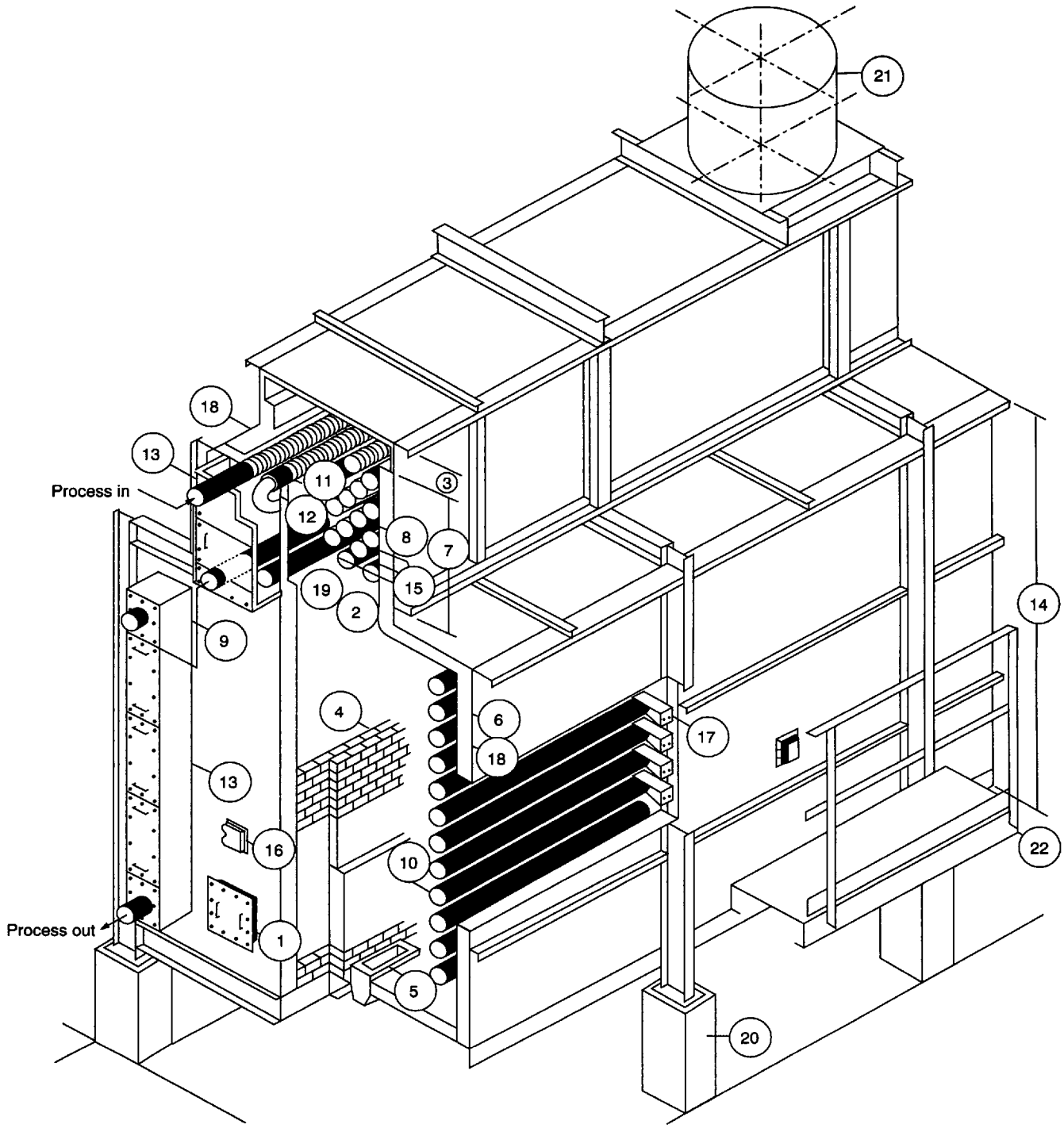
Metals Handbook, Volume 3, "Properties and Selection: Stainless Steels, Tool Materials and Special-Purpose Metals" (Ninth Edition, 1980)

ASME

B17.1 *1967 Keys and Keyseats (R-1989)*
Boiler and Pressure Vessel Code, Section V, "Nondestructive Examination," and Section VIII, "Rules for Construction of Pressure Vessels"

[10]American Society for Metals International, 9639 Kinsman Road, Metals Park, Ohio 44073.

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Notes:
1. Access door.
2. Arch.
3. Breeching.
4. Bridgewall.
5. Burner.
6. Casing
7. Convection section.
8. Corbel.
9. Crossover.
10. Tubes.
11. Extended surface.
12. Return bend.
13. Header box.
14. Radiant section.
15. Shield section.
16. Observation door.
17. Tube support.
18. Refractory lighting.
19. Tube sheet.
20. Pier.
21. Stack/duct.
22. Platform.

Figure 3—Heater Components

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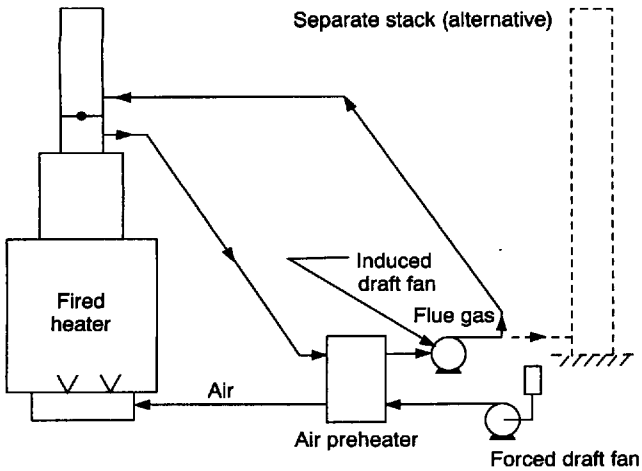


Figure 4—Air Preheat System Using Regenerative, Recuperative, or Heat Pipe Unit

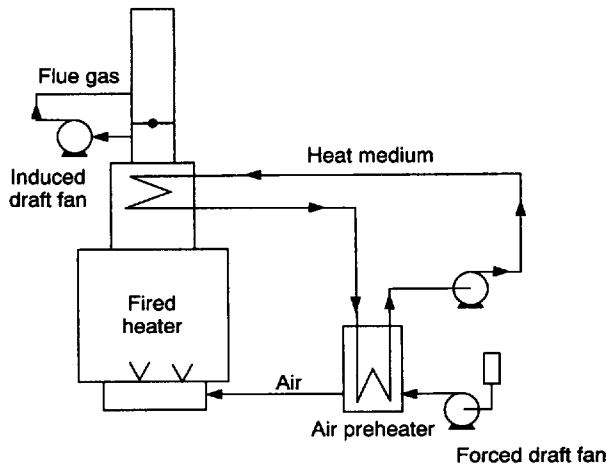


Figure 5—System Using Indirect Closed System Air Preheater with Mechanical Circulation

ASTM[11]

All materials used in the construction of fired heaters shall comply with appropriate ASTM material specifications.

A 216	*Carbon Steel Castings Suitable for Fusion Welding for High-Temperature Service*
A 217	*Martensitic Stainless and Alloy Steel Castings for Pressure-Containing Parts Suitable for High-Temperature Service*
A 240	*Heat-Resisting Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels*
A 283	*Low and Intermediate Tensile Strength Carbon Steel Plates, Shapes, and Bars*
A 297	*Heat-Resistant Iron-Chromium and Iron-Chromium-Nickel Steel Castings for General Application*
A 387	*Chromium-Molybdenum Alloy Steel Pressure Vessel Plates*
A 447	*Chromium-Nickel-Iron Alloy Steel Castings (25-12 Class) for High-Temperature Service*
A 515/A 515M	*1992 Standard Specification for Pressure Vessel Plates, Carbon Steel, for Intermediate- and Higher- Temperature Service*
A 560	*Chromium-Nickel Alloy Castings*
C 543	*Specification for Mineral Fiber Blanket and Felt Insulation (Industrial Type)*
C 612	*Specification for Block and Board Thermal Insulation*
DS 5	*Report on the Elevated-Temperature Properties of Stainless Steels*
DS 5S2	*An Evaluation of the Yield, Tensile, Creep, and Rupture Strengths of Wrought 304, 316, 321 and 347 Stainless Steels at Elevated Temperatures*
DS 6	*Report on the Elevated Temperature Properties of Chromium-Molybdenum Steels*
DS 6S2	*Supplemental Report on the Elevated Temperature Properties of Chromium-Molybdenum Steels*
DS 11S1	*An Evaluation of the Elevated Temperature Tensile and Creep-Rupture Properties of Wrought Carbon Steel*
DS 58	*Evaluation of the Elevated Temperature Tensile and Creep-Rupture Properties of 3 to 9 Percent Chromium-Molybdenum Steels*
E 94	*1993 Standard Guide for Radiographic Testing*
E 125	*1963 Standard Reference Photographs for Magnetic Particle Indications on Ferrous Castings (R-1993)*
E 142	*1992 Standard Method for Controlling Quality of Radiographic Testing*
E 709	*1994 Standard Guide for Magnetic Particle Examination*

NFPA[12]

70	*National Electric Code*

[11]American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103-1187.

[12]National Fire Protection Association, 1 Batterymarch Park, Quincy, Massachusetts 02269-9101.

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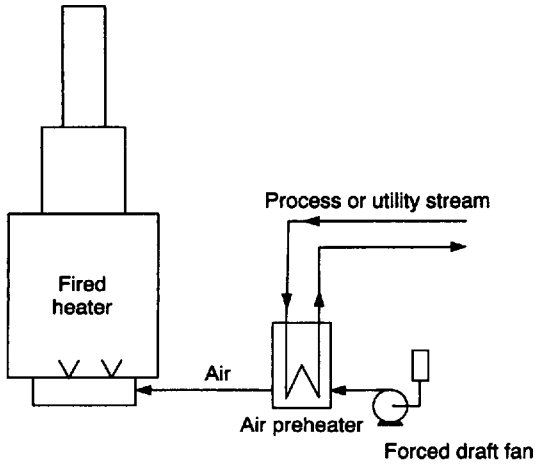


Figure 6—External Heat Source for Air Preheating

SFSA[13]
Steel Castings Handbook (Fifth Edition, 1980)

● **1.6.3** The purchaser and the vendor shall mutually determine the measures that must be taken to comply with any federal, state, or local codes, regulations, ordinances, or rules that may be applicable to the equipment.

1.7 Proposals

1.7.1 PURCHASER'S RESPONSIBILITIES

● **1.7.1.1** The purchaser's inquiry shall include data sheets, a check list, and other applicable information outlined herein. This information shall include any special requirements or exceptions to this standard.

1.7.1.2 The purchaser is responsible for the correct process specification to enable the vendor to prepare the fired heater design. The purchaser should complete, as a minimum, those items on the data sheet that are designated by an asterisk (*).

1.7.1.3 The purchase order shall state clearly the vendor's scope of work.

● **1.7.1.4** The purchaser's inquiry shall specify the number of copies of drawings, data sheets, specifications, data reports, operating manuals, installation instructions, spare parts lists, and other data to be supplied by the vendor, as required by 1.8 and 1.9.

[13]Steel Founders' Society of America, 455 State Street, Des Plaines, Illinois 60016.

1.7.2 VENDOR'S RESPONSIBILITIES

The vendor's proposal shall include:

a. Complete API fired heater and associated equipment data sheets for each heater (see Appendix A).
b. A drawing showing major dimensions in plan and elevation.
c. A full definition of the extent of shop assembly (see Appendix C), including the number, size, and weight of prefabricated parts and the number of field welds.
d. A detailed description of any exceptions to the specified requirements.
● e. A completed noise data sheet when the data sheet is supplied by the purchaser.
f. Curves shall be provided for heaters in vaporizing service showing pressure, temperature, vaporization, and bulk velocity as a function of the tube number.
g. The number of weeks the vendor shall require to submit outline drawings and foundation loading diagrams to the purchaser for review.
h. The equipment shipping date specified as the number of weeks after receipt of an order. The vendor shipping date should allow a specified period of time for purchaser's review and return of drawings.
i. List of utilities and quantities required.

1.8 Documentation

1.8.1 DRAWINGS FOR PURCHASER'S REVIEW

The vendor shall submit for review the general arrangement drawings for each heater. The final general arrangement drawings shall include the following information:

a. Heater service, the purchaser's equipment number, the project name and location, the purchase order numbers, and the vendor's reference number.
b. Coil terminal sizes, including flange ratings and facings; dimensional locations; direction of process flow; and allowable loads, moments, and forces on terminals.
c. Coil and crossover arrangements, tube spacings, tube diameters, tube wall thicknesses, tube lengths, material specifications, including grades for pressure parts only, and all extended surface data.
d. Coil design pressures, hydrostatic test pressures, design fluid and tube wall temperatures, and corrosion allowance.
e. Coil design code or recommended practice and fabrication code or specification.
f. Refractory and insulation types, thicknesses, and service temperature ratings.
g. Types and materials of anchors for refractory and insulation.
h. Location and number of access doors, observation doors, burners, sootblowers, dampers, and instrument and auxiliary connections.
i. Location and dimension of platforms, ladders, and stairways.
j. Overall dimensions including auxiliary equipment.

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1.8.2 FOUNDATION LOADING DIAGRAMS

The vendor shall submit for review foundation loading diagrams for each heater. The diagram shall include the following information:

a. Number and location of piers and supports.
b. Baseplate dimensions.
c. Anchor bolt locations, bolt diameters, and projection above foundations.
d. Dead loads, live loads, wind or earthquake loads, reaction to overturning moments, and lateral shear loads.

1.8.3 DOCUMENTS FOR PURCHASER'S REVIEW

The vendor also shall submit to the purchaser the following documents for review and comment before beginning fabrication:

a. Structural steel drawings; details of stacks, ducts and dampers; and structural calculations.
b. Burner assembly drawings and burner piping.
c. Tube support details.
d. Thermowell and thermocouple details.
e. Welding, examination, and test procedures.
f. Installation, dry out, and test procedures for refractories and insulation.
g. Refractory thickness calculations including temperature gradients through all refractory sections and sources of thermal conductivities.
h. Decoking procedures.

1.8.4 CERTIFIED DRAWINGS AND DIAGRAMS

After receipt of the purchaser's comments on the general arrangement drawings and diagrams, the vendor shall furnish certified general arrangement drawing and foundation loading diagrams. The vendor shall furnish design detail drawings, erection drawings, and an erection sequence. Drawings of auxiliary equipment shall also be furnished.

1.9 Final Records

Within 10 days after the completion of heater shipment, the vendor shall furnish the purchaser with the following documents:

● a. As-manufactured data sheets and drawings. In the event field-changes are made, as-built drawings and data sheets shall not be provided unless specifically requested by the purchaser.
b. Certified material reports, mill test reports, or ladle analysis for all pressure parts and alloy extended surfaces.
c. Installation, operation, and maintenance instructions for the heater and auxiliary equipment, such as air preheaters, fans, drivers, dampers, and burners.
d. Performance curves or data sheets for air preheaters, fans, drivers, and burners and other auxiliary equipment.
e. Bill of materials.
● f. Noise data sheets, when data sheets are supplied by the purchaser.
g. Refractory dry-out procedures.
h. Decoking procedures.

SECTION 2—DESIGN CONSIDERATIONS

2.1 Process

2.1.1 Heaters shall be designed for uniform heat distribution. Multi-pass heaters shall be designed for hydraulic symmetry of all passes.

2.1.2 The number of passes for vaporizing fluids shall be minimized. Each pass shall be a single circuit from inlet to outlet.

2.1.3 Average heat flux density in the radiant section is normally based on a single row of tubes spaced at 2 nominal tube diameters. The first row of shield section tubes shall be considered as radiant service in determining the average heat flux density when these tubes are exposed to direct flame radiation.

2.1.4 The maximum allowable inside film temperature for any process service shall not be exceeded in the radiant, shield, or convection sections anywhere in the specified coil.

2.2 Combustion

2.2.1 Calculated and actual efficiencies shall be based on the lower heating value of the primary fuel and shall include a minimum radiation loss of 1.5 percent of the calculated normal heat release. Heaters employing air preheat systems shall include a minimum radiation loss of 2.5 percent of the total heat input.

● **2.2.2** Unless otherwise specified by the purchaser, calculated efficiencies for natural draft operation shall be based upon 20 percent excess air when fuel gas is the primary fuel, and 25 percent excess air when oil is the primary fuel. In the case of forced-draft operation, calculated efficiencies shall be based on 15 percent excess air for fuel gas and 20 percent excess air for fuel oil.

2.2.3 The heater efficiency shall be calculated using the specified fouling resistances.

2.2.4 Volumetric heat release shall not exceed 12,000 British thermal units per hour per cubic foot (124 kilowatts per cubic meter) for oil-fired heaters and 16,000 British thermal units per hour per cubic foot (165 kilowatts per cubic meter) for gas-fired heaters based upon the design heat absorption.

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2.2.5 Stack and flue gas systems shall be designed so that a negative pressure of at least 0.10 inches of water column (25 pascals) is maintained in the radiant and convection sections at 120 percent of normal heat release with design excess air and maximum ambient temperature.

2.3 Mechanical

2.3.1 Provisions for thermal expansion shall take into consideration all specified operating conditions, including short-term conditions such as steam-air decoking.

2.3.2 The convection section tube layout shall include space for future installation of sootblowers, water washing, or steam-lancing doors.

● **2.3.3** When the heater is designed for heavy fuel-oil firing, sootblowers shall be provided for convection cleaning. When light fuel oils such as naphtha are fired, the purchaser shall specify if sootblowers are to be supplied.

2.3.4 The convection section design shall incorporate space for the future addition of two rows of tubes, including the end and intermediate tubesheets. Placement of sootblowers and cleaning lanes shall be based upon the addition of the future tubes. Holes in end tube sheets shall be plugged off to prevent flue gas leakage.

2.3.5 Vertical cylindrical heaters shall be designed with a maximum height-to-diameter ratio of 2.75, where the height is the radiant section height (inside refractory face) and the diameter is the tube circle diameter, both measured in consistent units.

2.3.6 Horizontal tube, floor-fired heaters shall have a maximum height-to-width ratio of 2.75, where the height is the dimension from the floor to the arch refractory or tubes on the centerline of the chamber, and the width is the distance between refractory walls.

2.3.7 Shield sections shall have at least three rows of bare tubes.

● **2.3.8** Convection sections shall be designed with corbels to minimize flue gas bypassing the heating surface. Corbels or baffles may be employed.

2.3.9 The minimum clearance from grade to burner plenum or register shall be 6 feet, 6 inches (2000 millimeters) for floor-fired heaters.

2.3.10 For radiant cylindrical heaters, the maximum radiant straight tube length shall be 60 feet (18,300 millimeters). For horizontal heaters fired from both ends, the maximum radiant straight tube length shall be 40 feet (12,200 millimeters).

2.3.11 Radiant tubes shall be installed with minimum spacing from refractory or insulation to tube centerline of 1.5 nominal tube diameters, with a clearance of not less than 4 inches (102 millimeters) from the refractory or insulation. For horizontal radiant tubes, the minimum clearance from floor refractory to tube outside diameter shall be not less than 12 inches (305 millimeters).

2.3.12 The heater arrangement shall allow for replacement of individual tubes or hairpins without disturbing adjacent tubes.

SECTION 3—TUBES

3.1 General

3.1.1 Tube wall thickness for coils shall be determined in accordance with the procedures set forth in API Recommended Practice 530. The practical limit to minimum thickness for new tubes shall be as specified in API Recommended Practice 530. For materials not included in API Recommended Practice 530, tube wall thickness shall be determined in accordance with procedures outlined in API Recommended Practice 530 using stress values mutually agreed upon between purchaser and supplier.

3.1.2 Calculations made to determine tube wall thickness for coils shall include considerations for erosion and corrosion allowances for the various coil materials The following corrosion allowances shall be used as a minimum:

a. Carbon steel through C-$^1/_2$Mo: 0.125 inch (3 millimeters).

b. Low alloys through 5Cr-$^1/_2$Mo: 0.100 inch (2.5 millimeters).

c. 7Cr-$^1/_2$Mo through austenitic steels: 0.050 inch (1.3 millimeters).

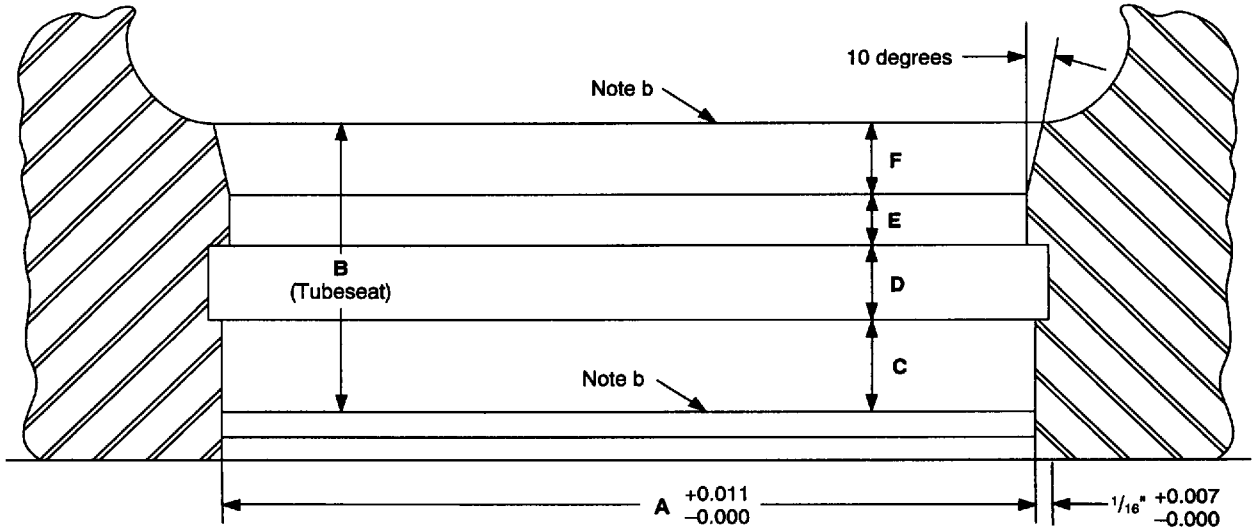
3.1.3 Maximum tube metal temperature shall be determined by following the procedures set forth in API Recommended Practice 530. The design tube metal temperature, as a minimum, shall be 25°F (14°C) above that temperature calculated by the procedure outlined in API Recommended Practice 530.

3.1.4 All tubes shall be seamless, preferably in single continuous lengths. Electric flash welding is not permitted for intermediate welds. Tubes furnished to an average wall thickness shall be in accordance with ASTM tolerances so that the required minimum wall thickness is provided.

3.1.5 Tubes, when projected into header box housings, shall extend a minimum of 6 inches (15 centimeters) (in the

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Table 1—Tubeseat Dimensions[a]



Tube Outside Diameter	A	B	C	D	E	F	Minimum Tube Wall	Maximum Tube Wall
2.375	2.405	1-1/8	3/8	3/8	3/16	3/16	3/16	7/16
2.875	2.905	1-1/4	3/8	3/8	1/4	1/4	3/16	7/16
3.500	3.530	1-3/8	7/16	7/16	1/4	1/4	3/16	7/16
4.000	4.030	1-5/8	7/16	7/16	1/4	3/8	1/16	7/16
4.500	4.530	1-5/8	7/16	7/16	3/8	3/8	3/16	1/2
5.000	5.030	1-3/4	1/2	1/2	3/8	3/8	1/4	1/2
5.563	5.593	1-7/8	9/16	9/16	3/8	3/8	1/4	1/2
6.000	6.030	2	3/8	5/8	3/8	3/8	1/4	9/16
6.625	6.655	2-1/8	11/16	11/16	3/8	3/8	1/4	9/16
7.625	7.655	2-1/2	3/4	3/4		1/2	1/4	5/8
8.625	8.655	3	1	1	1/2	1/2	1/4	11/16

Notes:
[a]All dimensions are in inches. Tube wall thicknesses shown in the table are nominal thicknesses. Tolerances shown are consistent with tubing specifications and are not suitable for piping specifications.
[b]Casting and machining details beyond this line (outside the tubeseat) shall be the manufacturer's standard.

cold position) beyond the face of the end tube sheet of which 4 inches (10 centimeters) must be bare. When rolled plug headers are used, the tube length shall include a bare tube projection equal to two times the tubeseat dimension, plus 2 inches (see Table 1).

3.1.6 Tube size (outside diameter, in inches) shall be selected from the following sizes: 2.375; 2.875; 3.50; 4.00; 4.50; 5.563; 6.625; 8.625; or 10.75. Other tube sizes should be used only if warranted by special process considerations.

3.1.7 When the shield and radiant tubes are in the same service, the shield tubes exposed to direct flame radiation shall be of the same material and thickness as the connecting radiant tubes.

3.2 Extended Surface

● **3.2.1** The extended surface in convection sections may be studded, where each stud is attached to the tube by arc or resistance welding; or finned, where helically wound fins are continuously welded to the tube.

3.2.2 Metallurgy for the extended surface shall be selected on the basis of maximum calculated tip temperature as listed in Table 2.

● **3.2.3** Extended surface dimensions shall be limited to those listed in Table 3.

3.3 Materials

Tube materials shall conform to ASTM specifications as listed in Table 4. For any material not listed in Table 4, refer to 3.1.

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December 27, 1999 12:18:25

API STD*560 95 ■ 0732290 0550151 407 ■

SECTION 4—HEADERS

4.1 General

4.1.1 The design stress for headers shall not be higher than that allowed for similar materials as shown in API Recommended Practice 530 and shall be reduced by casting factors where applicable. Casting factors shall be based on ASME B31.3.

4.1.2 Headers shall be of the same metallurgy as the tubes.

4.1.3 Headers shall be welded return bends or rolled and welded plug-type or plug headers, depending on the service and operating conditions.

4.1.4 All header wall thickness shall include an allowance for erosion and corrosion. This allowance shall not be less than that used for the tubes.

4.2 Plug Headers

4.2.1 Plug headers shall be located in a header box and shall be selected for the same design pressure as the connecting tubes and for a design temperature equal to the maximum fluid operating temperature at that location plus a minimum of 50°F (28°C).

4.2.2 Tubes and plug headers shall be arranged so that there is sufficient space for field maintenance operations such as rolling, welding, and stress relieving.

4.2.3 When plug headers are specified to permit mechanical cleaning of coked or fouled tubes, they shall consist of the two-hole type. Single-hole, 180-degree, plug headers may be installed only for tube inspection and draining.

Table 2—Extended Surface Materials

	Maximum Tip Temperature			Maximum Tip Temperature	
Stud Material	°F	°C	Fin Material	°F	°C
Carbon steel	950	510	Carbon steel	850	454
$2^1/_4$Cr-1Mo, 5Cr-$^1/_2$Mo	1100	593	—	—	—
11-13 Cr	1200	649	11-13 Cr	1100	593
18Cr-8Ni stainless steel	1500	815	18Cr-8Ni stainless steel	1500	815
25Cr-20Ni stainless steel	1800	982	25Cr-20Ni stainless steel	1800	982

4.2.4 When plug headers are specified and horizontal tubes are 60 feet (18.3 meters) or longer, two-hole plug headers shall be used for both ends of the coil assembly. For shorter length coils, plug headers shall be provided on one end of the coil with welded return bends on the opposite end.

4.2.5 When plug headers are specified for vertical tube heaters, two-hole plug headers shall be installed on the top of the coil and one-hole, Y-fittings at the bottom of the tubes.

4.2.6 Headers and corresponding plugs shall be match marked by $^1/_2$-inch (13 millimeter) permanent numerals and installed in accordance with a fitting location drawing.

4.2.7 Type 304 stainless steel thermowells, when required for temperature measurement and control, shall be provided in the plugs of the headers.

4.2.8 Tube center-to-center dimensions shall be as shown in Table 5.

4.2.9 Dimensions of the tubeseat shall conform to details shown in Table 1 for tube wall thickness within the limits shown. Dimensions for tube walls thinner or thicker than shown in Table 1 are not within the scope of this standard.

4.2.10 Plugs and screws shall be assembled in the fittings with an approved compound on the seats and screws to prevent galling.

4.3 Return Bends

4.3.1 Return bends are preferred for the following conditions:

a. In clean service, where coking or fouling of tubes is not anticipated.
b. Where leakage is a hazard.
c. Where steam-air decoking facilities are provided for decoking of furnace tubes.

4.3.2 Return bends inside the firebox shall be selected for the same design pressure and temperature as the connecting tubes. Return bends inside a header box shall be selected for the same design pressure as the connecting tubes and for a

Table 3—Extended Surface Dimensions

	Studs				Fins					
	Minimum Diameter		Maximum Height		Minimum Normal Thickness		Maximum Height		Maximum Density	
Fuel	Inches	mm	Inches	mm	Inches	mm	Inches	mm	Per Inch	Per mm
Gas	$^1/_2$	12.7	1	25.4	0.05	1.3	1	25.4	5	197
Oil	$^1/_2$	12.7	1	25.4	0.10	2.5	$^3/_4$	19.1	3	118

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API STD*560 95 ■ 0732290 0550152 343 ■

Table 4—Tube Materials and ASTM Specifications

Material	Pipe Specification	Tube Specification
Carbon steel	A 53, A 106, Gr B	A 161, A 192, A 210, Gr A-1
Carbon-$^1/_2$Mo	A 335 Gr P1	A 161, A 209, Gr T1
$1^1/_4$Cr-$^1/_2$Mo	A 335, Gr P11	A 200, A 213, Gr T11
$2^1/_4$Cr-1Mo	A 335, Gr P22	A 200, A 213, Gr T22
3Cr-1Mo	A 335, Gr P21	A 200, A 213, Gr T21
5Cr-$^1/_2$Mo	A 335, Gr P5	A 200, A 213, Gr T5
5Cr-$^1/_2$Mo-Si	A 335, Gr PS5b	A 213, Gr TSb
7Cr-$^1/_2$Mo	A 335, Gr P7	A 200, A 213, Gr T7
9Cr-1Mo	A 335, Gr P9	A 200, A 213, Gr T9
9Cr-1Mo-V	A 335, Gr P91	A 200, A 213, Gr T91
18Cr-8Ni	A 312, A 376, TP 304 and TP 304H	A 213, A 271, TP 304 and TP 304H
16Cr-12Ni-2Mo	A 312, A 376, TP 316 and TP 316H	A 213, A 271, TP 316 and TP 316H
18Cr-10Ni-Ti	A 312, A 376, TP 321 and TP 321H	A 213, A 271, TP 321 and TP 321H
18Cr-10Ni-Cb	A 312, A 376, TP347 and TP 347H	A 213, A 271, TP 347 and TP 347H
Alloy 800H /HT[a]	B 407	B 407
25Cr-20Ni	A 312, TP 310H	A 213, TP 310H
25Cr-20Ni	—	A608, Gr HK40

[a]With minimum grain size of ASTM #5 or coarser.

Table 5—Tube Center-to-Center Dimensions[a]

Tube Size Outside Diameter (Inches)	Header Center-to-Center Dimension			
	Preferred		Minimum for Rolling	
	Inches	mm	Inches	mm
2.375	4.00[b,c]	101.6	4.25	108.0
2.875	5.00[b,c]	127.0	4.75	120.7
3.50	6.00[b]	152.4	5.875	149.2
4.00	7.00[b]	177.8	6.50	165.1
4.50	8.00[b]	203.2	7.125	181.0
5.00	9.00	228.6	7.75	196.9
5.563	10.00[b]	254.0	8.50	215.9
6.00	11.00	279.4	9.00	228.6
6.625	12.00[b]	304.8	9.75	247.7
7.625	14.00	355.6	12.00	304.8
8.625	16.00[b]	406.4	14.00	355.6
10.75	20.00[b]	508.0	18.00	457.2

[a]Center-to-center dimensions are applicable only to manufacturer's standard header pressure ratings for 850 pounds per square inch gauge nominal fittings.
[b]This center-to-center dimension equals two times the corresponding nominal size and is based on the center-to-center dimension for short-radius welded return bends.
[c]This center-to-center dimension is inadequate for the tubeseat detail specified by Table 1 of this specification; hence this center-to-center dimension is applicable only to welded plug headers.

design temperature equal to the maximum fluid operating temperature at that location plus a minimum of 50°F (28°C). Return bends shall be at least the same thickness as the connecting tubes.

4.3.3 Regardless of the location of the welded return bends, the heater design shall incorporate means to permit convenient removal and replacement of tubes and return bends.

4.3.4 Longitudinally welded fittings are not permitted.

4.4 Materials

4.4.1 Plug header and return bend material shall conform to the ASTM specifications listed in Table 6. Other materials and alternative specifications are permitted when approved by the purchaser.

4.4.2 Cast fittings shall have the ASTM material identification permanent marking on the fitting, with raised letters or low-stress stamps.

SECTION 5—PIPING, TERMINALS, AND MANIFOLDS

5.1 General

5.1.1 The minimum corrosion allowance shall be in accordance with 3.1.2.

5.1.2 All flanges shall be welding-neck flanges.

5.1.3 Piping, terminals, and manifolds external to the heater enclosure shall be in accordance with ASME B31.3.

● **5.1.4** When inspection openings are required, terminal flanges may be used, provided that pipe sections are readily removable for inspection access.

5.1.5 Threaded connections are not acceptable.

● **5.1.6** When low-point drains and high-point vents are required, they shall be accessible from outside the heater casing.

5.1.7 Manifolds and external piping shall be located so as not to block access for the removal of single tubes or hairpins.

5.1.8 Manifolds inside a header box shall be selected for the same design pressure as the connecting tubes and for a design temperature equal to the maximum fluid operating temperature at that location plus a minimum of 50°F (28°C).

● 5.2 Allowable Movement and Loads

Heater terminals shall be designed to accept the moments and forces, or the movements listed in Table 7, unless otherwise specified by the purchaser.

5.3 Materials

The crossover piping shall be the same metallurgy as the preceding heater tube.

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API STD*560 95 ■ 0732290 0550153 28T ■

SECTION 6—TUBE SUPPORTS

6.1 General

6.1.1 The design temperature for tube supports and guides exposed to flue gas shall be based on design operation of the furnace as follows:

a. For the radiant and shield section, the bridgewall temperature plus 200°F (111°C). The minimum design temperature shall be 1600°F (871°C).
b. For the convection section, the temperature of the flue gas in contact with the support plus 100°F (56°C) for gas firing or plus 200°F (111°C) for oil firing.
c. The maximum flue gas temperature gradient across a single convection intermediate tube support shall be 400°F (222°C).
d. No credit shall be taken for the insulating effect of refractory coatings on intermediate supports or guides.

6.1.2 Guides, horizontal radiant section intermediate tube supports, and top supports for vertical radiant tubes shall be designed to permit their replacement without tube removal and with minimum refractory repair.

6.1.3 The unsupported length of horizontal tubes shall not exceed 35 times the outside diameter or 20 feet (6 meters), whichever is less.

6.1.4 The minimum corrosion allowance for all exposed surfaces of each tube support and guide contacting flue gases shall be 0.05 inch (1.3 millimeters) for austenitic materials and 0.10 inch (2.5 millimeters) for ferritic materials.

6.1.5 End tube sheets for tubes with external headers shall be designed as described in 6.1.5.1 through 6.1.5.4.

6.1.5.1 Tube sheets shall be structural plate. If the tubesheet design temperature exceeds 800°F (427°C), alloy materials shall be used.

6.1.5.2 Minimum thickness of tube sheets shall be $^1/_2$-inch (13 millimeters).

6.1.5.3 Tube sheets shall be insulated on the flue gas side with a castable having a minimum thickness of 3 inches (76 millimeters) for the convection section and 5 inches (127 millimeters) for the radiant section. As a minimum, Type 304 stainless steel anchors shall be used.

6.1.5.4 Sleeves with an inside diameter at least $^1/_2$-inch (13 millimeters) greater than the tube or the extended surface outside diameter shall be welded to the tube sheet at each tube hole, to prevent the refractory from being damaged by

Table 6—Plug Header and Return Bend Materials

Material	Forged	Wrought	Cast
Carbon steel	A 105, A 181 Class 60 or 70	A 234 Gr WPB	A 216 Gr WCB 1
C-$^1/_2$Mo	A 182 Gr F1	A 234 Gr WP1	A 217 Gr WCI
1$^1/_4$Cr-$^1/_2$Mo	A 182 Gr F11	A 234 Gr WP11	A 217 Cr WC6
2$^1/_4$Cr-1Mo	A 182 Gr F22	A 234 Gr WP22	A 217 Gr WC9
3Cr-1Mo	A 182 Gr F21	—	—
5Cr-$^1/_2$Mo	A 182 Gr F5	A 234 Gr WP5	A 217 Gr C5
7Cr-$^1/_2$Mo	A 182 Gr F7	A 234 Gr WP7	
9Cr-1Mo	A 182 Gr F9	A 234 Gr WP9	A 217 Gr C12
18Cr-8Ni			
Type 304	A 182 F304	A 403 WP304	A 351 Gr CF8
Type 304H	A 182 F304H	A 403 WP304H	A 351 Gr CF8
18Cr-10Ni-Ti			
Type 321	A 182 F321	A 403 WP321	A 351 Gr CF8C
Type 321H	A 182 F321H	A 403 WP321H	
18Cr-10Ni Cb			
Type 347	A 182 F347	A 403 WP347	A 351 Gr CF8C
Type 347H	A 182 F347H	A 403 WP347H	
16Cr-1 2Ni-2Mo			
Type 316	A 182 F316	A 403 WP316	A 351 Gr CF8M
Type 316H	A 182 F316H	A 403 WP316H	
Alloy 800H[a]	B 564 Alloy 800H	B 366 Alloy 800H	A-351 CT-15C
Alloy 800HT[a]	B 564 Alloy 800HT	B 366 Alloy 800HT	
25Cr-20Ni	A-182 F-310H	A-403 WP-310H	A-351 CK-20
	—		A-351 Gr HK40

[a] With minimum grain size of ASTM #5 or coarser.

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the tubes. The sleeve material shall be Type 304 stainless steel.

6.1.6 Extended surface tubes shall be supported as described in 6.1.6.1 through 6.1.6.3.

6.1.6.1 Intermediate supports shall be designed to prevent mechanical damage to the extended surface and shall permit easy removal and insertion of the tubes without binding.

6.1.6.2 For studded tubes, a minimum of three rows of studs shall rest on each support.

6.1.6.3 For finned tubes, a minimum of five fins shall rest on each support.

6.2 Loads and Allowable Stress

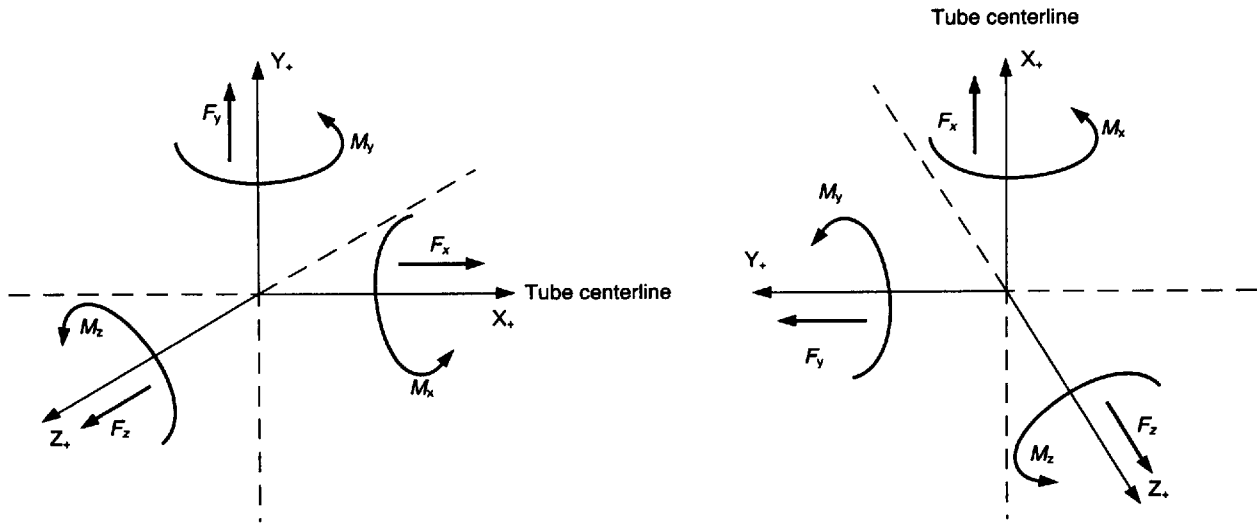
6.2.1 Tube support loads shall be determined as described in 6.2.1.1 through 6.2.1.3.

6.2.1.1 Loads shall be determined in accordance with AISC procedures for supporting continuous beams on multiple supports.

6.2.1.2 Friction loads shall be based on a minimum friction coefficient of 0.30.

6.2.1.3 Friction loads shall be based on all tubes expanding and contracting in the same direction. Loads shall not be

Table 7—Maximum Allowable Forces, Moments, and Movements



HORIZONTAL TUBES **VERTICAL TUBES**

Allowable Forces and Moments

Pipe Size (Inches)	F_x		F_y		F_z		M_x		M_y		M_z	
	pounds	N	pounds	N	pounds	N	ft-lb	N m	ft-lb	N m	ft-lb	N m
2	100	445.	200	890.	200	890.	350	475	250	339	250	339
3	150	667.	300	1334.	300	1334.	450	610	350	475	350	475
4	200	890.	400	1779.	400	1779.	600	813	450	610	450	610
5	225	1001.	450	2002.	450	2002.	660	895	500	678	500	678
6	250	1112.	500	2224.	500	2224.	730	990	550	746	550	746
8	300	1334.	600	2669.	600	2669.	860	1166	650	881	650	881
10	350	1557.	650	2891.	650	2891.	930	1261	700	949	700	949
12	400	1779.	700	3114.	700	3114.	1000	1356	750	1017	750	1017

Allowable Movements

	Horizontal Tubes			Vertical Tubes		
	Δx	Δy	Δz	Δx	Δy	Δz
	inches (mm)			inches (mm)		
Radiant terminals	0	+1(25)	±1(25)	0	±1(25)	±1(25)
Convection terminals	0	+1/2(13)	±1/2(13)	—	—	—

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Table 8—Maximum Design Temperatures for Tube Support Materials

Maximum Design Temperature				
°F	°C	Material	Casting	Plate
800	427	Carbon steel	A 216 Gr WCB	A 283 Gr C
1200	649	$2^1/_4$Cr-1Mo	A 217 Gr WC 9	A 387 Gr 22, C1.1
1200	649	5 Cr-$^1/_2$Mo	A 217 Gr C5	A 387 Gr 5, C1.1
1500	816	19Cr-9Ni	A 297 Gr HF	A 240, Type 304H
1600	871	25Cr-12Ni		A 240, Type 309H
1800	982	25Cr-12Ni	A 447 Type II	
1600	871	25Cr-20Ni		A240, Type 310H
2000	1093	25Cr-20Ni	A 297 Gr HK	
1800	982	50Cr-50Ni-Cb	A 560 Gr 50Cr-50Ni-Cb	

Note: For exposed radiant and shield section tube supports, the material shall be 25Cr-12Ni as a minimum.

considered to be cancelled or reduced due to tubes moving in opposite directions.

6.2.2 Tube support maximum allowable stresses at design temperature shall not exceed the following:

a. Dead load stress:
   1. One-third of the ultimate tensile strength.
   2. Two-thirds of the yield strength (0.2 percent offset).
   3. Fifty percent of the average stress required to produce 1 percent creep in 10,000 hours.
   4. Fifty percent of the average stress required to produce rupture in 10,000 hours.

b. Dead load plus frictional stress:
   1. One-third of the ultimate tensile strength.
   2. Two-thirds of the yield strength (0.2 percent offset).
   3. The average stress required to produce 1 percent creep in 10,000 hours.
   4. The average stress required to produce rupture in 10,000 hours.

6.2.3 For castings, a casting factor multiplier of 0.8 shall be applied to the allowable stress value.

6.2.4 Stress data are presented in Appendix D.

6.3 Materials

6.3.1 Tube support materials shall be selected for maximum design temperatures as shown in Table 8. Other materials and alternative specifications must be approved by the purchaser.

● **6.3.2** If the tube support design temperature exceeds 1200°F (649°C) and the fuel contains more than 100 parts per million total vanadium and sodium, the supports shall be one of the following design details:

a. Covered with 2 inches (50 millimeters) of castable refractory having a minimum density of 130 pounds per cubic foot (2081 kilograms per cubic meter).
b. Constructed of 50Cr-50Ni-Cb metallurgy, as a minimum, without any coating.

SECTION 7—REFRACTORIES AND INSULATION

7.1 General

● **7.1.1** The temperature of the outside casing of the radiant and convection sections and hot ductwork shall not exceed 180°F (82°C) at an ambient temperature of 80°F (27°C) in still air. Radiant floors shall not exceed 195°F (91°C).

7.1.2 Walls, arches, and floors shall be designed to allow for proper expansion of all parts. Where multilayer or multicomponent linings are used, joints shall not be continuous through the lining.

7.1.3 Any layer of refractory shall be suitable for a service temperature at least 300°F (167°C) above its calculated hot face temperature. Minimum service temperature for refractories shall be 1800°F (982°C) in the radiant and shield sections.

7.1.4 The floor hot surface shall be a $2^1/_2$-inch (6.4 centimeter) thick layer of high-duty fireclay brick or a 3-inch (7.6 centimeter) thick layer of castable of 2500°F (1371°C) service temperature and minimum cold crush strength of 500 pounds per square inch (3447 kilopascals) after drying at 230°F (110°C).

7.1.5 Burner blocks shall have a minimum service temperature of 3000°F (1650°C).

7.1.6 Target walls with flame impingement on both sides shall be constructed of high-duty firebrick of at least 2800°F (1538°C) rating. Bricks shall be laid dry or with mortared joints. Expansion joints shall be packed with ceramic fiber strips having a service temperature rating equal to or greater than the brick.

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Target walls with flame impingement on one side may be of brick or of plastic refractory of equivalent maximum service temperature. Either may be backed by a castable or ceramic fiber board.

7.1.7 Access doors shall be protected from direct radiation by a refractory system of at least the same thermal rating and resistance as the adjacent wall lining.

7.1.8 Refractory anchors are not mandatory for floor castable unless required for shipping considerations.

7.1.9 Maximum temperatures for anchor tips are listed in Table 9.

7.1.10 Recommended refractory dryout procedure shall be provided by the heater vendor.

Table 9—Maximum Temperatures for Anchor Tips

Anchor Material	Maximum Anchor Temperature °F	°C
Carbon steel	800	427
TP 304 stainless steel	1400	760
TP 316 stainless steel	1400	760
TP 309 stainless steel	1500	815
TP 310 stainless steel	1700	927
RA 330 stainless steel	1900	1038
Alloy 601	2000	1093
Ceramic studs and washers	> 2000	> 1093

7.2 Brick and Tile Construction

7.2.1 Brick construction can be used for gravity walls, floors, or as hot-face layers.

7.2.2 Radiant chamber walls of gravity construction shall not exceed 24 feet (7.3 meters) in height and shall be at least high-duty fireclay brick. The base width shall be a minimum of 8 percent of wall height. The height-to-width ratio of each wall section shall not exceed 5 to 1. The walls shall be self-supporting and the wall base shall rest directly on the steel floor, not on other refractory.

7.2.3 Gravity walls shall be of mortared construction. The mortar shall be non-slagging, air-setting, chemically compatible with adjacent refractory, including the rated temperature of the brick.

7.2.4 Vertical expansion joints shall be provided at gravity wall ends and required intermediate locations. All expansion joints shall be kept open and free to move. If the joint is formed with lapped brick no mortar shall be used, that is, a dry joint.

7.2.5 Floor brick shall not be mortared. A $^1/_2$-inch (1.3 centimeter) gap for expansion shall be provided on 6 foot (1.8 meter) intervals. This gap may be packed with fibrous refractory material in strip, not loose bulk, form.

7.2.6 Minimum service temperature for a hot face brick layer shall be 2600°F (1427°C) on walls with expected flame impingement and 2300°F (1260°C) for other exposed wall applications. Minimum service temperature for shielded walls shall be 2000°F (1093°C).

7.2.7 All brick linings on vertical flat casing shall be tied back to and supported by the structural steel framing members. All tie members shall be austenitic alloy material, except that pipe supports located in the backup layer may be carbon steel. A minimum of 15 percent of the bricks shall be tied back. Brick lining on cylindrical casing need not be tied back when the radius of the curvature of the casing keys the bricks.

7.2.8 Brick linings shall be supported by metal support shelves (lintels) attached to the casing on vertical centers not to exceed 6 feet (1.8 meters). Support shelves shall be slotted to provide for differential thermal expansion. Shelf material will be defined by the calculated service temperature; carbon steel is satisfactory up to 700°F (371°C).

7.2.9 Expansion joints shall be provided in both vertical and horizontal directions of the walls, at wall edges, and about burner tiles, doors, and sleeved penetrations.

7.3 Castable Construction

7.3.1 Hydraulic-setting castables are suitable as lining for all parts of fired heaters. Minimum castable construction is a 1:2:4 volumetric mix of lumnite-haydite-vermiculite, limited to a maximum service temperature rating of 1900°F (1038°C) and clean fuel applications. This castable shall be limited to 8-inch (20.3 centimeters) maximum thickness on arches and walls.

7.3.2 For dual layer castable construction, the hot face layer shall be a minimum of 3 inches (7.6 centimeters) thick. The anchoring systems shall provide independent support for each layer when in arch or other overhead position.

7.3.3 Anchoring penetration shall not be less than 70 percent of the individual layer being anchored for castable thickness greater than 2 inches (5 centimeters). The anchor shall not be closer than $^1/_2$ inch (1.3 centimeters) from the hot face.

7.3.4 The anchoring spacing shall be a maximum of three times the total lining thickness but shall not exceed 12 inches (30 centimeters) on a square pattern for walls and 9 inches (23 centimeters) on a square pattern for arches. The anchor orientation shall be varied to avoid creating continuous shear planes.

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7.3.5 Anchors for total castable thickness up to 6 inches (15.2 centimeters) shall be a minimum of $^3/_{16}$ inch (4.8 millimeter) diameter. Greater thickness requires a minimum of $^1/_4$ inch (6.3 millimeter) diameter anchors.

7.3.6 Castable linings in header boxes, breechings, and lined flue gas ducts and stacks shall not be less than 2 inches (5 centimeters) thick.

7.3.7 Anchors in 2-inch (5 centimeter) thick castable lining shall be held in place by 10 gauge minimum, bare carbon steel chain-link fencing, wire mesh, or linear anchors anchored to the steel.

7.3.8 Expansion joints shall be provided around burner blocks, brick, and prefired shapes.

7.3.9 When metallic fiber is added for reinforcement it shall only be used in castables of 55 pounds per cubic foot (969 kilograms per cubic meter) or greater density. Metallic fibers shall be limited to no more than 3 percent by weight of the dry mixture.

7.3.10 Low iron content materials shall be used when total heavy metals content within fuels exceeds 100 parts per million.

7.4 Ceramic Fiber Construction

7.4.1 Ceramic fiber in layered or modular construction may be used in all heater areas except stacks, ducts, and floors.

7.4.2 The hot face of layered ceramic fiber blanket installations shall be a minimum of 1 inch (2.5 centimeters) thick, 8 pounds per cubic foot (128 kilograms per cubic meter) density, needled material. Ceramic fiber board, when applied as a hot face layer, shall not be less than $1^1/_2$-inch (3.8 centimeters) thick nor less than 14 pounds per cubic foot (240 kilograms per cubic meter) density. Backup layer(s) of ceramic fiber blanket shall be a minimum of 6 pounds per cubic feet (96 kilograms per cubic meter) density, needled material. Ceramic fiber board size, when used as hot face layer, shall be limited to a maximum of 24 inches by 24 inches (61 centimeters by 61 centimeters) when temperatures are below 2000°F (1093°C) and 18 inches by 18 inches (46 centimeters by 46 centimeters) when temperatures exceed 2000°F (1093°C).

7.4.3 Any layer of ceramic fiber shall be suitable for a service temperature at least 500°F (260°C) above its calculated hot face temperature.

7.4.4 The hot face layer of a ceramic fiber blanket system shall be anchored a maximum of 3 inches (7.6 centimeters) from all edges.

7.4.5 The anchor spacing for arches shall not exceed the following rectangular pattern: 6 inches by 9 inches (15 centimeters by 23 centimeters) for 12-inch (30-centimeter) wide blankets; 9 inches by 9 inches (23 centimeters by 23 centimeters) for 24-inch (61-centimeter) wide blankets; 9 inches by 10 inches (23 centimeters by 25 centimeters) for 36-inch (91-centimeter) wide blankets; and 9 inches by $10^1/_2$-inches (23 centimeters by 27 centimeters) for 48-inch (122-centimeter) wide blankets.

7.4.6 The anchor spacing for walls shall not exceed the following rectangular pattern: 6 inches by 12 inches (15 centimeters by 23 centimeters) for 12-inch (30-centimeter) wide blankets; 9 inches by 12 inches (23 centimeters by 30 centimeters) for 24-inch (61-centimeter) wide blankets; and $10^1/_2$-inches by 12 inches (27 centimeters by 30 centimeters) for 48-inch (122-centimeter) wide blankets.

7.4.7 Metallic anchor parts that are not shielded by tubes shall be completely wrapped with ceramic fiber patches or be protected by ceramic retainer cups filled with moldable ceramic fiber.

7.4.8 Ceramic fiber blanket is unacceptable to be used as the hot face layer when flue gas velocities are in excess of 40 feet per second (12.2 meters per second). Wet blanket, ceramic fiber board, or ceramic fiber modules shall be used on hot face layers with velocities greater than 40 feet per second (12.2 meters per second) but less than 80 feet per second (24.4 meters per second). Hot face refractory with velocities greater than 80 feet per second (24.4 meters per second) require castable or external lining.

7.4.9 Ceramic fiber blanket shall be installed with its longest dimension in the direction of gas flow. The hot face layer of blanket shall be constructed with all joints overlapped. Overlaps shall be in the direction of gas flow. Hot face layers of ceramic fiber board shall be constructed with tight butt joints.

7.4.10 Ceramic fiber blanket used in backup layers shall be installed with butt joints with at least 1 inch (2.5 centimeter) compression on the joints. All joints in successive layers of blanket shall be staggered.

7.4.11 Ceramic fiber blanket modules shall be installed in soldier course (with batten strips) patterns.

7.4.12 Module systems shall be installed so that joints at each edge are compressed to avoid gaps due to shrinkage.

7.4.13 Modules applied in arches shall be designed such that anchorage is provided over at least 80 percent of the module width.

7.4.14 Anchors shall be attached to the casing before modules are installed.

7.4.15 Anchor assembly shall be located in the module at a maximum distance of 2 inches (5 centimeters) from the module cold face.

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7.4.16 Module internal hardware shall be Type 304 stainless steel as a minimum.

● **7.4.17** When ceramic fiber construction is used with fuels having a sulfur content exceeding 10 parts per million, the casing shall have an internal protective coating to prevent corrosion. The protective coating shall be rated for 350°F (177°C) service temperature.

7.4.18 A vapor barrier of Type 304 stainless steel foil shall be provided when the fuel sulfur content exceeds 500 parts per million. The vapor barrier must be located so that the exposure temperature is at least 100°F (56°C) above the calculated acid dew point for all operating cases. Vapor barrier edges shall be overlapped; edges and punctures shall be sealed.

7.4.19 Ceramic fiber systems shall not be applied for services where the total heavy metals content in the fuel exceeds 100 parts per million.

7.4.20 Ceramic fiber shall not be used in convection sections where soot blowers, steam lances, or waterwash facilities are initially provided.

7.4.21 Anchors shall be installed before applying protective coating to the casing. The coating shall cover the anchors so that uncoated parts are above the acid dew point temperature.

7.5 Multicomponent Lining Construction

7.5.1 Castable layers shall be a minimum of 3 inches (7.6 centimeters) thick.

7.5.2 The anchoring system shall provide retention and support for each component layer.

7.5.3 Anchor types and installation for individual lining components shall meet the applicable requirements of 7.2, 7.3, and 7.4.

7.5.4 The material used in any layer shall be suitable for service temperature in accordance with 7.1.3 and 7.4.3.

7.5.5 Brick may be used for hot face service or as a backup layer if the hot face layer is brick.

7.5.6 Block insulation shall be calcium silicate or mineral wool fiber with a minimum service temperature rating of 1800°F (983°C). Block insulation shall only be used as a backup material, but shall not be used when the fuel sulfur content exceeds 1 percent by weight in liquid fuel or 100 parts per million hydrogen sulfide in gas fuel. Block insulation shall not be used as backup material in floor construction.

● **7.5.7** When insulating block or ceramic fiber is used as backup insulation, the casing shall have a protective coating if the fuel sulfur content exceeds 10 parts per million. The protective coating shall be rated for 300°F (149°C) service temperature.

7.5.8 When used as backup for castable, block insulation or ceramic fiber shall be sealed to prevent water migration from the castable.

7.5.9 Minimum density of mineral wool block and ceramic fiber used as backup materials shall be 8 pounds per cubic foot (128 kilograms per cubic meter).

7.6 Materials

7.6.1 Materials shall conform to the following ASTM specifications:

a. Fireclay brick, ASTM C27.
b. Insulating firebrick, ASTM C155.
c. Castable refractory, ASTM C401, Class N, O, P, Q, or R.
d. Vermiculite sieve analysis, ASTM C332, Group 1 density.
e. Insulating block (mineral slag wool, neutral pH), ASTM C612 CL5.
f. Haydite, ASTM C332 Group II:
   1. Poured application: Fine aggregate No. 4.
   2. Gunned application: Combined fine and coarse $^3/_8$ inch (9.5 millimeters) to Fine Aggregate No. 0.

7.6.2 The following materials shall have a composition as shown:

a. Lumnite or calcium aluminate cement, 35 percent Al_2O_3 or better.
b. Ceramic fiber, at least 45 percent Al_2O_3; remainder primarily SiO_3 or ZrO_2.

SECTION 8—STRUCTURES AND APPURTENANCES

8.1 General

● **8.1.1** Structural steel shall be designed in accordance with the applicable provisions of the following codes and specifications (the purchaser shall set priorities if codes conflict):

a. International Conference of Building Officials, *Uniform Building Code.*
b. American Institute for Steel Construction, *Specification for Design, Fabrication, and Erection of Structural Steel for Buildings.*
c. ASCE standards.
d. Local codes as specified by purchaser.

● **8.1.2** Minimum design loads for wind and earthquake shall conform to requirements of the ICBO *Uniform Building*

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Code or ASCE 7-88, and/or local codes as specified by the purchaser.

8.1.3 Platform live loads shall be in accordance with the loads specified in the ICBO *Uniform Building Code* for heavy manufacturing facilities, or local codes as specified by the purchaser.

8.1.4 Structures and appurtenances shall be designed for all applicable load conditions expected during shipment, erection, and operation. Cold weather conditions shall be considered, particularly when the furnace is not in operation. These load conditions shall include, but are not limited to, dead load, wind load, earthquake load, live load, and thermal load.

● **8.1.5** Design metal temperature of structures and appurtenances shall be the calculated metal temperature plus 100°F (56°C), based on the maximum flue gas temperature expected for all operating modes with an ambient temperature of 80°F (27°C) in still air. The effect of elevated design temperature on yield strength and Modulus of Elasticity shall be modified in accordance with 9.3.5.

8.1.6 The material of the structures and appurtenances shall be adequate for all load conditions at the lowest specified ambient temperature when the furnace is not in operation.

8.2 Structures

8.2.1 All loads from the tubes and headers shall be supported by the structural steel and shall not be transmitted into the refractory.

8.2.2 Structural steel shall be designed to permit lateral and vertical expansion of all heater parts.

8.2.3 Heater casing shall be a minimum of $^3/_{16}$-inch (5-millimeter) thick plate, which shall be reinforced against warping. Casing, when used as part of the main heater structure, shall be $^1/_4$-inch (6 millimeter) minimum thickness. Floor and radiant roof plates shall be $^1/_4$-inch (6 millimeter) minimum thickness.

8.2.4 Heater casing plate shall be seal welded externally to prevent air and water infiltration.

● **8.2.5** The heater structure shall be capable of supporting ladders, stairs, and platforms in locations where installed or where specified for future use.

8.2.6 Flat roof design shall allow for runoff of rainwater. This can be accomplished by arrangement of structural members and drain openings, by sloping the roof or with a secondary roof for weather protection. When pitched roofs are provided for weather protection, eaves and gables shall prevent the entry of windblown rain.

● **8.2.7** When fireproofing is specified, the main structural columns from the base plate to the floor level plus the main floor beams shall allow for 2 inches (50 millimeters) of fireproofing.

8.2.8 Heaters with horizontal tubes that have return bends inside the firebox shall have removable end panels or panels in the sidewalls, to provide access to the return bend welds.

8.2.9 Duct structural systems shall support duct work independent of expansion joints during operation, when idle or with duct sections removed.

8.3 Header Boxes, Doors, and Ports

8.3.1 HEADER BOXES

8.3.1.1 Each header box shall allow for the total tube expansion. A minimum clearance of 3 inches (76 millimeters) shall be provided between the header box door refractory and the header in the hot position.

● **8.3.1.2** Header boxes enclosing plug headers shall have hinged doors or bolted end panels.

8.3.1.3 Header boxes, including doors, shall be of $^3/_{16}$-inch (5 millimeter) minimum steel plate reinforced against warping. Header boxes shall be removable.

8.3.1.4 When specified by the purchaser, to minimize flue gas bypassing, horizontal partitions shall be provided in convection section header boxes on a spacing not to exceed 5 feet (1.5 meters).

8.3.2 DOORS AND PORTS

8.3.2.1 Two access doors having a minimum clear opening of 24 inches by 24 inches (610 millimeters by 610 millimeters) shall be provided for each radiant chamber of a box or cabin heater.

8.3.2.2 One access door having a minimum clear opening of 18 inches by 18 inches (460 millimeters by 460 millimeters) shall be provided in the floor for vertical cylindrical heaters. A bolted and gasketed access door shall also be provided in any air plenum below the floor accessway.

8.3.2.3 One access door having a minimum clear opening of 24 inches by 24 inches (610 millimeters by 610 millimeters), or 24 inches (610 millimeters) in diameter shall be provided in the stack or breeching for access to the damper and convection sections.

8.3.2.4 One tube removal door having a minimum clear opening of 18 inches by 24 inches (460 millimeters by 610 millimeters) shall be provided in the arch of each radiant chamber of vertical tube heaters.

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8.3.2.5 Observation doors and ports shall be provided for viewing all radiant tubes and all burner flames for proper operation and light off.

8.3.2.6 Access doors shall be provided to ducts, plenums and at all duct connections to air preheaters and control dampers.

8.4 Ladders, Platforms, and Stairways

8.4.1 Platforms shall be provided as follows:

a. At burner and burner controls that are not accessible from grade.
b. At both ends of the convection section for maintenance purposes.
c. At damper and sootblower locations for maintenance and operation purposes.
d. At all observation ports and firebox access doors not accessible from grade.
e. At auxiliary equipment, such as steam drums, fans, drivers, and air preheaters as required for operating and maintenance purposes.

8.4.2 Vertical cylindrical heaters with shell diameters greater than 10 feet (3 meters) shall have a full circular platform at the floor level. Individual ladders and platforms to each observation door may be used when shell diameters are 10 feet (3 meters) or less.

8.4.3 Platforms shall have a minimum clear width as follows:

a. Operating platforms, 3 feet (915 millimeters).
b. Maintenance platforms, 3 feet (915 millimeters).
c. Walkways, 2 feet, 6 inches (760 millimeters).

● **8.4.4** Platform decking shall be $^1/_4$ inch (6 millimeter) checkered plate or 1 inch by $^3/_{16}$ inch (25 millimeter by 5 millimeter) open grating. Stair treads shall be open grating with checkered plate nosing.

8.4.5 Dual access shall be provided to each operating platform except when the individual platform length is less than 20 feet (6 meters).

8.4.6 An intermediate landing shall be provided when the vertical rise exceeds 30 feet (9 meters) for ladders and 15 feet (4.5 meters) for stairways.

8.4.7 Ladders shall be caged from a point 7 feet, 6 inches (2.3 meters) above grade. A safety gate shall be provided for all ladders serving platforms or landings.

8.4.8 Stairways shall have a minimum width of 2 feet, 6 inches (760 millimeters), a minimum tread width of $9^1/_2$ inches (240 millimeters), and a maximum riser of 8 inches (200 millimeters). The slope of the stairway shall not exceed a 9 (vertical) to 12 (horizontal) ratio.

8.4.9 Headroom over platforms, walkways, and stairways shall be a minimum of 7 feet (2.1 meters).

8.4.10 Handrails shall be provided on all platforms, walkways, and stairways.

8.4.11 Handrails, ladders, and platforms shall be arranged so as not to interfere with tube handling. Where interference exists, provide removable sections.

8.5 Materials

8.5.1 Materials used in the fabrication of fired heaters shall conform to the following specifications or purchaser's approved equivalent specifications:

a. Structural shapes, ASTM A 36.
b. Plate, ASTM A 36, A 283 Grade C, or A 285 Grade C.
c. Structural bolts, ASTM A 307, unfinished.
d. High-strength bolts, ASTM A 325 or ASTM A 490.
e. Pipe for columns and davits, ASTM A 53 Grade B.

● **8.5.2** Materials for service at design ambient temperatures below −20°F (−29°C) shall be as specified by the purchaser.

SECTION 9—STACKS, DUCTS, AND BREECHING

9.1 General

9.1.1 Stacks shall be self-supporting.

9.1.2 Stacks shall be bolted to their supporting structure.

9.1.3 Stacks, breeching, and ducts shall be of welded construction.

9.1.4 External attachments to stacks shall be seal welded.

9.1.5 Stacks, breechings, and ducts mounted on concrete shall be designed to prevent concrete temperatures in excess of 300°F (149°C).

9.1.6 Connections between stacks and flue gas ducts shall not be welded.

9.1.7 The top of stack linings shall be protected to prevent water penetration between the stack shell plate and the lining.

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9.1.8 All openings and connections on stacks, breechings, and ducts shall be sealed to prevent air or flue gas leakage.

9.1.9 Breechings shall have a minimum clear distance beyond the last (present and future) convection row of 2 feet, 6 inches (0. 8 meter) for access and flue gas distribution.

● **9.1.10** When operating with natural draft, control dampers shall be furnished on each heater stack or each inlet air plenum.

9.2 Design Considerations

9.2.1 The design shall be in accordance with the applicable provisions of the following codes and specifications:

a. ICBO *Uniform Building Code.*
b. American Institute for Steel Construction, *Specification for Design, Fabrication, and Erection of Structural Steel for Buildings.*
c. ASCE standards.
d. Local codes as specified by the purchaser.

● **9.2.2** Minimum design loads for wind and earthquake shall conform to the requirements of the ICBO *Uniform Building Code* or ASCE 7-88.

9.2.3 Stacks, breechings, and ducts shall be designed for all applicable load conditions expected during shipment, erection, and operation. Cold weather conditions shall be considered, particularly when the furnace is not in operation. These load conditions shall include, but not be limited to, dead load, wind load, earthquake load, live load, and thermal load.

9.2.4 The combination of loads that could occur simultaneously creating the maximum load condition shall be the design load, but in no case shall individual loads create stresses that will exceed those allowed by 9.3.5. Wind and earthquake loads shall not be considered as acting simultaneously. When critical wind velocities exceed 60 miles per hour (97 kilometers per hour), dynamic loads resulting from wind need not be included in the design load. Stacks located closer than 8 diameters may be subject to increased wind loads.

9.2.5 The minimum stack shell plate thickness shall be $^1/_4$-inch (6.0 millimeters) including corrosion allowance. The minimum corrosion allowance on stacks shall be $^1/_{16}$ inch (1.6 millimeters).

9.2.6 The minimum number of anchor bolts for any stack shall be eight.

9.2.7 Lifting lugs on stacks, when required, shall be designed for the lifting load as the stack is raised from a horizontal to a vertical position.

● **9.2.8** Design metal temperature of stacks, breechings, and ducts shall be the calculated metal temperature plus 100°F (56°C), based on the maximum flue gas temperature expected for all operating modes with an ambient temperature of 80°F (27°C) in still air.

9.2.9 The material of the stack, breeching, and duct shall be adequate for all load conditions at the lowest specified ambient temperature when the furnace is not in operation.

9.2.10 The minimum breeching and duct plate thickness shall be $^3/_{16}$ inch (4.8 millimeters).

9.2.11 Breechings and ducts shall be stiffened to prevent excessive warpage and deflection. Deflection of refractory lined breechings and ducts shall be limited to $^1/_{360}$th of the span. Deflection of other breechings and ducts shall be limited to $^1/_{240}$th of the span.

9.2.12 Butterfly dampers shall be limited to stacks and ducts having a maximum internal cross-sectional area of 13 square feet (1.2 square meters).

9.2.13 Louver dampers shall have a minimum of one blade for every 13 square feet (1.2 square meters) of internal cross-sectional area in the stack or duct. Each blade shall have approximately equal surface area. Blades shall have opposed movement.

9.2.14 Damper shafts and bolting shall be of the same material as the blade.

9.2.15 Damper bearings and control mechanisms shall be external. Bearings shall be aligning, self lubricating graphite bearings mounted in bearing manufacturer's standard housing.

9.3 Allowable Stresses

9.3.1 The maximum longitudinal stress in the stack shall not exceed the smaller of

$$0.5F_y \qquad (1)$$

or

$$\frac{0.56Et}{D\,[1+(0.004E/F_y)]}$$

Where:

E = material modulus of elasticity at design temperature, in pounds per square inch (megapascals).
t = corroded shell plate thickness, in inches (millimeters).
D = outside diameter of stack shell, in inches (millimeters).
F_y = material minimum yield strength at design temperature, in pounds per square inch (megapascals).

9.3.2 The combined membrane and bending stress in the stack shell or stiffening rings shall not exceed 90 percent of the minimum yield strength of the respective material at design temperature.

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9.3.3 The tensile stress on anchor bolts shall not exceed 20,000 pounds per square inch (136 megapascals) on the AISC tensile area for ASTM A 307 material and 33,000 pounds per square inch (227 megapascals) for A 320 L7 material.

9.3.4 The tensile stress on structural bolts shall not exceed 44,000 pounds per square inch (303 megapascals) on the AISC nominal area for ASTM A 325 and 33,000 pounds per square inch (227 megapascals) for ASTM A 193 B7 or A 320 L7 material.

9.3.5 The minimum yield strength and modulus of elasticity for structural steel shall be as listed in Table 10.

9.4 Static Design

9.4.1 All stacks shall be designed as cantilever beam columns.

9.4.2 Linings shall not be considered as contributing to the strength of the stack, breeching,or duct.

9.4.3 The minimum shape factor and effective diameter for wind loads shall be as listed in Table 11.

9.4.4 Discontinuities in the stack shell plate, such as cone-to- cylindrical junctions and non-circular transitions, shall be designed so that the combined membrane and bending stresses do not exceed the allowable stresses.

9.4.5 Openings in stacks, breechings, and ducts shall be reinforced to provide the required section properties. For single openings in cylindrical stacks, the chord shall not exceed 1.4 times the radius. For two openings, opposite each other, each chord shall not exceed the radius.

9.4.6 Changes in cylindrical stack diameters shall be made with cones having an apex angle of 60 degrees or less.

9.4.7 Stiffening rings are required when $t \leq (5M/9F_{ys})^{1/2}$ and shall be provided as follows:

a. Ring spacing limits: $1 \leq L/D \leq 3$.
b. Ring section modulus required: $S, \geq LM/0.6F_{yr}$.

Where:

M = maximum circumferential moment per unit length of shell, in inch-pounds per inch (newton-meter per meter).
F_{ys} = minimum yield strength of shell material at design temperature, in pounds per square inch (pascals).
t = corroded shell thickness, in inches (meters).
L = ring spacing, in inches (meters).
D = shell diameter, in inches (meters).
S = section modulus of ring, in cubic inches (cubic meters).
F_{yr} = minimum yield strength of ring material at the shell design temperature, in pounds per square inch (pascals).

9.4.8 Stack deflection due to static wind loads shall not exceed 6 inches (15 centimeters) per 100 feet (30 meters) of stack height, based on the shell plate thickness less 50 percent of the corrosion allowance.

9.5 Wind-Induced Vibration Design

9.5.1 Internal refractory lining shall be included in the mass calculation of the vibration design.

9.5.2 The critical wind velocity (V_c), for the first and second modes of vibration of the stack, shall be calculated for the new and corroded conditions according to the following equation:

$$V_{c1} = fD/S$$

Where:

V_{c1} = first mode, critical wind velocity, in feet per second (meters per second).
V_{c2} = second mode, critical wind velocity, = $V_{c1} \times 6.0$, in feet per second (meters per second).
f = frequency of transverse vibration for the stack, in cycles per second.
D = average stack shell diameter for its top 33 percent of height, in feet (meters).
S = Strouhal number 0.2 (dimensionless).

Table 10—Minimum Yield Strength and Modulus of Elasticity for Structural Steel

Design Temperature		ASTM A 36				ASTM A 242			
		F_y		E		F_y		E	
°F	°C	($\times 10^3$)psi	MPa	($\times 10^6$)psi	MPa($\times 10^3$)	($\times 10^3$)psi	MPa	($\times 10^6$)psi	MPa($\times 10^3$)
70	21	36.0	248	29.0	200	42.0	290	27.9	192
200	93	32.9	227	29.2	201	39.5	272	27.7	191
300	149	30.2	208	28.6	197	38.7	267	27.4	189
400	204	29.0	200	28.0	193	37.8	261	27.0	186
500	260	27.8	192	27.4	189	36.8	254	26.4	182
600	316	26.6	183	26.8	185	35.7	246	25.7	177
700	371	25.4	175	26.1	180	34.5	238	24.8	171
800	427	23.4	161	25.5	176	33.2	229	23.4	161

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Note: The determination of f requires a rigorous analysis of the stack and supporting structure. The equation for the frequency of transverse vibration (f) for a stack of uniform weight distribution and constant cross section of a rigid (fixed) base is as follows:

$$f = 0.5587\sqrt{EIg/WH^4}$$

Where:

E = modulus of elasticity at design temperature, in pounds per square inch.
I = moment of inertia of stack cross section, in inches^4.
g = acceleration due to gravity, 386 inches per second squared.
W = Weight per unit height of stack, in pounds per inch.
H = overall height of stack, in inches.

Equations for stacks not covered by this equation must be approved by the purchaser.

9.5.3 The stack design shall be such that its critical wind velocities (first and second modes) fall within an acceptable range as follows:

a. $0 \le V_c < 15$ miles per hour [22 feet (6.7 meters) per second]: Acceptable. When critical wind velocities occur in this range, consideration should be given to fatigue failure.
b. 15 miles per hour $\le V_c <$ 30 miles per hour [22 feet (6.7 meters) per second $\le V_c <$ 44 feet (13.4 meters) per second]: Acceptable if provided with strakes or vibration dampening.
c. 30 miles per hour $\le V_c <$ 60 miles per hour [44 feet (13.4 meters) per second $\le V_c <$ 88 feet (26.8 meters) per second]: Not acceptable.
d. 60 miles per hour $\le V_c$ [88 feet (26.8 meters) per second $\le V_c$]: Acceptable.

● **9.5.4** When strakes are required to disrupt wind-induced vibration, they shall be used on no less than the upper 33 percent of the stack height. These strakes shall be as described in either 9.5.4.1 or 9.5.4.2.

9.5.4.1 Staggered vertical plates not less than $^1/_4$ inch (6.3 millimeters) thick and not more than 5 feet (1.5 meters) long. Four strakes shall be spaced at 90 degrees around the stack and shall project 0.10 diameter from the outside surface of the stack. Adjacent levels of strakes shall be staggered 45 degrees from each other.

Table 11—Minimum Shape Factors and Effective Diameters for Wind Loads

Segments	Shape Factor	Effective Diameter
Stack Segments		
Smooth cylinder	0.6	D
Ladders, platforms and appurtenances	1.0	Width of total projected area
Strakes	1.0	Diameter circumscribing strakes
Ducts and Breeching		
Cylindrical	0.6	D
Flat sided	1.0	Width

Note: D = the outside shell diameter for the section considered.

9.5.4.2 Spiral spoilers with three rectangular, $^1/_4$ inch (6.3 millimeter) thick strakes at 120 degree spacing with a pitch of 5 diameters and a projection of 0.10 diameter.

9.5.5 Stiffening rings shall be used to prevent ovalling when the natural frequency of the free ring (f_r) is less than twice the vortex shedding frequency (f_v) at the level under consideration, where:

$$f_r = \frac{0.126\, t_r \sqrt{E}}{D_r^2}$$

Where:

f_r = natural frequency of free ring at level under consideration, in cycles per second.
t_r = corroded plate thickness at level under consideration, in inches (millimeters).
E = modulus of elasticity of stack plate material at design temperature, in pounds per square inch (kilopascals).
D_r = internal stack diameter at level under consideration, in feet (meters).

and

$$f_v = 13.2/D_r$$

Where:

f_v = vortex shedding frequency at level under consideration, in cycles per second.
D_r = internal stack diameter at level under consideration, in feet (meters).

Both of these frequencies should be calculated at each level,using the corresponding thickness (t_r) and diameters (D_r). The section modulus of required stiffeners (S_m) shall not be less than:

$$S_m = \frac{(7\times10^{-7})(V_c)^2(D_r)^2 H_r}{S_t}$$

Where:

$V_c = (60)(f_r)(D_r)/(2)(S)$.
f_r = natural frequency of free ring at level under consideration, in cycles per second.
t_r = corroded plate thickness at level under consideration, in inches (millimeters).
D_r = internal stack diameter at level under consideration, in feet (meters).
E = modulus of elasticity of stack plate material at design temperature, in pounds per square inch (kilopascals).
f_v = vortex shedding frequency at level under consideration, in cycles per second.
H_r = stiffening ring spacing, in feet (meters).
S_t = allowable tensile stress of stack plate material at design temperature, in pounds per square inch (kilopascals).
S = Strouhal number 0.2 (dimensionless).
V_c = critical wind velocity for ovalling at the level under consideration, in feet per minute (meters per minute).

Source: Mahajan, Kanti H., "Tall Stack Design Simplified," *Hydrocarbon Processing*, p. 217, September 1975.

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9.6 Materials

● **9.6.1** Stacks, ducts, and breechings shall be constructed of materials conforming to ASTM A 36 or A 242. Alternative materials may be selected when approved by purchaser.

● **9.6.2** When the minimum service temperature is above 0°F (−18°C), bolting material shall be per ASTM A 307, A 325, or A 193 B7. Below 0°F (−18°C), A 193 B7 bolts with A 194 2H nuts or A 320 L7 bolting shall be used. No welding is permitted on A 320 L7 or A 193 B7 materials.

9.6.3 Damper materials shall be limited to maximum service temperature as follows:

a. Carbon steel, 800°F (427°C).
b. 5Cr-$^1/_2$Mo, 1200°F (649°C).
c. 18Cr-8Ni, 1500°F (816°C).
d. 25Cr-12Ni, 1800°F (982°C).

SECTION 10—BURNERS AND AUXILIARY EQUIPMENT

10.1 Burners

10.1.1 Burner design and installation shall ensure against flame impingement on tubes and tube supports when burners are operating at maximum heat release.

10.1.2 Burners shall be designed in accordance with all local and national mandatory statutes and regulations.

10.1.3 For natural draft operation, with burners firing vertically or horizontally, the minimum clearances listed in Table 12 shall be provided.

● **10.1.4** Burners shall be sized for a maximum heat release at the design excess air based on the following:

a. Five or fewer burners, 125 percent of normal heat release.

Table 12—Minimum Clearance for Natural Draft Operation

Maximum Heat Release per Burner		Minimum Clearance A: Vertical to Centerline Roof Tubes or Refractory (Vertical Firing Only)		B: Horizontal to Centerline Wall Tubes from Burner Centerline		C: Horizontal from Centerline of Burner to Unshielded Refractory		D: Between Opposing Burner (Horizontal Firing)	
millionBtu/hr	Mw	feet	meters	feet	meters	feet	meters	feet	meters
				Oil Firing					
4	1.2	12	3.7	2′9″	0.8	2′0″	0.6	16	4.9
6	1.8	16	4.9	3′3″	1.0	2′6″	0.8	22	6.7
8	2.3	20	6.1	3′9″	1.1	3′0″	0.9	28	8.5
10	2.9	24	7.3	4′3″	1.3	3′6″	1.1	32	9.8
12	3.5	28	8.5	4′9″	1.4	4′0″	1.2	36	11.0
14	4.1	32	9.8	5′3″	1.6	4′6″	1.4	40	12.2
				Gas Firing					
2	0.6	7	2.1	2′0″	0.6	1′6″	0.5	8	2.4
4	1.2	10	3.0	2′6″	0.8	2′0″	0.6	12	3.7
6	1.8	13	4.0	3′0″	0.9	2′6″	0.8	16	4.9
8	2.3	16	4.9	3′6″	1.1	3′0″	0.9	20	6.1
10	2.9	19	5.8	4′0″	1.2	3′6″	1.1	24	7.3
12	3.5	22	6.7	4′6″	1.4	4′0″	1.2	26	7.9
14	4.1	25	7.6	5′0″	1.5	4′6″	1.4	28	8.5

Notes:
[a]For horizontal firing, the distance between the burner centerline and the roof tube centerline or refractory shall be 50 percent greater than the distances in Column B.
[b]For combination liquid and gas burners, the clearances will be based on liquid fuel firing, except when liquid fuel is used for start up only.
[c]For Low NOx gas burners, an increase in longitudinal clearance must be provided. This shall be achieved by multiplying dimensions in column A by a factor of 1.3, and D by a factor of 1.5. Low NOx burners are those which incorporate design features with the specific intent to reduce the formation of NOx below levels normally produced by conventional burners.
[d]For intermediate firing rates, the required clearances may be achieved by linear interpolation.

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b. Six or seven burners, 120 percent of normal heat release.
c. Eight or more burners, 115 percent of normal heat release.

● **10.1.5** For liquid fuel fired heaters with a maximum heat release greater than 15 million British thermal units per hour (4.4 megawatts), a minimum of three burners shall be used. Alternatively, a single burner with auxiliary guns may be used to permit gun maintenance without shutting down or upsetting the process.

● **10.1.6** Unless otherwise specified, gas pilots shall be provided for all burners.

10.1.7 When a pilot is provided, it shall meet the following requirements:

a. Pilot shall have a minimum heat release of 75,000 British thermal units per hour (.022 megawatts). The minimum heat release must be approved by the purchaser, when accompanying a high intensity burner whose heat release is 15 million British thermal units per hour (4.4 megawatts) or greater.
b. The pilot burner shall be provided with a continuous supply of air, under all operating conditions. This includes operation with the main burner out of service.
c. The pilot burner shall remain stable over the full firing range of the main burner. It shall also remain stable upon loss of main burner fuel, minimum draft, all combustion air rates and for all operating conditions.

10.1.8 Burner block installations shall be designed to expand and contract as a unit, independent of the heater refractory.

10.1.9 Burner tiles shall be prefired to no less than 500°F (260°C).

10.1.10 The materials used for construction of a burner shall be chosen for strength, as well as temperature and corrosion resistance, for the anticipated service conditions. Burner components shall be designed in accordance with the minimum requirements as shown in Table 13.

10.1.11 The burner shall have a turndown capability to at least 40 percent of the normal heat release without adjusting the air controls.

10.1.12 The burner shall be selected to use no less than 90 percent of the maximum draft available for the maximum specified heat release.

10.1.13 The burner fuel valve and air registers shall be operable from grade or platforms. A means shall be provided to view the burner and pilot flame during light off and operating adjustment.

● **10.1.14** When a natural draft burner is to be used in forced draft service, the purchaser shall specify the required heater capacity during natural draft operation, if required.

10.1.15 Natural draft oil burners shall include double refractory tiles to maintain flame stability when firing heavy fuels such as vacuum or atmospheric-reduced crude oils, tars, or other high viscosity fuels requiring preheat.

10.1.16 Oil burners shall be designed to operate on a maximum oil viscosity of 200 SSU (43 cs).

10.1.17 Atomizing steam shall be supplied dry at the burner, or with slight superheat.

10.1.18 When volatile fuels, such as naphtha or gasoline, are burned, a safety interlock shall be provided on each burner. The interlock design must (in sequence) shut off the fuel, purge the oil gun, and shut off the purge medium before the gun can be removed.

● **10.1.19** Gas manifolds and oil guns shall be removable while the heater is in operation. Purchasers must specify whether they require removal of the diffuser or the complete burner assembly.

10.2 Sootblowers

● **10.2.1** Unless otherwise specified, sootblowers shall be automatic, sequential, and fully retractable.

10.2.2 Individual sootblowers shall be designed to pass a minimum of 10,000 pounds (4535 kilograms) per hour of steam with a minimum steam pressure of 150 pounds per square inch gauge (1034 kilopascals) at the inlet flange.

10.2.3 Retractable sootblower lances shall have a maximum of two nozzles each. The minimum distance at any position between the lance outside diameter and the tube outside diameter shall be 9 inches (23 centimeters).

10.2.4 Spacing of retractable sootblowers shall be based upon a maximum horizontal or vertical coverage of 4 feet (1.2 meters) from the lance centerline, or five tube rows, whichever is less. The first (bottom) row of shield tubes may be neglected from sootblower coverage. Tube supports are considered as a limit to individual sootblower coverage.

● **10.2.5** Provide erosion protection of convection section walls, located within in soot blowing zones, using either high-duty fireclay brick or castable refractory with a minimum density of 125 pounds per cubic foot (2000 kilograms per cubic meter).

10.2.6 Protect retractable sootblower entrance ports (through the refractory wall) with stainless steel sleeves.

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10.3 Fans and Drivers

10.3.1 Fans and drivers for use with fired heaters shall be designed and built in accordance with the requirements in Appendix E of this standard.

10.3.2 The specified type of driver shall be adequate for the fan operating loads, including gear or coupling losses and for off-power drag of the start-up motor, if any. Service factors for the driver shall be determined using Table 14. Motor drivers shall be capable of starting the fan, with its control damper in the minimum fully open position and with 80 percent of the design voltage applied.

10.4 Damper Controls

● **10.4.1** Control dampers shall be designed to move to the position specified by the purchaser in the event of damper control signal failure or motive force failure.

Table 13—Materials of Construction

Component	Operation	Material
Fuel gas (burner and pilot)		
Fuel gas manifold and piping	Normal	Cast iron or carbon steel
	> 100 parts per million H_2S and either > 300°F fuel, or > 400°F combustion air	321L stainless steel
Fuel gas riser pipe	Normal	Carbon steel
	> 700°F combustion air	304 stainless steel
	> 100 parts per million H_2S and either > 300°F fuel, or > 400°F combustion air	321L stainless steel
Fuel gas tip	Normal	Cast iron or 300 series stainless steel
	> 100 parts per million H_2S and either > 300°F fuel, or > 400°F combustion air	321L stainless steel
Premix venturi	Normal	Cast iron or carbon steel
Fuel oil		
Oil gun receiver and body	Normal	Ductile iron
Oil gun tip	Normal	416 stainless steel
	Erosive oils	T-1 tool steel
Atomizer	Normal	Brass
	> 3 percent (weight) sulphur	303 stainless steel
Atomizer body only	Erosive oils	Nitride hardened nitralov
Other	Normal	Carbon steel
Burner housing		
Exterior casing	Normal	Carbon steel
	Preheated combustion air	Insulated carbon steel
Flame stabilizer or cone	Normal	300 series stainless steel
Insulation and noise reduction linings	≤ 700°F combustion air	mineral wool
	> 700°F combustion air	mineral wool covered with metal liner
Other interior metal parts	Normal	Carbon steel
	> 700°F combustion air	A242 or 304 stainless steel
Burner tile		
Normal	> 40% alumina refractory	
High intensity combustor	> 85% alumina castable refractory/firebrick	
Oil firing tile		
≤ 50 parts per million (weight) Va, Na	≥ 60% alumina refractory	
> 50 parts per million (weight) Va, Na	> 90% alumina refractory	

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10.4.2 Dampers shall be equipped with an external blade position visual indicator, on the damper shaft and on any remote control mechanism.

10.4.3 Dampers shall be furnished with a position control mechanism that is operable from grade and is capable of holding the damper blade in any position from fully open to fully closed. The damper controller shall provide positive action to translate the damper blade in either an open or closed direction.

10.4.4 Manual damper operators shall be designed so that one person can, without excessive effort, position the damper blade in any desired position. Wire rope damper operators shall be a minimum of $^1/_8$ inch (3 millimeter) diameter, Type 304 stainless steel wire rope with galvanized hardware such as thimbles, turnbuckles, and clamps.

SECTION 11—INSTRUMENT AND AUXILIARY CONNECTIONS

11.1 Flue Gas and Air

The heater vendor shall provide nozzles or connections as described in 11.1.1 through 11.1.3.

11.1.1 FLUE GAS AND COMBUSTION AIR TEMPERATURE

11.1.1.1 Provide one connection in the flue gas exit of each radiant section for each 30 feet (9.1 meters) of radiant box length or diameter. Provide a minimum of two connections.

11.1.1.2 One connection shall be provided in the convection section, preceding the first process or utility coil when multiradiant section heaters or multiple heaters have their flue gas combined to a common convection section, for each 30 feet (9.1 meters) of convection tube length.

11.1.1.3 One connection shall be provided in the convection section immediately after each process or utility coil for each 30 feet (9.1 meters) of convection tube length. Provide a minimum of two connections after the last convection coil.

11.1.1.4 Provide connections in each stack and each takeoff to a stack.

11.1.1.5 Provide connections in the inlet and outlet air and flue gas ductwork of an air heater.

11.1.1.6 The connections furnished shall be $1^1/_2$-inch NPS 3000-pound screwed forged steel couplings welded to the outside casing plate. When the refractory lining exceeds 3 inches (7.6 centimeters) in thickness, the opening shall be lined with stainless steel pipe, Schedule 80, Type 304. A hex-head forged steel screwed plug shall be furnished with each coupling.

Table 14—Service Factors

	Service Factor		
Power Requirement	Turbine	1.00 Motor	1.15 Motor
≤ 25 horsepower	1.10	1.25	1.14
26 to 75 horsepower	1.10	1.15	1.05
≥ 75 horsepower	1.10	1.10	1.00

11.1.2 FLUE GAS AND COMBUSTION AIR PRESSURE

11.1.2.1 Provide two connections in each radiant section located 1 foot to 2 feet (0.3 meter to 0.6 meter) above the top of the floor refractory.

11.1.2.2 For heaters with horizontal firing, provide one connection at the highest burner centerline on each burner wall.

11.1.2.3 Provide two connections in each radiant section at the point of minimum draft.

11.1.2.4 Provide a connection in the convection section outlet immediately after the final process or utility coil.

11.1.2.5 Provide connections upstream and downstream of the draft control dampers.

11.1.2.6 Provide connections in the inlet and outlet ductwork connected with a fan.

11.1.2.7 Provide connections in the inlet and outlet flue gas and combustion air ducting of a combustion air heater.

11.1.2.8 Provide a connection down-stream of any combustion air control damper in the burner windbox or plenum.

11.1.2.9 The connections furnished shall be $1^1/_2$-inch NPS 3000-pound screwed forged steel couplings welded to the outside casing plate. When the refractory lining exceeds 3 inches (7.6 centimeters) in thickness, the opening shall be lined with stainless steel pipe, Schedule 80, Type 304. A hex-head forged steel screwed plug shall be furnished with each coupling.

11.1.3 FLUE GAS SAMPLING

11.1.3.1 Provide connections in the flue gas exit from each radiant section.

11.1.3.2 Provide connections at the convection section outlet.

11.1.3.3 The connections shall be 4-inch NPS, Schedule 80 pipe with a 150-pound raised-face flange. The pipe shall

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be welded to the outside casing plate and project 8 inches (20 centimeters) to the face of the flange. A 150 pound class blind flange shall be furnished by the heater vendor for each connection, with appropriate gaskets for the temperature and corrosive conditions of the flue gas. The pipe shall extend into the heater to within $1^1/_2$ inches (3.8 centimeters) of the hot face of the refractory lining.

● **11.1.3.4** Additional connections to meet applicable governmental or local requirements shall be specified by the purchaser.

11.2 Process Fluid Temperature

● **11.2.1** The heater vendor shall provide fluid thermowell connections in the convection to radiant crossovers, if specified by the purchaser.

● **11.2.2** When process outlet thermowell connections are specified and individual outlets are provided by the heater vendor, the thermowell connections shall be furnished as part of the outlet piping system. When an outlet manifold is furnished, the specified thermowell connections shall be provided by the heater vendor.

11.2.3 Process fluid thermowell connections shall be $1^1/_2$-inch raised-face flanges with a rating adequate for the fluid design pressure and temperature. The material shall be the same as the tube or pipe to which it is connected.

11.3 Auxiliary Connections

The heater vendor shall provided auxiliary connections as described in 11.3.1 through 11.3.2.

11.3.1 PURGE STEAM CONNECTIONS

11.3.1.1 Purge connections may also be used as snuffing steam connections.

11.3.1.2 A minimum of two purge connections shall be provided for each firebox. The connections shall be $1^1/_2$- or 2-inch NPS, 3000-pound screwed forged steel pipe couplings, welded to the outside casing plate. The openings through the refractory shall be lined with a Schedule 80 stainless steel pipe, Type 304.

11.3.1.3 Purge connections shall provide a minimum of three firebox volume changes within 15 minutes.

11.3.1.4 Connections shall be located to preclude impingement on the heater coils and any ceramic fiber linings, and shall provide even distribution in the radiant section. The minimum size connection to header boxes shall be $^3/_4$-inch NPS. A minimum of one 1-inch NPS connection shall be provided for each burner plenum chamber.

11.3.1.5 For forced-draft systems, the forced draft fan can be used to purge the firebox in lieu of purge steam.

11.3.2 VENT AND DRAIN CONNECTIONS

11.3.2.1 Manifold or piping vents and drains shall be be a minimum 1-inch NPS, 6000-pound welded coupling, of the same metallurgy as the manifold or piping.

● **11.3.2.2** When water washing of either radiant or convection tubes is specified by the purchaser, provisions shall be made for draining water to the outside of the heater using at least one 4-inch NPS connection with a cap.

11.3.2.3 For header boxes containing flanged or plug fittings, a minimum $^3/_4$ inch NPS, 3000-pound screwed forged steel drain connection with hex plug shall be provided.

● 11.4 Tubeskin Thermocouples

Tubeskin thermocouples, when specified, shall be welded to a tube within a 60 degree arc on the tube wall, directly facing the flame envelope. Lead wire, insulators, and protective sheaths shall be designed to accommodate all anticipated tube movement.

Protective sheaths shall be made gastight and constructed of Type 310 stainless steel (as a minimum). Such sheaths are to be attached to the heater tubes by welded clips or bands. All thermocouple assemblies shall terminate on the exterior shell of the fired heater with a thermocouple head.

11.5 Access to Connections

11.5.1 All process thermocouple and flue gas sampling connections shall be accessible from grade, platforms, or ladders.

11.5.2 Connections considered as accessible from a platform shall be no more than 7 feet (2.1 meters) above the floor of the platform.

11.5.3 Connections considered as accessible from grade shall be no more than 10 feet (3 meters) above grade.

11.5.4 Connections considered as accessible from permanent vertical ladders shall be no more than 2 feet, 6 inches (0.8 meter) from the centerlines of such ladders and at least 3 feet (0.9 meter) below the top rung of such ladders.

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SECTION 12—SHOP FABRICATION AND FIELD ERECTION

12.1 General

● **12.1.1** The heater, all auxiliary equipment, ladders, stairs, and platforms shall be shop assembled to the maximum extent possible consistent with available shipping, receiving, and handling facilities. Individual sections shall be properly braced and supported to prevent damage during shipment. All blocking and bracing used for shipping purposes shall be clearly identified for field removal. Coil flange faces and other machined faces shall be coated with an easily removable rust preventive. Openings in pressure parts shall be covered to prevent entrance of foreign materials.

12.1.2 The vendor shall state the type of protection provided for refractory and insulation to avoid damage by handling or weather during shipment, storage, and erection.

12.1.3 All surfaces to be welded shall be free from scale, oil, grease, dirt, and other harmful agents. Welding operations shall be protected from wind, rain, and other weather conditions that may affect weld quality.

● **12.1.4** Heater steel shall be fabricated in accordance with the applicable provisions of the following codes and specifications:

a. ICBO *Uniform Building Code.*
b. American Institute for Steel Construction, *Specification for Design, Fabrication, and Erection of Structural Steel for Buildings.*
c. AWS D1.1, *Structural Welding Code.*
d. Local codes as specified by the purchaser.

12.1.5 Coils shall be fabricated in accordance with the applicable provisions of ASME B31.3 for piping, Chapter V.

12.2 Steel Fabrication

12.2.1 Welders for structural steel fabrication shall be qualified to the requirements of AWS D1.1, Section 5.

12.2.2 Seam welds between plates shall be continuous, full penetration welds.

12.2.3 Horizontal exterior welds between plates and structural members shall have a continuous fillet weld on the top side and 2 inch (50 millimeters) long fillet welds on 9 inch (230 millimeters) centers on the bottom side. Diagonal and vertical exterior welds shall have continuous fillet welds on both sides.

12.2.4 Fillet welds shall be of uniform size with full throat and legs.

12.2.5 Welding filler materials for the processes listed in 12.2.5.1 and 12.2.5.2 shall conform to the AISC *Specification for Design, Fabrication, and Erection of Structural Steel for Buildings* and Table 15.

● **12.2.5.1** Charpy impact test requirements shall be specified by the purchaser for all welds with design metal temperatures below –20°F (–29°C) and for submerged arc welds at design metal temperatures below 0°F (–18°C).

12.2.5.2 Welding filler materials shall have a chemical composition matching that of the base materials being joined.

12.2.6 Circular and slotted bolt holes in columns and baseplates shall be drilled or punched. Baseplates shall be shop welded.

12.2.7 The minimum thickness of gusset plates shall be $^1/_4$ inch (6 millimeters).

12.2.8 Shop connections shall be bolted or welded. Field joints between casing plates and stack intermediate joints shall be welded. All other field joints shall be bolted. Where field bolting is impractical, erection clips or other suitable positioning devices shall be furnished for field welded connections.

12.2.9 The minimum size of bolts shall be $^5/_8$ inch (16 millimeters) in diameter, except where the flange width prohibits use of $^5/_8$ inch (16 millimeter) bolts. In no case shall bolts be less than $^1/_2$ inch (13 millimeters) in diameter.

12.2.10 Drain holes in structural members shall be a minimum of $^1/_2$ inch (13 millimeters) in diameter. Checkered plate flooring shall be furnished with one $^1/_2$ inch (13 millimeter) diameter drain hole for every 15 square feet (1.4 square meters) of floor plate area.

12.2.11 Heater stacks shall be fabricated to the tolerances described in 12.2.11.1 through 12.2.11.6.

Table 15—Welding Filler Materials

Welding Process	Design Metal Temperature		
	–20°F to 0°F (–29°C to –18°C)	0°F to 50°F (–18°C to 10°C)	>50°F (>10°C)
Shielded metal arc	Low hydrogen electrodes only	All classifications of AWS A5.1 except E6012, E6013, E6020, E6022, E7014, and E7024	All classifications of AWS A5.1
Gas metal	—	ER70S-2, ER70S-3, and ER70S-6	All classifications of AWS A5.1
Flux core	—	E60T-8, E70T-8	All classifications of AWS A5.1
Arc welding		E70T-5 and E70T-6	

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12.2.11.1 The stack shall be sufficiently true so that the erected stack can be plumbed to within a maximum deviation of 1 inch (25 millimeters) per 50 feet (15.2 meters) of height.

12.2.11.2 The maximum deviation from a straight edge applied to the stack shell shall not exceed $^1/_8$ inch (3 millimeters) in any 10 feet (3 meters).

12.2.11.3 The variance between minimum and maximum diameters at any cross section along the stack length shall not exceed 2 percent of the nominal diameter for that section.

12.2.11.4 Plate misalignment at any joint shall not exceed the lesser of $^1/_8$ inch (3 millimeters) or 25 percent of the nominal plate thickness.

12.2.11.5 Vertical joint peaking shall not exceed a depth of $^3/_{16}$ inch (5 millimeters) when measured from a 24 inch (610 millimeter) circumferential template centered on the joint.

12.2.11.6 Circumferential joint banding shall not exceed a depth of $^5/_{16}$ inch (8 millimeters) when measured from a 36 inch (900 millimeter) straight edge centered on the joint.

12.2.12 The threads of bolts securing damper blades to the shaft shall be scored or tack welded after installation.

12.2.13 Attachment of refractory anchors or tieback to heater casing shall be by manual or stud-gun welding. When manual welding is employed, welds shall be "all around".

12.2.14 Suitable lifting lugs shall be provided for the erection of all sections where the section weight exceeds 4000 pounds (1814 kilograms). The lifting load used shall be 1.5 times the section weight to allow for impact.

12.2.15 All structural steel and subassemblies shall be clearly marked with 2 inch (50 millimeters) minimum high letters or numbers for field identification. All loose items such as rods, turnbuckles, clevises, bolts, nuts, and washers shall be shipped in bags, kegs, or crates. Bags, kegs, or crates shall be tagged with the size, diameter, and length of contents so that tags for each item are individually identifiable. Tags used for marking shall be metal and markings shall be applied by stamping.

12.2.16 The erection drawings and a bolt list shall be furnished prior to the shipping of heater steel. Erection marks, size, and length of field welds shown on erection drawings shall be a minimum of $^1/_8$ inch (3 millimeter) high lettering. The bolt list shall specify the number, diameter, length, and material for each connection. A bill of material shall also be furnished showing the weights of sections over 4000 pounds (1814 kilograms).

12.2.17 Furnish a minimum of 5 percent overage of each bolt and nut size and material used in the erection of the heater.

12.3 Coil Fabrication

● **12.3.1** The following welding processes are permitted, provided satisfactory evidence is submitted that the procedure is qualified in accordance with all applicable codes and standards:

a. Shielded metal arc with covered electrodes.
b. Gas tungsten-arc, manual and automatic.
c. Gas welding process for 2-inch NPS and smaller and for P-1 material.
d. Gas metal-arc welding in the spray transfer range.
e. Flux cored arc welding with external shielding gas.

12.3.2 Permanently installed backing rings shall not be used.

● **12.3.3** An argon or helium internal purge shall be used for gas tungsten arc root pass welding of $2^1/_4$Cr-1Mo and higher alloys except that nitrogen may also be used for austenitic stainless steels. The root pass in P-1, P-3, and P-4 steels may be welded with or without an internal purge.

12.3.4 Each weld shall be uniform in width and size throughout its full length. Each weld shall be smooth and free of slag, inclusions, cracks, porosity, lack of fusion, and undercut, except to the extent permitted by the referenced codes. In addition, the cover pass shall be free of course ripples, irregular surfaces, nonuniform head patterns, and high crowns and deep ridges or valleys between heads.

12.3.5 Butt welds shall be slightly convex and uniform in height as specified in the applicable codes. Limitations on weld reinforcement shall apply to the internal surface as well as the external surface.

12.3.6 Repair welds shall be done in accordance with a repair procedure approved by the purchaser. Repairs shall not damage the adjacent base material.

12.3.7 Preheat temperature, interpass temperature, and postweld heat treatment shall be in accordance with the provisions of the applicable codes.

12.4 Painting and Galvanizing

12.4.1 Heater steel shall be sandblasted to SSPC SP-6 and primed with one coat of inorganic zinc primer to a minimum dry film thickness (DFT) of 3 mils. Surfaces shall be painted in accordance with manufacturer's recommendations on temperature and relative humidity.

12.4.2 Uninsulated flue gas ducts and stacks shall be primed with an inorganic zinc primer. Surface preparation

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and dry film thickness shall be in accordance with the paint manufacturer's recommendation.

● **12.4.3** Platforms, handrails and toeboards, grating, stairways, fasteners, ladders, and attendant light structural supports shall be hot-dipped galvanized. Galvanizing shall conform to the applicable sections of ASTM A123, A143, A153, A384, and A385. Bolts joining galvanized sections shall be galvanized per ASTM A153 or zinc-coated per ASTM A164, Type LS coating.

12.5 Refractories and Insulation

12.5.1 Materials shall be stored in original containers, if possible, and shall be protected from moisture and from atmospheric and foreign contaminants. They shall be kept completely dry and at manufacturer's recommended storage temperature until used. Bricks shall be free of cracks, chips, spalling, or other defects.

12.5.2 Prior to installation of refractory all steel surfaces shall be cleaned to remove dirt, grease, paint, loose scale, or other foreign materials.

12.5.3 Water used to install refractories shall be of potable quality and the temperature shall be between 45°F (7°C) and 90°F (32°C), unless the refractory manufacturer specifies otherwise.

12.5.4 All material shall be prepared and installed in accordance with the manufacturer's recommendations.

12.5.5 The mortar joints in firebrick construction shall be as thin as possible. In applying the mortar, the brick shall be dipped or troweled on two edges. Expansion joints shall be mortar free. Brick should be placed against the mating surface and tapped gently to ensure uniform joints, no more than $^1/_{16}$ inch (1.5 millimeters) wide.

12.5.6 Anchors with circular bases shall be welded all around. Other anchors shall be welded to casing along both sides.

12.5.7 Chain link fence anchoring shall be pulled out and held in place after welding, and prior to castable application, to ensure proper position in the castable layer.

12.5.8 Requirements for castables are specified in 12.5.8.1 through 12.5.8.5.

12.5.8.1 The surfaces to which castable is applied shall be kept above 45°F (7°C) and below 100°F (38°C) during installation and curing.

12.5.8.2 For pneumatic application, the lining shall be applied in horizontal strips working upward from the bottom. It shall proceed continuously to the required thickness in given area. If the installation is interrupted, the lining shall be cut back immediately to the casing surface. This cut shall be full depth at a 90-degree angle to the casing surface.

12.5.8.3 Rebound materials shall not be re-used in applying linings.

12.5.8.4 Scoring of the castable surfaces will be in accordance with the vendor's specifications.

● **12.5.8.5** Each layer of the castable shall be properly cured after installation. A resin-base membrane curing compound shall be applied immediately after the initial set, with a curing period of at least 24 hours. Shop installed castable shall not be handled or tested for 72 hours after installation.

12.6 Field Erection

12.6.1 It shall be the responsibility of the erector to ensure that the heater is erected in accordance with the specifications and drawings furnished by the vendor. The heater shall be erected in accordance with the applicable sections of this standard.

12.6.2 Castable lined panels shall be handled to avoid excessive cracking or separation of the refractory from the steel.

12.6.3 Take care to avoid refractory damage due to weather. Standing water or saturation of the refractory shall be prevented. Protection shall include cover to avoid rain impingement and shall allow drainage, proper fit, and tightening of doors and header boxes.

12.6.4 Sections where refractory edges are exposed shall be protected against cracking of edges and corners. Avoid external blows to the steel casing.

12.6.5 Field joints between panels shall be sealed in accordance with the heater vendor's requirements.

12.6.6 Construction joints, resulting from panel or modular construction, shall have continuous refractory cover to the full thickness of the adjacent refractory.

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SECTION 13—INSPECTION, EXAMINATION, AND TESTING

13.1 General

13.1.1 Consistent with ASME B31.3 definitions, inspection applies to functions performed for the owner by the owner's inspector or the inspector's delegates. Examination applies to quality control functions performed by the manufacturer (for components only), fabricator, or erector.

13.1.2 The purchaser, his designated representative, or both, reserve the right to inspect all heater components and their assembled units at any time during the material procurement, fabrication, and shop assembly to ensure materials and workmanship are in accordance with applicable standards, specifications, codes, and drawings.

13.1.3 The vendor shall examine all individual heater components and their shop-assembled units to ensure that materials and workmanship are in accordance with applicable standards, specifications, codes, and drawings.

● **13.1.4** Preinspection meetings between purchaser and fabricator shall be held before start of fabrication.

13.2 Weld Inspection and Examination

13.2.1 Radiographic, ultrasonic, visual, magnetic particle, or liquid penetrant examination of welds in coils shall be in accordance with ASME B31.3 or the ASME *Boiler and Pressure Vessel Code,* if so specified.

13.2.2 The extent of examination of welds in coils, including return bends, fittings, manifolds, and crossover piping, shall conform to the requirements of 13.2.2.1 through 13.2.2.7.

13.2.2.1 The root passes of 10 percent of all austenitic welds for each welder shall be liquid-penetrant examined following weld surface preparation per ASME *Boiler and Pressure Vessel Code,* Section V, Article 6. When the required examination identifies a defect, progressive examination shall be required per ASME B31.3, Para. 341.3.4.

13.2.2.2 All chrome-moly or austenitic welds shall be 100 percent radiographed.

13.2.2.3 Ten percent of all carbon steel welds by each welder shall be 100 percent radiographed. When the required examination identifies a defect, progressive examination is required per ASME B31.3 Para 341.3.4. For each weld found to be defective, radiographs shall be made promptly on welds made by the same welder that produced the defective welds.

13.2.2.4 All radiographs shall show 2 percent (2-2T) sensitivity.

13.2.2.5 Acceptance criteria of welds shall be per ASME B31.3, table 341.2A for normal fluid service. Radiography requirements apply also to coils fabricated to the ASME *Boiler and Pressure Vessel Code.*

13.2.2.6 All longitudinal seam welds on manifolds shall be 100 percent radiographed. In addition, these welds shall be liquid-penetrant examined for austenitic materials or magnetic particle examined for ferritic materials.

13.2.2.7 In cases where weld or material configuration makes radiographic examination difficult to interpret or impossible to perform, such as nozzle (fillet) welds, ultrasonic examination may be substituted. When ultrasonic examination is impractical, liquid penetrant (for austenitic materials) or magnetic particle examination (for ferritic materials) may be substituted.

13.2.3 Post-weld heat treatment shall be performed in accordance with ASME B31.3, Para. 331.3, unless otherwise agreed to by the purchaser. Radiographic examination shall be performed upon completion of heat treatment. Hardness testing of the materials accessible to measuring equipment shall be performed in accordance with ASME B31.3, Para. 331.1.7.

13.2.4 Proposed welding procedures, procedure qualification records, and welding rod specifications for all pressure-retaining welds shall be in accordance with ASME IX and shall be submitted by fabricator for review, comment or approval by the purchaser.

13.2.5 Welder qualifications and applicable manufacturer's report forms shall be maintained. Examples include certified material mill test reports, AWS classification and manufacturer of electrode or filler material, welding specifications and procedures, positive materials identification documentation of alloy materials, and non-destructive examination procedures and results. Unless otherwise specified by the engineering design, records of examination procedures and examination personnel qualifications shall be retained for at least five years after the record is generated for the project.

13.3 Castings Examination

13.3.1 Shield and convection section cast tube supports shall be examined as described in 13.3.1.1 through 13.3.1.3. Material conformance shall be verified by review of chemical and physical test results submitted by the manufacturer. Positive materials identification may be requested to verify these results.

● **13.3.1.1** Tube supports shall be visually inspected (MSS SP-55) and dimensionally checked. Tube supports shall be adequately cleaned to facilitate examination of all surfaces. Intersections of all reinforcing ribs with the main member

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shall either be 100 percent liquid-penetrant examined (austenitic) or magnetic particle examined (ferritic). Testing shall be performed according to ASTM E 165 and ASTM A407, respectively. Acceptance levels shall be specified per E 433 for type and class.

Unacceptable surface defects and linear discontinuities exceeding the requirements specified in ASME *Boiler and Pressure Vessel Code,* Section VIII, Div. 1, Appendix 7, shall be removed and their removal verified by liquid-penetrant examination. Defects shall be considered major when the depth of the cavity, after preparation for repair, exceeds 20 percent of the section thickness or when the cavity exceeds 10 inches (250 millimeters) in length. All other defects shall be considered minor. Minor defects shall be repaired by welding.

● **13.3.1.2** Radiographic examination of critical sections shall be performed when specified by the purchaser with testing and acceptance criteria per ASME *Boiler and Pressure Vessel Code,* Section VIII.

13.3.1.3 All repairs shall be verified by liquid-penetrant examination. Liquid-penetrant examination shall be performed according to ASTM E 165 with ASTM E 433 as the referenced standard. In accordance with ASTM E 433, the purchaser shall specify acceptability criteria for type, class, and length, that is, maximum dimensions and quantity of indications per unit area. Major repairs shall be verified by radiography. Radiographic examination shall be in accordance with ASTM E 446 or ASTM E 186, to Severity Level 2, except for cracks and hot tears which shall be Level 0. Weld repairs shall be made using welding procedures and operators qualified in accordance with ASME *Boiler and Pressure Vessel Code,* Section IX.

● **13.3.2** Cast radiant tube supports, hangers, or guides shall be visually examined for surface imperfections using MSS SP-55 as a reference for categories and degrees of severity. Defects shall be marked for removal or repair, or to warrant complete replacement of the casting. Dimensions shall be verified with checks based on an agreed-upon sampling plan. Repairs shall be verified by liquid-penetrant examination.

13.3.3 Cast return bends and pressure fittings shall be examined as described in 13.3.3.1 through 13.3.3.4

13.3.3.1 All cast return bends and pressure fittings shall be visually examined per ASME B31.3, Para. 344.2 and MSS SP-55 for imperfections and to confirm dimensions in accordance with reference drawings and the agreed-upon sampling plan. Examination shall confirm proper and complete identification as specified in the purchase order.

13.3.3.2 All surfaces shall be suitably prepared for liquid penetrant examination (austenitic) or magnetic particle examination (ferritic). Testing shall be performed in accordance with ASTM specifications E 165 and E 433, and evaluated per agreed upon acceptance levels per MSS SP-93 and MSS SP-53, Table 1, respectively.

13.3.3.3 Radiographic examination of the cast return bends and pressure fittings shall be in accordance with ASME B31.3. Radiographic quality shall be in accordance with ASTM E446, Severity Level 2 minimum, for each category defect. Sampling quantities and coverage degree shall be specified by purchaser in addition to levels of acceptability for six categories listed.

● **13.3.3.4** Machined weld bevels shall be checked with liquid penetrant and examined for indications. Degrees of acceptability shall be set by purchaser.

13.3.4 Bearing surfaces of all castings shall be smooth and free from sharp edges and burrs.

13.4 Examination of Other Components

13.4.1 Examination of heater steel work shall be in accordance with the requirements of AWS D1.1.

13.4.2 Refractory linings shall be examined throughout for thickness variations during application and for cracks after curing. Thickness variations are limited to a range of $-1/4$ inch (-6 millimeters) to $1/2$ inch (13 millimeters). Cracks which are $1/8$ inch (3 millimeters) or greater in width and penetrate more than 50 percent of the castable thickness shall be repaired. Repairs shall be made by chipping out the unsound refractory to the backup layer interface and exposing a minimum of one tieback anchor or to the sound metal, making a joint between sound refractory that is perpendicular to the base metal, and then gunning, casting, or hand packing the area to be repaired.

13.4.3 Finned extended surface shall be examined to ensure fins are perpendicular to the tube within 15 degrees. The maximum discontinuity of the weld shall be $2 1/2$ inches (64 millimeters) in 100 inches (2.5 meters) of weld. The attachment weld shall provide a cross sectional area of not less than 90% of the cross sectional area of the root of the fin. Cross sectional area is the product of the fin width and the peripheral length.

13.4.4 Fins and studs shall be examined to verify conformity with specified dimensions.

13.4.5 For rolled joint fittings, the fitting tube hole inner diameter, the tube outer diameter and the tube inner diameter (before and after rolling) shall be measured and recorded in accordance with the fitting location drawing. These measurements shall be supplied to the purchaser.

13.4.6 Fabricated supports include both plate-fabricated and multicast techniques. Fabricated convection tube intermediate supports shall have support lug welds radiographed.

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API STD*560 95 ■ 0732290 0550174 T04 ■

Warping of the completed support shall be within AISC standards.

13.5 Testing

13.5.1 PRESSURE TESTING

13.5.1.1 All assembled pressure parts shall be hydrostatically tested to a minimum pressure equal to 1.5 times the coil design pressure, multiplied by the ratio of the allowable stress at 100°F (38°C) to the allowable stress at the design tube metal temperature. The following test requirements also apply:

a. The maximum test pressure shall be limited to the extent that the weakest component shall not be stressed beyond 90 percent of the material's yield strength at ambient temperature.
b. Hydrostatic test pressures shall be maintained for a minimum period of 1 hour to test for leaks.

13.5.1.2 If hydrostatic testing of pressure parts is not considered practical by agreement between the purchaser and the vendor, then pneumatic leak testing shall be substituted, using a nonflammable gas. The pneumatic test pressure shall be 60 pounds per square inch gage (430 kilopascals) or 15 percent of the maximum allowable design pressure, whichever is less. The pneumatic test pressure shall be maintained for a length of time sufficient to examine for leaks, but in no case for less than 15 minutes. A bubble surfactant will be applied to weld seams to aid visual leak detection.

13.5.1.3 Water used for hydrostatic testing shall be potable. For austenitic materials, the chloride content of the test water shall not exceed 50 parts per million.

13.5.1.4 Except when the test fluid is the process fluid, the test fluid shall be removed from all heater components upon completion of hydrostatic testing.

13.5.2 REFRACTORY TESTING

Installed castable linings shall undergo hammer tests to check for voids within the refractory material. For dual layer linings, the hammer tests shall be conducted on each layer, after curing. Linings shall be struck with a 1 pound (450 gram) ball peen machinist's hammer over the entire surface, using a grid pattern approximating the following:

a. For arch areas, 24 inch (610 millimeter) centers.
b. For sidewall and floor areas, 36 inch (920 millimeter) centers.

13.5.3 STUDDED TUBE TESTING

Each length of a studded tube assembly shall be randomly examined and inspected by hammer testing to verify the adequacy of the stud-to-tube weld.

13.5.4 POSITIVE MATERIALS IDENTIFICATION

13.5.4.1 Positive materials identification (PMI) is a procedure to ensure that specified metallic alloy materials are properly identified by their true elemental composition.

● **13.5.4.2** PMI program methods, degree of examination, PMI testing instruments, and tester qualifications shall be agreed upon between the purchaser and the vendor prior to manufacturing.

13.5.4.3 Unless superseded by the purchaser's requirements, 10 percent of all alloy components shall be PMI-tested. If random testing is done, PMI shall be made on components from different heat numbers. The purchaser may alternatively choose to specify that a PMI test be made on each component.

13.5.4.4 Tabulation of tested items shall be included within all final data books, keyed to weld maps on as-built drawings and mill certification document stampings. Tested items shall be immediately marked.

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APPENDIX A—EQUIPMENT DATA SHEETS

This appendix includes data sheets for the following equipment items:

a. Fired heater data sheets (6 sheets).
b. Burner data sheets (3 sheets).
c. Air preheater data sheets (2 sheets).
d. Fan data sheets (2 sheets).
e. Sootblower data sheets (1 sheet).

See 1.7 for instructions on using the equipment data sheets.

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PURCHASER / OWNER :				**ITEM NO. :**		
SERVICE :				**LOCATION :**		
1	UNIT :			* NUMBER REQUIRED :		REV
2	MANUFACTURER :			REFERENCE :		
3	TYPE OF HEATER :					
4	* TOTAL HEATER ABSORBED DUTY , MW :					
5	**PROCESS DESIGN CONDITIONS**					
6	* OPERATING CASE					
7	HEATER SECTION					
8	* SERVICE					
9	HEAT ABSORPTION, MW					
10	* FLUID					
11	* FLOW RATE, kg / s					
12	* FLOW RATE, B. P. D.					
13	* PRESSURE DROP, ALLOWABLE (CLEAN/FOULED), kPa					
14	PRESSURE DROP, CALCULATED (CLEAN / FOULED), kPa					
15	* AVG. RAD. SECT. FLUX DENSITY, ALLOW., W / m2					
16	AVG. RAD. SECT. FLUX DENSITY, CALC., W / m2					
17	MAX RAD. SECT. FLUX DENSITY, W / m2					
18	CONV. SECT. FLUX DENSITY (BARE TUBE), W / m2					
19	* VELOCITY LIMITATION, m / s					
20	PROCESS FLUID MASS VELOCITY, kg / s - m2					
21	* MAXIMUM ALLOW. / CALC. INSIDE FILM TEMPERATURE, C					
22	* FOULING FACTOR, m2 - K / W					
23	* COKING ALLOWANCE, mm					
24	**INLET CONDITIONS :**					
25	* TEMPERATURE, C					
26	* PRESSURE, (kPa.g) (kPa.abs)					
27	* LIQUID FLOW, kg / s					
28	* VAPOR FLOW, kg / s					
29	* LIQUID GRAVITY, (DEG API) (SP. GR. @ 15 C)					
30	* VAPOR MOLECULAR WEIGHT					
31	* VISCOSITY, (LIQUID / VAPOR), mPa.s					
32	* SPECIFIC HEAT, (LIQUID / VAPOR), kJ / kg - K					
33	* THERMAL CONDUCTIVITY, (LIQUID / VAPOR), W / m - C					
34	**OUTLET CONDITIONS:**					
35	* TEMPERATURE, C					
36	* PRESSURE, (kPa.g) (kPa.abs)					
37	* LIQUID FLOW, kg / s					
38	* VAPOR FLOW, kg / s					
39	* LIQUID GRAVITY, (DEG API) (SP. GR. @ 15 C)					
40	* VAPOR MOLECULAR WEIGHT					
41	* VISCOSITY, (LIQUID / VAPOR), mPa.s					
42	* SPECIFIC HEAT, (LIQUID / VAPOR), kJ / kg - C					
43	* THERMAL CONDUCTIVITY, (LIQUID / VAPOR), W / m - C					
44	**REMARKS AND SPECIAL REQUIREMENTS :**					
45	* DISTILLATION DATA OR FEED COMPOSITION:					
46	SHORT TERM CONDITIONS:					
47						
48	NOTES:					
49						
50						
51						
52						

FIRED HEATER DATA SHEET	**SI UNITS**		
API STD-560	REV:	DATE:	SHEET 1 OF 6

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COMBUSTION DESIGN CONDITIONS

					REV
1	OPERATING CASE				
2	* TYPE OF FUEL				
3	* EXCESS AIR, PERCENT				
4	CALCULATED HEAT RELEASE (LHV), MW				
5	FUEL EFFICIENCY CALCULATED, PERCENT (LHV)				
6	FUEL EFFICIENCY GUARANTEED, PERCENT (LHV)				
7	RADIATION LOSS, PERCENT OF HEAT RELEASE (LHV)				
8	FLUE GAS TEMPERATURE LEAVING: RADIANT SECTION, C.				
9	CONVECTION SECTION, C.				
10	AIR PREHEATER, C.				
11	FLUE GAS QUANTITY, kg / s				
12	FLUE GAS MASS VELOCITY THROUGH CONVECTION SECTION, kg / s - m2				
13	DRAFT: AT ARCH, Pa				
14	AT BURNERS, Pa				
15	* AMBIENT AIR TEMPERATURE, EFFICIENCY CALCULATION, C.				
16	* AMBIENT AIR TEMPERATURE, STACK DESIGN, C.				
17	* ALTITUTE ABOVE SEA LEVEL, m				
18	VOLUMETRIC HEAT RELEASE, (LHV), W / m3				
19	* REQUIRED EMISSIONS : PPMV (d) (CORRECTED TO 3% O2)	NOx :	CO :	SOx :	
20	kg / kJ (LHV) (HHV)	UHC :	PARTICULATES :		

FUEL CHARACTERISTICS :

22	* GAS TYPE		* LIQUID TYPE		* OTHER TYPE	
23	* LHV,	kJ / (kg) (Nm3)	* LHV,	kJ / kg	* LHV,	kJ / (kg) (Nm3)
24	* HHV,	kJ / (kg) (Nm3)	* HHV,	kJ / kg	* HHV,	kJ / (kg) (Nm3)
25	* PRESS. @ BURNER,	kPa.g	* PRESS @ BURNER,	kPa.g	* PRESS. @ BURNER,	kPa.g
26	* TEMP. @ BURNER,	C	* TEMP. @ BURNER,	C	* TEMP. @ BURNER,	C
27	* MOLECULAR WEIGHT		* VISCOSITY @	C, SSU		
28			* ATOMIZING STEAM TEMPERATURE,	C		
29			* PRESSURE,	kPa.g		

30	COMPOSITION	MOLE %	COMPOSITION	WT %	COMPOSITION	%
31						
32						
33						
34			* VANADIUM (PPM)			
35			* SODIUM (PPM)			
36			* SULFUR			
37			* ASH			

BURNER DATA :

39	MANUFACTURER:	SIZE / MODEL NO :		NUMBER :
40	TYPE:	LOCATION :		ORIENTATION :
41	HEAT RELEASE PER BURNER, MW	DESIGN:	NORMAL :	MINIMUM :
42	PRESSURE DROP ACROSS BURNER @ DESIGN HEAT RELEASE, Pa			
43	DISTANCE BURNR CENTER LINE TO TUBE CENTER LINE, HORIZONTAL, mm :			VERTICAL, mm :
44	DISTANCE BURNER CENTER LINE TO UNSHIELDED REFRACTORY, HORIZONTAL, mm :			VERTICAL , mm :
45	* PILOT, TYPE :	CAPACITY, MW:		FUEL :
46	* IGNITION METHOD :			
47	* FLAME DETECTION, TYPE:		NUMBER :	
48	NOTES :			
49				
50				
51				
52				

FIRED HEATER DATA SHEET	SI UNITS		
API STD-560	REV:	DATE:	SHEET 2 OF 6

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MECHANICAL DESIGN CONDITIONS

1	* PLOT LIMITATIONS :		* STACK LIMITATIONS:			REV
2	* TUBE LIMITATIONS :		* NOISE LIMITATIONS:			
3	* STRUCTURAL DESIGN DATA : WIND VELOCITY :		* WIND OCCURANCE:			
4	SNOW LOAD :		* SEISMIC ZONE:			
5	* MINIMUM / NORMAL / MAXIMUM AMBIENT AIR TEMPERATURE, C :		* RELATIVE HUMIDITY, %			
6	HEATER SECTION :					
7	SERVICE :					
8	**COIL DESIGN :**					
9	* DESIGN BASIS: TUBE WALL THICKNESS (CODE OR SPECIFICATION)					
10	RUPTURE STRENGTH (MINIMUM OR AVERAGE)					
11	* STRESS-TO-RUPTURE BASIS, HR.					
12	* DESIGN PRESSURE, ELASTIC / RUPTURE, kPa. g					
13	* DESIGN FLUID TEMPERATURE, C.					
14	* TEMPERATURE ALLOWANCE, C.					
15	CORROSION ALLOWANCE, TUBES / FITTINGS, mm					
16	HYDROSTATIC TEST PRESSURE, kPa.g					
17	* POSTWELD HEAT TREATMENT (YES OR NO)					
18	* PERCENT OF WELDS FULLY RADIOGRAPHED					
19	MAXIMUM (CLEAN) TUBE METAL TEMPERATURE, C.					
20	DESIGN TUBE METAL TEMPERATURE, C.					
21	INSIDE FILM COEFFICIENT, W / m2 - C					
22	**COIL ARRANGEMENT :**					
23	TUBE ORIENTATION: VERTICAL OR HORIZONTAL					
24	* TUBE MATERIAL (ASTM SPECIFICATION AND GRADE)					
25	TUBE OUTSIDE DIAMETER, mm					
26	TUBE WALL THICKNESS, (MINIMUM) (AVERAGE), mm					
27	NUMBER OF FLOW PASSES					
28	NUMBER OF TUBES					
29	NUMBER OF TUBES PER ROW (CONVECTION SECTION)					
30	OVERALL TUBE LENGTH, m					
31	EFFECTIVE TUBE LENGTH, m					
32	BARE TUBES: NUMBER					
33	TOTAL EXPOSED SURFACE, m2					
34	EXTENDED SURFACE TUBES: NUMBER					
35	TOTAL EXPOSED SURFACE, m2					
36	TUBE LAYOUT (IN LINE OR STAGGERED)					
37	TUBE SPACING, CENT. TO CENT.: HORZ. X DIAG (OR VERT.)					
38	SPACING TUBE CENT. TO FURNACE WALL (MIN.), mm					
39	CORBELS (YES OR NO)					
40	CORBEL WIDTH, mm					
41	**DESCRIPTION OF EXTENDED SURFACE :**					
42	TYPE: (STUDS) (SERRATED FINS) (SOLID FINS)					
43	MATERIAL					
44	DIMENSIONS (HEIGHT X DIAMETER / THICKNESS), mm					
45	SPACING (FINS / mm) (STUDS / PLANE)					
46	MAXIMUM TIP TEMPERATURE, (CALCULATED), C.					
47	EXTENSION RATIO (TOTAL AREA / BARE AREA)					
48	**PLUG TYPE HEADERS :**					
49	* TYPE					
50	MATERIAL (ASTM SPECIFICATION AND GRADE)					
51	NOMINAL RATING					
52	* LOCATION (ONE OR BOTH ENDS)					
53	WELDED OR ROLLED JOINT					
54	NOTES :					
55						
56						
57						
58						

FIRED HEATER DATA SHEET	**SI UNITS**		
API STD-560	REV:	DATE:	SHEET 3 OF 6

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MECHANICAL DESIGN CONDITIONS (Continued)

						REV
1	HEATER SECTION					
2	SERVICE					
3	**RETURN BENDS :**					
4	TYPE					
5	MATERIAL (ASTM SPECIFICATION AND GRADE)					
6	NOMINAL RATING OR SCHEDULE					
7	* LOCATION (F. B. = FIRE BOX, H. B. = HEADER BOX)					
8	**TERMINALS AND / OR MANIFOLDS :**					
9	* TYPE (BEV. = BEVELED, MANIF. = MANIFOLD, FLG. = FLANGED)					
10	INLET : MATERIAL (ASTM SPECIFICATION AND GRADE)					
11	SIZE / SCHEDULE OR THICKNESS					
12	NUMBER OF TERMINALS					
13	FLANGE MATERIAL (ASTM SPECIFICATION AND GRADE)					
14	FLANGE SIZE AND RATING					
15	OUTLET : MATERIAL (ASTM SPECIFICATION AND GRADE)					
16	SIZE / SCHEDULE OR THICKNESS					
17	NUMBER OF TERMINALS					
18	FLANGE MATERIAL (ASTM SPECIFICATION AND GRADE)					
19	FLANGE SIZE AND RATING					
20	* MANIFOLD TO TUBE CONNECTION (WELDED, EXTRUDED, ETC.)					
21	MANIFOLD LOCATION (INSIDE OR OUTSIDE HEADER BOX)					
22	**CROSSOVERS :**					
23	* WELDED OR FLANGED					
24	* PIPE MATERIAL (ASTM SPECIFICATION AND GRADE)					
25	PIPE SIZE / SCHEDULE OR THICKNESS					
26	* FLANGE MATERIAL					
27	FLANGE SIZE / RATING					
28	* LOCATION (INTERNAL / EXTERNAL)					
29	FLUID TEMPERATURE, C.					
30	**TUBE SUPPORTS :**					
31	LOCATION (ENDS, TOP, BOTTOM)					
32	MATERIAL (ASTM SPECIFICATION AND GRADE)					
33	DESIGN METAL TEMPERATURE, C.					
34	THICKNESS, mm					
35	TYPE AND THICKNESS OF INSULATION, mm					
36	ANCHOR (MATERIAL AND TYPE)					
37	**INTERMEDIATE TUBE SUPPORTS :**					
38	MATERIAL (ASTM SPECIFICATION AND GRADE)					
39	DESIGN METAL TEMPERATURE, C.					
40	THICKNESS, mm					
41	SPACING, m					
42	**TUBE GUIDES :**					
43	LOCATION :					
44	MATERIAL :					
45	TYPE / SPACING :					
46	**HEADER BOXES :**					
47	LOCATION :			HINGED DOOR / BOLTED PANEL :		
48	CASING MATERIAL :			THICKNESS, mm :		
49	LINING MATERIAL :			THICKNESS, mm :		
50	ANCHOR (MATERIAL AND TYPE):					
51	NOTES :					
52						
53						
54						

FIRED HEATER DATA SHEET	**SI UNITS**		
API STD-560	REV:	DATE:	SHEET 4 OF 6

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MECHANICAL DESIGN CONDITIONS (continued)

					REV
1	**REFRACTORY DESIGN BASIS :**				
2	AMBIENT, C.:	WIND VELOCITY, m / s:		CASING TEMPERATURE, C.:	
3	**EXPOSED VERTICAL WALLS :**				
4	LINING THICKNESS, mm:	HOT FACE TEMPERATURE , DESIGN / CALCULATED , C.:			
5	WALL CONSTRUCTION:				
6					
7	ANCHOR (MATERIAL & TYPE):				
8	CASING MATERIAL:	THICKNESS, mm :		TEMPERATURE, C. :	
9	**SHIELDED VERTICAL WALLS :**				
10	LINING THICKNESS, mm:	HOT FACE TEMPERATURE , DESIGN / CALCULATED , C. :			
11	WALL CONSTRUCTION:				
12					
13	ANCHOR (MATERIAL & TYPE):				
14	CASING MATERIAL:	THICKNESS , mm :		TEMPERATURE, C. :	
15	**ARCH :**				
16	LINING THICKNESS, mm:	HOT FACE TEMPERATURE , DESIGN / CALCULATED , C. :			
17	WALL CONSTRUCTION:				
18					
19	ANCHOR (MATERIAL & TYPE):				
20	CASING MATERIAL:	THICKNESS , mm :		TEMPERATURE, C. :	
21	**FLOOR :**				
22	LINING THICKNESS, mm:	HOT FACE TEMPERATURE , DESIGN / CALCULATED , C. :			
23	FLOOR CONSTRUCTION:				
24					
25	CASING MATERIAL:	THICKNESS , mm :		TEMPERATURE, C. :	
26	MINIMUM FLOOR ELEVATION, m :	FREE SPACE BELOW PLENUM, m :			
27	**CONVECTION SECTION :**				
28	LINING THICKNESS, mm:	HOT FACE TEMPERATURE , DESIGN / CALCULATED , C. :			
29	WALL CONSTRUCTION:				
30					
31	ANCHOR (MATERIAL & TYPE):				
32	CASING MATERIAL:	THICKNESS , mm :		TEMPERATURE, C. :	
33	**INTERNAL WALL :**				
34	TYPE:	MATERIAL :			
35	DIMENSION, HEIGHT / WIDTH:				

36	**DUCTS :**	FLUE GAS			COMBUSTION AIR	
37	LOCATION :	BREECHING				
38	SIZE : m OR NET FREE AREA, m2					
39	CASING MATERIAL:					
40	CASING THICKNESS, mm:					
41	LINING: INTERNAL / EXTERNAL					
42	THICKNESS, mm					
43	MATERIAL					
44	ANCHOR (MATERIAL & TYPE)					
45	CASING TEMPERATURE, C.					

46	**PLENUM CHAMBER (AIR) :**		
47	CASING MATERIAL :	THICKNESS, mm :	SIZE, m :
48	LINING MATERIAL :		THICKNESS, mm :
49	ANCHOR (MATERIAL & TYPE) :		
50	NOTES :		
51			
52			

FIRED HEATER DATA SHEET		**SI UNITS**	
API STD-560	REV:	DATE:	SHEET 5 OF 6

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MECHANICAL DESIGN CONDITIONS (continued)

							REV
1	**STACK OR STACK STUB :**						
2	NUMBER :	SELF - SUPPORTED OR GUYED			LOCATION :		
3	CASING MATERIAL :	* CORROSION ALLOW., mm :			MINIMUM THICKNESS, mm :		
4	INSIDE METAL DIAMETER, m :	HEIGHT ABOVE GRADE, m :			STACK LENGTH, m :		
5	LINING MATERIAL :				THICKNESS, mm :		
6	ANCHOR (MATERIAL AND TYPE):						
7	EXTENT OF LINING:	INTERNAL OR EXTERNAL:					
8	DESIGN FLUE GAS VELOCITY, m/s	FLUE GAS TEMP., C. :					
9	**DAMPERS :**						
10	LOCATION						
11	TYPE (CONTROL, TIGHT SHUT-OFF, ETC.)						
12	MATERIAL: BLADE						
13	SHAFT						
14	MULTIPLE / SINGLE LEAF						
15	PROVISION FOR OPERATION (MANUAL OR AUTOMATIC)						
16	TYPE OF OPERATOR (CABLE OR PNEUMATIC)						
17	**MISCELLANEOUS :**						
18	PLATFORMS : LOCATION	NUMBER	WIDTH	LENGTH / ARC	STAIRS/LADDER	ACCESS FROM	
19							
20							
21							
22							
23							
24	TYPE OF FLOORING :						
25	DOORS :		NUMBER	LOCATION	SIZE	BOLTED / HINGED	
26	ACCESS						
27							
28	OBSERVATION						
29							
30	TUBE REMOVAL						
31							
32	INSTRUMENT CONNECTIONS :			NUMBER	SIZE	TYPE	
33	FLUE GAS / COMBUSTION AIR TEMPERATURE						
34	FLUE GAS / COMBUSTION AIR PRESSURE						
35	FLUE GAS SAMPLE						
36	SNUFFING STEAM / PURGE						
37	O2 ANALYZER						
38	CO OR NOx ANALYZER						
39	VENTS / DRAINS						
40	PROCESS FLUID TEMPERATURE						
41	TUBESKIN THERMOCOUPLES						
42							
43							
44	PAINTING REQUIREMENTS :						
45	INTERNAL COATING :						
46	GALVANIZING REQUIREMENTS :						
47	ARE PAINTERS TROLLEY AND RAIL INCLUDED :						
48	SPECIAL EQUIPMENT: SOOTBLOWERS :						
49	AIR PREHEATER :						
50	FAN (S) :						
51	OTHER :						
52	NOTES :						
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FIRED HEATER DATA SHEET	**SI UNITS**		
API STD-560	REV:	DATE:	SHEET 6 OF 6

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PURCHASER / OWNER :		**ITEM NO. :**	
SERVICE :		**LOCATION:**	
1	**GENERAL DATA :**		REV
2	TYPE OF HEATER		
3	ALTITUDE ABOVE SEA LEVEL, m		
4	AIR SUPPLY :		
5	AMBIENT / PREHEATED AIR / GAS TURBINE EXHAUST		
6	TEMPERATURE, C. (MIN./MAX./DESIGN)		
7	RELATIVE HUMIDITY, %		
8	DRAFT TYPE: FORCED / NATURAL / INDUCED		
9	DRAFT AVAILABLE, Pa : ACROSS BURNER		
10	ACROSS PLENUM		
11	REQUIRED TURNDOWN		
12	BURNER WALL LINING THICKNESS, mm		
13	HEATER CASING THICKNESS, mm		
14	FIREBOX HEIGHT, m		
15	TUBE CIRCLE DIAMETER, m		
16	**BURNER DATA :**		
17	MANUFACTURER		
18	TYPE OF BURNER		
19	MODEL / SIZE		
20	DIRECTION OF FIRING		
21	LOCATION (ROOF / FLOOR / SIDEWALL)		
22	NUMBER REQUIRED		
23	MINIMUM DISTANCE BURNER CENTERLINE; m:		
24	TO TUBE CENTERLINE (HORIZONTAL / VERTICAL)		
25	TO ADJACENT BURNER CENTERLINE (HORIZONTAL / VERTICAL)		
26	TO UNSHIELDED REFRACTORY (HORIZONTAL / VERTICAL)		
27	BURNER CIRCLE DIAMETER, m		
28	PILOTS :		
29	NUMBER REQUIRED		
30	TYPE		
31	IGNITION METHOD		
32	FUEL		
33	FUEL PRESSURE, kPa.g		
34	CAPACITY, MW		
35	**OPERATING DATA :**		
36	FUEL		
37	HEAT RELEASE PER BURNER, MW (LHV)		
38	DESIGN		
39	NORMAL		
40	MINIMUM		
41	EXCESS AIR @ DESIGN HEAT RELEASE, (%)		
42	AIR TEMPERATURE, C.		
43	DRAFT LOSS, Pa		
44	DESIGN		
45	NORMAL		
46	MINIMUM		
47	FUEL PRESSURE REQUIRED, kPa.g		
48	FLAME LENGTH @ DESIGN HEAT RELEASE, m		
49	ATOMIZING MEDIUM / OIL RATIO, kg/kg		
50	NOTES :		
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BURNER DATA SHEET		**SI UNITS**	
API STD-560	REV:	DATE:	SHEET 1 OF 3

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GAS FUEL CHARACTERISTICS

					REV
1	FUEL TYPE				
2	HEATING VALUE (LHV) , (kJ / kg) (kJ / Nm3)				
3	SPECIFIC GRAVIRTY (AIR = 1.0)				
4	MOLECULAR WEIGHT				
5	FUEL TEMPERATURE @ BURNER, C.				
6	FUEL PRESSURE ; AVAILABLE @ BURNER, kPa.g				
7	FUEL GAS COMPOSITION (MOLE %)				
8					
9					
10					
11					
12					
13					
14					
15					
16					
17					
18					
19					
20	TOTAL				

LIQUID FUEL CHARACTERISTICS

22	FUEL TYPE			
23	HEATING VALUE (LHV) , kJ / kg			
24	SPECIFIC GRAVITY / DEG. API			
25	H / C RATIO (BY WEIGHT)			
26	VISCOSITY, @ C. (SSU)			
27	@ C. (SSU)			
28	VANADIUM, ppm			
29	POTASSIUM, ppm			
30	SODIUM, ppm			
31	NICKEL, ppm			
32	FIXED NITROGEN, ppm			
33	SULFUR, % WT.			
34	ASH, % WT.			
35	WATER, % WT.			
36	DISTILLATION : ASTM INITIAL BOILING POINT, C.			
37	ASTM MID-POINT, C.			
38	ASTM END POINT, C.			
39	FUEL TEMPERATURE @ BURNER, C.			
40	FUEL PRESSURE AVAILABLE @ BURNER, kPa.g			
41	ATOMIZING MEDIUM : AIR / STEAM / MECHANICAL			
42	TEMPERATURE, C.			
43	PRESSURE, kPa.g			

44	NOTES :
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BURNER DATA SHEET	**SI UNITS**		
API STD-560	REV:	DATE:	SHEET 2 OF 3

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MISCELLANEOUS

				REV
1	BURNER PLENUM :	COMMON / INTEGRAL		
2		MATERIAL		
3		PLATE THICKNESS, mm		
4		INTERNAL INSULATION		
5	INLET AIR CONTROL :	DAMPER OR REGISTERS		
6		MODE OF OPERATION		
7		LEAKAGE, %		
8	BURNER TILE :	COMPOSITION		
9		MINIMUM SERVICE TEMPERATURE, C.		
10	NOISE SPECIFICATION			
11	ATTENUATION METHOD			
12	PAINTING REQUIREMENTS			
13	IGNITION PORT :	SIZE / NO.		
14	SIGHT PORT :	SIZE / NO.		
15	FLAME DETECTION :	TYPE		
16		NUMBER		
17	SCANNER CONNECTION : SIZE / NO.			
18	SAFETY INTERLOCK SYSTEM FOR ATOMIZING MEDIUM & OIL			
19	PERFORMANCE TEST REQUIRED (YES OR NO)			
20	**EMISSION REQUIREMENTS :**			
21	FIREBOX BRIDGEWALL TEMPERATURE, C.			
22	NOx	* PPMV (D) or kg / kJ (LHV) (HHV)		
23	CO	* PPMV (D) or kg / kJ (LHV) (HHV)		
24	UHC	* PPMV (D) or kg / kJ (LHV) (HHV)		
25	PARTICULATES	* PPMV (D) or kg / kJ (LHV) (HHV)		
26	SOx	* PPMV (D) or kg / kJ (LHV) (HHV)		
27				
28	* CORRECTED TO 3% O2 (DRY BASIS @ DESIGN HEAT RELEASE)			
29	NOTES :			
30	1. AT DESIGN CONDITIONS, MINIMUM OF 90% OF THE AVAILABLE DRAFT WITH AIR REGISTER FULLY OPEN SHALL BE			
31	UTILIZED ACROSS THE BURNER. IN ADDITION, A MINIMUM OF 75% OF THE AIR SIDE PRESSURE DROP WITH AIR			
32	REGISTERS FULL OPEN SHALL BE UTILIZED ACROSS BURNER THROAT.			
33	2. VENDOR TO GUARANTEE BURNER FLAME LENGTH.			
34	3. VENDOR TO GUARANTEE EXCESS AIR, HEAT RELEASE AND DRAFT LOSS ACROSS BURNER.			
35				
36				
37				
38				
39				
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BURNER DATA SHEET	**SI UNITS**		
API STD-560	REV:	DATE:	SHEET 3 OF 3

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PURCHASER / OWNER :				**ITEM NO. :**			
SERVICE :				**LOCATION :**			
1	MANUFACTURER:						REV
2	MODEL:						
3	NUMBER REQUIRED:						
4	HEATING SURFACE, m2						
5	WEIGHT, kg						
6	APPROXIMATE DIMENSIONS: (H x W x L) m						
7	**PERFORMANCE DATA**						
8	OPERATING CASE						
9							
10	AIR SIDE: FLOW ENTERING, kg / s						
11	INLET TEMPERATURE, C.						
12	OUTLET TEMPERATURE, C.						
13	PRESSURE DROP: ALLOWABLE, mm H20						
14	CALCULATED, mm H20						
15	HEAT ABSORBED, MW						
16	FLUE GAS SIDE: FLOW, kg / s						
17	INLET TEMPERATURE, C.						
18	OUTLET TEMPERATURE, C.						
19	PRESSURE DROP: ALLOWABLE, mm H20						
20	CALCULATED, mm H20						
21	HEAT EXCHANGED, MW						
22	AIR BY-PASS, kg / s						
23	TOTAL AIR FLOW TO BURNERS, kg / s						
24	MIX AIR TEMPERATURE, C.						
25	FLUE GAS COMPOSITION, MOLE % (O2 / N2 / H2O / CO2 / SOx)						
26	FLUE GAS SPECIFIC HEAT, kJ / kg - C						
27	FLUE GAS ACID DEW POINT TEMPERATURE, C.						
28	MINIMUM METAL TEMPERATURE: ALLOWABLE, C.						
29	CALCULATED, C.						
30	**MISCELLANEOUS :**						
31	MINIMUM AMBIENT AIR TEMPERATURE, C.						
32	SITE ELEVATION ABOVE SEA LEVEL, m						
33	RELATIVE HUMIDITY						
34	EXTERNAL COLD AIR BY-PASS (YES / NO)						
35	COLD END THERMOCOUPLES (YES / NO) / NUMBER REQUIRED						
36	ACCESS DOORS : NUMBER / SIZE / LOCATION						
37	INSULATION (INTERNAL / EXTERNAL) :						
38	CLEANING MEDIUM : STEAM OR WATER						
39	PRESSURE, kPa.g						
40	TEMPERATURE, C.						
41							
42	**MECHANICAL DESIGN :**						
43	DESIGN FLUE GAS TEMPERATURE, C.						
44	DESIGN PRESSURE DIFFERENTIAL, mm H2O						
45	SEISMIC FACTOR						
46	PAINTING REQUIREMENTS						
47	LEAK TEST						
48	STRUCTURAL WIND LOAD, kg / m2						
49	AIR LEAKAGE (GUARANTEED), %						
50							
51	NOTES : (ALL DATA ON PER UNIT BASIS)						
52							
53							
54							

AIR PREHEATER DATA SHEET		**SI UNITS**	
API STD-560	REV:	DATE:	SHEET 1 OF 2

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CONSTRUCTION DATA

				REV
1	I. CAST IRON :			
2	NUMBER OF PASSES			
3	NUMBER OF TUBES PER BLOCK			
4	NUMBER OF BLOCKS			
5	TYPE OF SURFACE			
6	TUBE MATERIAL			
7	TUBE OF THICKNESS, mm			
8	GLASS BLOCK (YES / NO)			
9	NUMBER OF GLASS TUBES			
10	AIR CROSS-OVER DUCT: NUMBER			
11	BOLTED / WELDED			
12	SUPPLIED WITH CLIPS			
13	WATER WASH: YES / NO			
14	TYPE (OFF-LINE OR ON-LINE)			
15	LOCATION			
16				
17	II. PLATE TYPE :			
18	NUMBER OF PASSES			
19	NUMBER OF PLATES PER BLOCK			
20	NUMBER OF BLOCKS			
21	PLATE THICKNESS, mm			
22	WIDTH OF AIR CHANNEL, mm			
23	WIDTH OF FLUE GAS CHANNEL, mm			
24	AIR SIDE RIB PITCH, mm			
25	FLUE GAS SIDE RIB PITCH, mm			
26	MATERIAL: PLATE			
27	RIB			
28	FRAME			
29	AIR CROSS-OVER DUCT : NUMBER			
30	BOLTED / WELDED			
31	SUPPLIED WITH CLIPS			
32	WATER WASH: YES / NO			
33	TYPE (OFF-LINE OR ON-LINE)			
34	LOCATION			
35				
36	III. HEAT PIPE :			
37	NUMBER OF TUBES			
38	TUBES OD / WALL THICKNESS, mm			
39	TUBE MATERIAL			
40	TUBES PER ROW			
41	NUMBER OF ROWS			
42	TUBE PITCH (SQUARE / TRIANGULAR), mm			
43		AIR SIDE	GAS SIDE	
44	FINS : TYPE			
45	HEIGHT x THICKNESS x NO. / m			
46	MATERIAL			
47	EFFECTIVE LENGTH, m			
48	HEATING SURFACE, m2			
49	MAXIMUM ALLOWABLE SOAK TEMPERATURE, C.			
50	SOOTBLOWER: YES / NO			
51	TYPE			
52	LOCATION			
53	NOTES :			
54				
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56				
57				

AIR PREHEATER DATA SHEET		SI UNITS	
API STD-560	REV:	DATE:	SHEET 2 OF 2

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PURCHASER / OWNER :					**ITEM NO. :**				
SERVICE :					**LOCATION :**				
1	**FAN MANUFACTURER :**				**MODEL / SIZE :**		**ARRANGEMENT :**		REV
2	**SERVICE :**				**NO. REQUIRED :**				
3	**DRIVE SYSTEM :**				**FAN ROTATION FROM DRIVEN END :**		☐ CW	☐ CCW	
4	**GAS HANDLED :**				**MOLECULAR WEIGHT :**				
5	**SITE ELEVATION, m :**				**FAN LOCATION :**				
6	**OPERATING CONDITIONS :**								
7	**OPERATING CONDITION / CASE :**				**DESIGN**	**TEST BLOCK**	**OTHER CONDITIONS**		
8	CAPACITY, kg / s								
9	CAPACITY, Am3 / s								
10	DENSITY, kg / m3								
11	TEMPERATURE , C.								
12	RELATIVE HUMIDITY, %								
13	STATIC PRESSURE @ INLET, mm H2O.								
14	STATIC PRESSURE @ OUTLET, mm H2O.								
15	**PERFORMANCE :**								
16	kw @ TEMPERATURE (ALL LOSSES INCLUDED)								
17	FAN SPEED, RPM								
18	STATIC PRESSURE RISE ACROSS FAN, mm H2O								
19	INLET DAMPER / VANE POSITION								
20	DISCHARGE DAMPER POSITION								
21	FAN STATIC EFFICIENCY, %								
22	STEAM RATE, kg / kwh (TURBINE ONLY)								
23	**FAN CONTROL :**				DRIVE				
24	AIR SUPPLY				MAKE			TYPE	
25	FAN CONTROL FURNISHED BY				RATED kw			RPM	
26	METHOD	☐	INLET DAMPER	☐ OUTLET DAMPER	ELECTRICAL AREA CLASSIFICATION				
27		☐	INLET GUIDE VANES	☐ VARIABLE SPEED	CLASS	GROUP		DIVISION	
28	STARTING METHOD				POWER	VOLTS		PH	Hz
29	**CONSTRUCTION FEATURES :**								
30	**HOUSING :**				**BEARINGS :**				
31	MATERIAL	THICKNESS, mm			☐ HYDRODYNAMIC		☐ ANTI - FRICTION		
32	SPLIT FOR WHEEL REMOVAL	YES	NO		TYPE				
33	DRAINS, NO / SIZE				LUBRICATION				
34	ACCESS DOORS, NO / SIZE				COOLANT REQUIRED	m3 / s WATER @	C		
35	**BLADES :**				THERMOSTATICALLY CONTROLLED HEATERS		☐ YES	☐ NO	
36	TYPE				TEMPERATURE DETECTORS		☐ YES	☐ NO	
37	NO	THICKNESS, mm			VIBRATION DETECTORS		☐ YES	☐ NO	
38	MATERIAL								
39	**HUB :**				**SPEED DETECTORS :**				
40	☐	☐ SHRINK FIT	☐ KEYED		☐	☐ NON - CONTACT PROBE			
41	MATERIAL				☐	☐ SPEED SWITCH			
42	**SHAFT :**				☐	☐ OTHER			
43	MATERIAL				**COUPLINGS :**				
44	DIAMETER @ BRGS., mm				TYPE				
45	**SHAFT SLEEVES :**				MAKE			MODEL	
46	MATERIAL				SERVICE FACTOR				
47	**SHAFT SEALS :**				MOUNT COUPLING HALVES				
48	TYPE				☐	☐ FAN			
49					☐	☐ DRIVER			
50	**WR2, kg - m2**				SPACER	☐ YES	☐ NO	LENGTH, mm	
51	NOTES								
52									
53									

FAN DATA SHEET		**SI UNITS**	
API STD-560	REV:	DATE:	**SHEET 1 OF 2**

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CONSTRUCTION FEATURES (CONTINUED)

MISCELLANEOUS : REV

☐	COMMON BASEPLATE (FAN, DRIVER)	☐ SILENCER (INLET) (OUTLET)	☐ INLET (SCREEN) (FILTER)
☐	BEARING PEDESTALS / SOLEPLATES	☐ EVASE	☐ HOUSING DRAIN CONNECTION
☐	PERFORMANCE CURVES	☐ VIBRATION ISOLATION	☐ SPARK RESISTANT COUPLING GUARD
☐	SECTIONAL DRAWING	☐ TYPE	☐ INSULATION CLIPS
☐	OUTLINE DRAWING	☐ SPECIAL COATINGS	☐ INSPECTION ACCESS
☐	INLET BOXES	☐ CONTROL PANEL	☐ HEAT SHIELDS

NOISE ATTENUATION :

MAXIMUM ALLOWABLE SOUND PRESSURE LEVEL DBA @ m

PREDICTED SOUND PRESSURE LEVEL DBA @ m

ATTENUATION METHOD

FURNISHED BY

WEIGHTS, kg

FAN DRIVER BASE

SOUND TRUNK

EVASE

TOTAL SHIPPING WEIGHT

PAINTING :

☐ MANUFACTURERS STANDARD

SHIPMENT :

☐ DOMESTIC ☐ EXPORT ☐ EXPORT BOXING REQUIRED

ERECTION :

☐ ASSEMBLED

☐ PARTLY ASSEMBLED

☐ OUTDOOR STORAGE OVER 6 MONTHS

APPLICABLE SPECIFICATIONS :

CONNECTIONS :

	SIZE	RATING	ORIENTATION
INLET			
OUTLET			
DRAINS			

TESTS :

☐ MECHANICAL RUN IN (NO LOAD)

☐ WITNESSED PERFORMANCE

☐ ROTOR BALANCE

☐ SHOP INSPECTION

☐ ASSEMBLY AND FIT - UP CHECK

NOTES .

☐ ITEMS MARKED TO BE INCLUDED IN VENDOR SCOPE OF SUPPLY.

FAN DATA SHEET		**SI UNITS**	
API STD-560	REV	DATE:	**SHEET 2 OF 2**

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PURCHASER / OWNER :				**ITEM NO. :**		
SERVICE :				**LOCATION :**		
1	**OPERATING DATA :**					REV
2	FUEL OIL TYPE / SPECIFIC GRAVITY OR API.					
3	SULFUR, % WT.					
4	VANADIUM, PPM (WT.)					
5	NICKEL, PPM (WT.)					
6	ASH, % WT.					
7	LANE LOCATION					
8	FLUE GAS TEMPERATURE @ BLOWER, MAXIMUM C.					
9	FLUE GAS PRESSURE @ BLOWER, MAXIMUM C.					
10	BLOWING MEDIUM					
11	**UTILITY DATA :**					
12						
13	STEAM _____ kPa.g @ _____ C. _____ kg / s PER BLOWER					
14						
15	AIR _____ kPa.g _____ Nm3 / s PER BLOWER					
16						
17	POWER _____ VOLTS _____ PHASE _____ HZ					
18						
19	**LAYOUT DATA :**					
20	TUBE OUTSIDE DIAMETER, mm					
21	TUBE LENGTH, m					
22	TUBE SPACING, (STAG. / IN LINE), mm					
23	BANK WIDTH, m					
24	NO. OF INTERMEDIATE TUBE SHEETS					
25	LANE DIMENSION (MINIMUM CLEARANCE), mm					
26	MAXIMUM CLEANING RADIUS, m					
27	EXTENDED SURFACE TYPE					
28	NO. OF EXTENDED SURFACE ROWS					
29	LINING THICKNESS, mm					
30	**BLOWER DATA :**					
31	MANUFACTURER					
32	TYPE					
33	MODEL					
34	NUMBER REQUIRED					
35	NUMBER OF LANES (ROWS)					
36	NUMBER PER LANE					
37	ARRANGEMENT					
38	OPERATION					
39	CONTROL REQUIRED					
40	CONTROL PANEL LOCATION (LOCAL OR REMOTE)					
41	DRIVER TYPE (MANUAL, PNEUMATIC OR ELECTRICAL MOTOR)					
42	ELECTRICAL AREA CLASSIFICATION					
43	MOTOR STARTERS CLASSIFICATION					
44	MOTOR: kW					
45	ENCLOSURE					
46	RPM					
47	LANCE TRAVEL SPEED					
48	HEAD: MATERIAL & RATING					
49	WALL BOX ISOLATION					
50						
51						
52	NOTES:					
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SOOTBLOWER DATA SHEET	**SI UNITS**		
API STD-560	REV:	DATE:	SHEET 1 OF 1

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PURCHASER / OWNER :				**ITEM NO. :**		
SERVICE :				**LOCATION :**		
1	UNIT :			* NUMBER REQUIRED :		REV
2	MANUFACTURER :			REFERENCE :		
3	TYPE OF HEATER :					
4	* TOTAL HEATER ABSORBED DUTY , MM BTU / HR :					
5	**PROCESS DESIGN CONDITIONS**					
6	* OPERATING CASE					
7	HEATER SECTION					
8	* SERVICE					
9	HEAT ABSORPTION, MM BTU / HR.					
10	* FLUID					
11	* FLOW RATE, LB / HR.					
12	* FLOW RATE, B.P.D.					
13	* PRESSURE DROP, ALLOWABLE (CLEAN / FOULED), PSI.					
14	PRESSURE DROP, CALCULATED (CLEAN / FOULED), PSI.					
15	* AVG. RAD. SECT. FLUX DENSITY, ALLOW. BTU / HR - FT2.					
16	AVG. RAD. SECT. FLUX DENSITY, CALC., BTU / HR - FT2.					
17	MAX. RAD. SECT. FLUX DENSITY, BTU / HR - FT2.					
18	CONV. SECT. FLUX DENSITY, (BARE TUBE), BTU / HR - FT2					
19	* VELOCITY LIMITATION, FT/S.					
20	PROCESS FLUID MASS VELOCITY, LB / SEC - FT2.					
21	* MAXIMUM ALLOW. / CALC. INSIDE FILM TEMPERATURE, F.					
22	* FOULING FACTOR, HR - FT2 - F / BTU.					
23	* COKING ALLOWANCE, IN.					
24	**INLET CONDITIONS :**					
25	* TEMPERATURE, F.					
26	* PRESSURE, (PSIA) (PSIG)					
27	* LIQUID FLOW, LB / HR					
28	* VAPOR FLOW, LB / HR.					
29	* LIQUID GRAVITY, (DEG API) (SP GR @ 60 F.)					
30	* VAPOR MOLECULAR WEIGHT					
31	* VISCOSITY, (LIQUID / VAPOR), CP					
32	* SPECIFIC HEAT, (LIQUID / VAPOR), BTU / LB - F					
33	* THERMAL CONDUCTIVITY, (LIQUID / VAPOR), BTU / HR - FT F.					
34	**OUTLET CONDITIONS:**					
35	* TEMPERATURE, F.					
36	* PRESSURE , (PSIA) (PSIG)					
37	* LIQUID FLOW, LB / HR.					
38	* VAPOR FLOW, LB / HR.					
39	* LIQUID GRAVITY, (DEG API) (SP GR. @ 60 F.)					
40	* VAPOR MOLECULAR WEIGHT					
41	* VISCOSITY, (LIQUID / VAPOR), CP.					
42	* SPECIFIC HEAT, (LIQUID / VAPOR), BTU / LB - F					
43	* THERMAL CONDUCTIVITY, (LIQUID / VAPOR), BTU / HR - FT F					
44	**REMARKS AND SPECIAL REQUIREMENTS :**					
45	* DISTILLATION DATA OR FEED COMPOSITION:					
46	SHORT TERM OPERATING CONDITIONS:					
47						
48	NOTES					
49						
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51						
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FIRED HEATER DATA SHEET		**CUSTOMARY UNITS**	
API STD-560	REV:	DATE:	SHEET 1 OF 6

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COMBUSTION DESIGN CONDITIONS

					REV
1	OPERATING CASE				
2	* TYPE OF FUEL				
3	* EXCESS AIR, PERCENT,				
4	CALCULATED HEAT RELEASE (LHV), MM BTU / HR.				
5	FUEL EFFICIENCY CALCULATED, PERCENT (LHV)				
6	FUEL EFFICIENCY GUARANTEED, PERCENT (LHV)				
7	RADIATION LOSS, PERCENT OF HEAT RELEASE (LHV)				
8	FLUE GAS TEMPERATURE LEAVING: RADIANT SECTION, F.				
9	CONVECTION SECTION, F.				
10	AIR PREHEATER, F.				
11	FLUE GAS QUANTITY, LB / HR.				
12	FLUE GAS MASS VELOCITY THROUGH CONVECTION SECTION, LB / SEC - FT 2				
13	DRAFT: AT ARCH, IN H20.				
14	AT BURNERS, IN H20.				
15	* AMBIENT AIR TEMPERATURE, EFFICIENCY CALCULATION, F.				
16	* AMBIENT AIR TEMPERATURE, STACK DESIGN, F.				
17	* ALTITUDE ABOVE SEA LEVEL, FT.				
18	VOLUMETRIC HEAT RELEASE (LHV), BTU / HR - FT3.				
19	* REQUIRED EMISSIONS : PPMV (d) (CORRECTED TO 3% O2)	NOx :	CO :	SOx :	
20	LB / MMBTU. (LHV) (HHV)	UHC :	PARTICULATES :		

FUEL CHARACTERISTICS :

	* GAS TYPE		* LIQUID TYPE		* OTHER TYPE	
22						
23	* LHV,	BTU / (LB) (SCF)	* LHV,	BTU / LB	* LHV,	BTU / (SCF) (LB)
24	* HHV,	BTU / (LB) (SCF)	* HHV,	BTU / LB	* HHV,	BTU / (SCF) (LB)
25	* PRESS. @ BURNER,	PSIG	* PRESS @ BURNER,	PSIG	* PRESS. @ BURNER,	PSIG
26	* TEMP. @ BURNER,	F	* TEMP. @ BURNER,	F	* TEMP. @ BURNER,	F
27	MOLECULAR WEIGHT		* VISCOSITY @	F, SSU		
28			* ATOMIZING STEAM TEMPERATURE,	F		
29			* PRESSURE, PSIG.	PSIG		

	COMPOSITION	MOLE %	COMPOSITION	WT %	COMPOSITION	%
30						
31						
32						
33						
34			* VANADIUM (PPM)			
35			* SODIUM (PPM)			
36			* SULFUR			
37			* ASH			

BURNER DATA :

39	MANUFACTURER:	SIZE / MODEL NO :		NUMBER :
40	TYPE:	LOCATION :		ORIENTATION :
41	HEAT RELEASE PER BURNER, MMBTU / HR.	DESIGN:	NORMAL :	MINIMUM :
42	PRESSURE DROP ACROSS BURNER @ DESIGN HEAT RELEASE, IN H20:			
43	DISTANCE BURNER CENTER LINE TO TUBE CENTER LINE, HORIZONTAL:			VERTICAL, IN. :
44	DISTANCE BURNER CENTER LINE TO UNSHIELDED REFRACTORY, HORIZONTAL, IN :			VERTICAL, IN. :
45	* PILOT, TYPE :	CAPACITY (BTU / HR) :		FUEL :
46	* IGNITION METHOD :			
47	* FLAME DETECTION, TYPE :		NUMBER :	
48	NOTES :			
49				
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FIRED HEATER DATA SHEET	**CUSTOMARY UNITS**		
API STD-560	REV:	DATE:	SHEET 2 OF 6

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MECHANICAL DESIGN CONDITIONS

						REV
1	* PLOT LIMITATIONS :		* STACK LIMITATIONS :			
2	* TUBE LIMITATIONS :		* NOISE LIMITATIONS :			
3	* STRUCTURAL DESIGN DATA : WIND VELOCITY :		* WIND OCCURANCE :			
4	SNOW LOAD :		* SEISMIC ZONE :			
5	* MINIMUM / NORMAL / MAXIMUM AMBIENT AIR TEMPERATURE, F :		* RELATIVE HUMIDITY, % :			
6	HEATER SECTION :					
7	SERVICE :					
8	**COIL DESIGN :**					
9	* DESIGN BASIS : TUBE WALL THICKNESS (CODE OR SPECIFICATION)					
10	RUPTURE STRENGTH (MINIMUM OR AVERAGE)					
11	* STRESS - TO - RUPTURE BASIS, HR.					
12	* DESIGN PRESSURE, ELASTIC / RUPTURE, PSIG.					
13	* DESIGN FLUID TEMPERATURE, F.					
14	* TEMPERATURE ALLOWANCE, F.					
15	CORROSION ALLOWANCE, TUBES / FITTINGS, IN.					
16	HYDROSTATIC TEST PRESSURE, PSIG.					
17	* POST WELD HEAT TREATMENT (YES OR NO)					
18	* PERCENT OF WELDS FULLY RADIOGRAPHED					
19	MAXIMUM (CLEAN) TUBE METAL TEMPERATURE, F.					
20	DESIGN TUBE METAL TEMPERATURE, F.					
21	INSIDE FILM COEFFICIENT, BTU / HR FT2 - F.					
22	**COIL ARRANGEMENT :**					
23	TUBE ORIENTATION : VERTICAL OR HORIZONTAL					
24	* TUBE MATERIAL (ASTM SPECIFICATION AND GRADE)					
25	TUBE OUTSIDE DIAMETER, IN.					
26	TUBE WALL THICKNESS, (MINIMUM) (AVERAGE), IN.					
27	NUMBER OF FLOW PASSES					
28	NUMBER OF TUBES					
29	NUMBER OF TUBES PER ROW (CONVECTION SECTION)					
30	OVERALL TUBE LENGTH, FT.					
31	EFFECTIVE TUBE LENGTH, F.					
32	BARE TUBES : NUMBER					
33	TOTAL EXPOSED SURFACE, FT2.					
34	EXTENDED SURFACE TUBES : NUMBER					
35	TOTAL EXPOSED SURFACE, FT2.					
36	TUBE LAYOUT (IN LINE OR STAGGERED)					
37	TUBE SPACING, CENT. TO CENT.: HORZ. X DIAG. (OR VERT.)					
38	SPACING TUBE CENT. TO FURNACE WALL, (MIN.), IN.					
39	CORBELS (YES OR NO)					
38	CORBEL WIDTH, IN.					
41	**DESCRIPTION OF EXTENDED SURFACE :**					
42	TYPE : (STUDS) (SERRATED FINS) (SOLID FINS)					
43	MATERIAL					
44	DIMENSIONS (HEIGHT X DIAMETER / THICKNESS), IN.					
45	SPACING (FINS / IN) (STUDS / PLANE)					
46	MAXIMUM TIP TEMPERATURE, (CALCULATED), F.					
47	EXTENSION RATIO (TOTAL AREA / BARE AREA)					
48	**PLUG TYPE HEADERS :**					
49	* TYPE					
50	MATERIAL (ASTM SPECIFICATION AND GRADE)					
51	NOMINAL RATING					
52	* LOCATION (ONE OR BOTH ENDS)					
53	WELDED OR ROLLED JOINT					
54	NOTES:					
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FIRED HEATER DATA SHEET	**CUSTOMARY UNITS**		
API STD-560	REV:	DATE:	SHEET 3 OF 6

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MECHANICAL DESIGN CONDITIONS (Continued)

						REV
1	HEATER SECTION					
2	SERVICE					
3	**RETURN BENDS :**					
4	TYPE					
5	MATERIAL (ASTM SPECIFICATION AND GRADE)					
6	NOMINAL RATING OR SCHEDULE					
7	* LOCATION (F. B. = FIRE BOX, H. B. = HEADER BOX)					
8	**TERMINALS AND / OR MANIFOLDS :**					
9	* TYPE (BEV. = BEVELED, MANIF. = MANIFOLD, FLG. = FLANGED)					
10	INLET : MATERIAL (ASTM SPECIFICATION AND GRADE)					
11	SIZE / SCHEDULE OR THICKNESS					
12	NUMBER OF TERMINALS					
13	FLANGE MATERIAL (ASTM SPECIFICATION AND GRADE)					
14	FLANGE SIZE AND RATING					
15	OUTLET : MATERIAL (ASTM SPECIFICATION AND GRADE)					
16	SIZE / SCHEDULE OR THICKNESS					
17	NUMBER OF TERMINALS					
18	FLANGE MATERIAL (ASTM SPECIFICATION AND GRADE)					
19	FLANGE SIZE AND RATING					
20	* MANIFOLD TO TUBE CONNECTION (WELDED, EXTRUDED, ETC.)					
21	MANIFOLD LOCATION (INSIDE OR OUTSIDE HEADER BOX)					
22	**CROSSOVERS :**					
23	* WELDED OR FLANGED					
24	* PIPE MATERIAL (ASTM SPECIFICATION AND GRADE)					
25	PIPE SIZE / SCHEDULE OR THICKNESS					
26	* FLANGE MATERIAL					
27	FLANGE SIZE / RATING					
28	* LOCATION (INTERNAL / EXTERNAL)					
29	FLUID TEMPERATURE, F.					
30	**TUBE SUPPORTS :**					
31	LOCATION (ENDS, TOP, BOTTOM)					
32	MATERIAL (ASTM SPECIFICATION AND GRADE)					
33	DESIGN METAL TEMPERATURE, F.					
34	THICKNESS, IN.					
35	TYPE AND THICKNESS OF INSULATION, IN.					
36	ANCHOR (MATERIAL AND TYPE)					
37	**INTERMEDIATE TUBE SUPPORTS :**					
38	MATERIAL (ASTM SPECIFICATION AND GRADE)					
39	DESIGN METAL TEMPERATURE, F.					
40	THICKNESS, IN.					
41	SPACING, FT.					
42	**TUBE GUIDES :**					
43	LOCATION					
44	MATERIAL					
45	TYPE / SPACING					
46	**HEADER BOXES :**					
47	LOCATION ;			HINGED DOOR / BOLTED PANEL :		
48	CASING MATERIAL :			THICKNESS, IN. :		
49	LINING MATERIAL :			THICKNESS, IN. :		
50	ANCHOR (MATERIAL AND TYPE):					
51	NOTES :					
52						
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FIRED HEATER DATA SHEET	**CUSTOMARY UNITS**		
API STD-560	REV:	DATE:	SHEET 4 OF 6

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MECHANICAL DESIGN CONDITIONS (continued)

				REV
1	**REFRACTORY DESIGN BASIS :**			
2	AMBIENT, F:	WIND VELOCITY, MPH / FPS :	CASING TEMPERATURE, F :	
3	**EXPOSED VERTICAL WALLS :**			
4	LINING THICKNESS, IN :	HOT FACE TEMPERATURE , DESIGN / CALCULATED , F :		
5	WALL CONSTRUCTION :			
6				
7	ANCHOR (MATERIAL & TYPE) :			
8	CASING MATERIAL :	THICKNESS, IN :	TEMPERATURE, F :	
9	**SHIELDED VERTICAL WALLS :**			
10	LINING THICKNESS, IN :	HOT FACE TEMPERATURE , DESIGN / CALCULATED , F :		
11	WALL CONSTRUCTION :			
12				
13	ANCHOR (MATERIAL & TYPE) :			
14	CASING MATERIAL :	THICKNESS , IN :	TEMPERATURE, F :	
15	**ARCH :**			
16	LINING THICKNESS, IN :	HOT FACE TEMPERATURE , DESIGN / CALCULATED , F :		
17	WALL CONSTRUCTION :			
18				
19	ANCHOR (MATERIAL & TYPE) :			
20	CASING MATERIAL :	THICKNESS , IN :	TEMPERATURE, F :	
21	**FLOOR :**			
22	LINING THICKNESS, IN :	HOT FACE TEMPERATURE , DESIGN / CALCULATED , F :		
23	FLOOR CONSTRUCTION :			
24				
25	CASING MATERIAL :	THICKNESS , IN :	TEMPERATURE, F :	
26	MINIMUM FLOOR ELEVATION, FT. :	FREE SPACE BELOW PLENUM, FT. :		
27	**CONVECTION SECTION :**			
28	LINING THICKNESS, IN :	HOT FACE TEMPERATURE , DESIGN / CALCULATED , F :		
29	WALL CONSTRUCTION :			
30				
31	ANCHOR (MATERIAL & TYPE) :			
32	CASING MATERIAL :	THICKNESS , IN :	TEMPERATURE, F :	
33	**INTERNAL WALL :**			
34	TYPE :	MATERIAL :		
35	DIMENSION, HEIGHT / WIDTH :			

36	**DUCTS :**	FLUE GAS			COMBUSTION AIR	
37	LOCATION	BREECHING				
38	SIZE, FT. OR NET FREE AREA, FT2					
39	CASING MATERIAL					
40	CASING THICKNESS, IN.					
41	LINING : INTERNAL / EXTERNAL					
42	THICKNESS, IN.					
43	MATERIAL					
44	ANCHOR (MATERIAL & TYPE)					
45	CASING TEMPERATURE, F.					

46	**PLENUM CHAMBER (AIR) :**			
47	CASING MATERIAL :	THICKNESS, IN :	SIZE, FT :	
48	LINING MATERIAL :		THICKNESS, IN :	
49	ANCHOR (MATERIAL & TYPE) :			
50	NOTES :			
51				
52				
53				

FIRED HEATER DATA SHEET	**CUSTOMARY UNITS**		
API STD-560	REV:	DATE:	SHEET 6 OF 6

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December 27, 1999 12:18:25

MECHANICAL DESIGN CONDITIONS (continued)

							REV
1	**STACK OR STACK STUB :**						
2	NUMBER :	SELF - SUPPORTED OR GUYED		LOCATION :			
3	CASING MATERIAL :	* CORROSION ALLOW., IN. :		MINIMUM THICKNESS, IN. :			
4	INSIDE METAL DIAMETER, FT. :	HEIGHT ABOVE GRADE, FT. :		STACK LENGTH, FT. :			
5	LINING MATERIAL :			THICKNESS, IN. :			
6	ANCHOR (MATERIAL AND TYPE):						
7	DESIGN FLUE GAS VELOCITY, FT / S :	FLUE GAS TEMP., F. :					
8	EXTENT OF LINING:	INTERNAL OR EXTERNAL:					
9	**DAMPERS :**						
10	LOCATION						
11	TYPE (CONTROL, TIGHT SHUT-OFF, ETC.)						
12	MATERIAL: BLADE						
13	SHAFT						
14	MULTIPLE / SINGLE LEAF						
15	PROVISION FOR OPERATION (MANUAL OR AUTOMATIC)						
16	TYPE OF OPERATOR (CABLE OR PNEUMATIC)						
17	**MISCELLANEOUS :**						
18	PLATFORMS : LOCATION	NUMBER	WIDTH	LENGTH / ARC	STAIRS/LADDER	ACCESS FROM	
19							
20							
21							
22							
23							
24	TYPE OF FLOORING :						
25	DOORS :	NUMBER	LOCATION	SIZE	BOLTED / HINGED		
26	ACCESS						
27							
28	OBSERVATION						
29							
30	TUBE REMOVAL						
31							
32	INSTRUMENT CONNECTIONS :	NUMBER	SIZE	TYPE			
33	FLUE GAS / COMBUSTION AIR TEMPERATURE						
34	FLUE GAS / COMBUSTION AIR PRESSURE						
35	FLUE GAS SAMPLE						
36	SNUFFING STEAM / PURGE						
37	O2 ANALYZER						
38	CO OR NOx ANALYZER						
39	VENTS / DRAINS						
40	PROCESS FLUID TEMPERATURE						
41	TUBESKIN THERMOCOUPLES						
42							
43							
44	PAINTING REQUIREMENTS :						
45	INTERNAL COATING :						
46	GALVANIZING REQUIREMENTS :						
47	ARE PAINTERS TROLLEY AND RAIL INCLUDED :						
48	SPECIAL EQUIPMENT: SOOTBLOWERS :						
49	AIR PREHEATER :						
50	FAN (S) :						
51	OTHER :						
52	NOTES :						
53							
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FIRED HEATER DATA SHEET	**CUSTOMARY UNITS**		
API STD-560	REV:	DATE:	SHEET 6 OF 6

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December 27, 1999 12:18:25

PURCHASER / OWNER :		**ITEM NO. :**	
SERVICE :		**LOCATION:**	
1	**GENERAL DATA :**		REV
2	TYPE OF HEATER		
3	ALTITUDE ABOVE SEA LEVEL, FT.		
4	AIR SUPPLY :		
5	AMBIENT / PREHEATED AIR / GAS TURBINE EXHAUST		
6	TEMPERATURE, F. (MIN./MAX./DESIGN)		
7	RELATIVE HUMIDITY, %		
8	DRAFT TYPE: FORCED / NATURAL / INDUCED		
9	DRAFT AVAILABLE, : ACROSS BURNER, IN H20		
10	ACROSS PLENUM, IN H20		
11	REQUIRED TURNDOWN		
12	BURNER WALL LINING THICKNESS, IN.		
13	HEATER CASING THICKNESS, IN.		
14	FIREBOX HEIGHT, FT.		
15	TUBE CIRCLE DIAMETER, FT.		
16	**BURNER DATA :**		
17	MANUFACTURER		
18	TYPE OF BURNER		
19	MODEL / SIZE		
20	DIRECTION OF FIRING		
21	LOCATION (ROOF / FLOOR / SIDEWALL)		
22	NUMBER REQUIRED		
23	MINIMUM DISTANCE BURNER CENTERLINE; FT:		
24	TO TUBE CENTERLINE (HORIZONTAL / VERTICAL)		
25	TO ADJACENT BURNER CENTERLINE (HORIZONTAL / VERTICAL)		
26	TO UNSHIELDED REFRACTORY (HORIZONTAL / VERTICAL)		
27	BURNER CIRCLE DIAMETER, FT.		
28	PILOTS :		
29	NUMBER REQUIRED		
30	TYPE		
31	IGNITION METHOD		
32	FUEL		
33	FUEL PRESSURE, PSIG.		
34	CAPACITY, MMBTU / HR.		
35	**OPERATING DATA :**		
36	FUEL		
37	HEAT RELEASE PER BURNER, MMBTU / HR. (LHV)		
38	DESIGN		
39	NORMAL		
40	MINIMUM		
41	EXCESS AIR @ DESIGN HEAT RELEASE, (%)		
42	AIR TEMPERATURE, F.		
43	DRAFT (AIR PRESSURE) LOSS, IN H20		
44	DESIGN		
45	NORMAL		
46	MINIMUM		
47	FUEL PRESSURE REQUIRED, PSIG		
48	FLAME LENGTH @ DESIGN HEAT RELEASE, FT.		
49	FLAME SHAPE (ROUND, FLAT, ETC.)		
50	ATOMIZING MEDIUM / OIL RATIO, LB/LB		
51	NOTES :		
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BURNER DATA SHEET		**CUSTOMARY UNITS**	
API STD-560	REV:	DATE:	SHEET 1 OF 3

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December 27, 1999 12:18:25

GAS FUEL CHARACTERISTICS

					REV
1	FUEL TYPE				
2	HEATING VALUE (LHV), (BTU / SCF) (BTU / LB)				
3	SPECIFIC GRAVITY (AIR = 1.0)				
4	MOLECULAR WEIGHT				
5	FUEL TEMPERATURE @ BURNER, F.				
6	FUEL PRESSURE AVAILABLE @ BURNER, PSIG.				
7	FUEL GAS COMPOSITION (MOLE %)				
8					
9					
10					
11					
12					
13					
14					
15					
16					
17					
18					
19					
20	TOTAL				

LIQUID FUEL CHARACTERISTICS

22	FUEL TYPE			
23	HEATING VALUE (LHV), BTU / LB.			
24	SPECIFIC GRAVITY / DEG. API			
25	H / C RATIO (BY WEIGHT)			
26	VISCOSITY, @ F. (SSU)			
27	@ F. (SSU)			
28	VANADIUM, PPM			
29	SODIUM, PPM			
30	POTASSIUM, PPM			
31	NICKEL, PPM			
32	FIXED NITROGEN, PPM			
33	SULFUR, % WT.			
34	ASH, % WT.			
35	WATER, % WT.			
36	DISTILLATION : ASTM INITIAL BOILING POINT, F.			
37	ASTM MID-POINT, F.			
38	ASTM END POINT, F.			
39	FUEL TEMPERATURE @ BURNER, F.			
40	FUEL PRESSURE AVAILABLE @ BURNER, PSIG.			
41	ATOMIZING MEDIUM : AIR / STEAM / MECHANICAL			
42	TEMPERATURE, F.			
43	PRESSURE, PSIG.			

44	NOTES :
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BURNER DATA SHEET	**CUSTOMARY UNITS**		
API STD-560	REV:	DATE:	SHEET 2 OF 3

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December 27, 1999 12:18:25

MISCELLANEOUS

				REV
1	BURNER PLENUM :	COMMON / INTEGRAL		
2		MATERIAL		
3		PLATE THICKNESS, IN.		
4		INTERNAL INSULATION		
5	INLET AIR CONTROL :	DAMPER OR REGISTERS		
6		MODE OF OPERATION		
7		LEAKAGE, %		
8	BURNER TILE :	COMPOSITION		
9		MINIMUM SERVICE TEMPERATURE, F		
10	NOISE SPECIFICATION			
11	ATTENUATION METHOD			
12	PAINTING REQUIREMENTS			
13	IGNITION PORT :	SIZE / NO.		
14	SIGHT PORT :	SIZE / NO.		
15	FLAME DETECTION :	TYPE		
16		NUMBER		
17	SCANNER CONNECTION : SIZE / NO.			
18	SAFETY INTERLOCK SYSTEM FOR ATOMIZING MEDIUM & OIL			
19	PERFORMANCE TEST REQUIRED (YES OR NO)			
20	**EMISSION REQUIREMENTS :**			
21	FIREBOX BRIDGEWALL TEMPERATURE, F.			
22	NOx	* PPMV (D) or LB / MMBTU (LHV) (HHV)		
23	CO	* PPMV (D) or LB / MMBTU (LHV) (HHV)		
24	UHC	* PPMV (D) or LB / MMBTU (LHV) (HHV)		
25	PARTICULATES	* PPMV (D) or LB / MMBTU (LHV) (HHV)		
26	SOx	* PPMV (D) or LB / MMBTU (LHV) (HHV)		
27				
28	* CORRECTED TO 3% O2 (DRY BASIS @ DESIGN HEAT RELEASE)			
29	NOTES :			
30	1. AT DESIGN CONDITIONS, MINIMUM OF 90% OF THE AVAILABLE DRAFT WITH AIR REGISTER FULLY OPEN SHALL BE			
31	UTILIZED ACROSS THE BURNER. IN ADDITION, A MINIMUM OF 75% OF THE AIR SIDE PRESSURE DROP WITH AIR			
32	REGISTERS FULL OPEN SHALL BE UTILIZED ACROSS BURNER THROAT.			
33	2. VENDOR TO GUARANTEE BURNER FLAME LENGTH.			
34	3. VENDOR TO GUARANTEE EXCESS AIR, HEAT RELEASE AND DRAFT LOSS ACROSS BURNER.			
35				
36				
37				
38				
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BURNER DATA SHEET		**CUSTOMARY UNITS**	
API STD-560	REV:	DATE:	SHEET 3 OF 3

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December 27, 1999 12:18:25

PURCHASER / OWNER :				**ITEM NO. :**		
SERVICE :				**LOCATION :**		
1	MANUFACTURER :					REV
2	MODEL :					
3	NUMBER REQUIRED :					
4	HEATING SURFACE, FT2.					
5	WEIGHT, LBS.					
6	APPROXIMATE DIMENSIONS : (H x W x L), FT.					
7	**PERFORMANCE DATA**					
8	OPERATING CASE					
9						
10	AIR SIDE: FLOW ENTERING, LB/HR.					
11	INLET TEMPERATURE, F.					
12	OUTLET TEMPERATURE, F.					
13	PRESSURE DROP: ALLOWABLE, IN H20					
14	CALCULATED, IN H20					
15	HEAT ABSORBED, MMBTU/HR.					
16	FLUE GAS SIDE: FLOW, LB / HR.					
17	INLET TEMPERATURE, F.					
18	OUTLET TEMPERATURE, F.					
19	PRESSURE DROP: ALLOWABLE, IN H20					
20	CALCULATED, IN H20					
21	HEAT EXCHANGED, MMBTU / HR					
22	AIR BY-PASS, LB / HR.					
23	TOTAL AIR FLOW TO BURNERS, LB / HR.					
24	MIX AIR TEMPERATURE, F.					
25	FLUE GAS COMPOSITION, MOLE %. (O2 / N2 / H20 / CO2 / SOx)					
26	FLUE GAS SPECIFIC HEAT, BTU / LB - F.					
27	FLUE GAS ACID DEW POINT TEMPERATURE, F.					
28	MINIMUM METAL TEMPERATURE : ALLOWABLE, F.					
29	CALCULATED, F.					
30	**MISCELLANEOUS :**					
31	MINIMUM AMBIENT AIR TEMPERATURE, F.					
32	SITE ELEVATION ABOVE SEA LEVEL, FT.					
33	RELATIVE HUMIDITY					
34	EXTERNAL COLD AIR BY-PASS (YES / NO)					
35	COLD END THERMOCOUPLES (YES / NO) / NUMBER REQUIRED.					
36	ACCESS DOORS : NUMBER / SIZE / LOCATION					
37	INSULATION (INTERNAL / EXTERNAL) :					
38	CLEANING MEDIUM : STEAM OR WATER					
39	PRESSURE, PSIG.					
40	TEMPERATURE, F.					
41						
42	**MECHANICAL DESIGN :**					
43	DESIGN FLUE GAS TEMPERATURE, F.					
44	DESIGN PRESSURE DIFFERENTIAL, IN H2O					
45	SEISMIC FACTOR					
46	PAINTING REQUIREMENTS					
47	LEAK TEST					
48	STRUCTURAL WIND LOAD, PSF					
49	AIR LEAKAGE (GUARANTEED), %					
50						
51	NOTES : (ALL DATA ON PER UNIT BASIS)					
52						
53						
54						
55						

AIR PREHEATER DATA SHEET	**CUSTOMARY UNITS**		
API STD-560	REV:	DATE:	SHEET 1 OF 2

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December 27, 1999 12:18:25

CONSTRUCTION DATA

			REV
1	I. CAST IRON :		
2	NUMBER OF PASSES		
3	NUMBER OF TUBES PER BLOCK		
4	NUMBER OF BLOCKS		
5	TYPE OF SURFACE		
6	TUBE MATERIAL		
7	TUBE THICKNESS, IN.		
8	GLASS BLOCK (YES / NO)		
9	NUMBER OF GLASS TUBES		
10	AIR CROSS-OVER DUCT : NUMBER		
11	BOLTED / WELDED		
12	SUPPLIED WITH CLIPS		
13	WATER WASH : YES / NO		
14	TYPE (OFF-LINE OR ON-LINE)		
15	LOCATION		
16			
17	II. PLATE TYPE :		
18	NUMBER OF PASSES		
19	NUMBER OF PLATES PER BLOCK		
20	NUMBER OF BLOCKS		
21	PLATE THICKNESS, IN.		
22	WIDTH OF AIR CHANNEL, IN.		
23	WIDTH OF FLUE GAS CHANNEL, IN.		
24	AIR SIDE RIB PITCH, IN.		
25	FLUE GAS SIDE RIB PITCH, IN.		
26	MATERIAL : PLATE		
27	RIB		
28	FRAME		
29	AIR CROSS-OVER DUCT : NUMBER		
30	BOLTED / WELDED		
31	SUPPLIED WITH CLIPS		
32	WATER WASH : YES / NO		
33	TYPE (OFF-LINE OR ON-LINE)		
34	LOCATION		
35			
36	III. HEAT PIPE :		
37	NUMBER OF TUBES		
38	TUBE OD / WALL THICKNESS, IN.		
39	TUBE MATERIAL		
40	TUBES PER ROW		
41	NUMBER OF ROWS		
42	TUBE PITCH (SQUARE / TRIANGULAR), IN.		
43		AIR SIDE	GAS SIDE
44	FINS : TYPE		
45	HEIGHT x THICKNESS x NUMBER / IN.		
46	MATERIAL		
47	EFFECTIVE LENGTH, FT.		
48	HEATING SURFACE, FT.		
49	MAXIMUM ALLOWABLE SOAK TEMPERATURE, F.		
50	SOOTBLOWER : YES / NO		
51	TYPE		
52	LOCATION		
53	NOTES :		
54			
55			
56			
57			

AIR PREHEATER DATA SHEET	CUSTOMARY UNITS		
API STD-560	REV:	DATE:	SHEET 2 OF 2

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PURCHASER / OWNER :		**ITEM NO. :**		
SERVICE :		**LOCATION :**		
1	**FAN MANUFACTURER :**	**MODEL / SIZE :**	**ARRANGEMENT :**	REV
2	**SERVICE :**	**NO. REQUIRED :**		
3	**DRIVE SYSTEM :**	**FAN ROTATION FROM DRIVEN END :**	☐ CW ☐ CCW	
4	**GAS HANDLED :**	**MOLECULAR WEIGHT :**		
5	**SITE ELEVATION, FT. :**	**FAN LOCATION :**		

OPERATING CONDITIONS :

	OPERATING CONDITION / CASE :	DESIGN	TEST BLOCK	OTHER CONDITIONS	
8	CAPACITY, LB / HR				
9	CAPACITY, ACFM				
10	DENSITY, LB / FT3				
11	TEMPERATURE , F				
12	RELATIVE HUMIDITY, %				
13	STATIC PRESSURE @ INLET, IN H2O				
14	STATIC PRESSURE @ OUTLET, IN H2O				
15	**PERFORMANCE :**				
16	BHP @ TEMPERATURE (ALL LOSSES INCLUDED)				
17	FAN SPEED, RPM				
18	STATIC PRESSURE RISE ACROSS FAN, IN H2O				
19	INLET DAMPER / VANE POSITION				
20	DISCHARGE DAMPER POSITION				
21	FAN STATIC EFFICIENCY, %				
22	STEAM RATE, LB - HP / HR (TURBINE ONLY)				

23	**FAN CONTROL :**			DRIVE		
24	AIR SUPPLY			MAKE		TYPE
25	FAN CONTROL FURNISHED BY			RATED HP		RPM
26	METHOD	☐ INLET DAMPER	☐ OUTLET DAMPER	ELECTRICAL AREA CLASSIFICATION		
27		☐ INLET GUIDE VANES	☐ VARIABLE SPEED	CLASS	GROUP	DIVISION
28	STARTING METHOD			POWER	VOLTS	PH Hz

CONSTRUCTION FEATURES :

30	**HOUSING :**			**BEARINGS :**		
31	MATERIAL	THICKNESS , IN		☐ HYDRODYNAMIC	☐ ANTI - FRICTION	
32	SPLIT FOR WHEEL REMOVAL	YES	NO	TYPE		
33	DRAINS, NO / SIZE			LUBRICATION		
34	ACCESS DOORS, NO. / SIZE			COOLANT REQUIRED	GPM WATER @	F.
35	**BLADES :**			THERMOSTATICALLY CONTROLLED HEATERS	☐ YES	☐ NO
36	TYPE			TEMPERATURE DETECTORS	☐ YES	☐ NO
37	NO	THICKNESS, IN		VIBRATION DETECTORS	☐ YES	☐ NO
38	MATERIAL					
39	**HUB :**			**SPEED DETECTORS :**		
40	☐	☐ SHRINK FIT	☐ KEYED	☐	☐ NON - CONTACT PROBE	
41	MATERIAL			☐	☐ SPEED SWITCH	
42	**SHAFT :**			☐	☐ OTHER	
43	MATERIAL			**COUPLINGS :**		
44	DIAMETER @ BRGS , IN			TYPE		
45	**SHAFT SLEEVES :**			MAKE		MODEL
46	MATERIAL			SERVICE FACTOR		
47	**SHAFT SEALS :**			MOUNT COUPLING HALVES		
48	TYPE			☐	☐ FAN	
49				☐	☐ DRIVER	
50	**WR2, LB-FT2 :**			SPACER	☐ YES ☐ NO	LENGTH, IN.
51	NOTES (ALL DATA ON PER UNIT BASIS)					
52						
53						

FAN DATA SHEET		**CUSTOMARY UNITS**	
API STD-560	REV	DATE.	**SHEET 1 OF 2**

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CONSTRUCTION FEATURES (CONTINUED)

					REV
1	**MISCELLANEOUS :**				
2	☐ ☐ COMMON BASEPLATE (FAN, DRIVER)	☐ SILENCER (INLET) (OUTLET)	☐ INLET (SCREEN) (FILTER)		
3	☐ ☐ BEARING PEDESTALS / SOLEPLATES	☐ EVASE	☐ HOUSING DRAIN CONNECTION		
4	☐ ☐ PERFORMANCE CURVES	☐ VIBRATION ISOLATION	☐ SPARK RESISTANT COUPLING GUARD		
5	☐ ☐ SECTIONAL DRAWING	☐ TYPE	☐ INSULATION CLIPS		
6	☐ ☐ OUTLINE DRAWING	☐ SPECIAL COATINGS	☐ INSPECTION ACCESS		
7	☐ ☐ INLET BOXES	☐ CONTROL PANEL	☐ HEAT SHIELDS		
8	**NOISE ATTENUATION :**		**WEIGHTS, LB.**		
9	MAXIMUM ALLOWABLE SOUND PRESSURE LEVEL ____ DBA @ ____ FT		FAN ____ DRIVER ____ BASE		
10	PREDICTED SOUND PRESSURE LEVEL ____ DBA @ ____ FT		SOUND TRUNK		
11	ATTENUATION METHOD		EVASE		
12	FURNISHED BY		TOTAL SHIPPING WEIGHT		
13	**PAINTING :**		**CONNECTIONS :**		
14	☐ ☐ MANUFACTURERS STANDARD			**SIZE** / **RATING** / **ORIENTATION**	
15			INLET		
16	**SHIPMENT :**		OUTLET		
17	☐ ☐ DOMESTIC ☐ EXPORT ☐ EXPORT BOXING REQUIRED		DRAINS		
18					
19	**ERECTION :**				
20	☐ ☐ ASSEMBLED		**TESTS :**		
21	☐ ☐ PARTLY ASSEMBLED		☐ ☐ MECHANICAL RUN IN. (NO LOAD)		
22	☐ ☐ OUTDOOR STORAGE OVER 6 MONTHS		☐ ☐ WITNESSED PERFORMANCE		
23	**APPLICABLE SPECIFICATIONS :**		☐ ☐ ROTOR BALANCE		
24			☐ ☐ SHOP INSPECTION		
25			☐ ☐ ASSEMBLY AND FIT - UP CHECK		
26					
27					
28	NOTES				
29	☐ ☐ ITEMS MARKED TO BE INCLUDED IN VENDOR SCOPE OF SUPPLY.				
30					
31					
32					
33					
34					
35					
36					
37					
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51					
52					

FAN DATA SHEET		**CUSTOMARY UNITS**	
API STD-560	REV	DATE	**SHEET 2 OF 2**

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PURCHASER / OWNER :				**ITEM NO. :**		
SERVICE :				**LOCATION :**		
1	**OPERATING DATA :**					REV
2	FUEL OIL TYPE / SPECIFIC GRAVITY OR API.					
3	SULFUR, % WT.					
4	VANDADIUM, PPM (WT)					
5	NICKEL, PPM (WT.)					
6	ASH, % WT.					
7	LANE LOCATION					
8	FLUE GAS TEMPERATURE @ BLOWER, MAXIMUM F.					
9	FLUE GAS PRESSURE @ BLOWER, MAXIMUM F.					
10	BLOWING MEDIUM					
11	**UTILITY DATA :**					
12						
13	STEAM_____ PSIG @_____ F_____ LB / HR PER BLOWER					
14						
15	AIR_____ PSIG_____ SCFM PER BLOWER					
16						
17	POWER_____ VOLTS_____ PHASE_____HZ					
18						
19	**LAYOUT DATA :**					
20	TUBE OUTSIDE DIAMETER, IN.					
21	TUBE LENGTH, FT.					
22	TUBE SPACING, (STAG. / IN LINE), IN.					
23	BANK WIDTH, FT.					
24	NO. OF INTERMEDIATE TUBE SHEETS					
25	LANE DIMENSION (MINIMUM CLEARANCE), IN.					
26	MAXIMUM CLEANING RADIUS, FT.					
27	EXTENDED SURFACE TYPE					
28	NO. OF EXTENDED SURFACE ROWS					
29	LINING THICKNESS, IN.					
30	**BLOWER DATA :**					
31	MANUFACTURER					
32	TYPE					
33	MODEL					
34	NUMBER REQUIRED					
35	NUMBER OF LANES (ROWS)					
36	NUMBER PER LANE					
37	ARRANGEMENT					
38	OPERATION					
39	CONTROL REQUIRED					
40	CONTROL PANEL LOCATION (LOCAL OR REMOTE)					
41	DRIVER TYPE (MANUAL, PNEUMATIC OR ELECTRICAL MOTOR)					
42	ELECTRICAL AREA CLASSIFICATION					
43	MOTOR STARTERS CLASSIFICATION					
44	MOTOR : HP					
45	ENCLOSURE					
46	RPM					
47	LANCE TRAVEL SPEED					
48	HEAD : MATERIAL & RATING					
49	WALL BOX ISOLATION					
50						
51						
52	NOTES :					
53						
54						

SOOTBLOWER DATA SHEET	**CUSTOMARY UNITS**		
API STD-560	REV:	DATE:	SHEET 1 OF 1

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APPENDIX B—PURCHASER'S CHECKLIST

This checklist is to be used for indicating the purchaser's specific requirements when the standard provides a choice or designates that a decision must be made. These items are indicated by a bullet (●) in this standard. This completed checklist shall be considered part of the specification.

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PURCHASER'S CHECKLIST

Paragraph	Item	Requirement
1.3.1	Order of precedence of standards and specifications.	_____
1.6.1	Editions of reference publications applicable to this project.	_____
1.6.3	Applicable local rules and regulations.	_____
1.7.1.1	Special requirements or exceptions to API Standard 560.	_____
1.7.1.4	Number of copies of referenced drawings and data required.	_____
1.7.2, item e, 1.9, item f	Noise data sheets provided.	Yes No
1.9, item a	As-built data sheets and drawings required.	Yes No
2.2.2	Design excess air to be used as basis for calculated efficiencies.	Gas: _____ Liquid: _____
2.3.3	Sootblowers to be provided.	Yes No
2.3.8	Corbels acceptable. Baffles acceptable.	Yes No Yes No
3.2.1	Acceptable extended surface type: Studs Solid fins Segmented fins	 Yes No Yes No Yes No
3.2.3	Basis for extended surface dimensions.	Gas Oil
5.1.4	Inspection openings are required. If yes, terminal flanges are acceptable.	Yes No Yes No
5.1.6	Low-point drains shall be furnished by the vendor. High-point vents shall be furnished by the vendor.	Yes No Yes No
5.2	Alternative terminal loadings.	_____
6.3.2	Acceptable tube support corrosion protection: Refractory coating 50Cr-50Ni material	 Yes No Yes No
7.1.1	Alternative casing temperature design.	_____
7.4.17, 7.5.7	Specified internal protective coating.	_____
8.1.1, 12.1.4	Specify structural design code applicable to this project.	_____
8.1.2, 9.2.2	Code requirements for wind and earthquake loads.	_____
8.1.5, 9.2.8	Alternative ambient conditions for structural design.	_____
8.2.5	Locations for future platforms, ladders, and stairways.	_____
8.2.7	Allowance for fireproofing.	Yes No
8.3.1.2	Header box closures: Hinged doors Bolted panels	 Yes No Yes No
8.4.4	Platform decking requirements.	_____
8.5.2	Acceptable low temperature materials.	_____
9.1.10	Location of control dampers.	_____

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PURCHASER'S CHECKLIST—continued

Paragraph	Item	Requirement
9.5.4	Acceptable strake design: Staggered plates Spiral spoilers	 Yes No Yes No
9.6.1	Acceptable materials of construction: Stacks Ducts Breechings	 _____ _____ _____
9.6.2	Acceptable bolting materials.	_____
10.1.4	Alternative basis for burner sizing.	_____
10.1.5	Single burner with multiple guns is acceptable.	Yes No
10.1.6	Pilots for gas burners.	Yes No
10.1.14	Required heater capacity during forced draft outage and continued operation on natural draft.	_____ _____
10.1.19	On-stream removal of complete burner assembly is required.	Yes No
10.2.1	Acceptable sootblower type: Retractable Automatic Sequential	 Yes No Yes No Yes No
10.2.5	Acceptable erosion protection: Stainless shrouds High-duty brick Castable refractory	 Yes No Yes No Yes No
10.4.1	Damper position on failure.	Open Close
11.1.3.4	Additional flue gas sampling connections.	_____
11.2.1	Crossover thermowell connections required.	Yes No
11.2.2	Outlet thermowell connections required.	Yes No
11.3.2.2	Water washing is required: Radiant Convection	 Yes No Yes No
11.4	Tubeskin thermocouple requirements.	_____ _____ _____
12.1.1	Site receiving and handling limitations.	_____ _____
12.2.5.1	Charpy impact test requirements.	_____ _____
12.3.1	Acceptable referenced welding processes.	_____
12.3.3	Acceptable referenced purge gases. Root pass welding restrictions.	_____ _____
12.4.3	Acceptable bolt weather protection: Galvanizing Zinc-coating	 Yes No Yes No
12.5.8.5	Acceptable castable curing method: Resin membrane	 Yes No Yes No Yes No
13.1.4	Preinspection meetings required prior to the start of fabrication.	Yes No
13.3.1.1, 13.3.2	Prior approval of major defect repairs required. Criteria for rejection:	Yes No _____ _____ _____

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PURCHASER'S CHECKLIST—continued

Paragraph	Item	Requirement
13.3.1.2	Radiograph examination requirements:	_____ _____ _____
13.3.3.4	Acceptance criteria for machined weld bevels.	_____ _____
13.5.4.2	Positive materials identification requirements.	_____ _____
E.2.1.4	Electrical area classification for fired heater equipment/ system.	_____ _____
E.2.1.7	Weather and environmental requirements for outdoor installation.	_____ _____
E.2.1.15.2	Alternative requirement for air intake height above grade.	_____ _____
E.2.2.1	Corrision allowance required for fan scroll and housing.	Yes No _____
E.2.5.8	Corrosion-resistant shaft sleeves required for I.D. fans.	Yes No
E.2.7.4	Rotor response analysis required.	Yes No
E.2.8.3, E.4.2.1	Mechanical run test required required. Other tests?	Yes No _____
E.2.11.1.2, E.2.11.3	Environmental conditions of flue gas and air affecting fan material selections.	_____ _____ _____
E.3.1.1.1, E.3.1.1.2	Fan driver type and sizing basis.	_____ _____
E.3.3.1.2	Fan vendor is required to review overall control system for compatibility.	Yes No
E.3.3.3.2	Fan vendor to state maximum expected leakage through closed dampers and vanes.	Yes No
E.4.1.6	Shop fit-up and assembly of fan, drivers and other auxiliaries required prior to shipment. Hardness testing?	Yes No _____ _____
E.4.3.1	Equipment shall be special prepared for six months of outdoor storage.	Yes No

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APPENDIX C—PROPOSED SHOP ASSEMBLY CONDITIONS

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SHOP ASSEMBLY CONDITIONS

SERVICE _____ EQUIPMENT NO. _____
UNIT _____ PLANT LOCATION _____
TYPE _____ NO. REQUIRED _____
OWNER _____ REFERENCE NO. _____
PURCHASER _____ REFERENCE NO. _____
VENDOR _____ REFERENCE NO. _____
DATE _____ PAGE 1 OF _____

DEGREE OF ASSEMBLY

Complete Assembly (Number of Sections):	Radiant	Convection
Boxes:		
1. Refractory only		
2. With anchors only		
Panels:		
3. With tubes and refractory installed		
4. With refractory only		
5. With anchors only		
Coils:		
6. Number of coil assemblies		
7. Number of hairpins, canes, tubes		
8. Field welds, number/size		

Number of Pieces:	Lined	Unlined: With Anchors	Unlined: Without Anchors
9. Breeching			
10. Flue gas ducts			
11. Combustion air ducts			
12. Header boxes			
13. Plenum chamber			
14. Stack			

Installation:	Shop Installed	Field Installed
15. Tube supports		
16. Floor refractory		
17. Header boxes		
18. Plenum chambers		
19. Bridgewall		
20. Dampers		
21. Cages to ladders		
22. Platform flooring to framing		
23. Platform support clips to casing		
24. Handrails, midrails, and toe plates to posts		
25. Stair treads to stringers		
26. Doors		
27. Tube-skin thermocouples		
28. Internal coatings		
29. Burners		
30. Sootblowers		

Air Heater:

31. _____
32. _____
33. _____
34. _____
35. _____
36. _____
37. _____
38. _____
39. _____
40. _____

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SHOP ASSEMBLY CONDITIONS

SERVICE _____ EQUIPMENT NO. _____
UNIT _____ PLANT LOCATION _____
TYPE _____ NO. REQUIRED _____
OWNER _____ REFERENCE NO. _____
PURCHASER _____ REFERENCE NO. _____
VENDOR _____ REFERENCE NO. _____
DATE _____ PAGE 2 OF _____

DEGREE OF ASSEMBLY—continued

Fans:

1. _____
2. _____
3. _____

Drivers:

4. _____
5. _____
6. _____

Other:

7. _____
8. _____
9. _____

ESTIMATED SHIPPING WEIGHTS AND DIMENSIONS

10. Total heater weight, in tons _____
11. Total, ladders, stairs, platform, in tons _____
12. Total stack, in tons _____
13. Maximum radiant section, in tons _____
14. Maximum radiant section dimensions, length × width × height _____
15. Maximum convection section, in tons _____
16. Maximum convection section dimensions, length × width × height _____

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APPENDIX D—STRESS CURVES FOR USE IN THE DESIGN OF TUBE SUPPORT ELEMENTS

D.1 Scope

This appendix provides stress curves that can be used in the design of tube support elements. The range of tube support materials has been expanded from that presented in Table 8 of Standard 560 to include $2^1/_4$Cr-1Mo castings and plate and 50Cr-50Ni-Cb castings. The following stress curves are provided:

a. One-third of the ultimate tensile strength.
b. Two-thirds of the yield strength (0.2 percent offset).
c. Fifty percent of the average stress required to produce 1 percent creep in 10,000 hours.
d. Fifty percent of the average stress required to produce rupture in 10,000 hours.

Some of the stresses listed in items a. through d. were not available for carbon steel castings or plate or for 50Cr-50Ni-Cb castings.

The stress curves were plotted from data gathered over normal design ranges. All of the materials are suitable for application at lower temperatures.

D.2 Referenced Publications

The following specifications and other publications are referenced in this appendix:

ASM[1]

Metals Handbook, Volume 3, "Properties and Selection: Stainless Steels, Tool Materials and Special-Purpose Metals" (Ninth Edition, 1980)

ASTM[2]

A 216 *Carbon Steel Castings Suitable for Fusion Welding for High-Temperature Service*
A 217 *Martensitic Stainless and Alloy Steel Castings for Pressure-Containing Parts Suitable for High-Temperature Service*
A 240 *Heat-Resisting Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels*
A 283 *Low and Intermediate Tensile Strength Carbon Steel Plates, Shapes, and Bars*
A 297 *Heat-Resistant Iron-Chromium and Iron-Chromium-Nickel Steel Castings for General Application*
A 387 *Chromium-Molybdenum Alloy Steel Pressure Vessel Plates*
A 447 *Chromium-Nickel-Iron Alloy Steel Castings (25-12 Class) for High-Temperature Service*
A 560 *Chromium-Nickel Alloy Castings*
DS 5 *Report on the Elevated-Temperature Properties of Stainless Steels*
DS 5S2 *An Evaluation of the Yield, Tensile, Creep, and Rupture Strengths of Wrought 304, 316, 321 and 347 Stainless Steels at Elevated Temperatures*
DS 6 *Report on the Elevated Temperature Properties of Chromium-Molybdenum Steels*
DS 6S2 *Supplemental Report on the Elevated Temperature Properties of Chromium-Molybdenum Steels*
DS 11S1 *An Evaluation of the Elevated Temperature Tensile and Creep-Rupture Properties of Wrought Carbon Steel*
DS 58 *Evaluation of the Elevated Temperature Tensile and Creep-Rupture Properties of 3 to 9 Percent Chromium-Molybdenum Steels*

SFSA[3]

Steel Castings Handbook (Fifth Edition, 1980)

D.3 Casting Factor

For cast materials, the stresses shown in Figures D-1 through D-13 are actual stresses based on published data accepted by the industry. A casting-factor multiplier of 0.8 shall be applied to the allowable stress value in the calculation of the minimum thickness.

D.4 Minimum Cross Sections

When good foundry practice or casting methods or tolerances require the use of a cross section heavier than that based on the calculation specified in D.3 or the stress curves shown in Figures D-1 through D-13, the governing thickness must be specified.

D.5 Maximum Design Temperatures

The maximum design temperatures shown in Figures D-1 through D-13 were obtained from Table 8 of API Standard 560 and are based on resistance to oxidation, except for the maximum design temperatures shown in Figures D-10 and D-12 (Types 309H and 310H plate), which are based on available stress data. The stress curves for some materials extend beyond

[1]American Society for Metals International, 9639 Kinsman Road, Metals Park, Ohio 44073.
[2]American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.
[3]Steel Founders' Society of America, 455 State Street, Des Plaines, Illinois 60016.

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API STD*560 95 ■ 0732290 0550212 648 ■

the maximum design temperature because of the materials' possible use, with high oxidation rates, at higher temperatures.

D.6 Corrosion Resistance

ASTM A 560, Grade 50Cr-50Ni-Cb, material is generally selected for its resistance to vanadium attack; however, its resistance diminishes at temperatures above 1600°F.

D.7 Proprietary Alloys

Most low-chromium-alloy cast iron materials are proprietary. The allowable stresses for the design of castings that use these materials must therefore be obtained from the supplier and agreed upon by the purchaser.

D.8 Stress Curves

All the stress curves in Figures D-1 through D-13 are based on published data. Apparent anomalies in the shapes of the curves reflect the actual data points used to construct the curves.

D.9 Data Sources

Table D-1 lists the sources of the stress data presented in Figures D-1 through D-13.

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API STD*560 95 ■ 0732290 0550213 584 ■

Table D-1—Sources of Data Presented in Figures D-1 through D-13

Figure	Material	Curve	Data Source
D-1	Carbon steel castings	Tensile strength	SFSA *Steel Castings Handbook*
		Yield strength	SFSA *Steel Castings Handbook*
D-2	Carbon steel plate	Tensile strength	ASTM DS 11S1
		Yield strength	ASTM DS 11S1
D-3	$2^1/_4$Cr-1Mo castings	Tensile strength	ASTM DS 6
		Yield strength	ASTM DS 6S2
		Rupture stress	ASTM DS 6S2
		Creep stress	ASTM DS 6S2
D-4	$2^1/_4$Cr-1Mo plate	Tensile strength	ASTM DS 6S2
		Yield strength	ASTM DS 6S2
		Rupture stress	ASTM DS 6S2
		Creep stress	ASTM DS 6S2
D-5	5Cr-$^1/_2$Mo castings	Tensile strength	ASTM DS 6
		Yield strength	ASTM DS 58
		Rupture stress	ASTM DS 58
		Creep stress	ASTM DS 58
D-6	5Cr-$^1/_2$Mo plate	Tensile strength	ASTM DS 58
		Yield strength	ASTM DS 58
		Rupture stress	ASTM DS 58
		Creep stress	ASTM DS 58
D-7	19Cr-9Ni castings	Tensile strength	ASM *Metals Handbook*
		Yield strength	ASM *Metals Handbook*
		Rupture stress	ASM *Metals Handbook*
		Creep stress	ASM *Metals Handbook*
D-8	Type 304H plate	Tensile strength	ASTM DS 5S2
		Yield strength	ASTM DS 5S2
		Rupture stress	ASTM DS 5S2
		Creep stress	ASTM DS 5S2
D-9	25Cr-12Ni castings	Tensile strength	ASM *Metals Handbook*
		Yield strength	ASM *Metals Handbook*
		Rupture stress	ASM *Metals Handbook*
		Creep stress	ASM *Metals Handbook*
D-10	Type 309H plate	Tensile strength	ASTM DS 5
		Yield strength	ASTM DS 5
		Rupture stress	ASTM DS 5
		Creep stress	ASTM DS 5
D-11	25Cr-20Ni castings	Tensile strength	ASM *Metals Handbook*
		Yield strength	ASM *Metals Handbook*
		Rupture stress	ASM *Metals Handbook*
		Creep stress	ASM *Metals Handbook*
D-12	Type 310H plate	Tensile strength	ASTM DS 5
		Yield strength	ASTM DS 5
		Rupture stress	ASTM DS 5
		Creep stress	ASTM DS 5
D-13	50Cr-50Ni-Cb castings	Rupture stress	IN-657[a]
		Creep stress	IN-657[a]

[a] *IN-657 Cast Nickel-Chromium-Niobium Alloy for Service Against Fuel-Ash Corrosion-Engineering Properties,* Inco Alloy Products Ltd., Wiggin Street, Birmingham B16 0AJ, England, United Kingdom.

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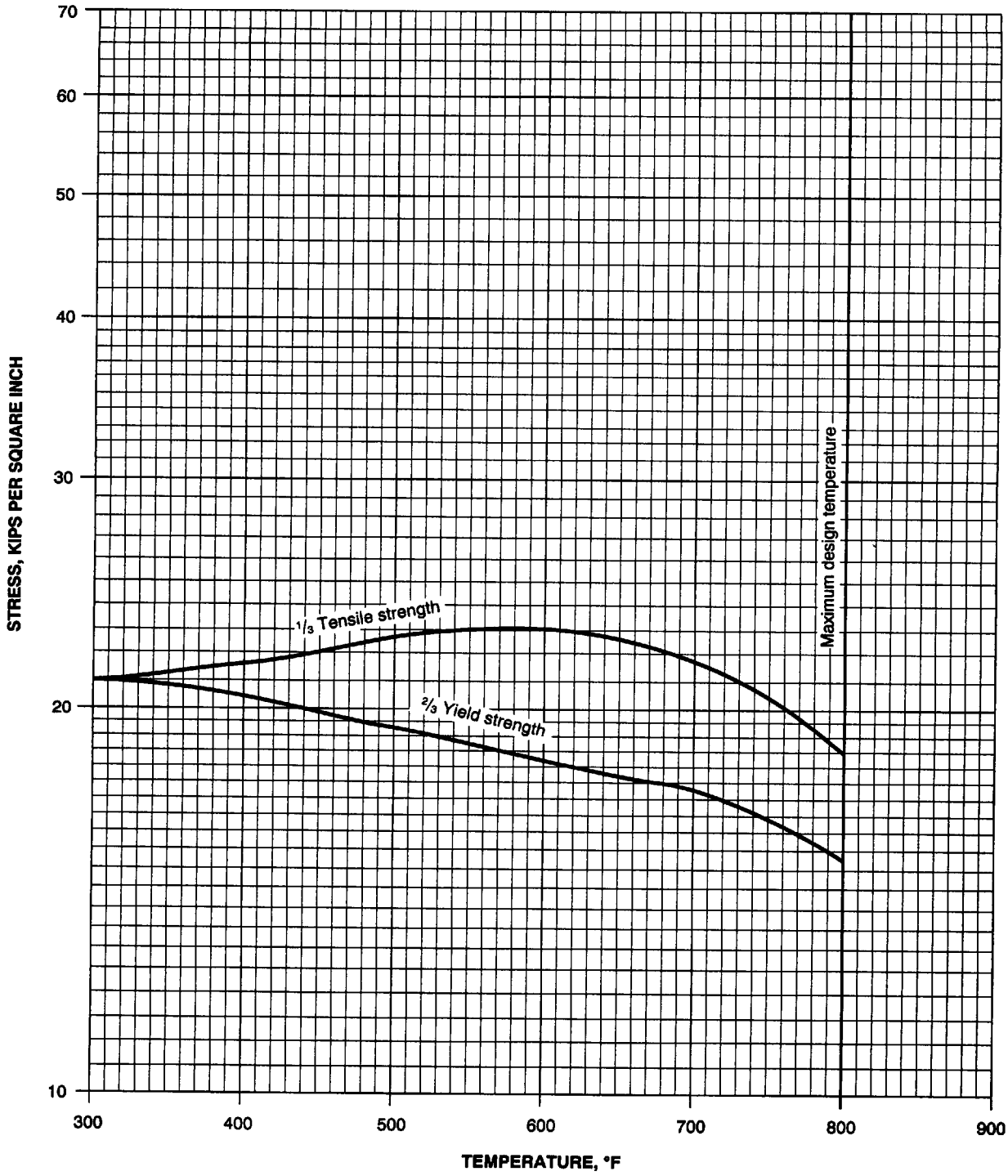


Figure D-1—Carbon Steel Castings: ASTM A 216, Grade WCB

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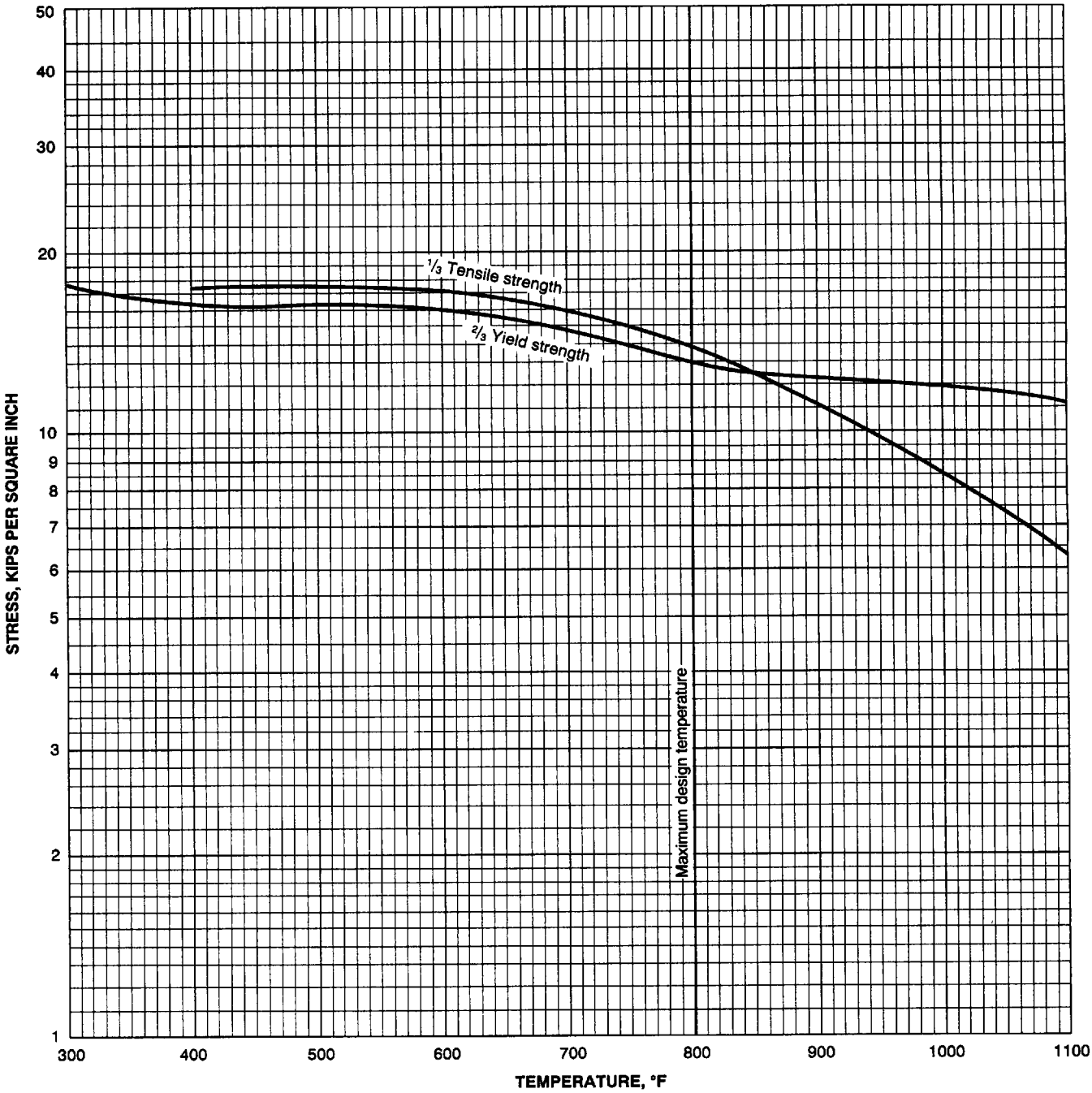


Figure D-2—Carbon Steel Plate: ASTM A 283, Grade C

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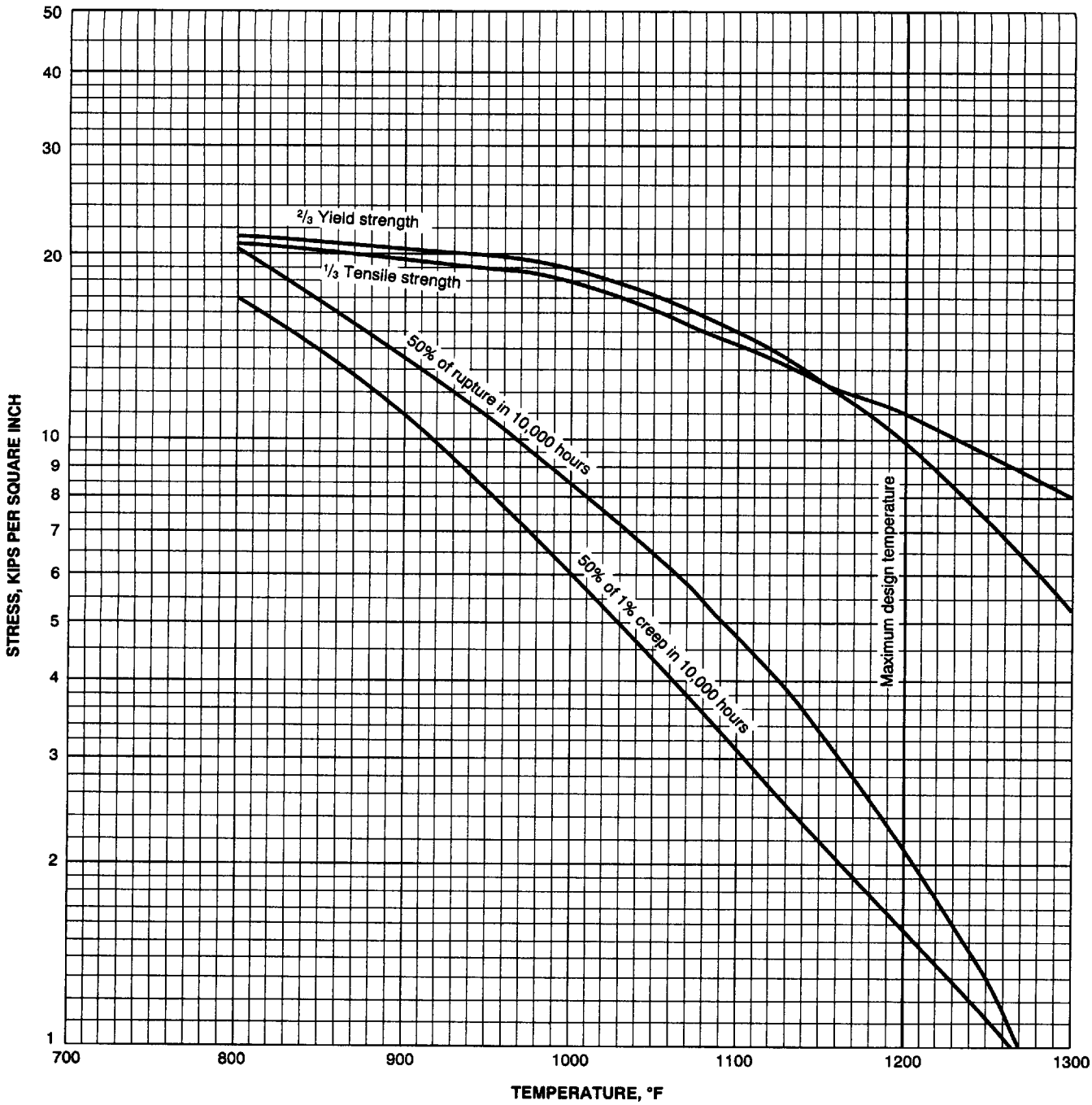


Figure D-3—$2^{1}/_{4}$Cr-1Mo Castings: ASTM A 217, Grade WC9

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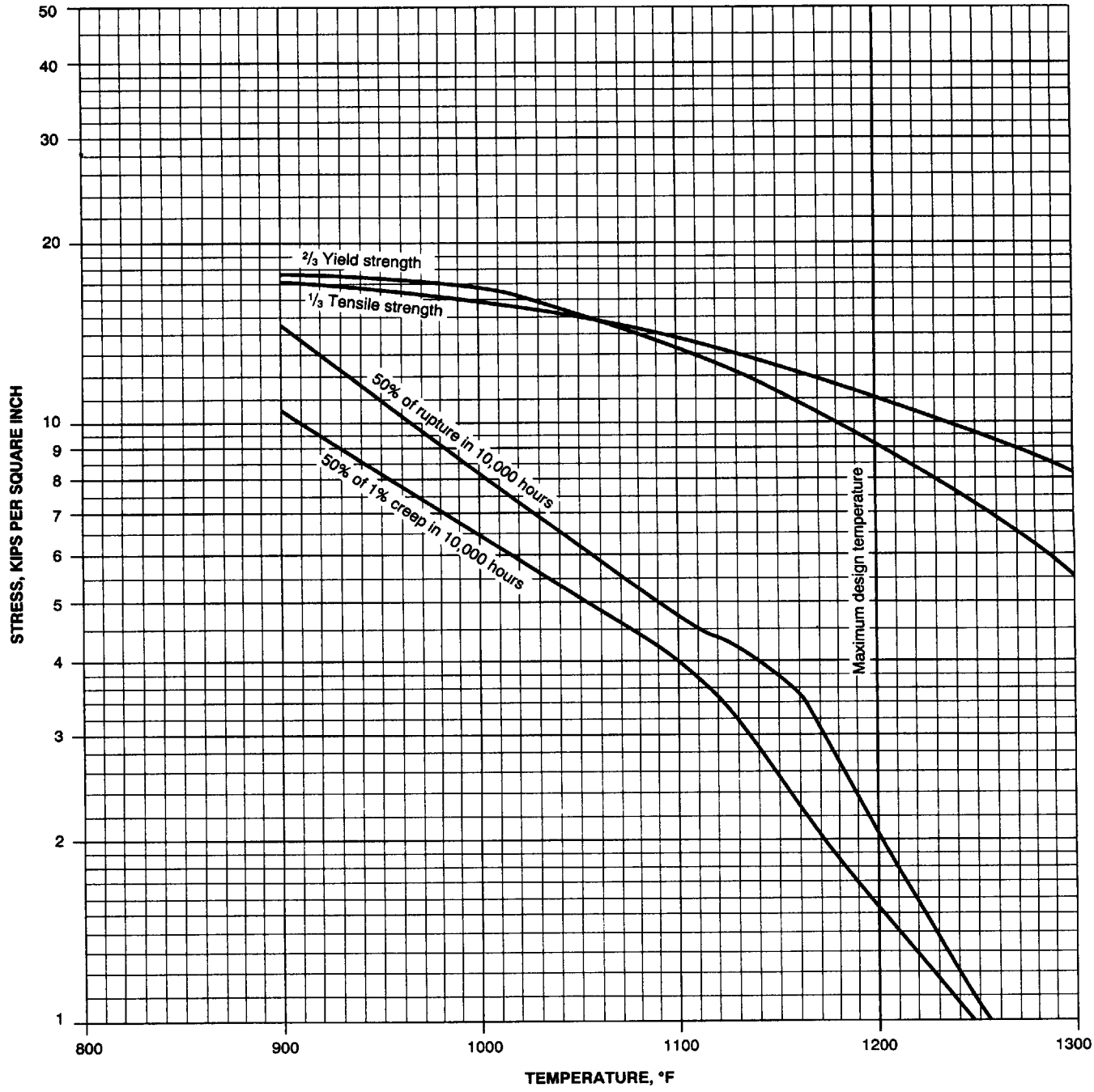


Figure D-4—$2^1/_4$Cr-1Mo Plate: ASTM A 387, Grade 22, Class 1

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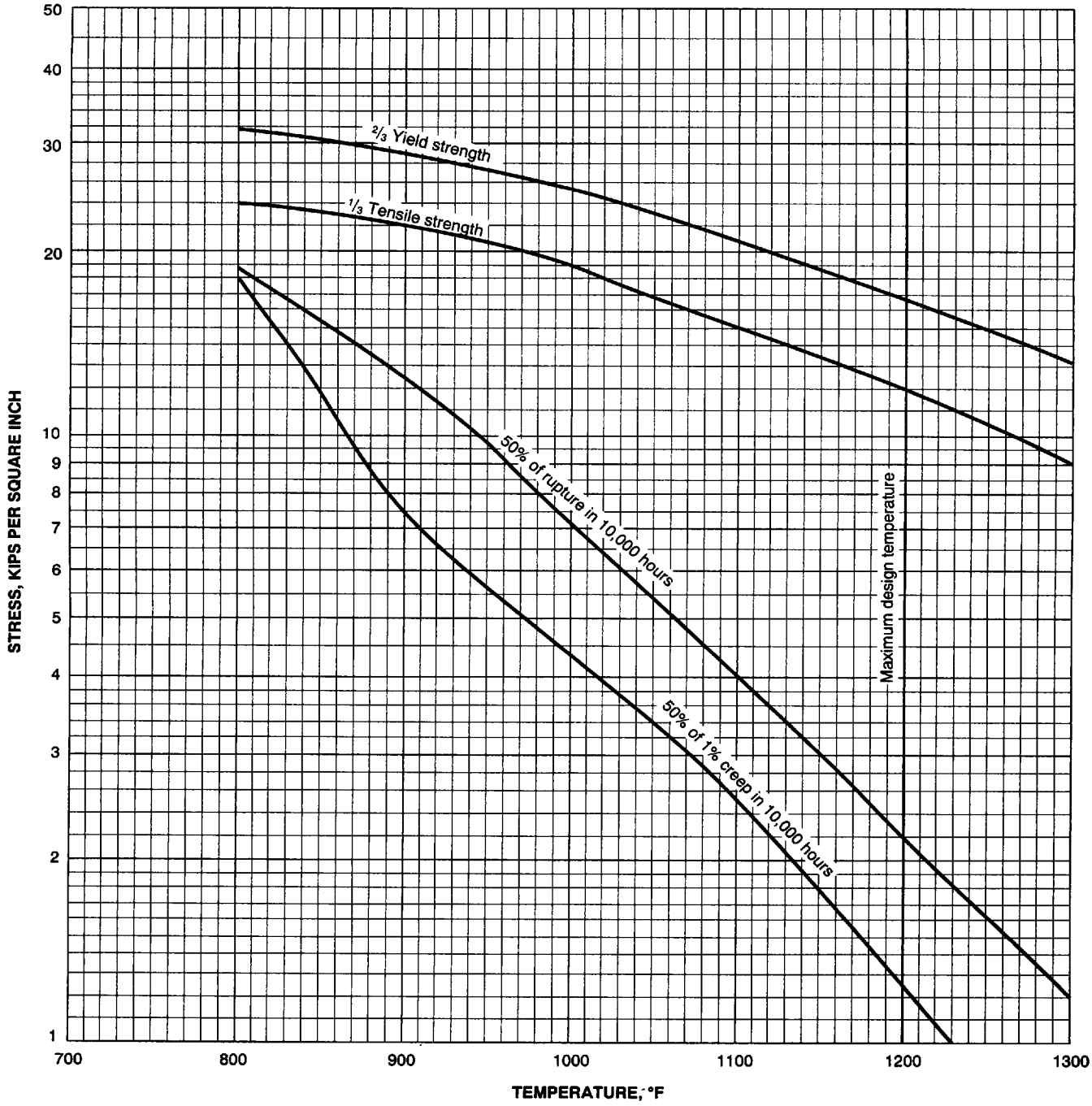


Figure D-5—5Cr-$^1/_2$Mo Castings: ASTM A 217, Grade C5

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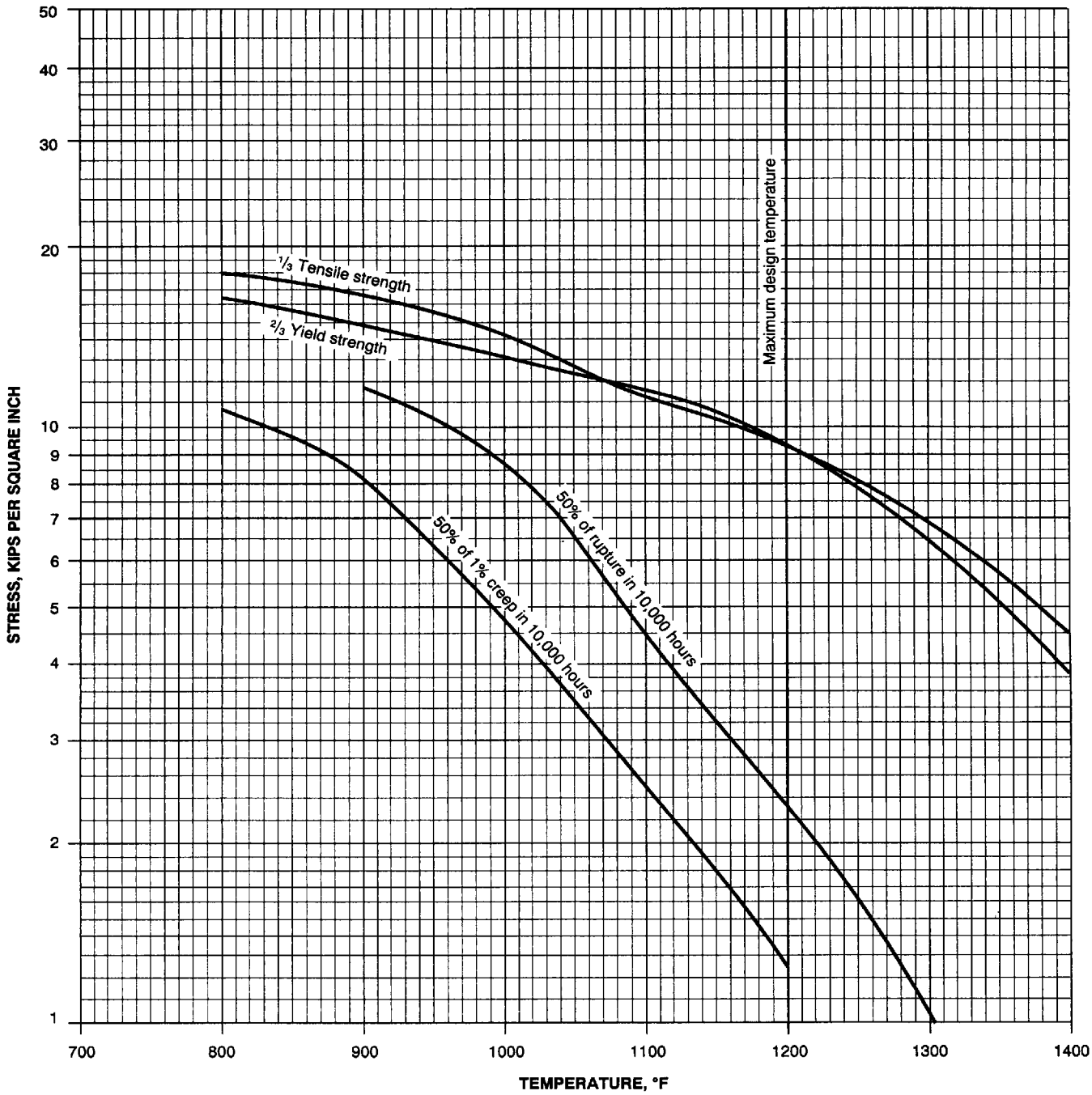


Figure D-6—5Cr-$^1/_2$Mo Plate: ASTM A 387, Grade 5, Class 1

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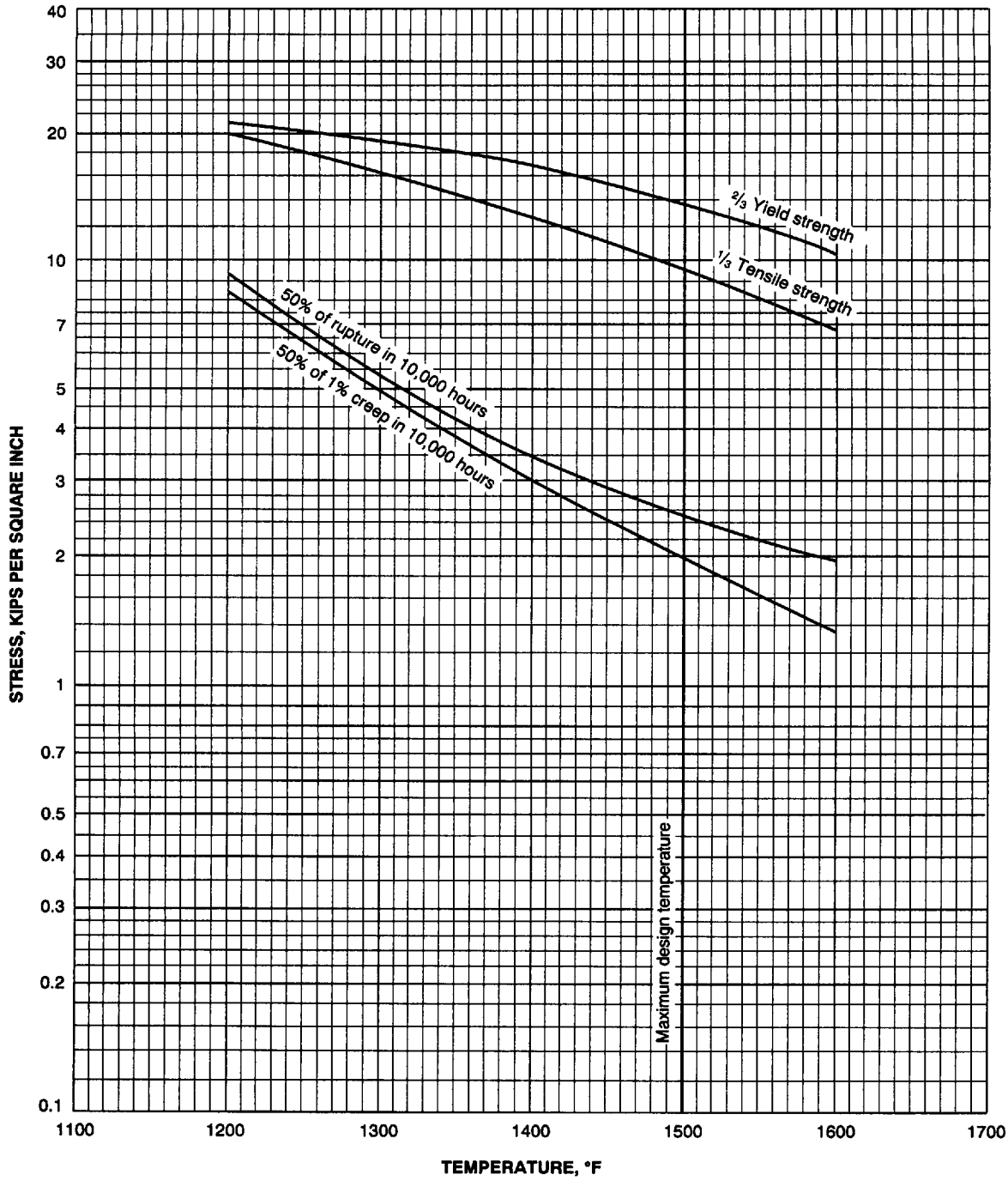


Figure D-7—19Cr-9Ni Castings: ASTM A 297, Grade HF

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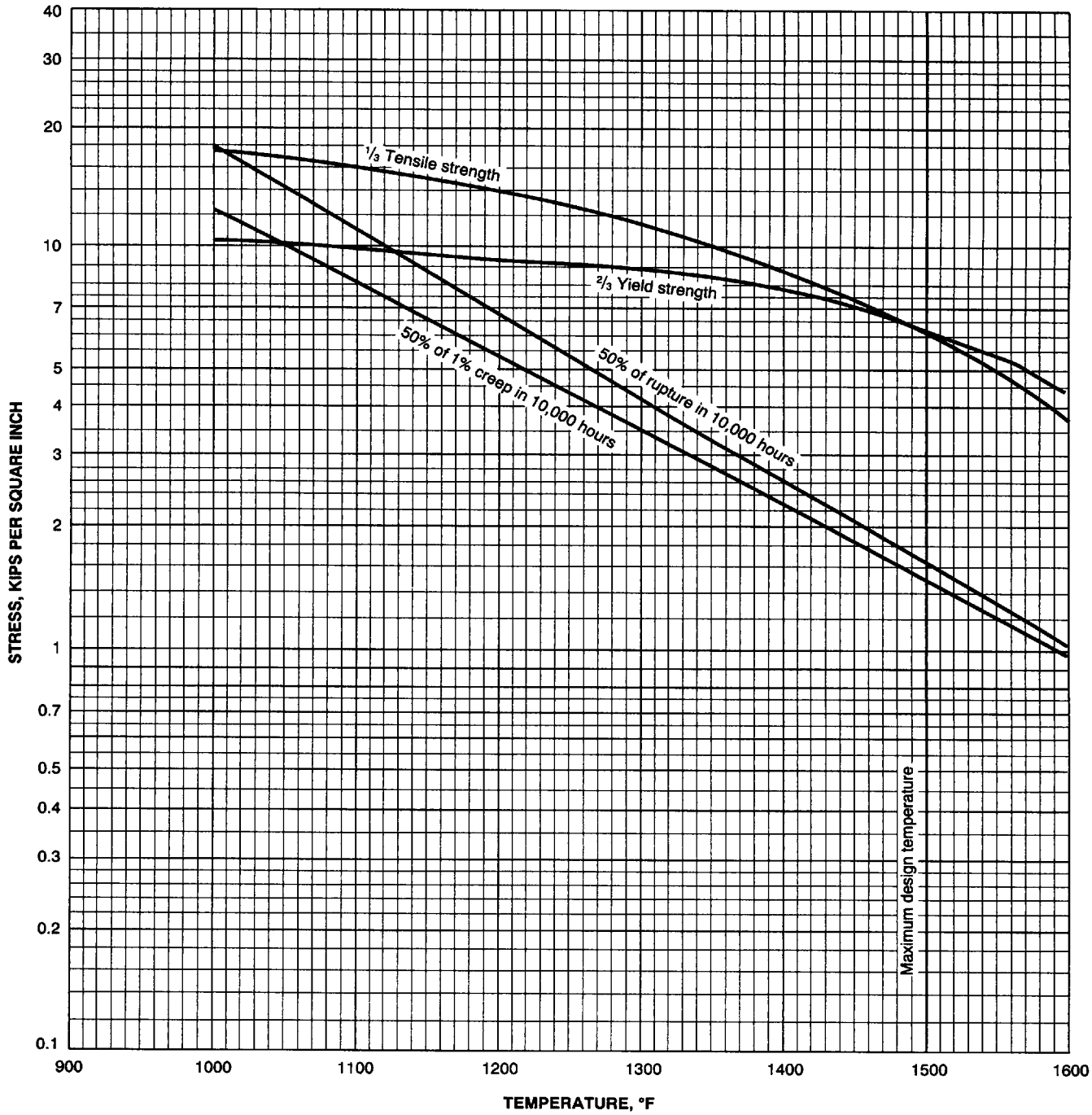


Figure D-8—Type 304H Plate: ASTM A 240, Type 304H

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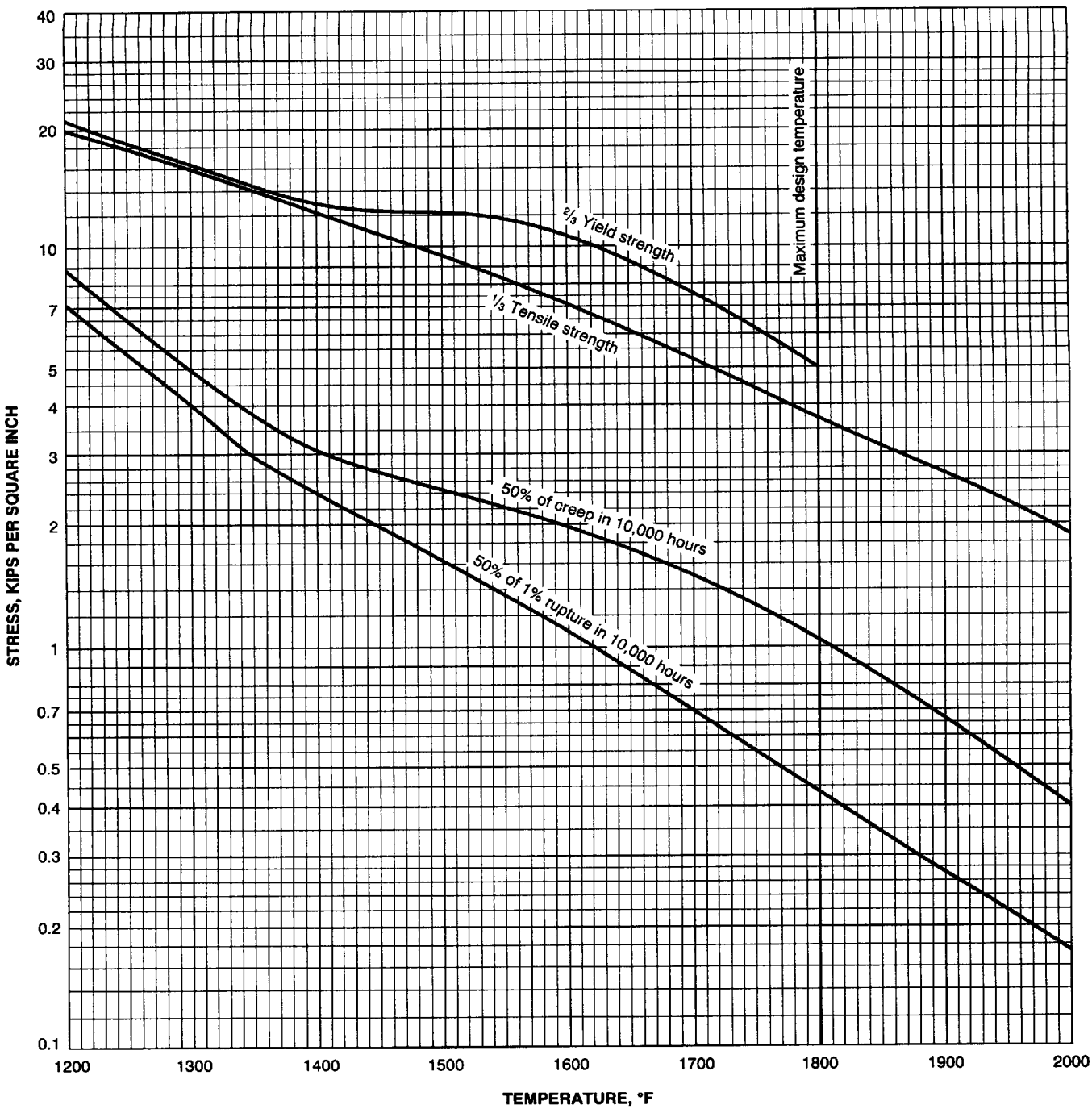


Figure D-9—25Cr-12Ni Castings: ASTM A 447, Grade HH, Type II

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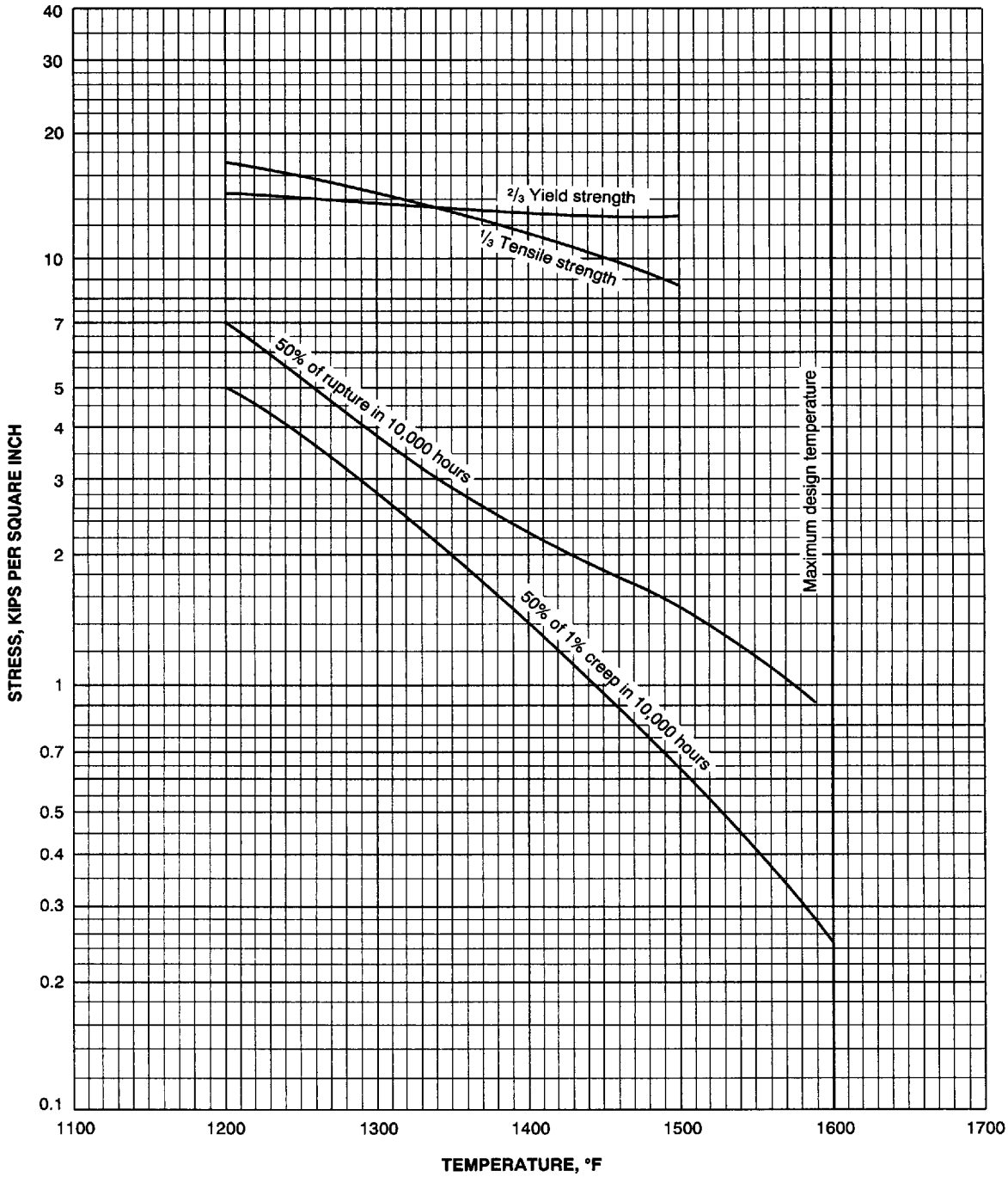


Figure D-10—Type 309H Plate: ASTM A 240, Type 309H

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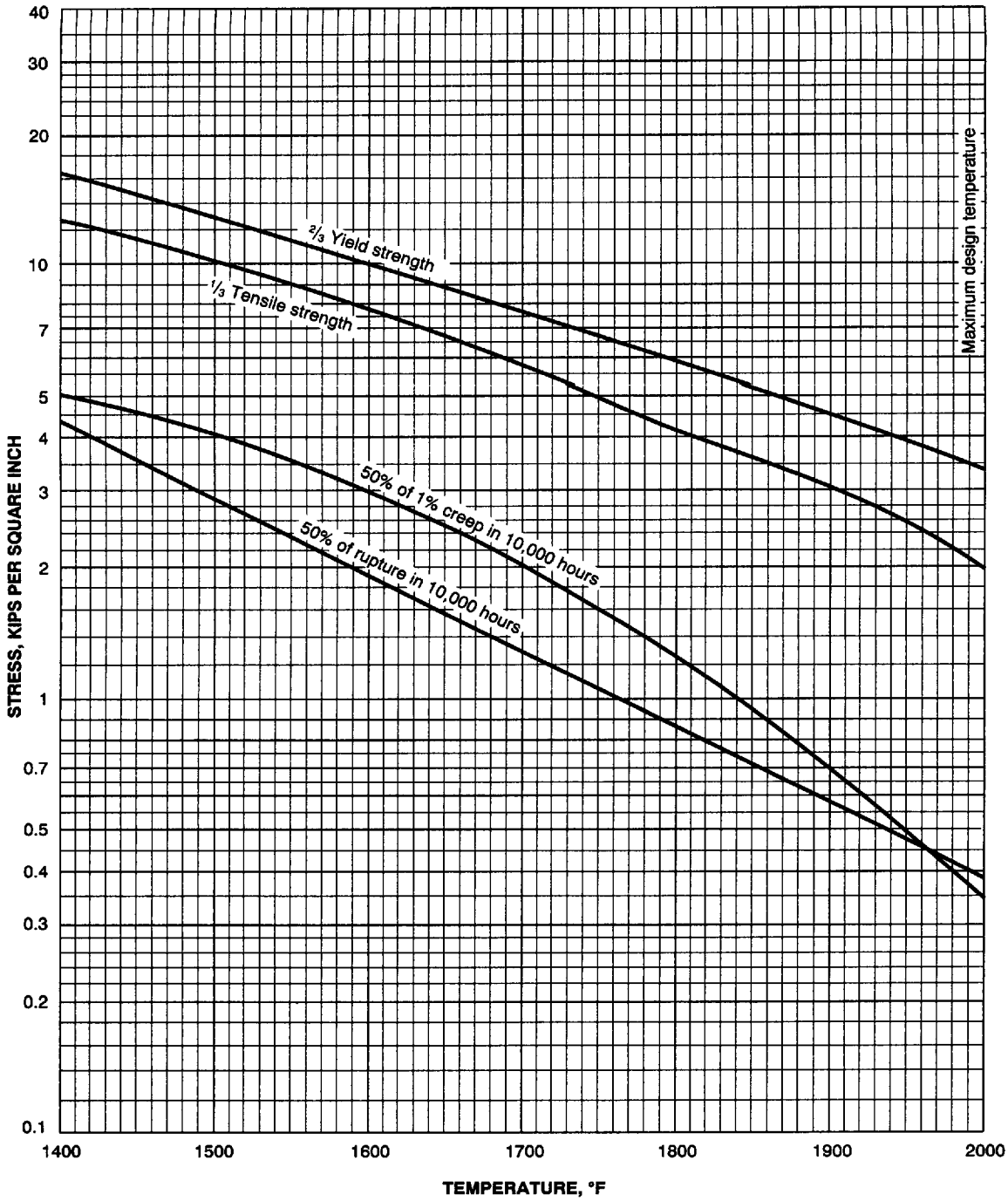


Figure D-11—25Cr-20Ni Castings: ASTM A 297, Grade HK

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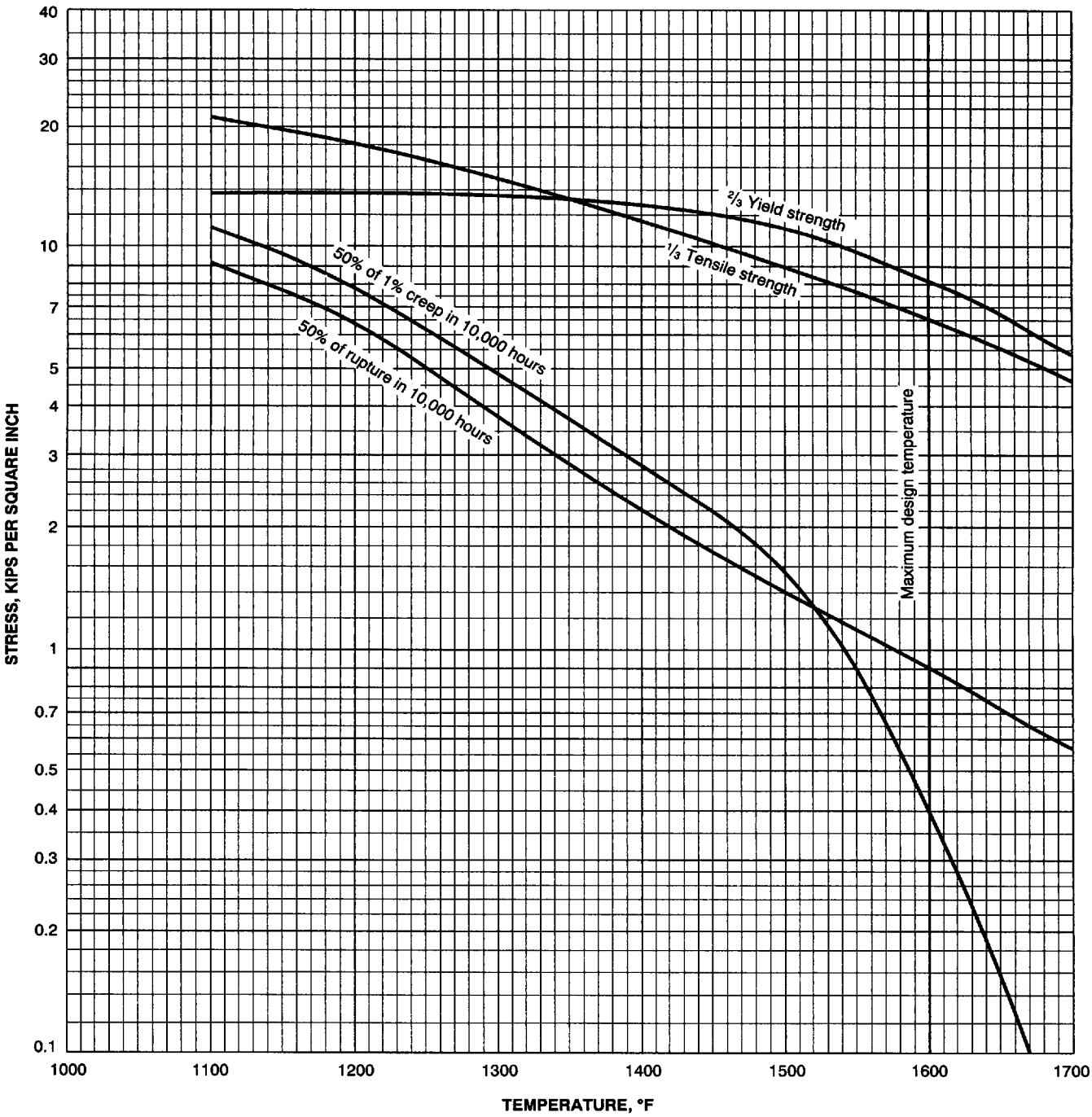


Figure D-12—Type 310H Plate: ASTM A 240, Type 310H

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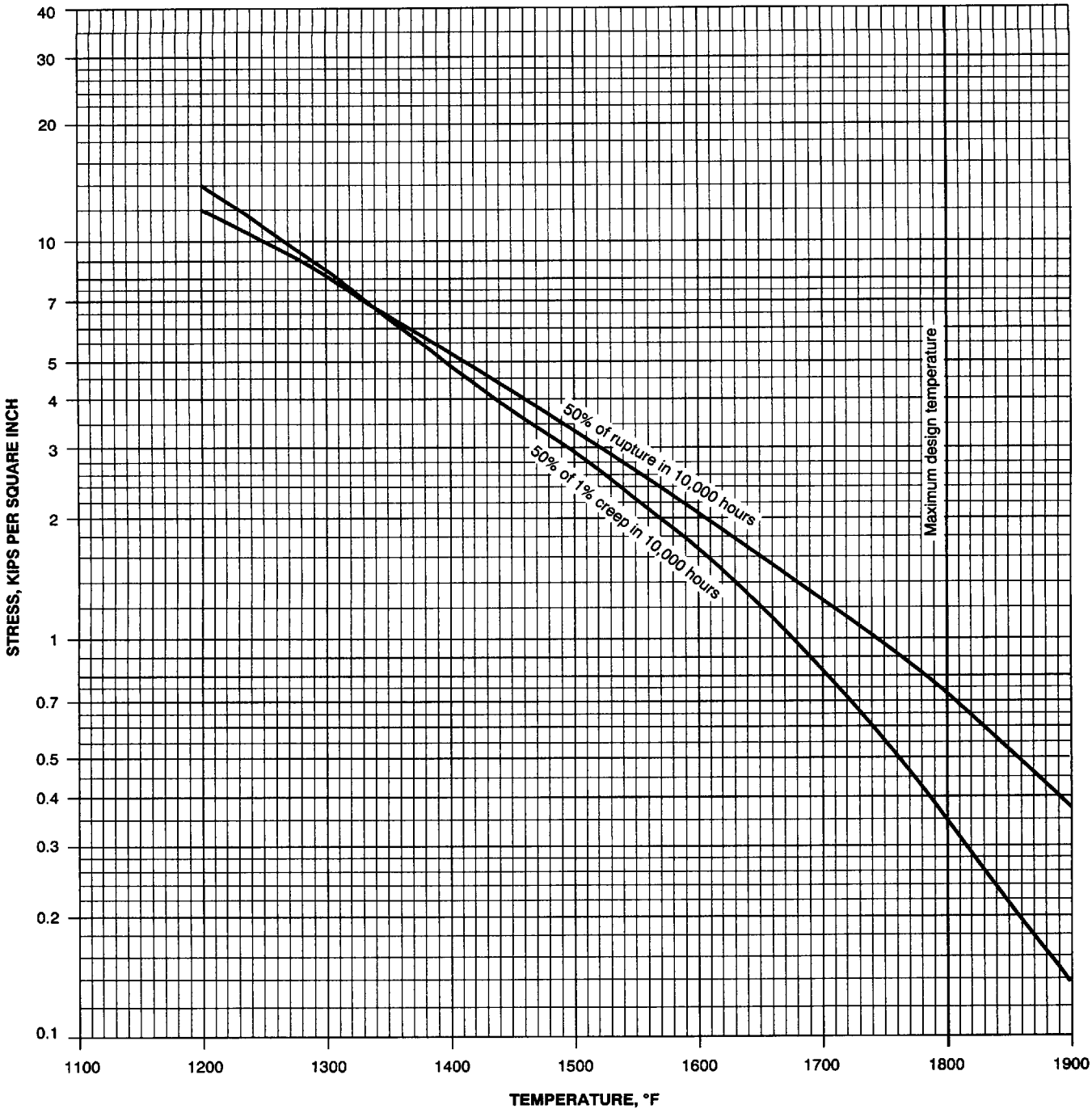


Figure D-13—50Cr-50Ni-Cb Castings: ASTM A 560, Grade 50Cr-50Ni-Cb

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APPENDIX E—CENTRIFUGAL FANS FOR FIRED HEATER SYSTEMS

E.1 General

E.1.1 SCOPE

This section of Appendix E covers the minimum requirements for centrifugal fans intended for continuous duty in fired heater systems.

E.1.2 DEFINITION OF TERMS

E.1.2.1 *Fan rated point* is defined as:

a. The highest speed necessary to meet any specified operating condition.
b. The rated capacity required by fan designs to meet all operating points. (The vendor shall select this capacity point to best encompass specified operating conditions within the scope of the expected performance curve.)

E.1.2.2 The *normal operating point* is the point at which usual operation is expected and optimum efficiency is desired. This point is usually the point at which the vendor certifies that performance is within the tolerances stated in this standard.

E.1.2.3 *Maximum allowable speed* (revolutions per minute) is the highest speed at which the manufacturer's design will permit continuous operation.

E.1.2.4 *Maximum allowable temperature* is the maximum continuous temperature for which the manufacturer has designed the equipment (or any part to which the term is referred) when handling the specified fluid at the specified pressure.

E.1.2.5 *Fan total pressure* is the difference between the total pressure at the fan outlet and the total pressure at the fan inlet.

E.1.2.6 *Fan velocity pressure* is the pressure corresponding to the average velocity at the specified fan outlet area.

E.1.2.7 *Fan static pressure* is the difference between the fan total pressure and the fan velocity pressure. Therefore, the fan static pressure is the difference between the static pressure at the fan outlet and the total pressure at the fan inlet.

E.1.2.8 *Static pressure rise* is often mistaken for fan static pressure. The value of static pressure rise is the static pressure at the fan outlet minus the static pressure at the fan inlet.

E.1.2.9 The *inlet velocity pressure* is the difference between fan static pressure and static pressure rise.

E.1.2.10 *Actual cubic feet per minute (ACFM)* refers to the flow rate determined at the conditions of static pressure, temperature, compressibility, and gas composition, including moisture, at the fan inlet flange.

E.1.2.11 The *fan vendor* is the manufacturer of the fan.

E.1.3 REFERENCED PUBLICATIONS

The following specifications and publications are referenced in this appendix:

ABMA[1]
Standard 9 *Load Ratings and Fatigue Life for Ball-Bearings*

AMCA[2]
99-86 *Standards Handbook*
99-2404-78 *Drive Arrangements for Centrifugal Fans*
201 *Fans and Systems*
203 *Field Performance Measurement of Fan Systems*
210 *Laboratory Methods of Testing Fans for Rating*
801 *Power Plant Fans—Specification Guidelines*
802 *Power Plant Fans—Establishing Performance using Laboratory Models*
803 *Power Plant and Industrial Fans—Site Performance Test Standard*

API
RP 500A *Classification of Locations for Electrical Installations in Petroleum Refineries, 4th edition 1982, Reaffirmed 1987*

ASME[3]
B17.1 *1967 Keys and Keyseats (R-1989)*
Boiler and Pressure Vessel Code, Section V, "Nondestructive Examination," and Section VIII, "Rules for Construction of Pressure Vessels"

ASTM[4]
A 515/A 515M *1992 Standard Specification for Pressure Vessel Plates, Carbon Steel, for Intermediate- and Higher- Temperature Service*
E 94 *1993 Standard Guide for Radiographic Testing*

[1]American Bearing Manufacturing Association, 1200 19th Street, N.W., Suite 300, Washington, D.C. 20036.
[2]Air Movement & Control Association, 30 West University Drive, Arlington Heights, Illinois 60004.
[3]American Society of Mechanical Engineers, 345 East 47th Street, New York, New York 10017.
[4]American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103-1187.

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E 125 *1963 Standard Reference Photographs for Magnetic Particle Indications on Ferrous Castings (R-1993)*
E 142 *1992 Standard Method for Controlling Quality of Radiographic Testing*
E 709 *1994 Standard Guide for Magnetic Particle Examination*

AWS[5]
D1.1 *Structural Welding Code—Steel*
D14.6 *Specification for Welding of Rotating Elements of Equipment*

NFPA[6]
70 *National Electric Code*

SSPC[7]
SP-6 *1991 Surface Preparation Specification No. 6 Commercial Blast Cleaning*

E.2 Design

E.2.1 GENERAL

E.2.1.1 The equipment (including auxiliaries) covered by the standard shall be designed and constructed for a minimum service life of 20 years and at least 3 years of uninterrupted operation. It is recognized that this is a design criterion.

E.2.1.2 Fans shall be designed to operate satisfactorily at all specified operating conditions. The two operating points of particular concern are the rated point and the normal operating point (see E.1.2.1 and E.1.2.2). It shall be the responsibility of the fan purchaser to provide complete required data (such as acfm, pressure, temperature, and inlet gas density) to the fan manufacturer. In developing this data the fan purchaser must consider the following:

a. The normal operating point is that point at which it is expected that the furnace will be operated the majority of the time. It shall be the fan manufacturer's responsibility to optimize the fan's efficiency as close to this point as practical.
b. The fan rated point shall include the flow required to meet the heater maximum design firing rate (including all overages for excess air, system leakage and design safety factor). In no case shall the rated point be less than 115 percent of the normal operating flow. The fan static pressure and temperature required for the rated point shall be specified by the purchaser subject to the requirements of the end user's specifications.

E.2.1.3 The arrangement of the equipment, including ducting and auxiliaries, shall be developed jointly by the purchaser and the vendor. The arrangement shall provide adequate clearance areas and safe access for operation, maintenance, and removal.

● **E.2.1.4** Motors, electrical components, and electrical installations shall be suitable for the area classification (class, group, and division) specified by the purchaser and shall meet the requirements of NFPA 70, Articles 500 and 501, as well as local codes specified and furnished by the purchaser. API Recommended Practice 500A provides guidance on area classification.

E.2.1.5 All equipment shall be designed to permit rapid and economical maintenance. Major parts such as fan housing, inlet cone, and bearing housings shall be designed (shouldered or doweled) and manufactured to ensure accurate alignment on reassembly. Field doweling by others may be required after final alignment.

E.2.1.6 The fan vendor shall formally review and approve or comment on the purchaser's inlet and outlet duct and equipment arrangement drawings. This review shall consider structural aspects, such as loading on fan parts, and configuration details that impact fan performance as described in AMCA 201. Foundation drawing review is not required unless specified by purchaser.

● **E.2.1.7** Installation will be outdoors with no roof. The purchaser will specify the weather and environmental conditions in which the equipment must operate (including maximum and minimum temperatures and unusual humidity or dust problems). The unit and its auxiliaries shall be suitable for operation under these specified conditions. For the purchaser's guidance, the vendor shall list in the proposal any special protection that the purchaser is required to supply.

E.2.1.8 Spare parts for the machine and all furnished auxiliaries shall meet all the criteria of this standard.

E.2.1.9 The selected operating speed of the fan shall not exceed 1800 revolutions per minute unless otherwise approved by the purchaser.

E.2.1.10 Fan arrangement and bearing support shall be in accordance with AMCA 99-86, AMCA Standard 99-2404-78, arrangement 3 or 7, with fan impeller located between bearings, the bearings mounted independently of fan housing on rigid pedestals and sole plates, and the bearings protected from the air or gas stream when any of the following conditions exist:

a. Driver rated power of 150 BHP (120 kilowatts) or greater.
b. Speed greater than 1800 revolutions per minute.

[5]American Welding Society, 550 N.W. LeJeune Road, Miami, Florida 33135.
[6]National Fire Protection Association, 1 Batterymarch Park, Quincy, Massachusetts 02269-9101.
[7]Steel Structures Painting Council, 4400 Fifth Avenue, Pittsburgh, Pennsylvania 15213-2683.

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c. Maximum specified operating temperature greater than 450°F (233°C).
d. Corrosive or erosive service.
e. Service subject to fouling deposits that could cause rotor unbalance.

For services not subject to the above conditions, AMCA arrangements 1, 8, and 9, all with bearings mounted independent of the fan housing, are acceptable with purchaser's specific approval.

Note: Additional considerations for fan selection include:

a. Reduced speed is desirable for erosive service and for units subject to fouling deposits on the rotor.
b. Belt drives should be limited to no more than 100 BHP (75 kilowatts) rated driver size.
c. When drivers are rated less than 40 BHP (30 kilowatts) and speeds greater than 1800 revolutions per minute, AMCA arrangements other than 3 and 7 may be specified on the data sheet.

E.2.1.11 Fan performance shall be based on fan static pressure rise not including discharge velocity pressure. When specifying required performance the purchaser is responsible for including the effect of inlet velocity pressure. To obtain the static pressure differential, the silencer and inlet losses, including control system losses, shall be added by the fan vendor to the purchaser's specified inlet and outlet static pressures.

E.2.1.12 Unless otherwise specified, fans shall have continuously rising pressure characteristic (pressure versus flow plot) from the rated capacity to 70 percent or less of rated flow.

Performance curves, corrected for the specified gas at the specified conditions, shall be based on performance tests in accordance with AMCA 210, including, where applicable, evase and inlet box(es).

E.2.1.13 The fan shall be mechanically designed for continuous operation at the following temperatures:

a. 100°F (56°C) above rated inlet flue gas temperature to induced draft fans.
b. 25°F (14°C) above maximum specified ambient air temperature to forced draft fans.

E.2.1.14 Fan, components, and accessories shall be designed to withstand all loads and stresses during rapid load changes, such as starting, failure of damper operator, or sudden position change of dampers. Considerations for driver sizing and starting operations are covered in E.3.1.3 and E.3.1.4.

E.2.1.15 Fan Inlets

E.2.1.15.1 For forced draft fans, provision of the inlet equipment and arrangements, including silencer(s) and transition piece(s), shall be coordinated between the purchaser and the fan vendor. (Portions may normally be supplied by each.)

● **E.2.1.15.2** Unless otherwise specified the air intake shall be at least 15 feet above grade. The purchaser shall evaluate air intake elevation requirements considering the possibility of dust entering the system and causing surface fouling, the area noise limitation requirement and the corresponding need for a silencer, the possibility of a combustible vapor entering the fan, and power penalties for inlet stack and silencer configuration.

E.2.1.15.3 The fan inlet equipment shall include intake cap or hood, trash screen, ducting and support, inlet damper or guide vanes, inlet boxes, and silencer, as required. All components shipped separately shall be flanged for assembly. The inlet equipment assembly shall be designed for the wind load shown on the fan data sheet.

E.2.2 FAN HOUSING

● **E.2.2.1** The fan scroll and housing sides shall be continuously welded plate construction. Minimum plate thickness shall be $^3/_{16}$ inches (4.8 millimeters) for forced draft and $^1/_4$ inches (6.4 millimeters) for induced draft fans. Where specified, a corrosion allowance may be required. Stiffeners shall be provided to form a rigid housing free of structural resonance and to limit vibration and noise. The external stiffeners may be intermittently welded to the fan housing.

For Arrangements 3 and 7 fans the housing and inlet box(es) shall be split at a bolted, flanged, and gasketed connection to allow assembled rotor removal and installation without disturbing duct connections. Other arrangements shall be similarly split where impeller diameter exceeds 42 inches (1067 millimeters).

The inlet cone shall be constructed so that it does not impede rotor removal or installation. The cone must either be split, separately removed as a whole, or removable in assembly with the rotor.

E.2.2.2 Bolted and gasketed access doors, of largest possible size up to 24 inch by 24 inch (610 millimeters by 610 millimeters), shall be provided in the scroll and inlet box(es) for access to the fan internals for inspection, cleaning, rotor balancing, and to any internal bolting necessary for rotor removal.

E.2.2.3 Adequate flanged sections shall be provided in the fan housing and inlet box(es) so that the rotor can be removed and installed without requiring personnel to enter the inlet box(es).

E.2.3 FAN HOUSING CONNECTIONS

E.2.3.1 Inlet and discharge connections shall be flanged and bolted. Facings, gaskets, and bolting of all connections shall prevent leakage.

E.2.3.2 Accessible flanged drain connections, 2 inch (50 millimeter) minimum size, shall be provided at the low point(s) of the housing and inlet boxes.

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E.2.4 EXTERNAL FORCES AND MOMENTS

Fan housings are generally designed to accept only low external forces and moments from the inlet and outlet connections. It shall be the responsibility of the purchaser to specify on the data sheets the expected external loads to be imposed on the fan housing from the ancillary equipment (that is, ducting, sound trunks, silencers, and filters), if this equipment is not supplied by the fan vendor. The vendor shall design the housing to accept the specified loads. The vendor shall provide the following information as required on the fan data sheets:

a. Maximum allowable external forces and moments.
b. Expansion joint information and recommendations if joints are required for thermal expansion, vibration isolation, or both.

E.2.5 ROTATING ELEMENTS

E.2.5.1 Fan impellers shall have a nonoverloading horsepower characteristic and shall be designed for the highest possible efficiency. Backward curved/backward inclined blades are available in the constructions detailed in items E.2.5.1.a. through E.2.5.1.c. Such configurations are nonoverloading.

Design and configurations available as options include:

a. Hollow airfoil construction of 0.10 inch (2.54 millimeters) minimum skin thickness material designed and constructed to prevent the internal accumulation of condensables, foulants, or corrosion products.
b. Solid blades with airfoil shape.
c. Single thickness, $^1/_4$ inch (6.35 millimeters) minimum nonairfoil shape.

E.2.5.2 Induced draft fan design shall consider operations in a possible dirty gas environment. Blade design shall be specified by the purchaser. Radial and radial-tipped configurations are considered non-fouling designs and have lower inherent efficiencies.

E.2.5.3 Welded construction of the impeller is required. Shrouds, back plates, and center plates shall normally be of one piece construction. They may be fabricated if the sections are joined by full penetration butt welds meeting the inspection requirements of E.4.1.1.2.1 and E.4.1.1.2.2. The vendor shall state whether post weld heat treatment of the fabricated wheel is required, after consideration of environmental and mechanical (residual stress) effects.

E.2.5.4 Gas temperature change rates, heating, and cooling, in excess of 15°F (8.3°C) per minute may be expected on induced draft fans. Fan vendors shall specify the maximum allowable rate of change to ensure that an adequate hub-to-shaft interference fit is maintained.

E.2.5.5 Impellers shall have solid hubs, be keyed to the shaft, and be secured with an interference fit. Unkeyed fits with appropriate interference are permissible with purchaser's approval. Cast or ductile iron hubs are acceptable below a mechanical design temperature of 300°F (149°C). When the impeller is to be bolted to the hub the manufacturer will, by design, preclude relative movement between the impeller and hub.

E.2.5.6 Shafts shall be of one piece, heat treated, forged steel; shafts 6 inches (15.2 centimeters) in diameter and smaller may be machined from hot rolled steel. Arrangements 3 or 7 shaft diameters shall be stepped on both sides of the impeller fit area to facilitate impeller assembly and removal. Fillets shall be provided at all changes in shaft diameters and in keyways. Keyways shall have fillet radii per ASME B17.1. Welding on the shaft is not permitted.

E.2.5.7 Shafts shall be capable of handling 110 percent of rated driver torque from rest to rated speed.

● **E.2.5.8** Induced draft fans shall be provided with corrosion-resistant shaft sleeves, when specified by the purchaser, to reduce the effect of dew point corrosion at shaft seals. Sleeves shall extend 6 inches (15 centimeters) into the fan housing.

E.2.6 SHAFT SEALING OF FANS

E.2.6.1 Shaft seals shall be provided to minimize leakage from or into fans over the range of specified operating conditions and during idle periods. Seal operation shall be suitable for variations in inlet conditions that may prevail during startup and shutdown or any special operation specified by the Purchaser.

E.2.6.2 Shaft seals shall be replaceable from the outside of the inlet box(es) without disturbing the shaft or bearings.

E.2.7 CRITICAL SPEEDS/RESONANCE

E.2.7.1 Unless otherwise specified, the separation margin of critical speeds from all lateral (including rigid and bending) modes shall be at least 25 percent over the maximum continuous speed. The separation margin is intended to prevent the overlapping of the resonance response envelope into the operating speed range.

Note: The term critical speed used herein considers the factors, defined by "design resonant speed" in AMCA Publication 801-92, Paragraph 3.2.3.

E.2.7.2 Resonances of support systems within the vendor's scope of supply shall not occur within the specified operating speed range or the specified separation margins, unless the resonances are critically damped.

E.2.7.3 Bearing housing resonance shall not occur within the specified operating speed range or specified separation margins.

● **E.2.7.4** When specified, critical speeds shall be determined analytically by means of a damped unbalanced rotor

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response analysis and, when specified, shall be confirmed by test-stand data.

E.2.7.5 The vendor who has unit responsibility shall determine that the drive-train critical speeds are compatible with the critical speeds of the machinery being supplied and that the combination is suitable for the specified operating speed range. A list of all undesirable speeds from zero to trip shall be submitted to the purchaser for his review and included in the instruction manual for his guidance.

E.2.8 VIBRATION AND BALANCING

E.2.8.1 Each rotating assembly shall be dynamically balanced.

E.2.8.2 Prior to rotor assembly, the shaft shall be inspected for mechanical runout and concentricity at impeller mounting surface seat and bearing journals. Runout shall not exceed the total indicator reading specified in Table E-1.

● **E.2.8.3** When specified, a mechanical running test shall be performed at the fan vendor's shop (see E.4.2.2). During the shop test of the assembled machine operating at maximum continuous speed or at any other speed within the specified operating speed range, the maximum allowable unfiltered peak vibration velocity, measured on the bearing housing in any radial plane, shall not exceed 0.2 inches per second (5.0 millimeters per second) or 0.1 inches per second (2.5 millimeters per second) at running frequency (1 ×). At the trip speed of the driver, the vibration shall not exceed 0.25 inches per second (6.3 millimeters per second) unfiltered velocity.

E.2.9 BEARINGS AND BEARING HOUSINGS

E.2.9.1 Bearing types shall be either anti-friction or hydrodynamic (sleeve). Unless otherwise specified, fans rated at 150 BHP (120 kilowatts) or greater shall have horizontally split, self-aligning hydrodynamic bearings.

E.2.9.2 Anti-friction bearings shall be self-aligning and the selection shall be based on the following ratings:

a. DN factor less than 200,000. (The DN factor is the product of bearing size, that is, bore, in millimeters and the rated speed in revolutions per minute.)

Table E-1—Maximum Shaft Runout Indicator Readings

Shaft Diameter (Inches)	Total Indicator Reading	
	Bearing Journal Area (Inches)	Wheel Mounting Area (Inches)
< 6	0.0010	0.002
6–14	0.0015	0.003
> 14	0.0020	0.004

b. L-10 life factor (see AFBMA Standard 9) of 100,000 hours or greater. (The rating life is the number of hours at rated bearing load and speed that 90 percent of the group of identical bearings will complete or exceed before the first evidence of failure.)

c. Load factor less than 2,700,000. (Load factor is the product of rated horsepower and rated speed in revolutions per minute.)

The use of "maximum load" (filling slot) antifriction bearings is specifically prohibited for any service, including drivers (motors, turbines, and gears).

E.2.9.3 Thrust bearings shall be sized for continuous operation under all specified conditions including double-inlet fans operating with one inlet cone 100 percent blocked. As a guide, thrust bearings shall be applied at no more than 50 percent of the bearing manufacturer's ultimate load rating.

E.2.9.4 Shaft bearings shall be accessible without dismantling duct work or fan casing. Overhung impeller designs shall have provisions for supporting the rotor during bearing maintenance.

E.2.9.5 All induced draft fans shall be supplied with a heat slinger (with safety guards), located between the fan housing and/or inlet box(es) and the adjacent bearing(s).

E.2.9.6 Sufficient cooling, including an allowance for fouling, shall be provided to maintain the oil temperature below 160°F (71°C) for pressurized systems and below 180°F (82°C) for ring-oiled or splash systems, based on the specified operating conditions and an ambient temperature of 110°F (43°C). Where water cooling is required, water jackets shall have only external connections between the upper and lower housing jackets and shall have neither gasketed nor threaded connection joints, which may allow water to leak into the oil reservoir. If cooling coils (including fittings) are used, they shall be of nonferrous material and shall have no internal pressure joints or fittings. Coils shall have a thickness of at least 19 Birmingham wire gauge [0.042 inch (1.07 millimeters)] and shall be at least 0.50 inch (12.7 millimeters) in diameter.

E.2.9.7 Bearing housings shall be drilled with pilot holes for use in final dowelling.

E.2.10 LUBRICATION

E.2.10.1 Unless otherwise specified, bearings and bearing housings shall be arranged for hydrocarbon oil lubrication in accordance with the bearing manufacturer's recommendations. Grease packed anti-friction bearings shall not be provided without purchaser's approval.

E.2.10.2 On dampers and variable inlet vanes all linkage, shaft fittings, and bearings shall be permanently lubricated.

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API STD*560 95 ■ 0732290 0550232 436 ■

Components requiring periodic lubrication shall be furnished with lubrication fittings which are accessible while the fan is in operation.

E.2.10.3 When required, forced feed oil system scope shall be mutually agreed upon by the purchaser and the vendor.

E.2.11 MATERIALS

E.2.11.1 General

E.2.11.1.1 Construction materials shall be the manufacturer's standard for the specified operating conditions, except as required by the purchaser.

● **E.2.11.1.2** The purchaser will specify any corrosive agents present in the flue gas and in the environment, including constituents that may cause stress corrosion cracking. The fan vendor shall recommend materials that are suitable for mechanical design and fabrication (see E.2.5.3).

E.2.11.1.3 Where mating parts such as studs and nuts of AISI[1] Standard Type 300 stainless steel or materials with similar galling tendencies are used, they shall be lubricated with an antiseizure compound rated for the specified temperatures.

E.2.11.1.4 Low-carbon steels can be notch sensitive and susceptible to brittle fracture at ambient or low temperatures. Therefore only fully killed, normalized steels made to fine-grain practice are acceptable. The use of ASTM A 515 steel is prohibited.

E.2.11.1.5 Internal bolting shall be ASME grade B as a minimum, but at least equivalent to the fan construction material.

E.2.11.2 Welding

E.2.11.2.1 All welding, including weld repairs, shall be performed by operators and procedures qualified in accordance with the specifications of AWS D14.6 for rotor welds and AWS D1.1 for housings and inlet boxes.

E.2.11.2.2 The vendor shall be responsible for the review of all welding, including weld repair, to ensure that the requirements of AWS D14.6, Section 7, have been satisfied.

E.2.11.2.3 All rotor component butt welds shall be continuous full penetration welds.

E.2.11.2.4 Intermittent welds, stitch welds, or tack welds are not permitted on any part of the fan or accessories furnished by the vendor, except as noted in E.2.2.1 and E.3.3.3.5. Such welds used for parts positioning during assembly shall be removed.

[1]AISI Standard 300 (out of print). American Iron & Steel Institute, 1101 17th Street, N.W., Suite 1300, Washington, D.C. 20036.

● **E.2.11.3 Low Temperature**

For operating temperatures below –20°F (–29°C) or when specified for other low ambient temperatures, steels shall have, at the lowest specified temperature, an impact strength sufficient to qualify under the minimum Charpy V-notch impact energy requirements of Section VIII, Division 1, UG-84, of the ASME Code. For materials and thickness not covered by the Code, the purchaser will specify the requirements on the data sheet.

E.2.12 NAMEPLATES AND ROTATION ARROWS

E.2.12.1 A nameplate shall be securely attached at an easily accessible point on the equipment and on any other major piece of auxiliary equipment.

E.2.12.2 The rated conditions and other data shall be clearly stamped on the nameplate and shall include, but be not be limited to, the following:

a. Vendor.
b. Model Number.
c. Serial Number.
d. Size.
e. Type.
f. Purchaser's equipment item number (may be listed on separate nameplate if space is insufficient).
g. Volume (actual cubic feet per minute or actual cubic meters per minute).
h. Static pressure differential (inches H_2O or millimeters H_2O).
i. Temperature, inlet.
j. Revolutions per minute, rated.
k. Revolutions per minute, maximum allowable (at maximum allowable temperature).
l. First critical speed.
m. BHP or kilowatts (rated).
n. WR^2, rated.
o. Rotor weight (pounds).
p. Design operating altitude (feet or meters above sea level).

The contract or data sheets shall specify customary, SI, or other units.

E.2.12.3 Rotation arrows shall be cast in or attached to each major item of rotating equipment.

E.2.12.4 Nameplates and rotation arrows (if attached) shall be of AISI Standard Type 300 stainless steel or of nickel-copper alloy (Monel or its equivalent). Attachment pins shall be of the same material. Welding is not permitted.

E.3 Accessories

The purchaser shall specify those accessories to be supplied by the fan vendor.

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API STD*560 95 ■ 0732290 0550233 372 ■

E.3.1 DRIVERS

E.3.1.1 General

● **E.3.1.1.1** The type of driver will be specified by the purchaser. The driver shall be sized to meet the fan rated point conditions, including external gear and/or coupling losses, and shall be in accordance with applicable specifications, as stated in the inquiry and order. The driver shall be sized and designed for satisfactory operation under the utility and site conditions specified by the purchaser.

● **E.3.1.1.2** Anticipated process variations that may affect the sizing of the driver (such as changes in the pressure, temperature, or properties of the fluid handled, as well as special plant startup conditions) will be specified by the purchaser.

E.3.1.1.3 Forced draft fan driver sizing shall consider fan performance at minimum ambient temperature.

E.3.1.1.4 Induced draft fan driver sizing shall consider possible variations in operating temperature and gas density (for example, a cold start).

Provisions for flow control, through dampering or speed variation, will allow for startup and operation to be at less than normal process operating temperature. With these features the need for greater driver size to handle low temperatures can be avoided. Operating instructions must cover the use of dampers or speed control for such cases, particularly at startup.

E.3.1.1.5 The starting conditions for the driven equipment will be specified by the purchaser, and the starting method shall be mutually agreed upon by the purchaser and the fan vendor. The driver's starting-torque capabilities shall exceed the speed-torque requirements of the driven equipment. The fan vendor shall verify that the starting characteristics of the fan and driver are compatible.

E.3.1.1.6 Unless otherwise specified, motor driven fans shall be direct connected.

E.3.1.1.7 For motor-driven units, the motor nameplate rating (exclusive of the service factor) shall be at least 110 percent of the greatest power required (including gear and coupling loses) for any of the specified operating conditions.

E.3.1.1.8 Full load and starting current, system WR^2, and motor curves showing speed-torque, speed-current, and speed-power factor shall be provided for each fan drive.

E.3.1.1.9 Motor drivers shall be capable of starting the fan, with the control damper in the minimum position, with 90 percent of the design voltage applied.

E.3.2 COUPLINGS AND GUARDS

E.3.2.1 Flexible couplings and guards between drivers and fans shall be supplied by the fan vendor unless otherwise specified on the data sheets.

E.3.2.2 Unless otherwise specified, all couplings shall be spacers with the spacer length sufficient to allow removal of the coupling hubs and allow maintenance of adjacent bearings and seals without removal of the shaft or disturbing the equipment alignment.

E.3.2.3 Each coupling shall have a coupling guard that sufficiently encloses the coupling and shafts to prevent any personnel access to the danger zone during operation of the equipment train. The guard shall be readily removable for inspection and maintenance of the coupling without disturbing the coupled machines.

E.3.3 CONTROLS AND INSTRUMENTATION

E.3.3.1 General

E.3.3.1.1 Unless otherwise specified, controls and instrumentation shall be designed for outdoor installation.

● **E.3.3.1.2** The fan vendor shall provide fan performance data (in accordance with E.5) to enable the purchaser to properly design a control system for startup and for all specified operating conditions. When specified by the purchaser, the fan vendor shall review the purchaser's overall fan control system for compatibility with fan vendor-furnished control equipment (see E.3.1.1.5).

E.3.3.2 Control Systems

E.3.3.2.1 The fan may be controlled on the basis of inlet pressure, discharge pressure, flow, or some combination of these parameters. This may be accomplished by suction or discharge throttling, or speed variation. The purchaser will specify the type and source of the control signal, its sensitivity and range, and the equipment scope to be furnished by the vendor.

E.3.3.2.2 For constant speed drive, the control signal shall actuate an operator which positions the inlet or outlet damper.

E.3.3.2.3 For a variable-speed drive, the control signal shall act to adjust the set point of the driver's speed-control system. Unless otherwise specified, the control range shall be from the maximum continuous speed to 95 percent of the minimum speed required for any specified operating case or 70 percent of the maximum continuous speed, whichever is lower.

E.3.3.2.4 The full range of the purchaser's specified control signal shall correspond to the required operating range of the driven equipment. Unless otherwise specified, the maximum control signal shall correspond to the maximum continuous speed or the maximum flow.

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API STD*560 95 ■ 0732290 0550234 209 ■

E.3.3.2.5 Unless otherwise specified, facilities shall be provided to automatically open or close (as specified) the dampers or variable inlet vanes on loss of control signal and to automatically lock or brake the dampers or vanes in their last position on loss of motive force (such as air supply or electric power). This is a specific system consideration and the associated controls shall be arranged to avoid creating hazardous or other undesirable conditions.

E.3.3.2.6 Unless otherwise specified, the fan vendor shall furnish and locate the operators, actuator linkages, and operating shafts for remote control of the dampers or variable inlet vanes. Operator output shall be adequate for the complete range of damper or variable inlet vane positions. The proposed location of operators linkages, and shafts shall be reviewed with the purchaser for consideration of maintenance access and safety.

E.3.3.2.7 External position indicators shall be provided for all dampers or variable inlet vanes.

E.3.3.2.8 Unless otherwise specified, pneumatic activators shall be mechanically suitable for 125 pounds per square inch gauge air pressure and shall provide the required output with air pressure as low as 60 pounds per square inch gauge.

E.3.3.3 Dampers or Variable Inlet Vanes

● **E.3.3.3.1** Frames for inlet dampers (unless integral with the inlet box) and outlet dampers shall be flanged and drilled air tight steel frames for tight fitting bolting to the fan or duck work. Dampers shall be either parallel or opposed blades, as appropriate to the specified control requirements. Damper blades shall be supported continuously by the shafts. No stub shafts are allowed. Damper shafts shall be sealed or packed to limit leakage, except for atmospheric air inlet dampers.

● **E.3.3.3.2** When specified, the fan vendor shall state the maximum expected leakage through the damper or vanes when closed at purchaser's specified operating temperature and pressure. Leakage stated shall correspond to pressure and temperature differentials expected with the fan operating.

E.3.3.3.3 Unless otherwise specified, the damper or variable inlet vane mechanisms shall be interconnected to a single operator. The operating mechanism shall be designed so the dampers or variable inlet vanes can be manually secured in any position.

E.3.3.3.4 Variable inlet vane operating mechanisms shall be located outside the flowing gas stream. The mechanism shall be readily accessible for in-place inspection and maintenance and be of bolted attachment construction to permit removal if necessary. Provision shall be furnished for lubrication of the mechanism during operation.

E.3.3.3.5 Variable inlet vanes shall be continuously welded to the spindle or intermittently welded on the back side of the blade with full slot welds along the full length of the front side.

E.3.4 PIPING AND APPURTENANCES

E.3.4.1 Inlet Trash Screens

Inlet trash screen(s) to prevent entry of debris shall be provided for forced draft fans handling atmospheric air. This screen shall be fabricated from 0.125 inch (3.18 millimeters) minimum diameter wire, with a mesh of $1^1/_2$ inch (38 centimeters) nominal opening. The screen shall be suitably supported by cross members. Rain hood(s) shall be provided on vertical inlets. Screen supports and rain hoods shall be of galvanized carbon steel or coated per E.3.5.1.1. Trash screen shall be of 300 series stainless steel.

E.3.4.2 Silencers and Inlet Ducts

E.3.4.2.1 The differential pressure across each inlet or exhaust silencer shall not exceed 0.8 inches (20.3 millimeters) water column.

E.3.4.2.2 Silencers shall be designed to prevent internal damage from acoustic or mechanical resonances.

E.3.4.2.3 Mineral wool fiber insulation shall not be used in silencer construction.

E.3.4.2.4 Carbon steel construction shall be of $^3/_{16}$ inch (4.8 millimeter) minimum thickness plate. Corrosion allowance and alternate material, if required, shall be specified by the purchaser.

E.3.4.2.5 Main inlet duct and silencer connections shall be flanged.

E.3.5 COATINGS, INSULATION, AND JACKETING

E.3.5.1 Coatings

E.3.5.1.1 Unless otherwise specified and when constructed of carbon or low-alloy steel or cast iron, the following areas shall be cleaned per SSPC SP 6 to a NACE[1] 3 condition and then painted with a 3 mil dry film thickness coat of inorganic zinc:

a. Internal surfaces of forced draft fan intake ducts and accessories, fan housing, and internals.
b. Internal surfaces of induced draft fan housing, inlet box(es), discharge connection, and accessories.
c. External nonmachined surfaces of all bearing pedestals and bearing housings, fan housings, inlet and discharge

[1]National Association of Corrosion Engineers, P.O. Box 218340, Houston, Texas 77218-8340.

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API STD*560 95 ■ 0732290 0550235 145 ■

connections and accessories on both insulated and uninsulated units. Apply after all external shop weldments are complete.

E.3.5.1.2 Coatings shall be selected to resist deterioration and fume generation at the maximum specified inlet gas temperature.

E.3.5.2 Insulation and Jacketing

E.3.5.2.1 Insulation clips or studs shall be shop welded on all fan housings, inlet boxes, and discharge connections where normal operating temperature is 180°F (83°C) or higher or if acoustic insulation of fans is required. Unless otherwise specified, the clips or studs will be designed and installed for a 2 inch (50 millimeter) minimum insulation thickness.

E.3.5.2.2 The insulation shall maintain a maximum jacket surface temperature of 180°F (83°C) at zero wind and 80°F (27°C) ambient conditions. Purchaser shall specify type of insulation and jacketing. This material to be supplied and field installed by other than the fan vendor unless otherwise specified.

E.4 Inspection, Testing, and Preparation for Shipment

E.4.1 INSPECTION

E.4.1.1 Material Inspection

When radiographic, ultrasonic, magnetic particle, or liquid penetrant inspection of welds, cast steel, and wrought materials is specified, the criteria in E.4.1.2 through E.4.1.5 shall apply unless other criteria are specified by the purchaser. Cast iron may be inspected in accordance with E.4.1.4 and E.4.1.5. Refer to E.2.11.1.2.

E.4.1.2 Radiography

Radiography shall be in accordance with ASTM E 94 and ASTM E 142.

The acceptance standard used for welded fabrications shall be Section VIII, Division 1, UW-52, of the ASME Code. The acceptance standard used for castings shall be Section VIII, Division 1, Appendix 7, of the ASME Code.

E.4.1.3 Ultrasonic Inspection

Ultrasonic inspection shall be in accordance with Section V, Article 5, of the ASME Code.

The acceptance standard used for welded fabrications shall be Section VIII, Division 1, Appendix 12, of the ASME Code. The acceptance standard used for castings shall be Section VIII, Division 1, Appendix 7, of the ASME Code.

E.4.1.4 Magnetic Particle Inspection

Both wet and dry methods of magnetic particle inspection shall be in accordance with ASTM E 709.

The acceptance standard used for welded fabrications shall be Section VIII, Division 1, Appendix 6, of the ASME Code. The acceptability of defects in castings shall be based on a comparison with the photographs in ASTM E 125. For each type of defect, the degree of severity shall not exceed the limits specified in Table E-2. Regardless of these generalized limits, it shall be the vendor's responsibility to review the design limits of all castings in the event that more stringent requirements are necessary. Defects that exceed the limits imposed in Table E-2 shall be removed to meet the quality standards cited above, as determined by the inspection method specified. Cracks and hot tears are unacceptable and cause for rejection.

E.4.1.5 Liquid Penetrant Inspection

Liquid penetrant inspection shall be in accordance with Section V, Article 6, of the ASME Code.

The acceptance standard used for welded fabrications shall be Section VIII, Division 1, Appendix 8 of the ASME Code. The acceptance standard used for castings shall be Section VIII, Division 1, Appendix 7 of the ASME Code.

E.4.1.6 Mechanical Inspection

● When specified, centrifugal fans shall be shop assembled prior to shipment. Drivers (if provided) and other auxiliaries shall be included in the shop assembly as specified. The purchaser shall be notified prior to completion of shop assembly to permit inspection prior to disassembly (when required) and shipment. If disassembly is required for shipment, all mating parts shall be suitably match-marked and tagged for field assembly. All equipment shall be furnished completely assembled to the maximum extent, limited only by the requirements of shipping.

During assembly of the system and before testing, each component (including cast-in passages of these components) and all piping and appurtenances shall be cleaned to remove foreign materials, corrosion products, and mill scale.

● When specified, the hardness of parts and heat-affected zones shall be verified as being within the allowable values by testing. The method, extent, documentation, and

Table E-2—Maximum Severity of Defects in Castings

Type	Defect	Maximum Severity Level
I	Linear discontinuities	1
II	Shrinkage	2
III	Inclusions	2
IV	Chills and chaplets	1
V	Porosity	1
VI	Welds	1

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witnessing of the testing shall be mutually agreed upon by the purchaser and the vendor.

E.4.2 TESTING

E.4.2.1 General

● When specified by the purchaser, the centrifugal fan equipment shall be tested. Minimum test requirements are listed in E.4.2.2. Additional requirements for a shop or field test shall be provided by the purchaser. The basis for testing may be contained in AMCA Standard 210 and AMCA Publications 203, 802, and 803.

Note: Many fan manufacturers do not have the capability to perform shop mechanical run tests except on the smaller units. The need for a shop test, along with the capability of vendors to perform the test, should be carefully considered before imposing such a requirement.

At least six weeks before the first scheduled test, the fan vendor shall submit to the purchaser, for his review and comment, detailed procedures for all running tests, including acceptance criteria for all monitored parameters.

The fan vendor shall notify the purchaser not less than five working days before the date the equipment will be ready for testing.

Equipment for specified tests shall be provided by the fan vendor.

Acceptance of shop tests does not constitute a waiver of requirements to meet field performance, under specified operating conditions, nor does inspection relieve the vendor of any required responsibilities.

E.4.2.2 Mechanical Running Test

When other test details are not specified, the testing shall include the following as a minimum:

a. The fan shall be operated from 0 percent to 115 percent of design speed for turbine drives and at 100 percent or rated speed for other drives. Operation shall be for an uninterrupted period of two hours, with stabilized bearing temperatures, to check bearing performance and vibration.
b. Operation and function of fan instrumentation and controls shall be demonstrated to the extent practical.
c. The vendor shall maintain a record of all final tests including vibration and bearing oil temperature data. Vibration measurements shall be recorded throughout the specified speed range.
d. Bearings shall be removed, inspected, and when required, reassembled in the fan after completion of a satisfactory mechanical run test.
e. All oil pressures, viscosities, and temperatures shall be within the range of operating values recommended in the vendor's operating instructions for the specified unit being tested. Oil flow rates for each bearing housing shall be determined.

All bearings shall be prelubricated.

E.4.2.3 Rotor Response Analysis

When specified, the rotor response analysis defined in E.2.7.4 shall be confirmed on the test stand.

E.4.3 PREPARATION FOR SHIPMENT

● **E.4.3.1** Equipment shall be suitably prepared for the type of shipment specified, including blocking of the rotor when necessary. When specified, preparation shall be made to make the equipment suitable for six months of outdoor storage from the time of shipment. If storage for a longer period is contemplated, the vendor shall provide recommended protection procedures.

E.4.3.2 Preparation for shipment shall be made after all testing and inspection of the equipment has been accomplished and the equipment has been approved by the purchaser. The shipping preparations shall be specified by the purchaser.

E.5 Vendor's Data

E.5.1 DATA REQUIRED WITH PROPOSALS

The following data is required with the vendor's proposals:

a. Copies of the purchaser's data sheets with complete fan vendor's information entered thereon.
b. Utility requirements, including lubricant.
c. Net and maximum operating and erection weights and maximum normal maintenance weights with item identification.
d. Typical drawings and literature to fully describe offering details.
e. Preliminary performance curves as described in E.5.2.1.

E.5.2 DATA REQUIRED AFTER CONTRACT

E.5.2.1 The fan vendor shall provide complete performance curves to encompass the map of operations, with any limitations indicated thereon. The fan vendor shall provide, as a minimum, fan static pressure/capacity and horsepower/capacity curves for 100 percent, 80 percent, 60 percent, 40 percent, and 20 percent damper position settings; and fan static efficiency/capacity curve. Where gas temperature variations are specified, separate curves shall be provided for maximum, minimum, and normal operating temperatures.

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E.5.2.2 For variable speed fan systems, the performance curves shall illustrate the degree of speed control necessary to attain rated, normal, and 50 percent of normal flows. If additional turndown is specified, an illustrative curve shall be provided.

E.5.2.3 The curves for dampered and variable speed systems shall contain a system resistance curve to illustrate the degree of control necessary to attain each operating point and shall correspond to the geometry of equipment as installed.

E.5.2.4 Fan static efficiency versus speed curves for variable speed fan systems (including fan and drivers), within the vendor's scope of supply, shall be provided.

E.5.2.5 Unless otherwise specified, the fan vendor shall provide fan and drive WR^2. For each motor-driven fan under full voltage, across-the-line starting conditions, the fan vendor shall provide:

a. Full load and starting currents.
b. Curves for motor speed versus torque, versus current, and versus power factor.
c. Allowable number of cold starts, hot restarts, or both per hour, and any at-rest period required.
d. System acceleration time versus current curve.
e. For each motor-driven fan under controlled frequency starting conditions, the fan vendor shall provide the recommended acceleration or deceleration rate for the variable frequency controller.
f. Preliminary outline and arrangement drawings and schematic diagrams.
g. Startup, shutdown, or operating restrictions recommended to protect equipment.
h. Spare parts recommendations, including drawings, part numbers, and materials.
i. List of special tools included or required.
j. Shaft seal details.
k. Certified drawings, including outline and arrangement drawings and schematic diagrams.
l. Shaft coupling details.
m. Cold-alignment setting data and data on expected thermal growth.
n. Details of damper linkages and control systems, including torque or power requirements.
o. Completed as-built data sheets.
p. Parts lists for all equipment supplied.
q. Instruction manuals covering installation, final tests and checks, startup, shutdown, operating limits, and recommended operating and maintenance procedures.

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APPENDIX F—AIR PREHEAT SYSTEMS FOR FIRED PROCESS HEATERS

F.1 General

F.1.1 SCOPE

This appendix provides guidelines for the selection or evaluation of air preheat systems applied to fired process heaters. Details of fired heater design are considered only where they interact with the air preheat system design. This appendix does not provide rules for design but indicates areas that need attention and offers calculation methods to check the suitability of systems or equipment.

The systems discussed in detail are those currently in common use in the industry and it is not intended to imply that other systems are not recommended. Many of the individual features dealt with in these guidelines will be applicable to any type of air preheat system.

F.1.2 REFERENCED PUBLICATIONS

The editions of the following standards, codes, and specifications that are in effect at the time of publication of this appendix shall, to the extent specified herein, form a part of this appendix. The applicability of changes in standards, codes, and specifications that occur after the inquiry shall be mutually agreed upon by the purchaser and the vendor.

AISC[1]

Specification for Design, Fabrication, and Erection of Structural Steel for Buildings

API

RP 530 *Calculation of Heater Tube Thickness in Petroleum Refineries*

Std 610 *Centrifugal Pumps for General Refinery Services*

Std 673 *Special Purpose Centrifugal Fans for General Refinery Services*

ASME[2]

B31.3 *Chemical Plant and Petroleum Refinery Piping*

ASTM[3]

C 543 *Specification for Mineral Fiber Blanket and Felt Insulation (Industrial Type)*

C 612 *Specification for Block and Board Thermal Insulation*

[1]American Institute of Steel Construction, 1221 Avenue of the Americas, New York, New York 10020.
[2]American Society of Mechanical Engineers, 345 East 47th Street, New York, New York 10017.
[3]American Society for Testing and Materials, 1916 Race.Street, Philadelphia, Pennsylvania 19103-1187.

F.2 General Considerations in Selecting an Air Preheat System

A number of general factors must be considered in selecting an air preheat system. Those general factors are discussed here. F.4 covers specific factors for selecting particular types and features of air preheat systems.

F.2.1 FACTORS AFFECTING SYSTEM APPLICATIONS

Combustion air preheating is usually applied to increase fired heater efficiency, and the effectiveness of air preheating must be compared with providing additional convection section surface. Air preheating becomes more attractive with increased Fred heater process inlet temperature or duty. An air preheat system allows recovery of heat that would otherwise be lost with the stack gases.

In some cases, air preheat application may allow an increase in fired heater capacity or duty. For example, when fired heater operation is limited by a large flame envelope or poor flame shape (flame impingement on tubes) or by inadequate draft (flue gas removal limitations), the addition of an air preheat system can increase heater capacity. The air preheat system equipment can overcome these limitations; however, mechanical and process factors must be carefully considered as a result of the increase in radiant heat flux. These factors include higher tube and firebox hardware temperatures and higher process film temperatures, possibly leading to unacceptable degradation or coking.

An economic analysis is generally required to justify an air preheat system, but consideration must also be given to the effect of adding air preheat on operating conditions. The economic analysis should compare the installed cost, the cost of required power, and maintenance costs with the value of expected fuel savings and/or increased capacity. In the case of a retrofit, the installed cost should include the cost of downtime of the heater for the installation to take place.

Application of combustion air preheat may provide the following advantages:

a. Improved control of combustion air flow.
b. Reduced oil burner fouling.
c. Better flame pattern control.
d. More complete combustion of difficult fuels.

Application of combustion air preheat has the following disadvantages:

a. Increased potential of corrosion of components in the flue stream due to sulfur trioxide (SO_3) attack.
b. Formation of acid mists, resulting in stack plume, if fuel sulfur content is high.

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c. Increased maintenance requirements for mechanical equipment.
d. Increased nitrogen oxide concentration in the flue gas.
e. Reduced stack effluent velocity and thermal rise of flue gases, when the existing stack is utilized in a retrofit.

F.2.2 TYPES OF AIR PREHEAT SYSTEMS

Many types of air preheat systems and designs are available, including both direct and indirect. The regenerative and recuperative designs are examples of direct air preheat systems where heat transfer takes place from the existing flue gas to the incoming combustion air. Indirect systems use an intemmediate fluid to absorb heat from the exiting flue gas and then to release it to the incoming combustion air. One other system uses an external source to heat the combustion air without cooling the flue gas.

Air preheat equipment always imposes a resistance to air flow and usually to the flue gas flow. It is normally necessary to use mechanical equipment, such as fans or blowers, to overcome these resistances. Forced draft (FD) fans are used to supply the combustion air and, where required, induced draft (ID) fans are used to remove the flue gas.

Some types of air preheaters are available as proprietary systems or designs that often provide specific arrangements that may be advantageous in certain processes.

F.2.2.1 Direct Air Preheaters

F.2.2.1.1 Regenerative Air Preheater

A regenerative air preheater contains a matrix of metal or refractory elements, which may be stationary or moving. For fired process heater applications, the commonly used regenerative air preheater has the elements housed in a rotating wheel. The elements are alternately heated in the exiting flue gas and cooled by the incoming combustion air. Figure 4 shows flows and arrangements for a typical system.

F.2.2.1.2 Recuperative Air Preheater

A recuperative air preheater has separate passages for the flue gas and the air, and heat flows from one to the other through the walls of these passages. The configuration is typically in the fomm of a tubular or plate heat exchanger in which the passages are fommed of tubes, plates, or a combination of tubes and plates clamped together in a casing. The system shown in Figure 4 is also typical of a recuperative preheater.

F.2.2.1.3 Heat Pipe Air Preheater

A heat pipe air preheater involves a number of banks of sealed heat pipes in which a heat transfer fluid vaporizes in the parts of the tubes in the flue gas and condenses in the parts of the tubes in the air. Figure 4 is system diagram for a heat pipe unit.

F.2.2.2 Indirect Air Preheaters

An indirect system typically has separate tube banks in the flue gas and combustion air streams, and pipes carrying a heat transfer fluid join one to the other. The heat transfer fluid absorbs heat from the flue gas, flows to the tubes in the air stream, and releases heat to the combustion air there. The heat transfer fluid is contained in a closed system and moves naturally or is moved by pumps. A forced circulation (or pumped) system is shown in Figure 5.

F.2.2.3 External Heat Source Air Preheater

An external heat source air preheater design uses a "once through" flow of utility or process fluid in a tubular coil located in the air stream. Figure 6 illustrates such a system.

F.2.3 FACTORS AFFECTING AIR PREHEAT SYSTEM CHOICE

Some of the factors, other than cost and efficiency, that should be considered when selecting the most appropriate type of air preheat system include:

a. Available plot area for the ducts, fans, and heat transfer device or devices.
b. The ability to service the air preheat system with minimum impact on the operating schedule of the fired heater.
c. Design features that minimize fouling or its effects or that provide onstream cleaning capability.
d. Air leakage into the flue gas as it affects corrosion of downstream equipment, consumption of power, and provision of adequate air for design combustion conditions.
e. Leakage of heat transfer fluid as it affects the safety of fired heater operation.
f. Limitations on maximum exposure temperatures.
g. Burner location and arrangement.
h. The possible need for alternate natural draft operation and the heater duty required in the natural draft mode.
i. Ease of switching safely to an alternate operating condition in the event of a component failure.
j. The potential severity of cold and corrosion and the methods available to reduce the impact of corrosion.
k. The feasibility of enlarging the air preheat system capacity to handle future increases in process requirements.
l. The effect of terminal temperatures on the available system efficiency.
m. The effect of burner type (forced versus natural draft).

F.3 Interaction of the Air Preheat System with the Fired Heater

F.3.1 PROCESS DESIGN

The interaction of the air preheater with the fired heater, in process terms, is particularly important in the case of a

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retrofit, but the same considerations apply in the case of a new installation. If an existing heater with unaltered heat transfer surface is to be retrofitted with an air preheat system and yet perform the same duty, the average flux in the radiant section will be increased as will the bridgewall temperature. A process check should be carried out to ascertain what the new conditions will be, and a revised data sheet should be prepared that reflects the new design conditions.

In redoing the data sheets and checking the process design, the excess air and radiation loss values should be reviewed. The excess air may be reduced by taking advantage of the better control afforded by forced draft combustion air supply. If effective insulation is employed, it may be more realistic to reduce the radiation loss value. If the additional surface area in the form of duct work increases the heat loss, it may be more realistic to increase the radiation loss value. The process check involves an iterative procedure since the hot air temperature will depend on the flue gas exit temperature and vice versa. As a result of these interactive features, it is clear that one cannot simply decide to preheat some air.

F.3.2 BURNERS

Even in the case of a retrofit, replacement burners must nearly always be selected. This selection should be used as an opportunity to achieve other objectives as well, for example, greater flexibility in fuel use, noise reduction, better control, and nitrogen oxides (NO_x) reduction.

If natural draft operation is required (and at what level) upon equipment failure, the type of burner is selected and sized on this basis; then combustion air pressure must be established for the preheat condition. If natural draft operation is is not a requirement, then there is a greater flexibility in the choice of bumers.

If the air preheater will have to function at high air temperatures, burners may need to be constructed of exotic alloys. In addition, difficulties may arise with the damper or burner register operation unless special arrangements are made. If a limitation on NO_x emissions must be met, air preheat temperature may have to be limited.

F.3.3 DRAFT

Most air preheat systems employ forced/induced draft (balanced draft) operation. For both operational and safety reasons, some alternative means of providing draft must be available upon loss of operation of the fans or the air preheater. The natural draft alternative for the combustion air supply has already been mentioned, but there are other techniques that do not place so great a restriction on burner selection. These include the provision of standby drivers, standby fans, and shutdown systems.

In the case of induced draft removal of the exhaust gases, a damper system that bypasses the air preheater and induced draft fan and allows the exhaust gases to escape directly is normally provided. The stack height should be selected to maintain a negative firebox pressure at a specified percentage of design load.

F.3.4 LININGS AND HEAT LOSS

Since air preheat is generally justified on fuel savings, sources of heat loss should be identified. The addition of ducts, fans, and the air preheater increases the surface from which heat loss can occur. The linings should be checked and possibly renewed since reducing heat losses also improves efficiency and reduces costs.

F.3.5 CONVECTION SECTION

In some cases, the efficiency of an existing heater is so low that other problems may arise. These problems may be solved by adding convection surface area. Additional convection surface area may be justified if:

a. Air temperature must be limited to control NO_x emissions, to handle high radiant flux, or for mechanical considerations.
b. The air preheater selected cannot tolerate high flue gas temperature leaving the heater. Additional convection surface may be added to reduce this temperature to acceptable levels.

In either of these two cases, the process design considerations are similar to those for a new heater and freedom can be exercised in choosing flue gas temperature. This flexibility in the choice of hot air temperature arises because the final exit flue gas temperature can be selected without the preheated air temperature becoming a limiting factor.

F.3.6 MECHANICAL DESIGN

In the case of a retrofit, checks should be carried out on the mechanical components of the heater to:

a. Determine if mechanical components are suitable for the new conditions.
b. Determine if significant deterioration has taken place.
c. Consider changes in design codes that may have taken place since the original heater was designed and installed.

Components that may be affected by these checks include tube supports and guides.

F.4 Selecting an Air Preheat System

General factors in the selection of an air preheat system are considered in F.2. This section describes these factors in more detail for each of the system types.

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F.4.1 REGENERATIVE AIR PREHEAT SYSTEM

F.4.1.1 Plot Area

The required plot area is comparatively large because of the need to connect air and flue gas ducting to the air preheater and to the forced and induced draft fans. If air flow measurements are to be made, air ducting length must usually be increased to improve accuracy. The air preheater and fans should be mounted independently of the fired heater structure; that is, integration is generally impractical.

Means to reduce the plot area and ductwork at grade include:

a. Stacking fans.
b. Using a vertical shaft air preheater.
c. Using duct over duct or side by side air preheater arrangements.
d. Providing axial flow fans.

F.4.1.2 Serviceability

The air preheater is mounted independently of the fired heater; therefore, the system is easily designed so that maintenance of the preheater has little impact on heater operation if natural drab capability is provided. The heat transfer surface can readily be reversed or replaced. Access should be provided at all four duct connections for service and maintenance.

F.4.1.3 Fouling and Cleanability

Heat transfer surface designs that minimize fouling are available. When fuels other than clean gas are fired, sootblowing and water washing facilities can be provided. Regular online sootblowing is recommended whenever liquid fuels are fired.

Fouling of the heat transfer surface affects thermal performance, and results in an increase in pressure drop across the unit.

F.4.1.4 Effects of Air Leakage into the Flue Gas

Air leakage into the flue gas is an inherent problem in the regenerative air preheater. The predicted leakage data at the design conditions will be provided by the air preheater manufacturer. The reduction in flue gas temperature resulting from this leakage must be considered when evaluating means to handle low temperature corrosion potential downstream of the air preheater and the air preheat system.

F.4.1.5 Limitations on Maximum Exposure Temperature

The manufacturer will provide recommended service temperature limits. The limits are generally set by metallurgical considerations.

F.4.1.6 Cold-End Corrosion of the Air Preheater

Cold-end seal corrosion may result in increased air leakage. Other effects can include a change in pressure drop and a small reduction in heat recovery. To overcome these effects, the cold-end heat transfer surface can be reversed or replaced on a scheduled basis.

Techniques to reduce cold-end corrosion include:

a. Cold air bypassing.
b. Cold air preheating ahead of the air preheater.
c. Hot air recirculation.
d. Use of low alloy corrosion resistant steel.
e. Use of ceramic coated heat transfer surface.

As with all air preheat systems that recover heat from flue gas, the protection of the downstream equipment from corrosion must be considered. Failure of the heat transfer elements, whether due to cold-end corrosion or other causes, has little impact on the operability of the fired heater. The increase of combustion air leakage to the flue gas stream, unless.significant, should not be sufficient to force firing reductions on the fired heater.

F.4.1.7 Burner Location and Arrangement

Systems using ducts to transport combustion air from the air preheater to the burners are most attractive when the burners are grouped and can be arranged in one or two adjacent plenums. Widely separated burner groups require extensive duct work.

F.4.1.8 Effects of Terminal Temperature

If air preheater elements corrode and foul, the reduction in air preheater performance, generally, will not be as much as in other systems. Fired heater capacity ususally will not be affected. Normal regenerative air preheater design includes an allowance for leaking air flows, and the additional leakage caused by leaking of the seals is readily handled by the fans without having an impact on the fired heater duty.

F.4.1.9 Increasing Air Preheater Capacity

If an increase in the fired heater capacity or a fuel change is anticipated in the future, the following options should be considered:

a. Providing a regenerative air preheater that has additional rotor depth capacity suitable for future operations.
b. Using variable speed drivers on the fans to conserve horsepower during initial operation. The fan operating curves must be satisfactory for all cases.
c. Designing ducts for future flow, temperature, and pressure requirements.

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F.4.1.10 Operation Without Fans

If natural draft burners, with low draft loss, are provided, it is possible and practical to achieve the design fired heater duty without the fans in service. Flue gas and air must bypass the air preheater. The stack height and cross-sectional area must be adequate to provide the necessary draft for the natural draft case.

F.4.1.11 Burners

The pressure differential across the air preheater causes air to leak into the flue gas. Thus additional fan capacity and horsepower are required. The use of forced draft burners with high pressure drop will increase that pressure differential and the rate of air leakage.

F.4.2 RECUPERATIVE AIR PREHEAT SYSTEM

F.4.2.1 Plot Area

The required plot area is comparatively large because of the need to connect air and flue gas ducting to the air preheater and to the forced and induced draft fans. If air flow measurements are to be made, air dusting length must usually be increased to improve accuracy.

Means to reduce the plot area and duct work at grade include:

a. Stacking fans.
b. Locating the recuperative air preheater with the fired heater convection section.
c. Using top connections on gradelocated recuperative air preheaters to minimize duct work at grade.

F.4.2.2 Serviceability

When the air preheater is mounted independently of the fired heater, the system is easily designed so that maintenance has little impact on the fired heater operation if natural draft capability is provided.

Access should be provided at all four duct connections for service and maintenance. There are no motors, moving parts, seals, or elements to be maintained.

When the air preheater is located with the fired heater convection section, maintenance cannot be performed during fired heater operation.

F.4.2.3 Fouling and Cleanability

The combustion air and flue gas are completely separated so fouling is usually limited to the flue gas side. An ample flow path is required to minimize loss of draft or air flow should fouling occur. Cleaning and water washing facilities can be provided. The manufacturer should be consulted on the advisability of online cleaning.

Fouling of the heat transfer surfaces reduces thermal performance and may increase the pressure drop across the unit. Access for offline cleaning must be considered in the duct design.

F.4.2.4 Effects of Air Leakage into the Flue Gas

Recuperative air preheat systems are usually designed for no air leakage. If substantial leakage does occur, the loss of air to the burners will reduce the fired heater capacity and the flue gas temperature to the stack.

F.4.2.5 Limitations on Maximum Exposure Temperature

The manufacturer will provide recommended service temperature limits. The limits are generally set by metallurgical and thermal expansion considerations.

F.4.2.6 Cold-End Corrosion of the Air Preheater

Metal temperatures at the coldest point must be maintained above the acid dew point to prevent corrosion; alternatively, corrosion resistant materials may be used.

Severe corrosion of the heat transfer surface can result in a change in pressure drop, a reduction in heat recovery, and a loss of fired heater capacity.

Techniques to reduce cold-end corrosion include:

a. Cold air bypassing.
b. Cold air preheating ahead of the air preheater.
c. Hot air recirculation.
d. Use of ceramic coated or glass cold-end heat transfer surface.
e. Increasing the ratio of total surface on the flue gas side to that on the air side.

As with all air preheating systems that recover heat from flue gas, the protection of the downstream equipment from corrosion must be considered.

F.4.2.7 Burner Location and Arrangement

Systems using ducts to transport combustion air from the air preheater to the burners are most attractive when the burners are grouped and can be arranged in one or two adjacent plenums. Widely separated burner groups require extensive ductwork.

F.4.2.8 Effect of Terminal Temperatures

If elements corrode and foul as a result of low flue gas temperature, air preheater capacity will be reduced. Fired heater capacity may also be reduced.

F.4.2.9 Increasing Air Preheater Capacity

If an increase in the fired heater capacity or a fuel change is anticipated in the future, the following options should be considered:

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a. Selecting a recuperative air preheater that can have elements added. Provision for adding elements usually has to be included in the original air preheater design.
b. Using variable speed drivers on the fans to conserve horsepower during initial operation. The fan operating curves must be satisfactory for all cases.
c. Designing ducts for future flow, temperature, and pressure requirements.

F.4.2.10 Operation without Fans

If natural draft burners, with low draft loss, are provided, it is possible and practical to achieve the design fired heater duty without the fans in service. Air must bypass the air preheater and be induced by the heater stack draft. When the air preheater is located below the heater flue gas outlet, flue gas must bypass the air preheater and be exhausted by the available stack draft. When the air preheater is located with the convection section, operation without the air to cool the air preheater may not be possible. Alternatively, higher draft loss may limit operation to a reduced capacity.

F.4.3 HEAT PIPE AIR PREHEAT SYSTEM

F.4.3.1 Plot Area

The required plot area is comparatively large as a result of the need to connect air and flue gas ducting to the air preheater and to the forced and induced draft fans. If air flow measurements are to be made, the air ducting length must usually be increased to improve accuracy.

Means to reduce the plot area and duct work at grade include:

a. Stacking fans.
b. Locating the heat pipe air preheater with the fired heater convection section.
c. Providing a geometry that accommodates different duct arrangements.
d. Using axial flow fans.

F.4.3.2 Serviceability

When the air preheater is mounted independently of the fired heater, the system is easily designed so that maintenance has little impact on fired heater operation if natural draft capability is provided.

Access should be provided at all four duct connections for service and maintenance.

There are no motors, moving parts, seals, or elements to be maintained.

When the air preheater is located with the fired heater convection section, maintenance cannot be conducted during fired heater operation.

F.4.3.3 Fouling and Cleanability

The combustion air and flue gas are completely separated so that fouling is limited to the flue gas side. An ample flow path is required to minimize loss of draft or air flow should fouling occur.

Sootblowing lanes should be provided and sootblowers installed whenever they are also installed in the convection coil of the fired heater. Online cleaning of the air preheater should be carried out whenever the convection coil is cleaned.

Fouling of the heat transfer surfaces reduces the thermal performance and may increase the pressure loss across the unit.

Access for offline cleaning must be considered in the duct design.

F.4.3.4 Effects of Air Leakage into the Flue Gas

Heat pipe air preheat systems are usually designed for no air leakage. If substantial leakage does occur, the loss of air to the burners will reduce the fired heater capacity and the flue gas temperature to the stack.

F.4.3.5 Limitations on Maximum Exposure Temperature

The manufacturer will provide recommended service temperature limits. The limits are generally set by metallurgical and thermal expansion considerations. Additionally, for the heat pipe, the working fluid temperature must be limited to avoid degradation and/or overpressure.

F.4.3.6 Cold-End Corrosion of the Air Preheater

Metal temperatures at the coldest point must be maintained above the acid dew point to prevent corrosion.

Severe corrosion of the heat transfer surface can result in a change of pressure drop, a reduction in heat recovery, heat transfer fluid leaks, and a loss of fired heater capacity. If a leak of working fluid occurs, a safety hazard may result.

Techniques to reduce cold-end corrosion include:

a. Cold air bypassing.
b. Cold air preheating ahead of the air preheater.
c. Hot air recirculation.
d. Increasing the ratio of total surface on the flue gas side to that on the air side.

As with all air preheat systems that recover heat from the flue gas, the protection of the downstream equipment from corrosion must be considered.

F.4.3.7 Burner Location and Arrangement

Systems using ducts to transport combustion air from the air preheater to the burners are most attractive when burners

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are grouped and can be arranged in one or two adjacent plenums. Widely separated burner groups require extensive duct work.

F.4.3.8 Effect of Terminal Temperatures

If elements corrode and foul as a result of low flue gas temperature, there will be a reduction of air preheater capacity and possibly a reduction of fired heater capacity.

F.4.3.9 Increasing Air Preheat Capacity

If an increase in the fired heater capacity or a fuel change is anticipated in the future, the following options should be considered:

a. Selecting a heat pipe air preheater that can have elements added. Provision for adding elements usually has to be included in the original air preheater design.
b. Using variable speed drivers on the fans to conserve horsepower during initial operation. The fan operating curves must be satisfactory for all cases.
c. Designing ducts for future flow, temperature, and pressure requirements.

F.4.3.10 Operation Without Fans

If natural draft burners, with low draft loss, are provided, it is possible and practical to achieve design fired heater duty without the fans in service. Air must bypass the air preheater and be induced by the heater stack draft. When the air preheater is located below the heater flue gas outlet, the flue gas must bypass the air preheater and be exhausted by the available stack draft. When the air preheater is located with the convection section, it may not be possible to operate without the air to cool the air preheater and the working fluid. Alternatively, draft losses may make it possible to operate only at a reduced capacity.

F.4.4 INDIRECT CLOSED CIRCULATING AIR PREHEAT SYSTEM

F.4.4.1 Plot Area

The overall area will be smaller than most other systems if the reheat coil is integrated with the convection section. As a minimum, plot area is required for the circulation system (surge tank, pumps, and piping) and, possibly, fans.

Means to reduce the plot area include:

a. Mounting the induced draft fan on top of the convection section.
b. Locating the forced draft fan. if required, close to the burner group. Two small fans located at widely separated burner groups could replace one large fan and extensive ducting.
c. Locating the reheat coil in the convection section.

F.4.4.2 Serviceability

Maintenance cannot be performed on the reheat coil during fired heater operation, unless it is located outside the fired heater; in which case an induced draft fan is usually required. When an induced drab fan is located on the convection section, it is less accessible for inspection and maintenance. There are no motors or moving parts in the air preheater but those on the circulation system require maintenance.

F.4.4.3 Fouling and Cleanability

The combustion air and flue gas are completely separated, so fouling is usually limited to the reheat coil.

Cleaning of the reheat coil can be handled in the same manner as the convection coil; that is by the provision of sootblowing lanes and the use of sootblowers.

Fouling of the heat transfer surfaces reduces the thermal performance and may increase the pressure loss across the unit.

F.4.4.4 Effects of Air Leakage into the Flue Gas

There is no leakage of air into the flue gas stream.

F.4.4.5 Limitation of Maximum Exposure Temperature

The heat transfer medium and the coil arrangement must be chosen so that service temperatures do not exceed the coking or degradation limits.

F.4.4.6 Cold-End Corrosion of the Air Preheat System

Metal temperatures at the coldest point in the reheat coil must be maintained above the acid dew point to prevent corrosion and possible leakage. Heat transfer fluid can be bypassed around the air coil or coils to maintain a higher fluid temperature and hence a higher reheat coil skin temperature.

Severe corrosion of heat transfer coils can result in a reduction of heat recovery and a loss of heat transfer fluid.

F.4.4.7 Burner Location and Arrangement

Widely separated burner groups do not have to have extensive ductwork to transport combustion air from the air preheater to the burners. Separate air heating coils can be used at each widely separated burner group since piping, rather than ductwork, is used to transport the heating medium.

F.4.4.8 Effect of Terminal Temperatures

If elements corrode and foul as a result of low flue gas temperature, air preheater capacity may be reduced, and a reduction of fired heater capacity is possible.

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F.4.4.9 Increasing Air Preheat Capacity

If an increase in fired heater capacity or a fuel change is anticipated in the future, the following options should be considered:

a. Leaving space for additional tubes in the reheat and air heating coils.
b. Designing the circulation system (surge tank, pumps, and piping) for the anticipated future conditions.
c. Using variable speed drivers on the fans to conserve horsepower during initial operation. The fan operating curves must be satisfactory for all cases.
d. Designing ducts for future flow, temperature, and pressure requirments.

F.4.4.10 Operation Without Fans

If natural draft burners, with low draft loss, are provided, it is possible to operate at design fired heater duty. In some configurations, air must bypass the air heating coil. Systems with the reheat coil in the convection sections may have limited duty depending on the available draft. Degradation of the working fluid may occur unless the reheat coil is drained.

F.4.4.11 Effect of Working Fluid Leakage

If a leak of working fluid occurs, there may be a safety hazard.

F.4.5 EXTERNAL HEAT SOURCE PREHEAT SYSTEM

F.4.5.1 Plot Area

The plot area required is less than that required for the other systems. An air heating coil, ductwork, and in some designs, a forced draft fan are used.

F.4.5.2 Serviceability

Maintenance has little impact on fired heater operation if natural draft capability is provided.

F.4.5.3 Fouling and Cleanability

The air heating coils can be fouled by airborne dust, pollen, and so forth unless a filter is fitted. This type of fouling is not usually significant.

F.4.5.4 Effect of Air Leakage Into the Flue Gas

There is no such leakage.

F.4.5.5 Limitation on Maximum Exposure Temperature

Generally, there is no problem.

F.4.5.6 Cold-End Corrosion of the Air Preheater

Since flue gas is not involved there is no problem.

F.4.5.7 Burner Location and Arrangement

Widely separated burner groups do not have to have extensive ductwork to transport combustion air from the air preheater to the burners. Separate air heating coils can be used at each widely separated burner group since piping rather than ductwork is used to transport the heating medium.

F.4.5.8 Effect of Terminal Temperatures

Fluid stream properties must be considered.

F.4.5.9 Increasing Air Preheat Capacity

If an increase in fired heater capacity or a fuel change is anticipated in the future, the following options should be considered:

a. Leaving space for additional tubes in the air heating coil or coils or for an additional exchanger.
b. Using variable speed drivers on a forced draft fan to conserve horsepower during initial operation. The fan operating curves must be satisfactory for both cases.
c. Designing ducts for future flow, temperature, and pressure requirements.

F.4.5.10 Operation Without Fans

If natural draft burners, with low draft loss, are provided, it is possible to operate at design fired heater duty without the fan in service. In some configurations, the air must bypass the air heating coil or coils.

F.4.5.11 Effect of Working Fluid Liquid

If a leak of working fluid occurs, there may be a safety hazard.

F.5 Safety, Operability, and Maintenance

The following paragraphs cover the basic considerations for providing an operable and safe air preheat system.

F.5.1 SAFETY

Air preheat system components that may be entered for maintenance while the fired heater is in operation should be isolated from the fired heater. Isolation may be by means of slide gates, guillotine blinds, or specially designed dampers. Consideration must be given to the tightness of closure required and to means of locking the actuator. Personnel exposure, the effects of leakage on fired heater operation, and the accessibility for actuation should be evaluated.

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When more than one process heater is connected to a common air preheat system, it is important to monitor the flue gas conditions at each heater to ensure that each has adequate air supply.

Emergency air inlets should be arranged so that a hot air blast will not harm personnel if the doors open when the forced draft fan is operating. Automatically operated air doors should be located so that they do not contact personnel when activated.

Thermal rise and effluent velocity must be evaluated so that personnel on adjacent structures will not come into contact with stack flue gas.

Periodic operational checks of the emergency air inlets, the stack damper, the spare fan or fans, and other pieces of equipment are recommended.

Temperature measuring points should be provided in the ducts to and from the air preheater to indicate overheating or possible presence of fire, resulting from a tube rupture, in the air preheater.

F.5.2 OPERABILITY

The flow element for measuring combustion air flow should be located so that only the combustion air to the burners is measured. No leakage air should be included in the measurement.

If the fired heater is to be fired over a wide operating range, the use of a variable speed or multispeed fan driver should be considered. These drivers can provide improved control, reduce noise, and conserve power.

When forced draft burners are used, operation on natural draft may not be possible. Personnel should be alerted to this operating fact.

Cleaning facilities should be provided at the air preheater whenever liquid fuels are fired. Online cleaning of the induced daft fan may also be desirable.

F.5.3 ACCESSIBILITY FOR MAINTENANCE

The most desirable location for duct blinds and dampers is near grade to limit work on or over an operating fired heater.

When locating the fans and the air preheater, accessibility for maintenance should be considered.

F.5.4 PERFORMANCE CHECKS

Temperature and pressure measuring points in all flue gas and air streams entering and leaving air preheater elements should be provided to monitor performance and fouling.

Pressure measuring points upstream and downstream of the fans should be provided to assist in monitoring performance.

Air preheater performance may be adversely affected by maldistribution of air or flue gas flows. Provision of ports to allow pitot tube checks of entering air and flue gas flow profiles should be considered.

Where continuous flue gas analysis is required, properly sized and oriented analyzer mounting ports must be provided.

F.5.5 AIR PREHEAT SYSTEM EQUIPMENT FAILURE

The preheat system and auxiliary equipment selected along with the required operational reliability of the fired heater will indicate the proper action in the event of equipment failure. Choices include:

a. Changing to natural draft.
b. Bypassing the air preheater.
c. Activating a spare fan or fans.
d. Shutting down the fired heater.

It should be remembered that not only the means to change the operation must be provided, but also the means to check that such change has been safely and successfully executed.

F.6 Selecting System Temperatures

F.6.1 INTRODUCTION

The design objective common to most air preheat systems is to obtain the maximum economic heat recovery consistent with availability and reasonable initial and maintenance costs. To achieve this objective, it is important to select cold-end design temperatures that limit fouling and corrosion and to provide a means for controlling cold-end temperatures at or above acceptable values.

The temperature at which corrosion and fouling become excessive is affected by:

a. Fuel sulfur or other contaminant content.
b. Fuel or flue gas additives.
c. Flue gas oxygen content.
d. Flue gas moisture content.
e. Combustion temperature.
f. Furnace cleanliness.
g. Burner design.
h. Air preheater design.
i. Ash content from heavy residual oils.

It is important to provide means for controlling cold-end temperatures in an air preheat system design because:

a. Ambient conditions change daily and seasonally.
b. Combustion conditions change with regard to furnace cleanliness, excess air, and duty.
c. Fuel conditions may change.
d. The temperature at which excessive corrosion and fouling will occur is difficult to predict with precision.

The system designer is responsible for assessing the factors that affect cold-end corrosion and fouling for specific applications. The designer should:

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a. Select design temperatures that provide a maximum economic level of heat recovery for the normal operating conditions.
b. Provide a system of cold-end temperature control that minimizes levels of corrosion and fouling for all anticipated operating conditions.

In general, the selection of hot-end temperatures requires knowledge of equipment and material temperature limitations and an evaluation of equipment costs and alternative equipment arrangements.

F.6.2 SELECTING COLD-END TEMPERATURES

F.6.2.1 Corrosion and Fouling

Corrosion of air preheater coldend surfaces is generally caused by the condensation of sulfuric acid vapor formed from the combustion products of fuel containing sulfur. The acid deposit also provides a moist surface that is ideal for collecting panicles that foul the air preheater heat transfer surface. Consequently, it is desirable to operate air preheaters at temperatures above the acid dew point level.

Appendix F-A includes a bibliography and a summary of some of the significant test work reported in the literature.

F.6.2.2 Cold-End Temperature Control

The control of the metal temperature in the cold-end of the air preheater must accommodate changes in entering air temperature or increases in sulfur trioxide concentration resulting from a change in fuel or combustion conditions. Three methods of cold-end temperature control for use with regenerative, recuperative, and heat pipe air preheat systems are discussed in F.6.2.2.1 through F.6.2.2.3. The fourth method, reheat of fluid inlet temperature control, is only applicable to indirect air preheat systems and is covered in F.6.2.2.4.

F.6.2.2.1 Cold Air Bypass

The simplest type of cold-end temperature control is the cold air bypass in which a portion of the combustion air is bypassed around the air preheater. The reduction of combustion air flow through the air preheater results in a reduction in the cooling of the flue gas. This allows the flue gas exit temperature to be maintained at one level while other conditions vary or to be raised, if this is necessary, to avoid too low a metal temperature. Control of the flue gas exit temperature can be used to compensate for low entering air temperature or for increased sulfur trioxide concentration. Note that if the control results in higher flue gas exit temperature than the design temperature, control is achieved at the expense of furnace efficiency.

F.6.2.2.2 External Preheat of Cold Air

In this system, the desired coldend metal temperature is maintained by heating the combustion air, before it enters the air preheater, with low pressure steam or some other source of low level heat. Consideration must be given to preventing fouling and plugging of the low-level heat unit with atmospheric dust that may be entrained in the combustion air and to preventing freeze-up of the coil in cold weather.

F.6.2.2.3 Recirculation of Hot Air

In this approach, hot combustion air is recirculated to the cold end of the exchanger through the forced draft fan. This system increases the size and power consumption of the forced draft fan.

F.6.2.2.4 Reheat Fluid Inlet Temperature Control

In the circulating fluid or once through air preheat systems, the temperature of the heat transfer surfaces exposed to the flue gases is regulated by controlling the inlet temperature of the fluid being reheated. Depending on the system design and configuration, reheat fluid temperature is increased either by bypassing a portion of the fluid around the air heating coil or by decreasing the reheat fluid flow rate.

F.6.2.3 Stack Temperature Control

In most applications, the primary emphasis of coldend temperature control is directed to the temperature of the heat transfer surfaces in the flue gas stream. These surfaces are generally lower in temperature than the surfaces of downstream equipment such as the draft fan and stack. Attention must, however, be given to conditions such as cold air leakage. insulation failures, stack downdrafts resulting from low exit gas velocities, and gusty wind conditions that may result in lower metal temperatures and lead to corrosion. A recommended minimum metal temperature curve for surfaces exposed to flue gas is provided in Figure F-1. Any of the four methods to control coldend temperatures (see F.6.2.2.1 through F.6.2.2.4) may also be used to control the stack temperature.

F.6.2.4 Flue Gas Dew Point Monitoring

For air preheat systems with the capacity for reducing the stack temperatures below the design temperature, a program of local, periodic flue gas dew point testing may offer considerable economic advantages. Dew point determinations can be used as a guide for varying the cold-end temperatures. It must be recognized that the cold-end metal temperature is substantially lower than the exit gas temperature so care must be exercised when exit gas temperature measurement is the only measurement used to indicate cold-end conditions.

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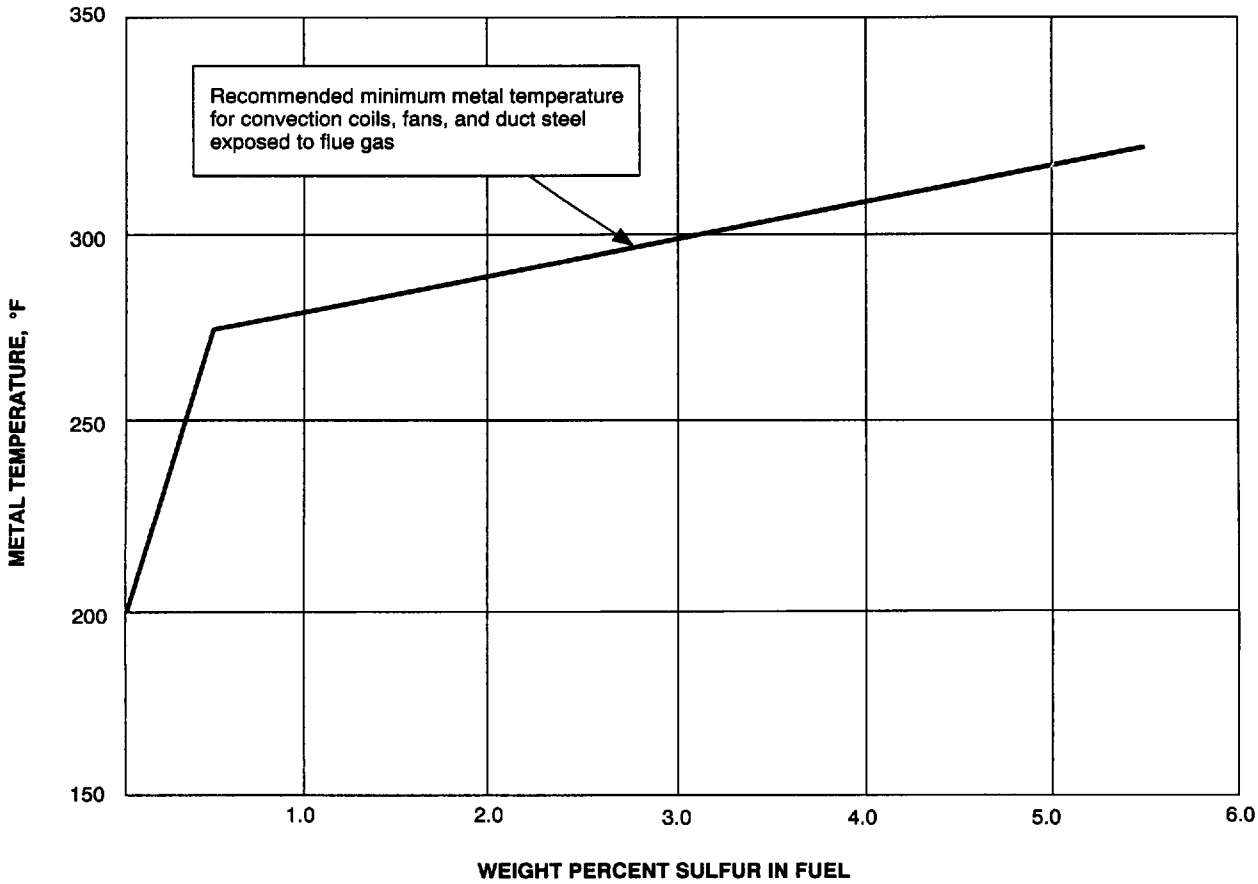


Figure F-1—Recommended Minimum Metal Temperature

F.6.2.5 Design and Materials Options

F.6.2.5.1 Regenerative Air Preheaters

The cold-end heat transfer surfaces of a regenerative air preheater are not required to serve as pressure parts confining a fluid, are designed to accommodate moderate corrosion, and should be replaced periodically. As a result, regenerative air preheaters may operate at lower metal temperatures than most other types of air preheaters. However, consideration must be given to effects on downstream equipment of the inherent air leakage and the periodic removal of acidic soot particles during sootblowing.

Regenerative air preheaters are commercially available in standard combinations of carbon steel, low-alloy steel, and corrosion-resistant enameled steel construction. The manufacturer should be consulted for recommended cold-end temperature limits.

F.6.2.5.2 Recuperative Air Preheaters

Recuperative air preheaters are commercially available with carbon steel, cast iron, and glass elements. The finning normally provided in the cast iron construction may be modified on the air side of the coldend elements to increase the metal temperatures.

Units equipped with glass elements will accommodate moderate acid condensation and fouling, but consideration must be given to requirements for the removal of deposits by sootblowing and for water washing without adversely affecting downstream equipment. In addition, the risk of breaking glass elements, particularly during cleaning operations, must be considered. The manufacturer should be consulted for minimum recommended cold-end temperatures and materials of construction for specific applications.

F.6.2.5.3 Indirect Systems

The heat transfer surfaces exposed to the fired heater flue gas in indirect systems are generally similar in construction to and located in the fired heater convection section. The service and construction of these coils make corrosion and fouling very undesirable.

Cold-end temperatures should be selected so that coils always operate with metal temperatures above the flue gas dew point. A recommended minimum metal temperature curve is shown in Figure F-1.

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F.6.3 SELECTING HOT-END TEMPERATURE

F.6.3.1 Regenerative Air Preheaters

Regenerative air preheaters are generally suitable for maximum inlet flue gas temperatures up to 1000°F (538°C). By using special materials and constructions, these air preheaters can be designed for maximum flue gas temperatures up to 1250°F (677°C). The manufacturer should be consulted for specific recommendations.

F.6.3.2 Recuperative Air Preheaters

The standard cast-iron recuperative air preheater is generally suitable for maximum inlet flue gas temperatures up to 1000°F (538°C). By using special materials and constructions, these air preheaters can be designed for maximum flue gas temperatures up to 1800°F (982°C). The manufacturer should be consulted for specific recommendations.

F.6.3.3 Indirect Systems

The coils of fluid systems, whether heat pipes or circulating, must be designed to avoid degradation of the contained fluid. For heat transfer fluids, the manufacturer's recommendation for the maximum film temperature should be followed. In the case of the heat pipe, the manufacturer should be consulted for specific recommendations.

F.6.4 COMBUSTION CONSIDERATIONS

F.6.4.1 Radiant Section Heat Flux

In fired heater coil configurations in which both a radiant and a convection surface are present, the preheating of the combustion air will increase the average radiant section flux and reduce the convection section flux for the same overall absorbed duty. Excessive tube wall temperatures may result.

F.6.4.2 NO_x Formation

If all other factors, such as burner configuration and excess air values, are held constant, the preheating of combustion air will increase the concentration of NO_x in the flue gas. It is important to select the combustion air temperatures that, together with the fuels, burners, and combustion conditions, will result in acceptable levels of NO_x formation.

F.6.4.3 Burner Design

The burner design must be appropriate for maximum design combustion air temperatures, minimum start-up or air preheater bypass operation temperature, and natural draft operation, if required.

F.7 Combustion Air and Flue Gas Quantities

F.7.1 INTRODUCTION

An important consideration in improving fired heater efficiency is the control of combustion air quantities to the lowest level while maintaining complete combustion, stable flames, and stable heater operation. (The addition of an air preheat system to improve fired heater efficiency may employ a forced and/or induced draft fan or fans.)

The discussion in this section is limited to establishing minimum combustion air and flue gas quantities for new and existing installations equipped with air preheat systems.

F.7.2 NEW INSTALLATIONS

F.7.2.1 Combustion Air Design Requirements

To establish design combustion air requirements, the type of burner system can usually be classified into two categories:

a. Natural/forced draft burner—Burners suitable for specified operation with and without the forced draft fan.
b. Forced draft or high intensity burner—Burner heat release can only be achieved with the fan in service. (It is generally recommended that this type of burner not be operated if the forced draft fan is not in service.)

The use of forced draft/high intensity burners usually permits fired heater operation at lower excess air than natural draft burners and improves fired heater operation, capacity, and efficiency without sacrificing combustion and flame stability characteristics.

F.7.2.1.1 Excess Air

The following minimum design burner excess air quantities are recommended for negative pressure fired heaters:

a. Natural draft burners designed to operate with forced draft hot/cold air and/or natural draft ambient air.
   1. Fuel gas fired, 10 percent.
   2. Fuel oil fired, 15 percent.
b. Forced draft/high intensity burners.
   1. Fuel gas fired, 5 percent.
   2. Fuel oil fired, 10 percent.

Note that these are minimum suggested values. Fired heater operation at lower excess air is not recommended without special control instrumentation or design features. Where operating or user experience dictates. Fired heaters may be designed to operate at higher excess air values. Design efficiencies of fired heaters should be at excess air quantities stated in 2.2.2.

F.7.2.1.2 Minimum Air Flow for Forced Draft Fans

If the air preheater is not airtight, the amount of leakage should be specified by the manufacturer and added directly

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to the design (100 percent) flow quantity to determine overall flow requirements. The fan selection shall be made for the calculated quantity of air for design (100 percent) operation (including any defined leakage from the preheater and other system losses) at design excess air, all multiplied by a factor of 1.15, a safety margin.

The range of ambient air conditions shall be specified by the purchaser. The forced draft fan shall be sized using maximum ambient air temperature and humidity at site elevation. Provision shall be made to allow the forced draft fan to be started and operated at minimum ambient air temperature and humidity.

F.7.2.2 Flue Gas Design Requirements for Induced Draft Fans

If the air preheater is not airtight, the amount of leakage should be specified by the manufacturer and added directly to the design (100 percent) flow quantity to determine overall flow requirements. The fan selection shall be made for the calculated quantity of flue gas for the design (100 percent) operation (including any defined leakage from the preheater and other units in a multiheater system) at design excess air, all multiplied by a factor of 1.15, a safety margin.

To allow for variation in heater operation from the design condition, an allowance for an increase in flue gas temperature should be considered. Provision shall be made to allow the induced draft fan to be started and run when the gas temperature is low, even as low as ambient. These recommended guidelines are minimum requirements, and the system should be designed to meet the user's specified requirements.

F.7.2.3 Fan Sizing

The design condition (100 percent) for the fired heater often includes allowances for safety, future process increases, or a general overage dictated by experience. As a consequence, the resulting air system may be much larger than that required for normal operation, and the heater operation at turndown may be difficult. The designer should consider the range of operation required when preparing specifications to obtain an optimum arrangement.

F.7.3 RETROFITS

Where air preheat systems are added to existing fired heater installations, flexibility in designing the most economical system is usually limited. The system designer must work closely with the user to achieve optimum results. To compensate for the possibility of greater leakage in an existing fired heater, increases in minimum design flow requirements should be considered.

Preheater air leakage may contribute to combustion requirements but leaking air does not pass through the air side of the preheater. If this fact is not taken into consideration, incorrect calculations of the exiting gas and air temperatures may result.

F.8 Duct Design and Damper Selection

F.8.1 INTRODUCTION

This section is intended to provide engineering procedures for the design and analysis of complex air preheat systems with regard to pressure drops and pressure profiles. It has been developed and based on commonly used correlations and procedures. While the calculation procedures are relatively simple, their application to duct systems common to fired air preheaters can be confusing. Comments on some specific applications have been included to provide guidance. This is not intended as a primer on fluid flow. A summary of references that provide a more complete treatment of the subject is included in F.8.9.

The basic assumption of this section is that all of the pertinent design data such as flows, temperatures, and pressure drops for equipment are available from the fired heater and air preheater designers. This data should be compiled in a usable form (see Figure F-2 as an example). In addition, the spatial relationships between the basic pieces of equipment must be known or laid out in the process of duct design.

F.8.2 PRESSURE DROP CALCULATION

The following equations and figures are a distillation of a mass of available literature on the subject of fluid flow. This material has been used successfully in the design of duct systems and it is thought to be particularly useful in that type of calculation. Two forms are presented, linear or mass velocity. Use of either form remains the preference of the designer.

F.8.2.1 Calculating Pressure Drop in a Straight Duct

$$\Delta P/100 = 3.587(f\rho V^2/d) \qquad \text{(F-1)}$$

or

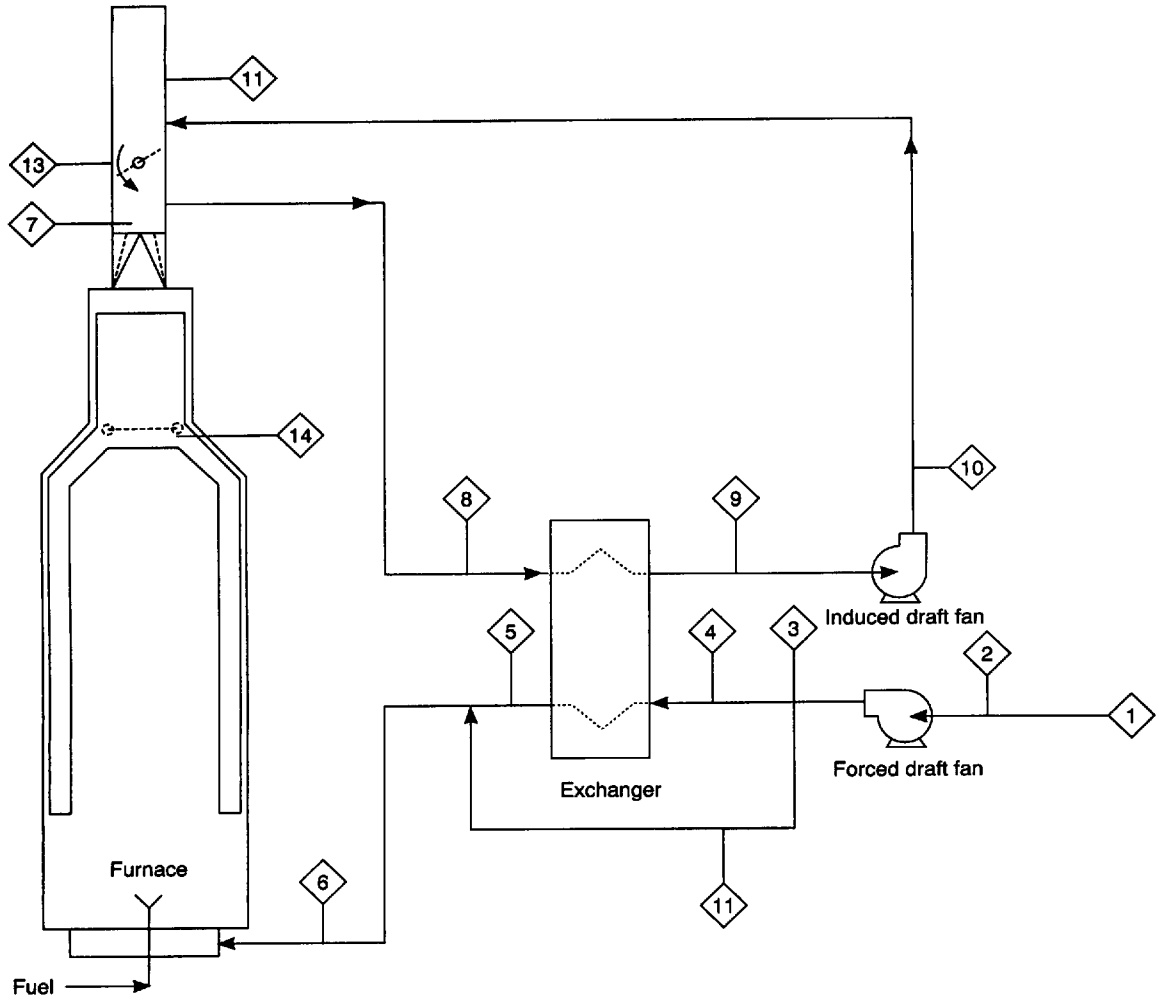
$$\Delta P/100 = 3.587\,(fg^2/d\rho) \qquad \text{(F-1)}$$

Where:

$\Delta P/100$ = pressure drop per 100 feet, in inches of water column (gage).
f = Moody's friction factor (see curve 1 on Figure F-3 and Equation F-2).
ρ = flowing density, in pounds per cubic foot.
V = linear velocity, in feet per second.
g = mass velocity, in pounds per square foot per second.
d = duct inside diameter, in inches.

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POINT NUMBER	FLOW RATE POUNDS/HOUR	TEMPERATURE °F	PRESSURE INCHES OF WATER
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			

Figure F-2—Sample Flow Sheet for Duct Design and Damper Selection

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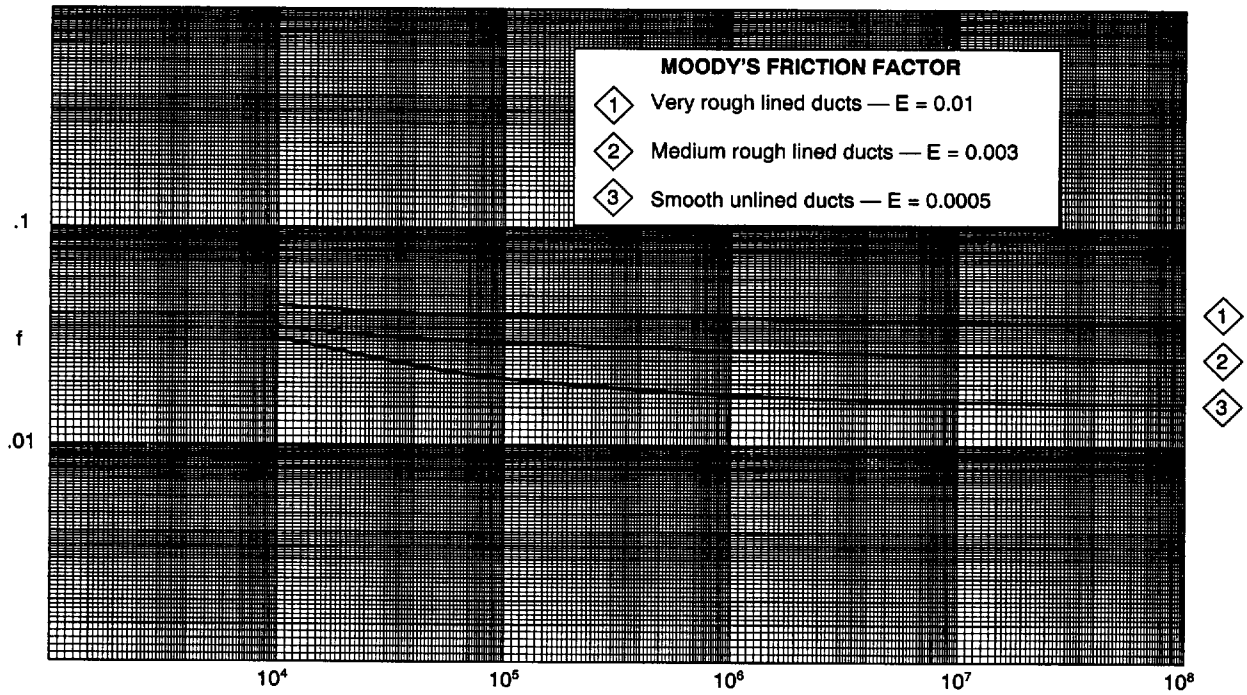


Figure F-3—Moody's Friction Factor

In SI units:

$$\Delta P/100 = 5.098 \times 10^3 (f\rho V^2/d)$$

or

$$\Delta P/100 = 5.098 \times 10^3 (fg^2/d)$$

Where:

$\Delta P/100$ = pressure drop per 100 meters, in millimeters of water column (gage).
f = Moody's friction factor.
ρ = flowing density, in kilograms per cubic meter.
V = linear velocity, in meters per second.
g = mass velocity, in kilograms per second per square meter.
d = duct inside diameter, in millimeters.

F.8.2.1.1 Calculating Reynolds Number

$$R_e = 123.9\, d\mathrm{V}\rho/\mu \qquad \text{(F-2)}$$

or

$$R_e = 123.9\, dg/\mu \qquad \text{(F-2)}$$

Where:

μ = viscosity, in centipoises.

In SI units:

$$R_e = 1.0\, d\mathrm{V}\rho/\mu$$

or

$$R_e = 1.0\, dg/\mu$$

Where:

μ = viscosity, in centipoises.

The following generalized equation may be used for viscosities for both air and flue gas without introducing any significant error into the pressure drop calculations.

$$\text{Viscosity} = 0.0162\,(T/460)^{0.691} \qquad \text{(F-3)}$$

Where:

T = temperature, in degrees Rankine.

In SI units:

$$\text{Viscosity} = 0.0162\,(T/255.6)^{0.691}$$

Where:

T = temperature, in degrees Kelvin.

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F.8.2.1.2 Calculating Hydraulic Mean Diameter

Equations F-1 and F-2 employ a diameter dimension (d) and hence are for round ducts. To use these equations for rectangular ducts, an equivalent circular duct diameter, referred to as the hydraulic mean diameter, should be calculated. A useful correlation for the hydraulic mean diameter is:

$$d_e = 2ab/(a + b) \qquad \text{(F-4)}$$

Where:

d_e = hydraulic mean diameter, in inches (millimeters).
a = length of one side of rectangle, in inches (millimeters).
b = length of other side of rectangle, in inches (millimeters).

Note: When using d in Equation F-4, use the actual velocity calculated for the rectangular duct.

F.8.2.1.3 Approximate Solution of Pressure Drop in Straight Duct

By making several assumptions, the calculation of pressure drop in straight ducts can be reduced to a simplifying chart, presented for convenience in Figure F-4. Any error introduced is not significant for most cases.

Note: When using a hydraulic mean diameter in Figure F-4, use the correlation shown on the curve rather than the one in Equation F-4.

F.8.2.2 Calculating Pressure Drop in Fittings and Cross-Section Changes

$$\Delta P = C(2.989 \times 10^{-3})\rho V^2 \qquad \text{(F-5)}$$

or

$$\Delta P = C(2.989 \times 10^{-3})g^2/\rho \qquad \text{(F-5)}$$

Where:

ΔP = pressure drop in fittings, in inches water gage.
C = fitting loss coefficient from Table F-1.
ρ = flowing density, in pounds per cubic foot.
V = linear velocity, in feet per second.
g = mass velocity, in pounds per square foot per second.

In SI units:

$$\Delta P = C(5.1 \times 10^{-2})\rho V^2$$

or

$$\Delta P = C(5.1 \times 10^{-2})g^2/\rho$$

Where:

ΔP = pressure drop in fittings, in millimeters water gage.
C = fitting loss coefficient from Table F-1.
ρ = flowing density, in kilograms per cubic meter.
V = linear velocity, in meters per second.
g = mass velocity, in kilograms per second per square meter.

Consideration should be given to the use of turning or splitter vanes to improve the characteristics of high pressure drop fittings. A complete treatment of this subject can be found in the references cited in F.8.9.

F.8.2.3 Calculating Pressure Drop in Branch Connections as in Header Ducts

$$H_v = (2.989 \times 10^{-3})\rho V^2 \qquad \text{(F-6)}$$

or

$$H_v = (2.989 \times 10^{-3})g^2/\rho \qquad \text{(F-6)}$$

Where:

H_v = velocity head, in inches water gage.

In SI units:

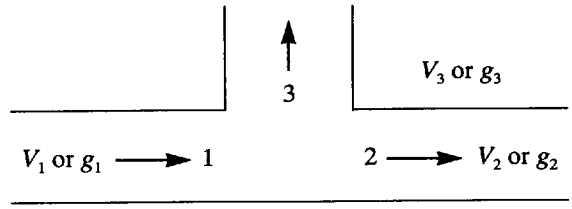
$$H_v = (5.1 \times 10^{-2})\rho V^2$$

or

$$H_v = (5.1 \times 10^{-2})g^2/\rho$$

Where:

H_v = velocity head, in millimeters water gage.



(F-7)

Where:

P Point 1 to Point 2 = 0.5 $(H_{v1} - H_{v2})$

Note: Loss coefficient of 0.5 is the net of loss and regain. It may be a lower value for a well designed branch.

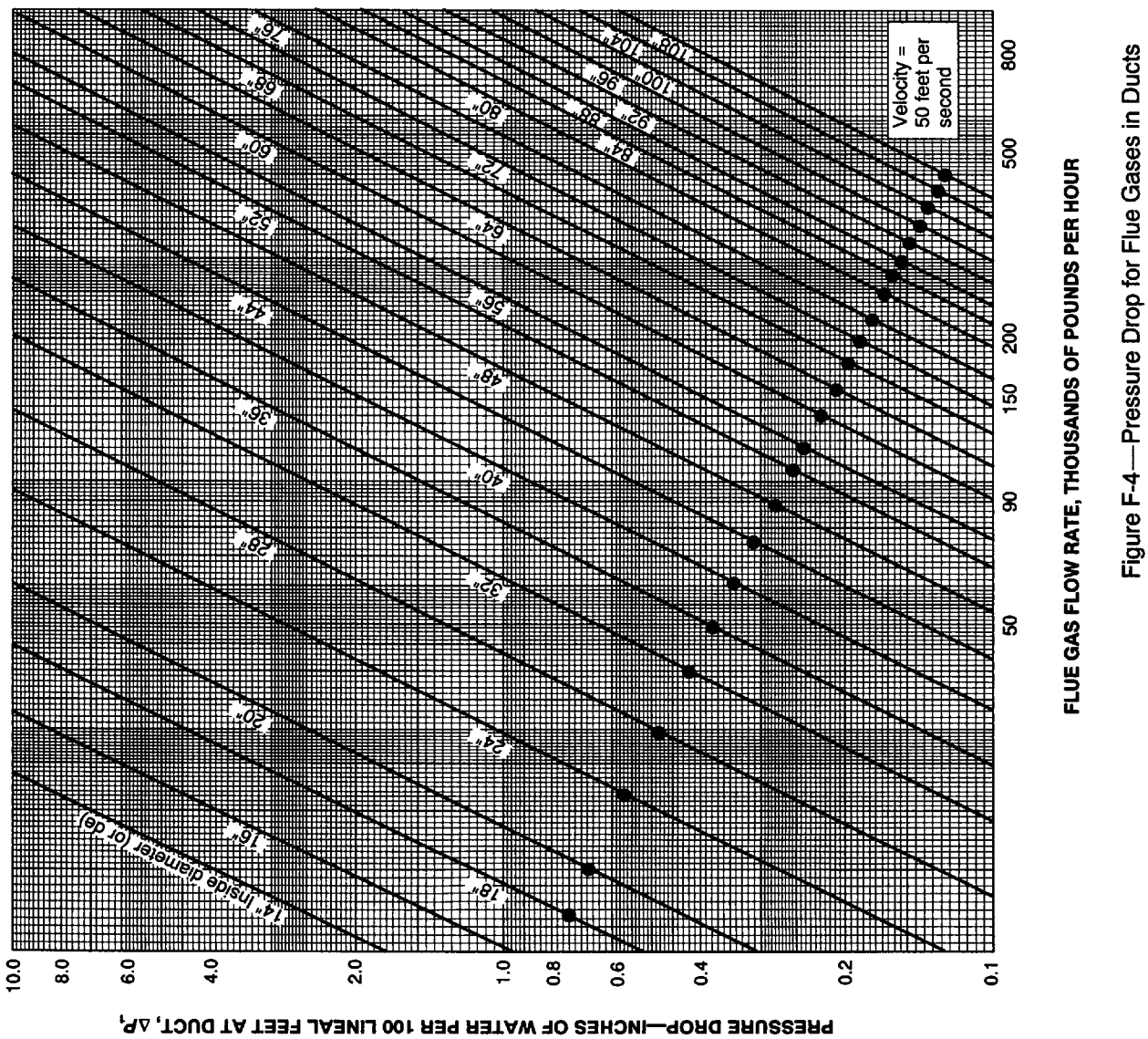
$$\Delta P \text{ Point 1 to Point 3} = H_{v1}(C_b - 1) + H_{v3} \qquad \text{(F-8)}$$

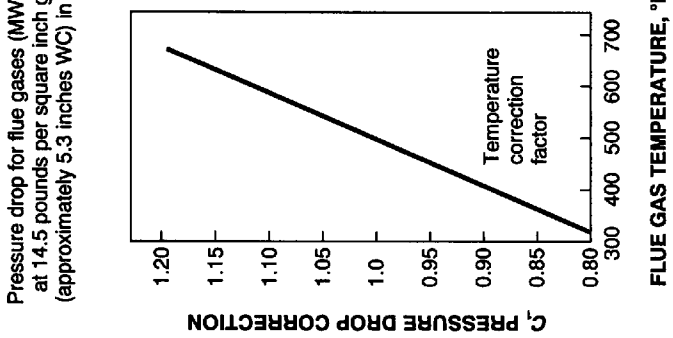
Where:

C_b = branch loss coefficient (see Figure F-6).
V_3 or g_3 = branch velocity.
V_2 or g_2 = downstream velocity.
V_1 or g_1 = upstream velocity.
V_x = linear velocity, in feet per second.
g_x = mass velocity, in pounds per second per square foot.
H_{vx} = velocity head, in inches water gage.
ΔP = pressure drop, in inches water gage.

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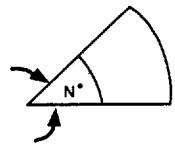
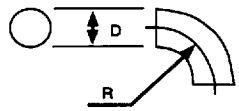
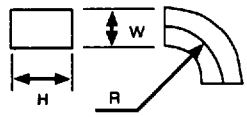
Notes:
1. Roughness correction, C_2.
   a. Very rough (for example, brick)—1.0.
   b. Medium rough (for example, castable refractory)—0.68.
   c. Smooth (for example, unlined steel)—0.45.
2. For rectangular ducts use:
   de = 1.3/ { [(ab) 0.65]/[(a+b) 0.25]}
3. $\Delta P\left[\frac{"WC}{100'}\right] = \Delta P_1(C_1)(C_2)$

Figure F-4—Pressure Drop for Flue Gases in Ducts

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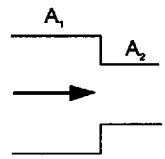
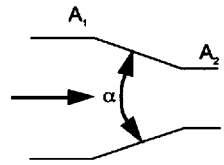
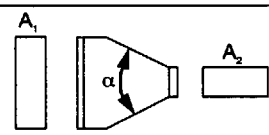
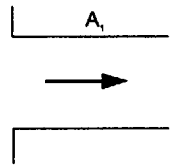
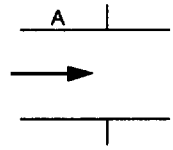
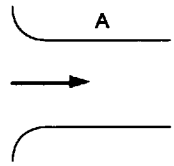
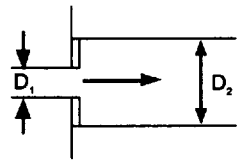
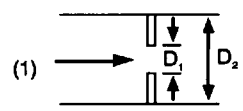
Table F-1—Fittings

Fitting Type	Fitting Illustration	Dimensional Condition	Loss Coefficient	*L/D* or *L/W*
Elbow of "N" degree turn (rectangular or round)		No vanes	*N*/90 times the value for a similar 90° elbow	
90° round section elbow		Miter (Note a) 0.5 (*R/D*) 1.0 1.5 2.0	1.30 0.90 0.33 0.24 0.19	65 45 17 12 10
90° rectangular section elbow		Miter 0.25 (*H/W*) 0.5 (*R/W*) 1.0 1.5 Miter 0.5 (*H/W*) 0.5 (*R/W*) 1.0 1.5 Miter 1.0 (*H/W*) 0.5 (*R/W*) 1.0 1.5 Miter 4.0 (*H/W*) 0.5 (*R/W*) 1.0 1.5	1.25 1.25 0.37 0.19 1.47 1.10 0.28 0.13 1.50 1.00 0.22 0.09 1.35 0.96 0.19 0.07	25 25 7 4 49 40 9 4 75 50 11 4.5 110 85 17 6
90° miter elbow with vanes			*C* = 0.1 to 0.25	
Mitered tee with vanes		Equal to an equivalent elbow (90°) (base loss on the entering velocity)		
Formed tee		Equal to an equivalent elbow (90°) (base loss on the entering velocity)		

Notes:
a. This value is for a two-piece miter. For three-, four-, or five-piece miters, see Figure F-8.
b. For permanent loss in venturis, use a loss coefficient of 0.05 based on throat area.

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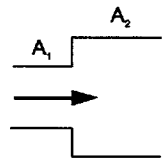
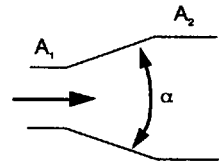
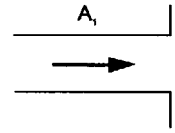
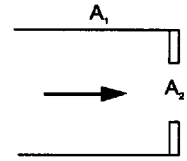
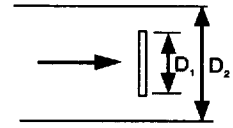
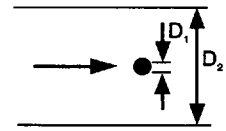
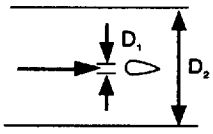
Table F-1—Fittings—continued

Fitting Type	Fitting Illustration	Dimensional Condition	Loss Coefficient Based on Velocity in Smaller Area
Sudden contraction		0.2 (A_2/A_1) 0.4 0.6 0.8	0.32 0.25 0.16 0.06
Gradual contraction		30° α 45° 60°	0.02 0.04 0.07
No contraction change of axis		$A_1 = A_2$ α ≤ 14°	0.15
Flanged entrance			0.34
Entrance to larger duct			0.85
Bell or formed entrance			0.03
Square edged orifice at entrance		0.2 (D_1/D_2) 0.4 0.6 0.8	1.90 1.39 0.96 0.61
Square edged orifice in duct	 Note b	0.2 (D_1/D_2) 0.4 0.6 0.8	1.86 1.21 0.64 0.20

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Table F-1—Fittings—continued

Fitting Type	Fitting Illustration	Dimensional Condition	Loss Coefficient Based on Velocity in Smaller Area
Sudden enlargement		0.1 (A_1/A_2) 0.3 0.6 0.9	0.81 0.49 0.16 0.01
Gradual enlargement		5° α 10° 20° 30° 40°	0.17 0.28 0.45 0.59 0.73
Sudden exit		$A_1/A_2 \cong 0$	1.0
Square edged orifice at exit		0.2 (A_1/A_2) 0.4 0.6 0.8	2.44 2.26 1.96 1.54
Bar in duct		0.10 (D_1/D_2) 0.25 0.50	0.7 1.4 4.0
Pipe or rod in duct		0.10 (D_1/D_2) 0.25 0.50	0.2 0.55 2.0
Streamlined object in duct		0.10 (D_1/D_2) 0.25 0.50	0.07 0.23 0.90

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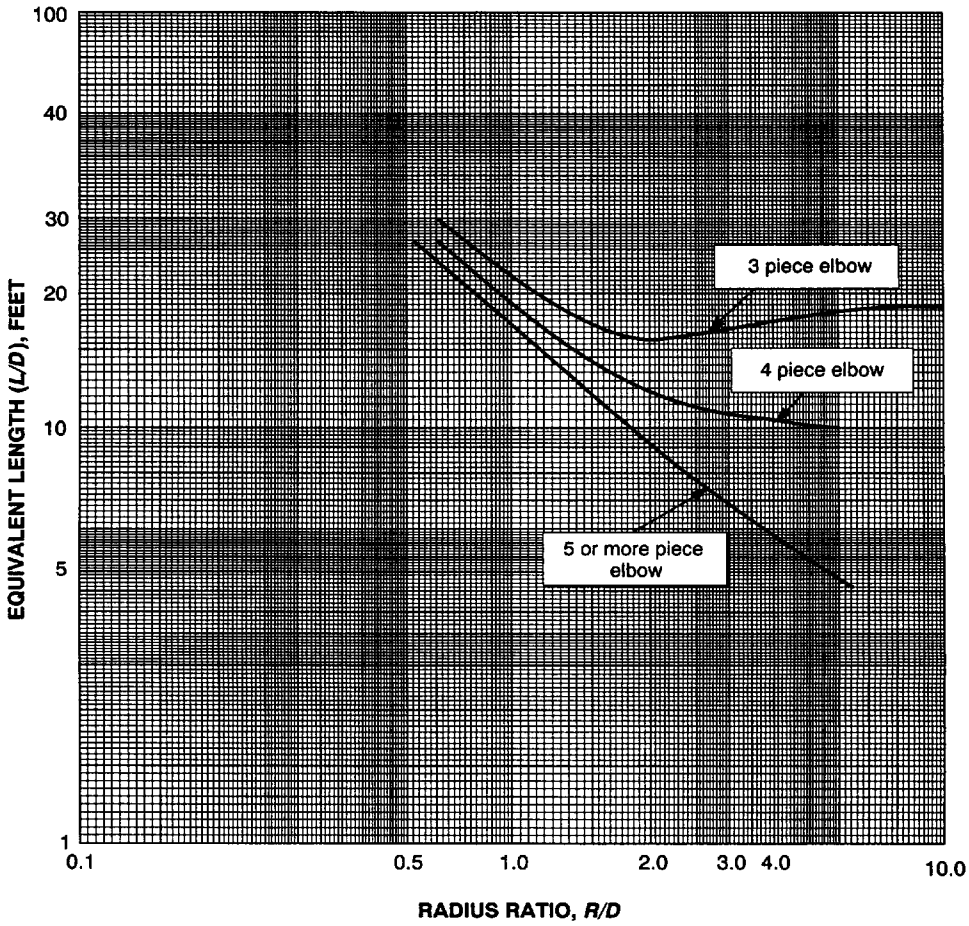


Figure F-5—Equipment Lengths (*L/D*) for Multiple Piece Miter Elbows of Round Cross Section

F.8.2.4 Calculating Differential Pressure Resulting from Differential Temperature (Draft)

$$DP = 0.0179\ (P_A)\ [(29/T_A) - (MW/T_G)]\ (Z_2 - Z_1) \quad \text{(F-9)}$$

Where:

DP = draft, in inches water gage.
P_A = atmospheric pressure at site grade, in pounds per square inch absolute.
T_A = temperature of ambient air, in degrees Rankine.
T_G = temperature of flue gas or air in duct, in degrees Rankine.
MW = molecular weight of flue gas.
Z_1 = elevation of Point 1 above grade, in feet.
Z_2 = elevation of Point 2 above grade, in feet.

In SI units:

$$DP = 0.1203\ P_A\ [(29/T_A) - (MW/T_G)]\ (Z_2 - Z_1)$$

Where:

DP = draft, in millimeters water gage.
P_A = atmospheric pressure at site grade, in kilopascals.
T_A = temperature of ambient air, in degrees Kelvin.
T_G = temperature of flue gas or air in duct, in degrees Kelvin.
MW = molecular weight of flue gas.
Z_1 = elevation of Point 1 above grade, in meters.
Z_2 = elevation of Point 2 above grade, in meters.

F.8.3 ZONE CONCEPT

Regardless of which type of air preheat system is used, one or more of the duct zones shown in Figure F-10 will probably be involved. The knowledge of the basic flows, temperatures. and pressure drops for equipment pointed out in F.8.1 as well as a knowledge of the basic spatial relationships of the components of the system are mandatory to begin meaningful calculations.

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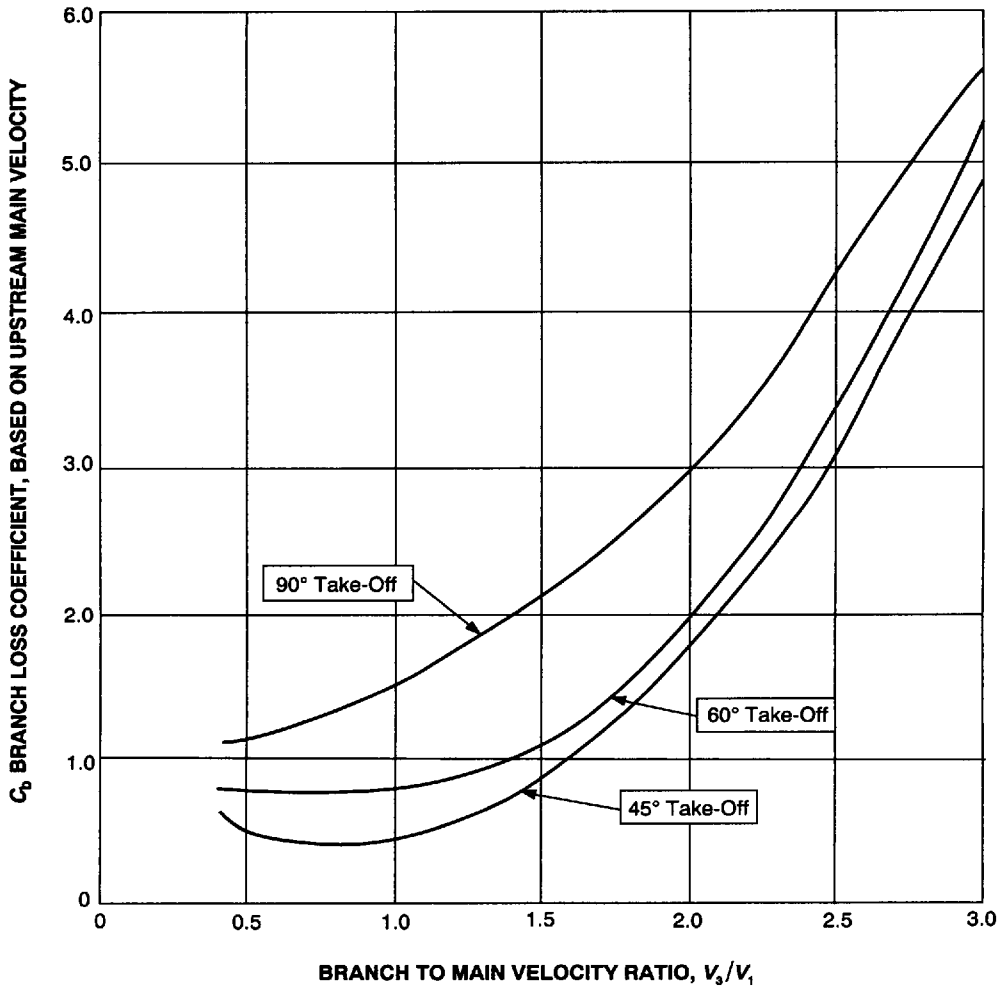


Figure F-6—Branch Loss Coefficients

F.8.3.1 Forced Air Zone

The basic elements in this zone will be the fired heater burners and plenum or heater duct, the air preheater the forced draft fan, and the inlet trunk or silencer. The nature of this zone makes it necessary that the calculations commence at the downstream terminus or burner exit.

The pressure at the burner location inside the fired heater will be the starting point. The pressure drop across the burner must be added to this pressure (whether it be negative or positive) to obtain the plenum or burner duct pressure. This may be negative, positive, or zero, depending on the combination of heater pressures and burner pressure drop.

At this point some form of branch connection or header duct calculation may be necessary depending on the configuration of the burners and the ductwork. Some allowance should be made for any dampers or flow measurement devices between the fired heater and the air preheater.

The air preheater air-side pressure drop should be known from the performance data provided by the manufacturer. Any air leakage should also be known since this must be added to the burner air flow to determine the flow rate between the air preheater and the forced draft fan.

Clearly, the sum of the heater plenum or header duct pressure, the duct pressure drop including dampers and flow devices, and the preheater air-side pressure drop will determine the forced draft fan discharge pressure. Equally clear is that the forced draft fan suction pressure will be atmospheric pressure less the inlet trunk or silencer pressure drop. This pressure drop would normally be available from the manufacturer.

F.8.3.2 Induced Draft Zone (Furnace to Induced Draft Fan)

The elements in this zone normally consist of the fired heater convection section. the stack breeching and lower

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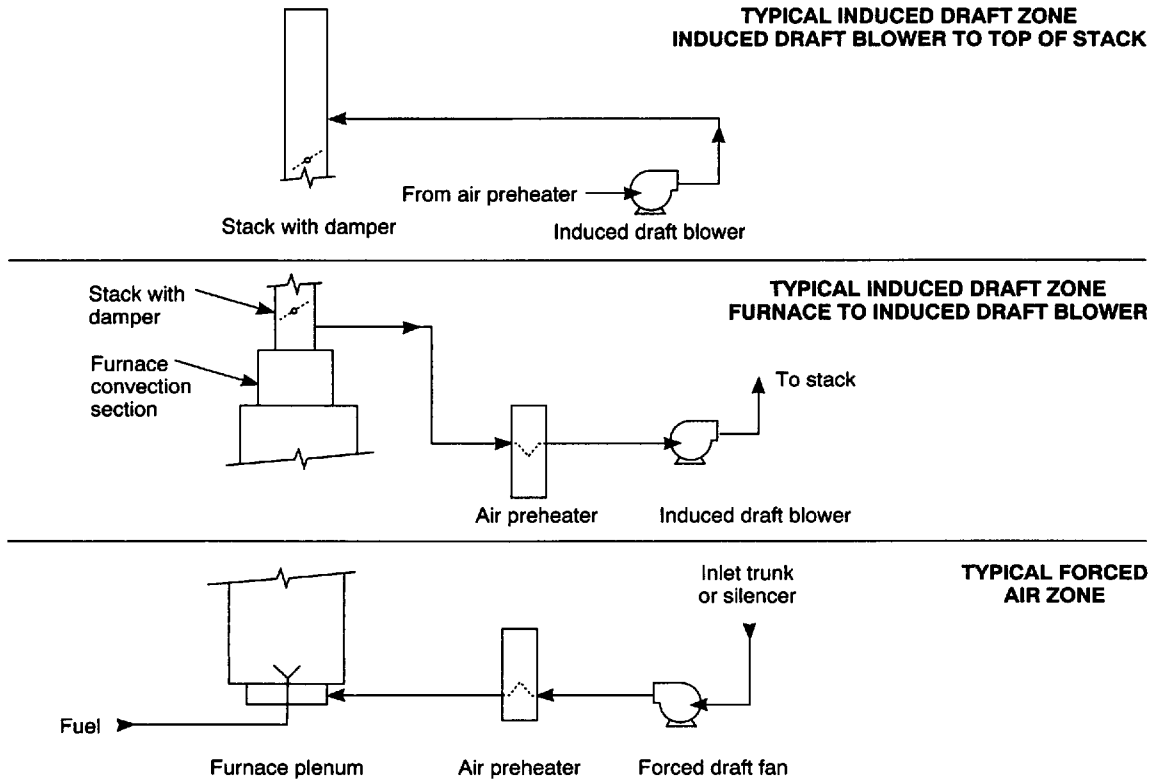


Figure F-7—Duct Zones

stack section, the isolation damper, the air preheater. and the induced draft fan. For the normal negative pressure heater setting, it is desirable to maintain the pressure immediately below the convection section at some predetermined slightly negative value. For this reason, the calculation for this zone should start at that point.

The pressure drop through the convection section should be available from the heater manufacturer. The convection section pressure drop taken in the direction of flow is a negative value and must be numerically added to the starting pressure. All pressures leading to the induced draft fan will be increasingly negative. Having determined the pressure at the exit from the convection section, the pressure drops from this point on are for ducts (including lower stack section and breeching) and the flue gas side of the preheater.

One point must be considered in any system of this type. Some pressure differential will exist across the isolation damper, and assuming that no bypassing will be permitted, some recycle leakage may occur. That is, some cooled flue gas in the stack above the isolation damper may leak through and flow along with the hot flue gas to the air preheater and induced draft fan and simply become recycle. Such recycle will reduce the effectiveness of the air preheater, and if the amount is large, may overload the fan. Unless positive steps are taken to avoid this, some allowance should be made in the flow rate and temperature of the flue gas going to the air preheater and the induced draft fan.

The pressure drop across the flue gas side of the air preheater should be known from the manufacturer. If there is any air leakage across the air preheater, it must be added at this point to determine the flow rate to the induced draft fan.

In summary, the numerical sum of the pressure required at the entry to the convection section; the convection section pressure drop; and the pressure drop in the breaching, lower stack, ductwork, including any dampers, and the preheater gas side will be the negative suction pressure for the induced draft fan. (Corrections for draft effects will be required. A discussion of this will follow the zone comments.)

Note: Pressure drops are sometimes referred to as draft losses.

F.8.3.3 Induced Draft Zone (Induced Draft Fan to Top of Stack)

The elements in this zone consist of the induced draft fan, the ductwork, and the top of the stack. It should be noted that a separate stack could be utilized so that the flue gas is not returned to the original stack.

Zone calculation must commence from the downstream exit or the atmosphere at the top of the stack. It is useful to

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treat the stack as simply an extension of the ductwork so that the calculation is reduced to a series of duct pressure drops. The sum of these pressure drops, when corrected for draft effects, becomes the induced draft fan discharge pressure. Any comments made about leakages (see F.8.3.1) must be considered in this zone.

F.8.4 DRAFT EFFECTS

All duct calculations must account for differential pressure resulting from thermal differences, commonly known as the draft effect. This effect can produce either positive or negative pressure depending on location and conditions. It must be accounted for in determining net pressure losses or gains in any system. It is often only referred to for the calculation involving the stack, but draft effects are present for any situation involving air or gas having a temperature different from the ambient temperature.

F.8.5 DUAL DRAFT SYSTEM

In those systems with burners intended to be operated on natural draft as well as in the forced or induced draft mode, take care in sizing and arranging ducts, plenums, and air door components to accommodate both types of operations. The draft on the heater side of the burners must be adequate to overcome the friction loss of the system between the burner and the atmosphere. To facilitate swift conversion to natural draft, it is common practice to operate the burner plenum at zero or slightly negative pressure when in the forced or induced draft mode. This should be accounted for in the basic design of the system.

F.8.6 VELOCITY GUIDELINES

In the absence of specific values, the designer may consider the following as a guideline to recommended duct velocities:

a. Straight duct—Recommended velocity, 50 feet (15.2 meters) per second.
b. Turns or tees—Recommended velocity, 50 feet (15 meters) per second (lower velocity may be justified by lower energy costs).
c. Burner air supply ducts—Recommended velocity, 25 feet to 35 feet (7.6 meters to 10.7 meters) per second. (An alternative approach is to set the velocity head in these ducts equal to 10 percent of the burner pressure drop.)

F.8.7 DAMPER CONSIDERATIONS AND SELECTION

F.8.7.1 General

In any attempt at duct system design, consider the placement and selection of dampers for control and isolation of various elements of the system. When selecting a damper, consider the operating differential pressure and the temperature across it. Some situations will require a much sturdier design than others.

If balancing dampers are deemed necessary in the air supply system, they should operate manually and should lock positively in the chosen position. Consider what measurements will be made or criteria set for selecting the damper position before judging that a damper is required.

Guillotine blinds or slide gates may be used to isolate equipment, either after a change to natural draft or when isolating one of several heaters served by a common preheat system. Consider exposure of personnel, the effects of leakage on heater operation, the tightness of damper shutoff, and the location of the damper (close to or remote from the affected heater).

Multiple louver, opposed blade dampers are preferred for control applications. They provide better control characteristics. Parallel blade or single blade dampers should not be applied where the flowdirecting feature inherent in their design may impair fan performance or provide an unbalanced flow distribution in the preheater.

Actuation linkage systems for dampers used for control or tight shutoff should have a minimum number of parallel or series arms. The potential for asymmetrical blade movement and leakage increases with linkage complexity.

When natural draft air inlets are required, they should be located where the flow of air to the burners will not be restricted.

The expected leakage or the leakage to be tolerated must be stated in specifying damper requirements. With the exception of isolation damper designs, the amount of leakage will vary with type and operating conditions.

F.8.7.2 Damper Function and Selection

Table F-2 provides a listing of equipment by function and recommended damper type that can be used for damper selection.

F.8.8 DUCT STRATIFICATION

Duct configurations in the vicinity of the fans, air preheater, and burners should not allow significant stratification of air or flue gas. Stratification may affect performance adversely. In addition, measurement point locations in these areas should be carefully selected to avoid errors resulting from stratification. A measurement point for traversing should be included.

F.8.9 ADDITIONAL REFERENCES

American Society of Heating, Refrigeration, and Air Conditioning Engineers, *Handbook of Fundamentals,* Second Edition, New York, 1974.

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Table F-2—Recommended Damper Types

Equipment	Function	Recommended Damper Type
Forced draft		
Inlet	Control	Radial vane damper, blade louver, or inlet box damper
Outlet	Isolation for personnel safety	Zero leakage slide gate or guillotine blind
Outlet	Control	Multiblade louver
Induced draft		
Inlet	Control	Radial vane damper, multiblade louver, or inlet box damper
Inlet	Isolation for personnel safety	Zero leakage slide gate or guillotine blind
Outlet	Isolation for personnel safety	Zero leakage slide gate or guillotine blind
Stack	Quick response, isolation. and control	Multiblade louver or butterfly damper
Combustion air bypass	Quick response, isolation, and control	Multiblade louver or butterfly damper
Emergency natural draft/air inlet	Quick response and isolation	Low leakage damper or door
Fired heater	Burner control	Multiblade or butterfly damper
	Isolation	Zero leakage slide gate or guillotine blind

Crane Co., Crane Technical Paper No. 410, New York, New York, 1957.

Buffalo Forge Co., *Fan Engineering,* Seventh Edition, Buffalo, New York, 1970.

Chemical Engineer's Handbook, Fifth Edition, Perry & Chilton, McGraw-Hill Book Co., New York, New York, 1973.

Trane Co., *Trane Air Conditioning Manual,* The Trane Co., La Crosse, Wisconsin, 1965 revision.

F.9 Guidelines for Specifying Equipment

F.9.1 INTRODUCTION

This section covers the requirements for the design and fabrication of the various connecting components of an air preheat system. The preferred choice of materials where applicable is also included.

This appendix primarily covers external interconnecting components between the fired process heater and the air preheater. For considerations concerning the fired heater or components internal to the fired heater, refer to Sections 1 through 13.

F.9.2 DUCTWORK AND STACK

F.9.2.1 General

The difference between this appendix and Sections 1 through 13 lies in the design of ductwork. The ductwork requirements for air preheat systems can be separated into two classifications: flue gas ductwork and air ductwork. Generally, the mechanical and structural design principles are the same for both. The flue gas ductwork is normally subjected to the corrosive stream (that is, flue gas). Cold-end temperatures may cause dew point corrosion, so it is important to select proper liners or ductwork material envelopes. The choice of round or rectangular duct, and internal insulation/refractory as opposed to external insulation is an economic decision that should be considered early in the design. Where space permits, round sections of ducts are recommended. Ducts shall be gastight and of flange-and-gasket or seal-welded construction.

Ductwork should be designed to permit replacement of components, such as dampers, blowers, heat exchangers, and expansion joints, within the duct system.

Air and flue gas ducts are required to provide uniform fluid flow distribution to the air preheater heat transfer surface. Failure to achieve uniform flow may cause a reduction in performance of the air heating device. Internal duct bracing, if used, should not be installed within three diameters of equipment since disruption or restriction of the flow may occur. Use of turning vanes or air straighteners should be considered to ensure uniform distribution.

F.9.2.2 Cross Section

Round ductwork is structurally simpler to design than rectangular ductwork and requires less material to contain the duct flow area. It can be reinforced with simple stiffening rings and generally requires less material for structural support. It can be designed for maximum flow area per unit of duct weight.

Rectangular ducts need to be reinforced in a manner that will keep the deflections and stresses within acceptable limits. Also, the designer should avoid having the flat side of ducts coincidently resonant with blower or fan speeds. Designing for possible buckling of flat walls may require additional bracing for stiffness.

F.9.2.3 Plenums or Wind Box

The plenum design and layout shall be such that there is a clearance around and under the plenum to permit withdrawal of burner parts without dismantling the plenum. The plenum shall not enclose the structural supports of the fired process heater without providing for structural integrity. Plenum design should be such that the process heater floor structure does not fail in the event of a fire in the plenum.

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In retrofit situations, the design of floor support beams in the existing process heater shall be verified during the design for the effects of preheated air on structural integrity. Separate insulated plenum boxes may be required. The use of air spaces between main structural supports and preheated air plenums should be considered during the design.

F.9.2.4 External or Internal Duct Linings

Internal insulation or refractory should be considered for flue gas ducts to reduce the metal temperature of the duct envelope, thereby reducing the duct thermal expansion. Consider internal coating for corrosion protection. In the event of a fire in the duct system, internal linings are desirable. Internal refractory may break loose from the duct wall and result in clogged ductwork, plugged air preheaters, and possible damage to fans. Loss of internal linings also will expose ductwork to corrosive attack and temperatures higher than design.

External insulation of ductwork may be desirable to maintain metal temperature and to prevent dew point corrosion. The ductwork will develop greater thermal expansion however, since the metal temperature is higher. External insulation can be applied after the ductwork has been set in place; it is not subject to shipping damage that may occur when insulation or refractory has been shop applied.

F.9.2.5 Mechanical Considerations

All ductwork subject to thermal expansion shall be analyzed for thermal stresses encountered at the design pressure and design metal temperature. All ductwork subject to thermal expansion shall have supports designed to freely accommodate the expected movement resulting from thermal effects or to accept the forces and stresses. The use of rollers, graphite slides, or polytetrafluoroethylene slide plates may be required to prevent binding of support shoes.

Ductwork shall be structurally designed for the maximum expected shut-in pressure of the blower or the differential pressure [that is. the outside atmospheric pressure minus the internal maximum operating pressure in absolute units of not less than 0.5 pounds per square inch (13.85 inches of water) (3.4 kilopascals)] whichever is greater. If the design defaults to 0.5 pound per square inch (3.4 kilopascals) minimum design differential pressure, it shall be assumed that the fluid pressure is positive within the duct. Flat surfaces on the rectangular ductwork, if operating at less than atmospheric pressure inside the duct, shall be designed for the expected vacuum. All flat surfaces of ductwork shall be reinforced whether under positive or negative pressure. Additional reinforcement may be required for transient conditions or resonant fan conditions.

Duct and supports should be designed to accommodate all thermal and mechanical loads that may be imposed, including erection (including the weight of wet refractory during startup, operation, or shutdown of the system). Where duct sections may be removed for maintenance activities, the effect of existing loads and new forces results in changes of deflection or stress; the entire system design shall again be mechanically verified in accordance with codes or procedures agreed to by the user and the vendor.

Where flue gas ducts tie into a stack mounted on a heater, provide a structural anchor on the duct at a point close to the outside perimeter of the heater stack. Provide an expansion joint between the fixed point (anchors) and the stack to control the significant bending moment resulting from duct thermal expansion forces reacting into the stack.

When multiple furnaces are serviced by a single air preheater, a single flue gas exhaust stack from the air preheater is recommended.

In multicell furnace design cases, or where widely spaced burners are provided that require multiple air ducts to a single furnace, consider manually adjustable and lockable dampers in each duct. These dampers will allow for balancing the flow and compensating for differences in duct hydraulic losses.

All duct sections shall be equipped with low point drain connections. These connections shall not be less than $1^1/_2$ inch nominal pipe size.

The load and thermal effects of cold weather design conditions (that is, snow and ice) during shutdowns shall be considered in the analysis of ductwork.

Manways shall be sized to a minimum of 18 by 18 inches (46 by 46 centimeters) and so located in ductwork (if size permits) to provide for internal access to the entire duct system.

Vertical, self-supporting cylindrical ducts shall be treated as stacks. They shall be designed to safely withstand wind loads and wind-induced (vortex shedding) vibrations in accordance with 9.5.

Structural force shall not be imposed on expansion joints.

Duct system expansion characteristics for lined ducts shall be based on a calculated shell temperature plus 100°F (38°C).

F.9.3 EXPANSION JOINTS

All ductwork subject to thermal expansion shall be furnished with metallic bellows or flexible fabric bellows expansion joints suitable for gas temperatures expected in the ductwork and resistant to any corrosion products in the gas stream. Internal sleeve liners to protect the bellows of the expansion joint should be considered. Stiffening rings may be installed on either end of expansion joints in the ductwork to prevent ovalling of the ductwork or other distortion of the ductwork in the event of replacement of the expansion joint.

Flexible fabric joints of suitable materials for the temperature of the flowing gas in the ductwork may be used to avoid deforming and stressing of adjoining equipment. These expansion joints are generally of layered insulated

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construction and can be constructed to withstand the flowing gas temperature.

When soft (fabric) expansion joints are used adjacent to components requiring steam cleaning or water washing, the use of internal sleeves is recommended to prevent water damage to the fabric joint.

All ducts having expansion joints at both ends shall be suitably anchored or restrained between the joints to ensure absorption of ductwork thermal growth in the expansion joints in the desired manner.

If duct thermal expansion is deliberately controlled to cause lateral deflection in the expansion joint, the expansion joint shall be specified to absorb lateral deflection or angulation without overstressing the bellows material at design temperature. Expansion joints subject to lateral deflection only shall be provided with tie rods across the bellows. The tie-rod connections to the duct work shall be gimbaled to allow lateral displacement in the expansion joint without bending or shearing the tie rods or tierod connections.

Do not use a tied expansion joint to absorb both axial and lateral deflections. Only internal pressure thrusts are contained by tie rods.

Packed slip expansion joints may be considered for negative pressure applications and shall be designed to provide positive retention of the packing to permit packing replacement from the outside while the duct is in service. These joints should be between solid anchor points in hot ductwork. They are subject to binding because of dirt, paint, or corrosion. Avoid using slip-type joints adjacent to blower/fan inlet or outlet flanges. Slide bars or guide pins shall be provided to prevent angulation or cocking in the gland if stress or friction within the gland is not consistent around the circumference. Packed expansion joints can be designed to take horizontal movements if used as two hinged joints.

F.9.4 DAMPERS

F.9.4.1 General

Dampers may be classified into four types based upon the amount of internal leakage across the closed damper at operating pressures.

a. Tight shutoff—low leakage.
b. Isolation or guillotine (slide gate)—no leakage.
c. Flow control or distribution—medium to high leakage.
d. Natural draft air inlet doors—low leakage to full open.

Tight shutoff dampers may be of single blade or multiblade construction. Leakage rates of 0.5 percent or less of flow at operating conditions are expected.

Isolation or guillotine (slide gate) dampers are designed to have no internal leakage when closed and may include double-gate with air purge or double-block-and-bleed designs consisting of one or more dampers in series with an air purge between. Internal leakage rates of 0 percent are expected with this type of damper. Dampers may have insulated blades to allow personnel to safely enter ductwork (downstream of the damper) during operation of connected equipment.

Natural draft air inlet doors shall be designed as fail-open devices in the event of loss of mechanical draft furnished by combustion air fan.

F.9.4.2 Design and Construction

Damper frames shall be channels using either rolled structural steel or formed plate. Material and weight of the frames shall be determined on the basis of any combined stress or individual stress, whichever is the maximum resulting from the following loads:

a. Seismic.
b. Wind.
c. Shipping or erection loads.
d. Actuator loading.
e. System failure or thermal or dead weight load.
f. Corroded condition load.

Dampers shall be considered structural members and as such shall meet all structural design criteria of fired heater structural members outlined in Section 8.

Damper blade deflections shall be less than $^{1}/_{360}$ of the blade span. Stress of each blade assembly component, based on maximum system static pressure, temperature, and seismic loading and the moment of inertia through the cross section of the blade assembly, shall not exceed those levels specified in AISC *Specification for Design, Fabrication, and Erection of Structural Steel for Buildings.* The torsional and bending stress shall be considered if the gas stream temperature is equal to or greater than 750°F (399°C). Allowable bending stress should be limited to 60 percent of the yield stress at the specified operating temperature. If the metal temperature is in the creep range, the allowable stress will be based upon 1 percent of the rupture stress at the 100,000 hour life span.

Each damper shall be equipped with an actuator mounted and linked by the damper manufacturer and tested in his shop before shipment. The actuator and linkage shall be installed outside of the flowing gas stream. The strength of the actuator mount on the damper frame shall be based on siesmic loading and required actuator torque. Its strength shall not exceed 10 percent of the yield strength of the damper in any mode of stress. Actuators and all drive system components shall be sized with a 3.0 safety factor.

F.9.4.3 Isolation/Guillotine Damper

The slide gate damper shall be a complete, self-sufficient structure not requiring additional integral support or bracing. The actuator for slide gate dampers shall be electric, manual, pneumatic, or hydraulic and shall be operated by sprockets,

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chains, jack screws, or a direct drive piston. If chains are used, a minimum of two chains should be used and arranged to drive evenly on each side of the blade to prevent binding. In the event of chain failure, the remaining chain or chains must support the entire blade load.

The time consumed in complete operation of the slide gate damper from full open to full closed must be specified by the user.

Operator and drive system sizing shall incorporate a 300 percent dead load plus a 200 percent live load (push-pull open/close) safety factor as a minimum. For installations that are safe for personel to enter, incorporate double block-and-bleed or double block-and-purge designs. The space between dual closed damper blades or the space between two rows of edge seals is normally purged with clean air of sufficiently greater pressure than duct stream or outside air pressure to ensure a clean air barrier to gas leaks into the duct system past the guillotine damper.

F.9.4.4 Louver Dampers

Louver dampers consist of a series of parallel damper blades. The blade construction may be a solid blade with a central axial round shaft. If the blade of the damper is of airfoil composite design, the central shaft may consist of a structural member as a central axial support of the airfoil blade. At each end, round stub shafts are splined into the axial structural member with suitable clearances to prevent buckling of the shaft as it thermally expands as a result of heat. The stub shafts pass through the bearings mounted on the damper frame. The edges of the blades are fitted with metal seals to minimize leakage past the damper edges when the damper blade is closed. These seals are often of proprietary design.

Air foil blade designs shall have blade skins provided with elongated bolt holes to compensate for thermal growth of the shaft and blade skin. Consider the use of heating holes in one side of sandwich (air foil) blade designs when excessive temperatures are encountered across closed dampers. This will reduce thermal stresses and warping of the blades. Blades and shafts shall be of thermally compatible material of similar thermal growth rates. If possible, provide for thermal growth of the damper blade away from the actuator or drive side of the damper.

Louver-style multiple damper blades shall be linked together exterior to the damper frame. Linkage shall consist of structural bar hinged with shoulder bolts, complete with lock nuts set in self-lubricating bearings of a type specified by the user. Other designs consisting of adjustable linkage to compensate for the differential expansion between the damper frame and the linkage to ensure tight shutoff at the operating temperature should be considered. Completed linkages shall be tested and fixed in position at the damper manufacturer's facility.

The link bars of each individual blade shall be welded to set collars fastened to the damper shaft with shear pins. Linkage shall be tight and vibration free and shall prevent independent action of the blade. The position of the damper on its shaft shall be scribed on the end of the shaft visible from outside the duct.

Other designs incorporating stainless steel stub shafts and linkage pins and hardware consisting of cast steel clevis arms attached to stub shaft may eliminate corrosion and may facilitate rapid removal. These designs should also be considered in situations where dampers may not be used open and tend to freeze.

Bearings shall be mounted in pillow block assemblies furnished by the bearing manufacturer and shall be bolted to bearing mounts welded to the damper frame. Each bearing and bearing mount, including welds holding the mount, shall have a duty factor capable of withstanding 200 percent of the stress transmitted as a result of the system load acting on the blade plus the operator output torque. If removable bearings are specified, linkage cranks shall be removable also. Do not weld linkage cranks to shafts.

A packing gland, if specified, shall be welded to the damper frame at each shaft clearance hole and shall be filled with packing adequate for the service. Design of the packing gland shall allow removal and replacement without removal of bearings or linkage. Packing glands are recommended in negative pressure flue gas service when sulfur bearing fuels are used.

F.9.4.5 Miscellaneous Construction Details

Dampers constructed integral to ducts, if permitted by the user, shall be of a bolted design to allow replacement of parts.

Damper bearings shall not be covered by insulation.

Damper shafts shall be of Type 304 stainless steel as a minimum.

F.9.5 DUCTWORK INSULATION AND REFRACTORY

F.9.5.1 General

Externally insulated air and flue gas duct sections shall be covered with weatherproofing and/or metal covers. The insulating material or any layer shall be suitable for a service temperature of at least 300°F (149°C) above its calculated hotface interface temperature.

Where the burner plenum is internally lined and fuel oil is fired in the heater, consider the choice of oil-resistant linings. High density refractory should be considered. Provide lining on the floor of the plenum and for at least 4 inches (10 centimeters) up the side walls.

Minimum internal lining thickness shall be 2 inches (5 centimeters). The minimum service temperature of the

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castable refractory shall be 300°F (149°C) above the maximum calculated temperature of the material.

All insulation except castable (internal or external) should be covered for weather protection during erection. Internal exposed insulation should be treated for stability or rigidity.

Duct internal casing surfaces covered by a blanket or block insulation shall have either a protective coating applied prior to application of insulating material or a vapor barrier if the flue gas contains products of combustion from fuels containing more than 1 percent by weight of sulfur in fuel oil or 1.5 percent by volume hydrogen sulfide in the fuel gas.

F.9.5.2 Ceramic Fiber Duct Lining

The use of ceramic fiber duct lining for hot flue gas or combustion air ducts shall be a minimum of 1 inch (2.5 centimeter) thick, 8 pound (metric?) per cubic foot density needled material. The back-up layer of ceramic fiber shall be a minimum of 1 inch (2.5 centimeter) thick, 4 pound (metric?) per cubic foot density needled material. When ceramic fiber construction is used, the casing shall have an internal protective coating to prevent corrosion of metal ductwork. Do not use unprotected ceramic fiber at duct bends, baffles, elbows, or constrictions.

F.9.5.3 Block and Board Insulation

Block insulation is defined as rigid, and board insulation as semirigid. Insulation should be specified as Class 3, ASTM C 612. If such insulation is not to be shielded by other materials, single layers may be used below 500°F (260°C) hot face. It may be used as a backup layer with other insulations whenever the sulfur content does not exceed 1 percent by weight in liquid fuel or 100 parts per million of hydrogen sulfide in gas fuel.

The velocity of the flowing gas stream must not exceed 20 feet per second (6 meters) unless the surface is protected with wire mesh, expanded metal, or sheet metal. Two layers of insulation are preferred.

F.9.5.4 Fiber Blanket Insulation (Mineral Wool)

Blanket insulation is a flexible material specified per ASTM C 553. Do not use unprotected insulation adjacent to water or steam cleaning devices. Surface protection consisting of wire mesh, expanded metal mesh, or chemical rigidizers shall be provided for areas where flue gas or air velocities exceed 40 feet per second (12 meters per second). Two layers are preferred. Materials shall be overlapped in the hot face on the first layer to ensure that no exposure of casing or duct envelope to lower temperature insulating materials occurs.

F.9.6 FANS AND DRIVERS

F.9.6.1 General

All fan and driver design and performance is to be in accordance with Appendix E of this standard.

F.9.6.2 Wheel Types

Maximum aerodynamic efficiency for fans can be achieved with backward inclined (nonoverloading) blades. The blade construction may be of single thickness or air foil design. On applications where the fan provides induced draft service, avoid air foil designs that have hollow cross-section blades consisting of metal skin on ribs if they are not furnished with wheel cleaning facilities. Induced draft fans handling elevated temperature flue gas containing significant particulates should be consid,ered and specified as radial or modified radial blades on the fan wheel.

F.9.6.3 Construction

Fans in combustion gas service shall have all seams continuously welded.

F.9.6.4 Shafts

Fan wheel shafts shall be capable of handling 110 percent of rated driver torque from rest to design speed.

F.9.6.5 Elimination of Induced Draft Fan

A stack of greater height than normally required might replace an induced draft fan on some systems, thereby improving the mechanical reliability of a system.

F.9.7 AIR PREHEATERS

F.9.7.1 Design Considerations for Direct Preheaters

In a fixed bundle air preheater, consider making the bundle removable if it will be subject to corrosion. Pressure parts of coils or tube bundles handling a combustible fluid shall be of all welded construction. Do not permit circumferential welds to be located in the air stream.

In rotating exchangers with metallic elements, the heating surface shall be provided in two or more layers. The coldend layer of elements shall be in baskets for radial removal through a housing. Other layers may be in baskets for removal through hot-end ductwork. Regenerative systems using revolving elements may be mechanically damaged if rotation stops while flue gas and air flow continue. An auxiliary drive on the preheater is recommended to protect against loss of rotation resulting from a power failure or other cause. An alternative action is to revert to natural draft, bypassing the preheater, until rotation can be reestablished.

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F.9.7.2 Design Considerations for Indirect Heaters

Fluid pressure-retaining circumferential field welds on the air heating element of systems employing a pumped, circulating, combustible heat medium shall be external to the air duct. Electric resistance welded tubing however is permitted for coil designs where the coil is internal to duct.

Tubular coils shall meet the requirements of Section 3, except for the extended surface in the air duct, which shall be as required by the application.

Tube wall thickness shall be in accordance with Recommended Practice 530, assuming a design life of 100,000 hours and a minimum corrosion allowance of $^{1}/_{16}$ inch (1.5 millimeters).

The extended surface shall not be included in designs intended for heavy oil fuel firing unless cleaning facilities are also included in the system.

Each pass of multiple pass coils shall be symmetrical and equal in length to all other passes. The design pressure of the coils in heated liquid service shall be based upon a pressure greater than the vapor pressure of the heating fluid at the operating temperature. This will ensure that the coil design pressure will be great enough to allow selection of pumping pressures sufficient to prevent possible two-phase (liquid/vapor) flowing regimes in the coils and to contain and hold the fluid should the blower fail with no reduction in heat input.

Recirculating reheat coils shall not be oriented to view direct radiation from the firebox or from high temperature refractory surfaces.

The performance of recirculating reheat coils is directly related to and dependent on the characteristics of the recirculating heat transfer medium in the coil. Some characteristics of the medium deteriorate with extreme service conditions. Systems with pumped closed circulating heat medium loops should incorporate provisions to drain the heat medium fluid from the heating coil in the event of low fluid flow or high flue gas temperature. Drainage should be manually or automatically actuated. Failure to drain the heating coil under these conditions may lead to thermal degradation or coking of the fluid in the coil.

All heating coils shall be drainable and shall include a high point vent and a low point drain unless specifically deleted by the user because of more appropriate locations in adjacent piping. All connection flanges should be located outside the duct periphery.

F.9.7.3 Two-Phase Operation

To ensure against "vapor lock" of the heat transfer fluid in the coils, elevate the system pressure to a level above the vapor pressure of the liquid, which ensures that the coil contains all liquid, and then reduce the pressure directly in a vapor "flash" drum downstream of the coil.

F.9.7.4 Pump Design for Circulating Systems

Pumps shall be designed in accordance with Standard 610.

Head capacity curves shall rise continuously to shut off. Rated pump capacity shall fall to the left or on the peak efficiency line.

Net positive suction head correction factors may not be required.

Pumps handling flammable or toxic liquids shall have flanged suction and discharge nozzles.

Spare pumps shall be provided unless used in a system that can be completely bypassed without detriment to the normal heater service.

F.9.7.5 Interconnecting Piping

Piping used to interconnect various components in an air preheat system shall be designed and fabricated according to ASME B31.3.

F.9.8 BURNER

Burner front plates shall be insulated to achieve a skin temperature equal to or less than the temperature specified for the air duct and plenum. Where practical, the insulation specifications for the burner shall be the same as for the duct and plenum.

A peep sight 1 inch (2.5 centimeters) or more in diameter shall be provided at each burner front plate to provide visibility of both the ignitor or pilot burner and the main flame. Consider heat resistant glass in pressurized systems.

Safe removal of oil guns, gas tips, and pilots that may be removed during heated air operation and that are located in a positive pressure plenum must be considered in the design.

Vertically fired liquid fuel burners should be designed to catch drips from burner tips and convey the drips outside the plenum. In positive pressure plenums, a drain tube can be provided with a pet cock to allow periodic blowdown by hand.

Horizontally fired liquid fuel burners shall be designed to permit drainage of oil into the firebox and prevent drainage into the plenum.

Individual burner air adjustment and air shut-off shall be provided.

Burner systems shall be designed so that light off can be accomplished without exposing the operator to positive pressure heated air. Electric ignition pilots may be required in a pressurized plenum for light-off of pilots that cannot be accomplished in a negative pressure zone of the fire box (pressurized plenum).

All modes of process heater operation such as mechanical or natural draft or turndown shall be considered in burner design and selection. The maximum combustion air pressure drop that is consistent with all modes of operation, if required, shall be used. Note that if natural draft operation is

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required, the natural draft available shall determine the burner size for the capacity specified in this mode.

Burner and burner parts intended for use with heated combustion air shall be fabricated from materials suitable for the temperature exposure.

F.10 Environmental Impact of Air Preheat Systems

There are five basic ways the the use of an air preheat system can have an environmental impact (see F.10.1 through F.10.5). In general, the environmental impact of a properly designed air preheat system is positive.

F.10.1 ENERGY CONSERVATION

The use of an air preheat system is one of the best techniques available to process plant operators for energy conservation. An air preheat system frequently provides adequate savings through reduced fuel cost to allow other pollution control systems to become economically feasible.

The use of any system of efficiency improvement results in a lower flue gas exit temperature, which will increase the possibility of an exhaust stack plume and can produce "acid rain." Although not specific to the use of an air preheat system versus any other system of efficiency improvement, the problem should be recognized. The normal way to eliminate any adverse effect is to increase the stack exit height above grade and increase the effluent velocity so that natural diffusion and wind currents will prevent acid fallout.

Since the air preheat system incorporates one or more fans, the energy is available to achieve high stack effluent velocities, and energy savings more than justify the taller stack.

In summary, the air preheat system actually provides a means to eliminate the problem of "acid rain" that can occur in many installations.

F.10.2 STACK EMISSIONS

The environmental impact of stack emissions from a fired heater is affected by the installation of an air preheat system. There are four separate pollutants that are normally monitored.

F.10.2.1 Sulfur Oxides

The sulfur oxide fraction of the flue gas is dependent solely on the composition of the gas or oil burned and is not affected to any extent by the air preheat system. However, since fuel consumption is reduced when an air preheat system is used, the pounds of sulfur dioxide (SO_2) emitted are reduced for the same process duty. This results in a net reduction in pollution and an improved environment.

F.10.2.2 Nitrogen Oxides

The oxides of nitrogen produced depend on the time, temperature, and oxygen concentration during combustion of any specific fuel. The reactions involved are many and complex. The following can be stated in general:

a. NO_x produced increases with increases in firebox or combustion temperature.
b. NO_x produced decreases with decreases in excess air.

This would indicate that preheating the combustion air can increase the NO_x formed. This has been shown to be true when expressed as concentration in the flue gas. However, as in the case of the sulfur oxides, the weight of nitrogen oxides expressed as nitrogen dioxide emitted is reduced as efficiency is increased. This tends to diminish the adverse effect of air preheat and can, in those cases where efficiencies are substantially improved, actually reduce the quantity of NO_x emitted.

Excess air appears to be the most significant factor in the control of NO_x formation. Since air preheat systems most often utilize forced draft burners, it is not only possible to operate at extremely low excess air levels, but also to more accurately control the fuel/air ratio. On natural draft burners, the minimum excess air levels previously considered necessary to provide for operational variations have been 20 percent on gas fuels and 25 percent on oil fuels. Most burner manufacturers represent that using their forced draft designs and preheated air requires minimum excess air levels of only 5 percent on gas and 10 percent on oil fuels, and controls can be furnished to ensure that these levels are maintained (see F.7.2.1 for recommendations). This low excess air operation will significantly reduce the NO_x level.

Burner manufacturers have developed special combustion type units that further reduce NO_x formation on natural draft or forced draft systems using preheated or ambient air. This allows the use of a low NO_x burner on most air preheat system applications to achieve the required NO_x levels.

F.10.2.3 Particulates

The formation of particulates during combustion is normally a function of burner application and the specific fuel burned. The use of air preheat and the forced draft systems involved have allowed burner manufacturers to reduce the formation of carbon when burning normal fuels. This can reduce the particulates formed to essentially the ash content of the fuel. Therefore, the use of an air preheat system reduces the total solids emission from many heater applications since the amount of fuel burned, and hence of ash emitted, is reduced.

F.10.2.4 Combustibles

The presence of combustibles, such as unburned hydrocarbons and carbon monoxide, in the flue gases from fired

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heaters is related to the incomplete combustion of the fuel. This in turn may result from insufficient excess air. The application of an air preheat system enhances the ability to burn fuels completely at the lowest possible excess air level. As a result, whether the burner is of the low NOx or standard type, unburned hydrocarbons and carbon monoxide can be controlled at acceptable levels (from a pollution control viewpoint) at lower excess air levels than previously possible without the use of an air preheat system.

F.10.3 NOISE

The main source of noise associated with a fired heater is normally the burners. The application of an air preheat system requires that the burners be housed in an insulated enclosure. In addition, high efficiency heater systems employ thermally more effective insulation and linings. Both of these measures reduce the noise level from the burners. Consequently, an air preheat system normally attenuates the noise from the burners below the statutory level.

However, the use of fans associated with an air preheat system introduces a new noise source, which must be considered in the initial design. Since adequate silencing techniques have been developed, it is only necessary for the designer to establish the noise level limit required so that the fan manufacturer can offer the appropriate selection.

In summary, although noise must be considered in the design of an air preheat system, it should not have an adverse impact on the overall environment of a typical heater installation.

F.10.4 HEAT

The installation of an air preheat system results in a lower flue gas exit temperature thereby reducing thermal pollution.

F.10.5 EFFLUENT

The air preheater can collect small quantities of solids combined with sulfur. During washdown cycles, if required, the liquid effluent can contain particulates with a weak acid that should be handled in an appropriate disposal system. Normally, the additional quantities produced as a consequence of the air preheat system are negligible.

F.11 Preparing an Inquiry

F.11.1 INTRODUCTION

The purpose of this section is to provide guidance and a checklist for obtaining sufficient information and data for selecting the most economical air preheat system and for preparing the required bid inquiry.

Prior to preparing a bid inquiry, it is recommended that an economic study be conducted to justify the installation of an air preheat system.

F.11.2 INQUIRY

Final selection of the air preheat system often requires cost and technical information on more than one system. This information is usually obtained from suppliers responding to the bid inquiry.

A bid inquiry for an air preheat system should include:

a. Data on the existing or proposed fired heater or heaters to be provided with an air preheat system. This data is available from the completed fired heater data sheet and should include at least the information on combustion design conditions. The fired heater operating data must represent the intended heater operation, which in the case of retrofit may vary from the original design data. If so, both the original and the intended operating data must be supplied.
b. Air preheater specification sheet.
c. Any information on space restrictions around the fired heater or heaters where the air preheat system is to be located.

F.11.3 AIR PREHEAT SYSTEM CHECKLIST

The following is a checklist of information and data to be included in the bid inquiry.

a. Applicable local building rules and regulations.
b. Environmental restrictions (noise, oxides of nitrogen and sulfur, and others).
c. Space limitations.
d. Number of fired heaters to be served by the proposed air preheat system.
e. Required reliability and service factor of the fired heater or heaters and operating modes in the event of equipment failure.
f. Fired heater data sheets.
   1. Total duty per heater, million British thermal units per hour (heat absorbed) for each operating mode: normal, maximum, and minimum.
   2. All items designated by an asterisk (*).
   3. excess air, percent.
   4. Radiation loss (lower heating value).
   5. Fuel evaluation cost.
   6. Power and other utility evaluation costs.
g. Air preheater data sheets.
   1. All performance data.
   2. List of accessories required.
   3. Other special requirements should be indicated on the data sheets at the time of inquiry.

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APPENDIX F.A—FLUE GAS DEW POINT DISCUSSION AND BIBLIOGRAPHY

F.A.1 INTRODUCTION

The furnace designer must be aware of the various design and operational factors that affect flue gas dew point and corrosion rates even though the designer only has control over a few of these variables. This summary of some of the more significant published test work describing the potential impact of these factors is intended to:

a. Broaden the designer's understanding of this complex phenomenon.
b. Serve as a starting point for further individual study.

Wherever possible, results from commercial-size equipment are reported. The results from laboratory-size equipment may not be directly applicable to commercial equipment design.

Flue gas dew point and corrosion are primarily related to the amount of sulfur trioxide present, not to the predominant amount of sulfur, which is present as sulfur dioxide. The factors that encourage or inhibit the oxidation of sulfur dioxide to sulfur trioxide are the significant factors. Many of these factors will be recognized as similar to the significant factors in the production of oxides of nitrogen. The significant factors are:

a. Fuel sulfur content.
b. Fuel and flue gas additives.
c. Flue gas oxygen content.
d. Flue gas moisture content.
e. Combustion temperature (firing rate).
f. Furnace cleanliness.
g. Burner design.

F.A.2 FUEL SULFUR CONTENT

Various investigators have differed on the impact of sulfur content of the fuel on the flue gas dew point. Corbett [1] in his tests of a commercial-size oil-fired boiler with fuel sulfur content varying from 0.75 to 3.5 weight percent found no direct relationship between flue gas dew point and the sulfur content of the fuel. Corbett's test results along with those of other investigators are plotted in Figures F.A-1 through F.A-3. Corbett's test results, along with those of other investigators, using commercial-size boilers are tabulated in Table F.A-l.

In tests with laboratory-size combustors and fuel sulfur contents in the range of 1 to 5 weight percents Rendle and Wilson [2] report an increase of the flue gas dew point of approximately 7.2°F (4°C) for each I percent increase in fuel sulfur. Taylor and Lewis [3] report an increase of the flue gas dew point of approximately 16.2°F (9°C) for each 1 percent increase in fuel sulfur.

F.A.3 FUEL AND FLUE GAS ADDITIVES

In their laboratory-size combustor, Rendle and Wilson [2] reported that by injecting ammonia vapor into the partially-cooled flue gas at a rate of 0.06 pounds of ammonia per pound of fuel containing 3.2 weight percent sulfur, they were able to reduce the dew point 160°F (89°C) to very close to the water dew point.

In tests of commerical-size boilers, Clark and Childs [4] report a 25°F (14°C) drop in dew point with magnesium hydroxide fuel additives.

F.A.4 FLUE GAS OXYGEN CONTENT

If the flue gas oxygen content can be controlled to less than 0.5 percent, the flue gas dew point can be dramatically lowered. In tests of two operating industrial boilers, Bunz, Niepenberg, and Rendle [5] demonstrated reductions in dew point of 300°F (149°C) in one boiler and 100°F (38°C) in another boiler as the flue gas oxygen was reduced from 1.4 percent to 0.2 percent oxygen.

By reducing the excess air from 15 percent (approximately 3 percent oxygen) to 5 percent (approximately 1 percent oxygen), Clark and Childs [4] report a dew point reduction of 30°F (17°C) in a commerical boiler. With flue gas oxygen content reductions from 8 percent to 3 percent at constant boiler load, Corbett's data [3] indicates little change in dew point.

F.A.5 FLUE GAS MOISTURE CONTENT

Flue gas moisture is produced by the fuel hydrogen content, ambient humidity, and atomizing steam. Martin [6] in analytical laboratory tests reports an increase in dew point of up to 15°F (8°C) as the flue gas moisture content increased from 10 percent (typical fuel oil) to 18 percent (typical fuel gas).

F.A.6 COMBUSTION TEMPERATURE (FIRING RATE)

The commercial boiler data of Draaijer and Pel [7] indicates a 20°F (11°C) increase in flue gas dew point with a 50 percent increase in firing rate and the boiler data of Bunz. Niepenberg, and Rendle 15] indicates a 45°F (25°C) increase in dew point with a 100 percent increase in firing rate.

F.A.7 FURNACE CLEANLINESS

In a commercial-size boiler fired with heavy fuel oil, Clark and Childs [4] report that the flue gas dew point is reduced 30°F (17°C) after each annual furnace cleaning.

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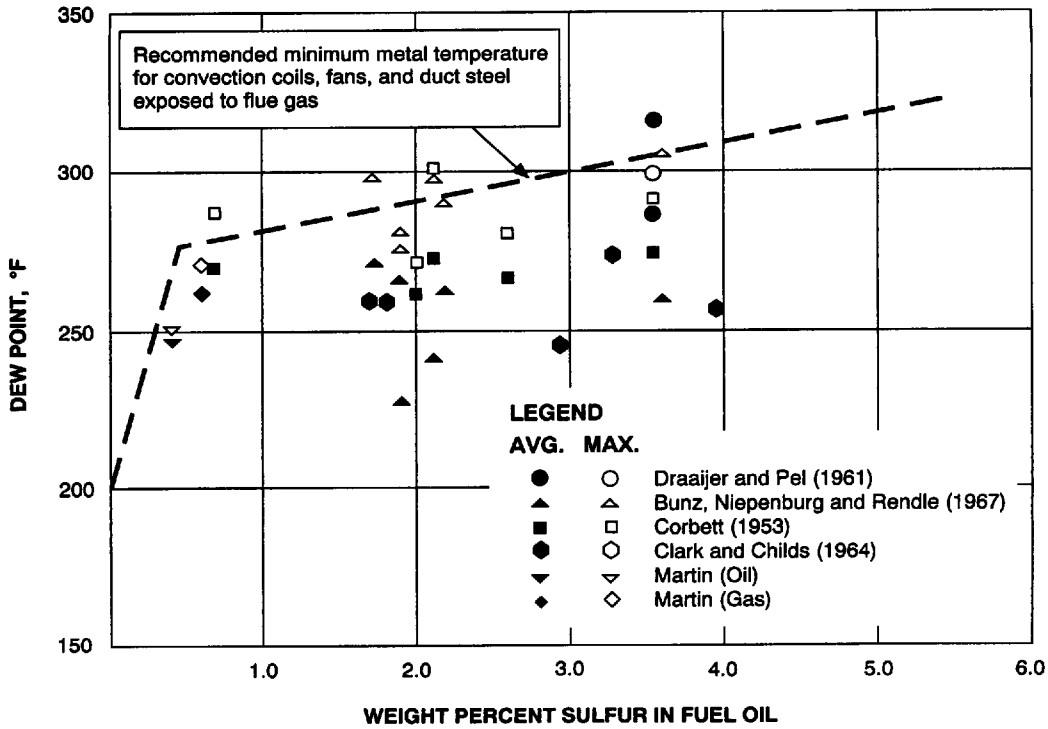


Figure F.A-1—Dew Point of Flue Gas Versus Sulfur in Fuel Oil (Test Data from Industrial Boilers)

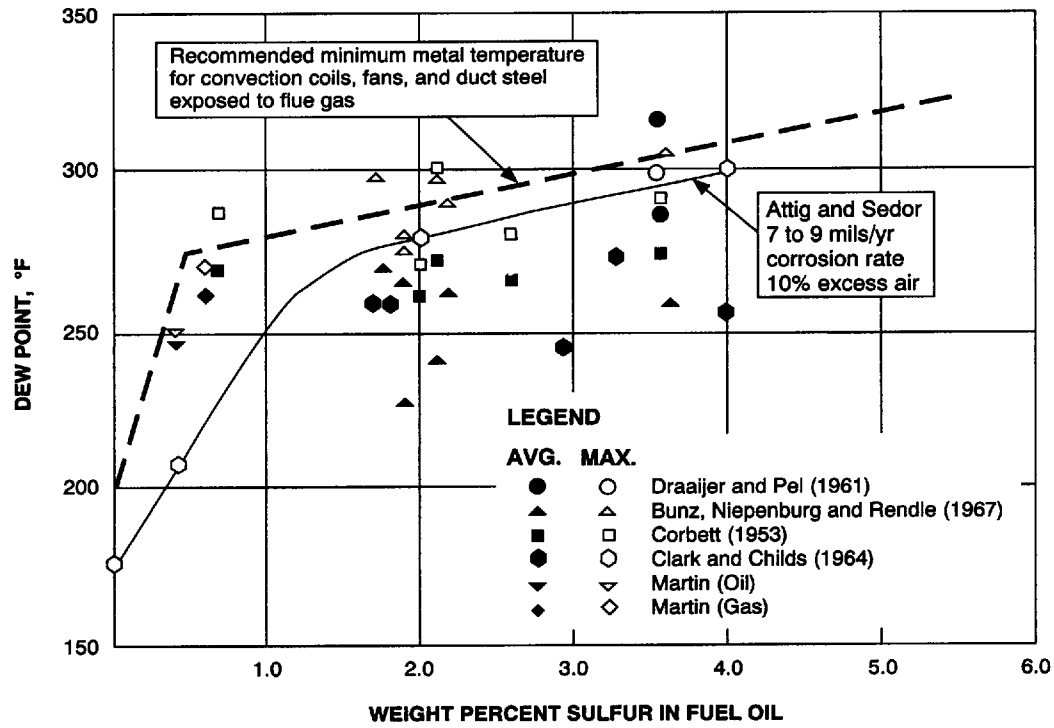
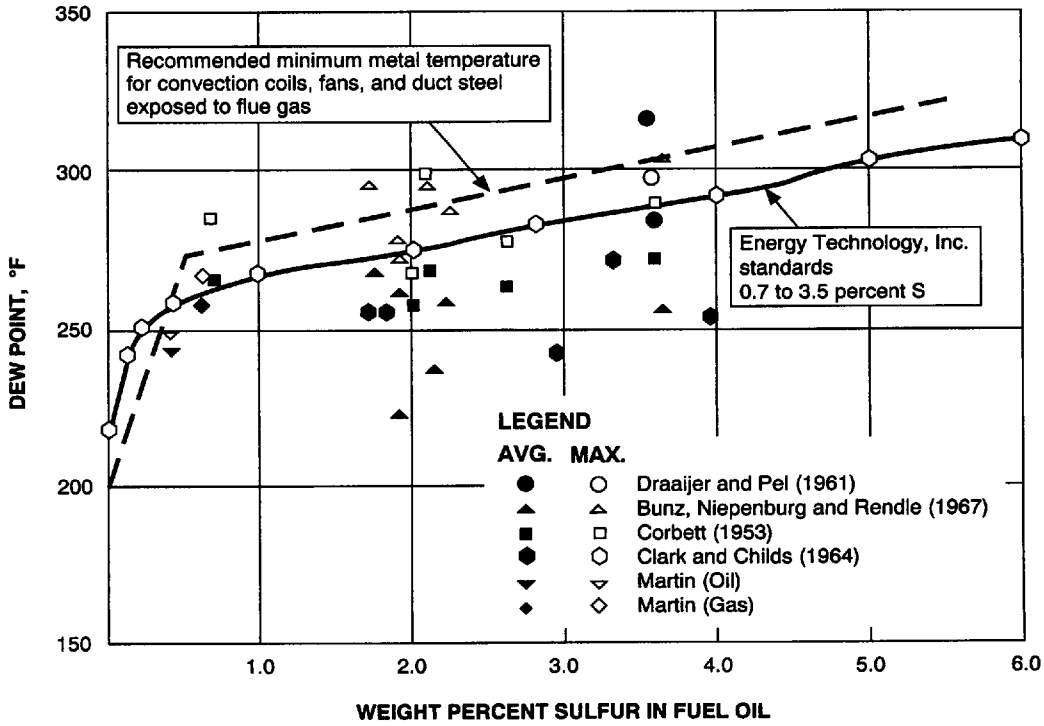


Figure F.A-2—Dew Point of Flue Gas Versus Sulfur in Fuel Oil
(Test Data from Industrial Boilers and Data of Attig and Sedor)

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F.A-3—Dew Point of Flue Gas Versus Sulfur in Fuel Oil
(Test Data from Industrial Boilers and Data of Energy Technology, Inc.)

Table F.A-1—Flue Gas Dew Point Data from Oil-Fired Industrial Boilers

Investigator	Boiler No.	Fuel Sulfur weight percent	Average Dew Point °F	Minimum Dew Point °F	Maximum Dew Point °F	Test Points	Steam Load 1000 kg/hour	Percent Excess O_2
Draaijer & Pell (1961)	1	3.55	317	315	318	4	40	—
	2	3.55	287	273	300	15	9–14	—
Bunz, Neipenberg, & Rendle (1967)	3	1.78	270	250	297	8	24 & 32	0.1–0.5
	3	1.90	264	244	280	6	24 & 32	0.1–1.5
	4	2.10	240	165	297	10	24 & 32	0.1–1.5
	4	2.18	261	187	289	10	24 & 32	0.1–1.5
	4	1.90	226	167	275	8	24 & 32	0.1–1.4
	4	3.61	260	158	306	16	16, 24, & 32	0.1–1.4
Corbett (1953)	5	2.60	266	245	280	49	9–23	3.4–13.2
	5	2.00	260	245	270	25	9–23	4.3–9.8
	5	2.10	271	243	300	27	9–22	3.1–10.6
	5	3.55	274	257	292	39	9–22	2.7–6.3
	5	0.75	267	250	285	22	13–18	3.6–8.2
Clark & Childs (1964)	6	3.97	257	—	—	—	—	—
	7	1.70	258	—	—	—	—	—
	8	1.82	258	—	—	—	—	—
	9	2.94	245	—	—	—	—	—
	10	3.29	274	—	—	—	—	—
R. Martin (1971)	11 (oil)	0.4	240[a]	287[a]	248[a]	8	100	2.7–3.6
	11 (gas)	0.6	260[a]	252[a]	268[a]	15	100	2.5–3.3

[a] Indirect measurements from SO_3 concentration. Electrical conductivity probe method would probably give lower values.

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F.A.8 BURNER DESIGN

In a laboratory combustor, Attig and Sedor [8] demonstrated that recirculation of 25 percent of the flue gases to the burners reduced flue gas sulfur trioxide concentration by one half (equivalent to a dew point reduction of at least 10°F [6°C]) and reduced corrosion rates by more than one third. The very low excess air operations reported by Bunz, Niepenberg, and Rendle [5] were achieved with a special low excess air design burner.

F.A.9 REFERENCES

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APPENDIX G—MEASUREMENT OF THE THERMAL EFFICIENCY OF FIRED PROCESS HEATERS

G.1 General

G.1.1 INTRODUCTION

This appendix is intended to establish a standard approach for measuring the thermal efficiency of fired process heaters and presents a comprehensive step-by-step procedure to be followed when measuring the thermal efficiency of fired heaters. This appendix is intended to be used for fired heaters burning liquid or Gaseous fuels. This appendix is not recommended for determining thermal efficiency when a solid fuel is being burned.

G.1.2 PURPOSE AND SCOPE

This appendix establishes a standard test procedure for determining the thermal efficiency of fired process heaters and provides instructions for conducting the necessary tests and reporting the results.

The test procedure considers only stack and radiation heat losses and the total heat input. Process data are obtained for the purposes of reference and comparison only. Any modifications of the procedure and any assumptions required for testing should be established before testing.

G.1.3 NOMENCLATURE AND DEFINITIONS

G.1.3.1 Nomenclature

The following symbols are used in this appendix:

e = net thermal efficiency, expressed as a percentage.
e_{gross} = gross thermal efficiency, expressed as a percentage.
LHV = lower heating value of the fuel burned, in British thermal units per pound of fuel.
HHV = higher heating value of the fuel burned, in British thermal units per pound of fuel.
C_{pair} = specific heat of the air, in British thermal units per pound of air per degree Fahrenheit.
Cp_{fuel} = specific heat of the fuel, in British thermal units per pound of fuel per degree Fahrenheit.
Cp_{medium} = specific heat of the atomizing medium, in British thermal units per pound of medium per degree Fahrenheit.
H_a = air sensible heat correction, in British thermal units per pound of fuel.
H_f = fuel sensible heat correction, in British thermal units per pound of fuel.
H_m = atomizing medium sensible heat correction, in British thermal units per pound of fuel.
Q_r = radiation heat loss, in British thermal units per pound of fuel.
Q_s = stack heat loss, in British thermal units per pound of fuel.
T_a = ambient air temperature, in degrees Fahrenheit.
T_d = design datum temperature, in degrees Fahrenheit.
T_e = exit flue gas temperature, in degrees Fahrenheit .
T_f = fuel temperature, in degrees Fahrenheit.
T_m = atomizing medium temperature, in degrees Fahrenheit .
T_t = air temperature, in degrees Fahrenheit.

G.1.3.2 Definitions

The following terms are used in this appendix:

Thermal efficiency is equal to the total heat absorbed divided by the total heat input. This definition differs from the traditional definition of fired heater efficiency, which considers only the net heat of combustion of the fuel (LHV) as heat input.

Total heal absorbed is equal to the total heat input minus the total heat losses.

Total heat input is equal to the sum of the net heat of combustion of the fuel (LHV) and the sensible heat of the air, fuel, and atomizing medium to the system.

Total heat losses are equal to the sum of the radiation heat loss and the stack heat loss.

Radiation heat loss is a defined percentage of the net heat of combustion of the fuel.

Stack heat loss is the total sensible heat of the flue gas components at the temperature of flue gas when it leaves the last heat exchange surface.

Sensible heat correction is the sensible heat differential at test temperatures when compared with a datum temperature of 60°F (16°C) for air, fuel, and the atomizing medium. [For steam as an atomizing medium, the datum enthalpy is 1087.7 British thermal units per pound (2530 kilojoules per kilogram).]

G.1.4 INSTRUMENTATION

The instrumentation specified in G.1.4.1 and G.1.4.2 is required for the collection of data and the subsequent calculations necessary to determine the thermal efficiency of a heater (see Figure G-1).

G.1.4.1 Temperature

A multishielded aspirating (high-velocity) thermocouple (see Figure G-2) shall be used to measure all temperatures of the flue gas and temperatures of the preheated combustion air above 500°F (189°C). Thermocouples with thermowells may be used to measure temperatures at or below 500°F (189°C).

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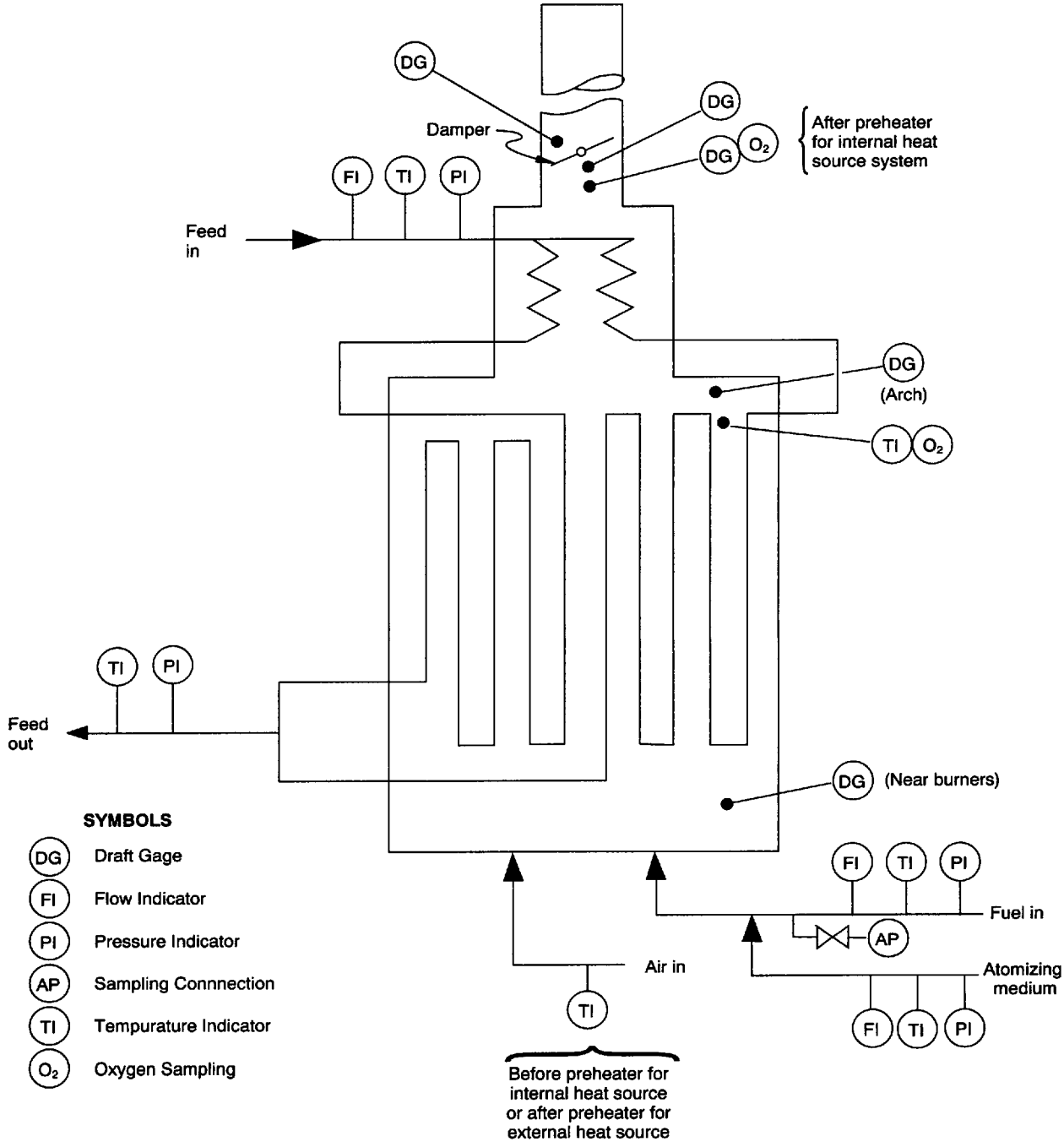
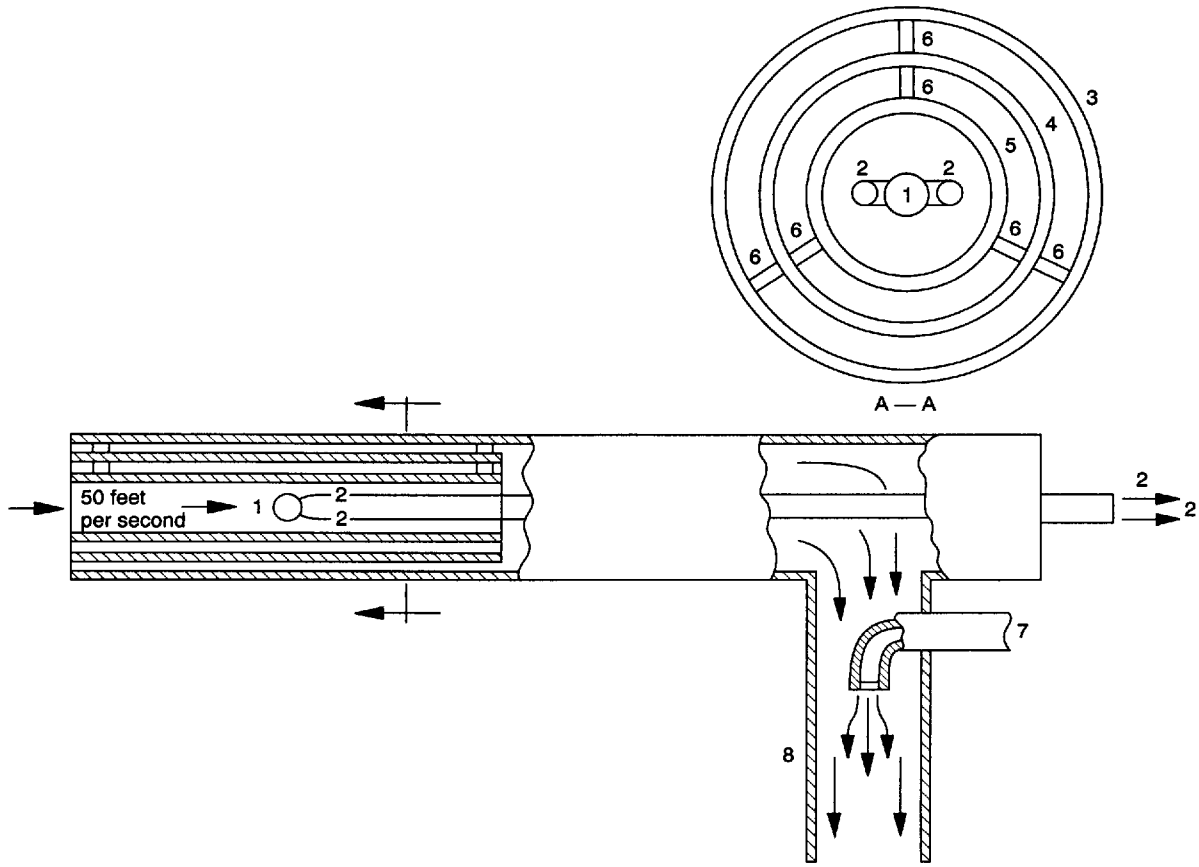


Figure G-1—Instrument and Measurement Locations

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Note: This figure is taken from *Furnace Operations* by Robert Reed, p. 46, Gulf Publishing Company, Houston, Texas, 1973. Reprinted with permission. 1 = thermocouple junction; 2 = thermocouple wires to temperature-indicating instrument; 3 = outer thin-wall 310 stainless steel tube; 4 = middle thin-wall 310 stainless steel tube; 5 = center thin-wall 310 stainless steel tube; 6 = centering tripods; 7 = air or steam at 10 pounds per square inch gage or more, in increments of 10 pounds per square inch until stable; 8 = hot gas eductor.

Figure G-2—Typical Aspirating (High-Velocity) Thermocouple

Conventional measuring devices may be used to measure the temperatures of the ambient air, the fuel, and the atomizing medium. For a discussion of conventional temperature measurements, refer to API Recommended Practice 554.

G.1.4.2 Flue Gas Analysis

A portable or permanently installed analyzer shall be used to analyze for oxygen and combustible gases in the flue gas. The analysis of the flue gas may be made on either a wet or a dry basis, but the calculations shall be consistent with the basis used. For a discussion of sampling systems and flue gas analyzers, refer to API Recommended Practice 555.

G.1.5 MEASUREMENT

The following measurements are to be taken for reference purposes and for identification of heater operating conditions. If more than one process service or auxiliary stream is present, the data should be taken for all services:

a. Fuel flow rate
b. Process flow rate.
c. Process fluid inlet temperature.
d. Process fluid outlet temperature.
e. Process fluid inlet pressure.
f. Process fluid outlet pressure.
g. Fuel pressure at the burner.
h. Atomizing medium pressure at the burner.
i. Flue gas draft profile.

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G.2 Testing

G.2.1 Preparation for Testing

G.2.1.1 The following ground rules must be established in preparation for the test, prior to the date of the actual test run:

a. The operating conditions that will prevail during the test.
b. Any rerating that will be necessary to account for differences between the test conditions and the design conditions.
c. The acceptability of the fuel or fuels to be fired.
d. The selection of instrumentation types, methods of measurement, and specific measurement locations.

G.2.1.2 All instrumentation that is to be used during the test shall be calibrated before the test.

G.2.1.3 Immediately before the actual test, the following items must be verified:

a. That the fired process heater is operating at steady-state conditions.
b. That the fuel to be fired is acceptable.
c. That the heater is operating properly with respect to the size and shape of the flame, excess air, flue gas draft profile, cleanliness of the heating surfaces, and balanced burner firing.

G.2.2 TESTING

G.2.2.1 The operation of the heater shall be maintained at a uniform rate throughout the test.

G.2.2.2 The test shall last for a minimum of four hours. Data shall be taken at the start of the test and every two hours thereafter.

G.2.2.3 The duration of the test shall be extended until three consecutive sets of collected data fall within the prescribed limits listed in Table G-1.

G.2.2.4 The data shall be collected in the manner specified in G.2.2.4.1 through G.2.2.4.7.

Table G-1—Limits on Variability of Data

Datum	Limit
Heating value of fuel	±5 percent
Fuel rate	±5 percent
Flue gas combustibles content	<0.1 percent
Flue gas temperature	±10°F
Flue gas oxygen content	±1 percent
Process flow rate	±5 percent
Process temperature in	±10°F
Process temperature out	±10°F
Process pressure out	±5 percent

G.2.2.4.1 All-of the data in each set shall be collected as quickly as possible, preferably within 30 minutes.

G.2.2.4.2 The quantity of fuel gas shall be measured and recorded for each set of data, and a sample shall be taken simultaneously for analysis.

G.2.2.4.3 For gaseous fuels, the net heating value shall be obtained by composition analysis and calculation.

G.2.2.4.4 The quantity of liquid fuel shall be measured and recorded for each set of data. It will only be necessary to take one sample for analysis during the test run.

G.2.2.4.5 For liquid fuels, the net heating value shall be obtained by calorimeter test. Liquid fuels shall also be analyzed to determine the hydrogen-carbon ratio, sulfur content, water content, and the content of other components.

G.2.2.4.6 Flue gas samples shall be analyzed to determine the content of oxygen and combustibles. Samples shall be taken downstream of the last heat-exchange (heat-absorbing) surface. When an air heater is used, samples shall be taken after the air heater. A traverse of the crosssectional area shall be made to obtain representative samples. A minimum of four samples shall be taken, not more than three feet apart.

G.2.2.4.7 The flue gas temperature shall be measured at the same location used to extract samples of flue gas for analysis. Systems designed to operate on natural draft upon loss of preheated air shall also measure the flue gas temperature above the stack damper. If the measured temperature reveals leakage (that is, if the stack temperature is higher than the temperature at the exit from the air heater), then flue gas samples must also be taken at this location to determine the correct overall thermal efficiency. A traverse of the cross-sectional area shall be made to obtain the representative temperature. A minimum of four measurements shall be taken, not more than three feet apart.

G.2.2.5 The thermal efficiency shall be calculated from each set of valid data. The accepted final results will then be the arithmetic average of the calculated efficiencies

G.2.2.6 All of the data shall be recorded on the standard forms presented in Appendix G.A.

G.3 Determination of Thermal Efficiency

G.3.1 CALCULATION OF THERMAL EFFICIENCY

Figures G-3, G-4, and G-5 illustrate heat inputs and heat losses for typical arrangements of fired process heater systems.

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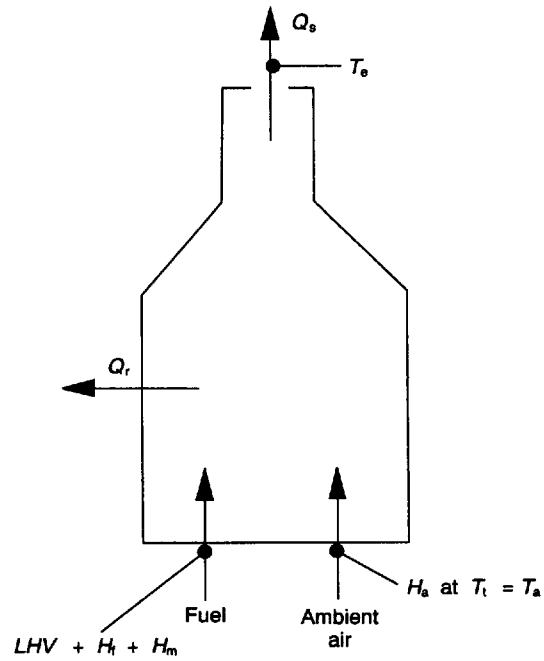


Figure G-3—Typical Heater Arrangement with Nonpreheated Air

G.3.1.1 Net Thermal Efficiency

For the arrangements in Figures G-3, G-4, and G-5, the net thermal efficiency (based on the lower heating value of the fuel) can be determined by the following equation:

$$\text{Efficiency} = \frac{\text{Total heat absorbed}}{\text{Total heat input}} \times 100$$

Also,

$$\text{Efficiency} = \frac{\text{Total heat input} - \text{Total heat losses}}{\text{Total heat input}}$$

Therefore,

$$e = \frac{(LVH + H_a + H_f + H_m) - (Q_r + Q_s)}{(LHV + H_a + H_f + H_m)} \times 100 \qquad \text{(G-1)}$$

Where:

e = net thermal efficiency, expressed as a percentage.

LHV = lower heating value of the fuel burned, in British thermal units per pound of fuel.

H_a = air sensible heat correction, in British thermal units per pound of fuel,
= $Cp_{air} \times (T_t - T_d) \times$ (Pounds of air per pound of fuel), or the enthalpy difference multiplied by pounds of air per pound of fuel.

H_t = fuel sensible heat correction, in British thermal units per pound of fuel,
= $Cp_{fuel} \times (T_f - T_d)$

H_m = atomizing medium sensible heat correction, in British thermal units per pound of fuel,
= $Cp_{medium} \times (T_m - T_d) \times$ (Pounds of medium per pound of fuel), or the enthalpy difference multiplied by pounds of medium per pound of fuel.[1]

Q_r = assumed radiation heat loss, in British thermal units per pound of fuel.

Q_s = calculated stack heat losses (see Stack Loss Work Sheet, Appendix B), in British thermal units per pound of fuel.

[1] "Pounds of medium per pound of fuel" may be an assumed value if it is not measured.

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G.3.1.2 Gross Thermal Efficiency

The gross thermal-efficiency of a fired process heater system is determined by substituting, in equation (G-1), the higher heating value, HHV, in place of LHV and adding to Q_s a value equal to 1059.7 British thermal units per pound of H2O multiplied by the number of pounds of H_2O formed in the combustion of the fuel:

$$e_{gross} = \frac{(HHV + H_a + H_f + H_m) - (Q_r + Q_s + H_20 \text{ formed} \times 1059.7)}{(HHV + H_a + H_f + H_m)} \times 100$$

However,

$$HHV = LHV + H_20 \text{ formed} \times 1059.7$$

Making this substitution, the equation reduces to the following:

$$e_{gross} = \frac{(LHV + H_a + H_f + H_m) - (Q_r + Q_s)}{(LHV + H_a + H_f + H_m) + H_2O \text{ formed} \times 1059.7} \times 100$$

Reducing further,

$$e_{gross} = \frac{(LHV + H_a + H_f + H_m) - (Q_r + Q_s)}{(HHV + H_a + H_f + H_m)} \times 100 \qquad \text{(G-2)}$$

Where:

e_{gross} = gross thermal efficiency, expressed as a percentage.

HHV = higher heating value of the fuel burned, in British thermal units per pound of fuel.

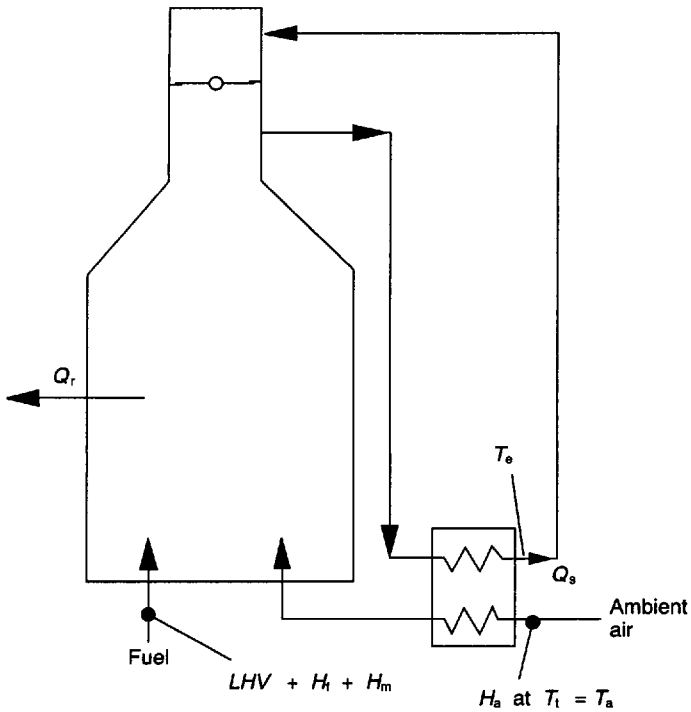


Figure G-4—Typical Heater Arrangement with Preheated Air from an Internal Heat Source

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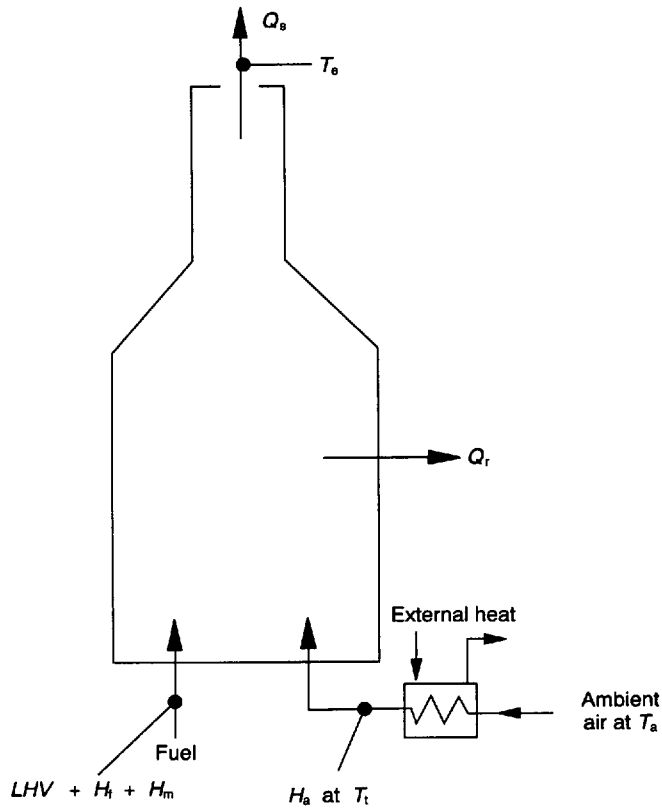


Figure G-5—Typical Heater Arrangement with Preheated Air from an External Heat Source

G.3.2 SAMPLE CALCULATIONS

The examples in G.3.2.1 through G.3.2.3 illustrate the use of the preceding equations to calculate the thermal efficiency of three typical heater arrangements.

G.3.2.1 Oil-Fired Heater with Natural Draft

In this example (see Figure G-3), the ambient air temperature (T_a) is 80°F (26.67°C), the air temperature (T_t) is 80°F (26.67°C), the flue gas temperature to the stack (T_e°) is 450°F (232.22°C), the fuel oil temperature (T_f) is 350°F (176.67°C), and the relative humidity is 50 percent. The flue gas analysis indicates that the oxygen content (on a wet basis) is 5 volume percent and that the combustibles content is nil. The radiation heat loss is 1.5 percent of the lower heating value of the fuel. The analysis of the fuel indicates that the fuel's gravity is 10°API, its carbon-hydrogen ratio is 8.06, its higher heating value (by calorimeter) is 18,300 British thermal units per pound (42,566 kilojoules per kilogram), its sulfur content is 1.8 weight percent, and its inerts content is 0.95 weight percent. The temperature of the atomizing steam (T_m) is 366°F (185.56°C) at a pressure of 150 pounds per square inch gage; the quantity of atomizing steam is 0.5 pound of steam per pound of fuel. Appendix G.C contains the work sheets from Appendix G.B filled out for this example.

The fuel's carbon content and the content of the other components are entered as weight fractions in Column 3 of the Combustion Work Sheet (see Appendix G.C) to determine the flue gas components. By entering the fuel's higher heating value (*HHV*) and its components on the Lower Heating Value (Liquid Fuels) Work Sheet (see Appendix G.C), the fuel's lower heating value (*LHV*) and carbon content (as a percentage) can be determined. Using

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this method, LHV = 17,277 British thermal units per pound of fuel (40,186 kilojoules per kilogram of fuel).

The radiation heat loss, Q_r, is determined by multiplying LHV by the radiation loss expressed as a percentage. Therefore, Q_s = 0.015 x 17,277 = 259.2 British thermal units per pound of fuel (602.9 kilojoules per kilogram of fuel).

The stack heat loss, Q_s, is determined from a summation of the heat content of the flue gas components at the exit flue gas temperature, T_e (see Stack Loss Work Sheet, Appendix G.C). Therefore, Q_s = 2058.5 British thermal units per pound of fuel at 450°F (4788.4 kilojoules per kilogram of fuel at 232.22°C).

The sensible heat corrections—H_a for combustion air, H_f for fuel, and H_m for atomizing steam—are determined as follows:

$$H_a = Cp_{air} \times (T_i - T_d) \text{ (Pounds of air per pound of fuel)}$$

Where:

Pounds of air per pound of fuel = the sum of the values from equations (b) and (e) on the Excess Air and Relative Humidity Work Sheet (see Appendix G.C).

H_a = 0.24 (80 − 60) (13.86 + 4.896)
= 90.0 British thermal units per pound of fuel
= 209.3 kilojoules per kilogram of fuel

H_f = Cp_{fuel} × (Tf − Td)
= 0.48 (350 − 60)
= 139.2 British thermal units per pound of fuel
= 323.8 kilojoules per kilogram of fuel

H_m = Enthalpy difference × (Pounds of steam per pound of fuel)
= (1195.5 − 1087.7) × 0.5
= 53.9 British thermal units per pound of fuel
= 125.4 kilojoules per kilogram of fuel

The net thermal efficiency can then be calculated as follows [see equation (G-1)]:

$$e = \frac{(17,277 + 90.0 + 139.2 + 53.9) - (259.2 + 2058.5)}{(17,277 + 90.0 + 139.2 + 53.9)} \times 100$$

= 86.8 percent

The gross thermal efficiency is determined as follows [see equation (G-2)]:

$$e_{gross} = \frac{(17,277 + 90.0 + 139.2 + 53.9) - (259.2 + 2058.5)}{(18,300 + 90.0 + 139.2 + 53.9)} \times 100$$

= 82.0 percent

G.3.2.2 Gas-Fired Heater with Preheated Combustion Air From an Internal Heat Source

In this example (see Figure G-4), the ambient air temperature (T_a) is 28°F (–2.22°C), the air temperature (T_i) is 28°F (–2.22°C), the flue gas temperature at the exit from the air heater is 300°F (148.89°C), the fuel gas temperature is 100°F (37.78°C), and the relative humidity is 50 percent. The flue gas analysis indicates that the oxygen content (on a wet basis) is 3.5 volume percent and that the combustibles content is nil. The radiation heat loss is 2.5 percent of the lower heating value of the fuel. The analysis of the fuel indicates that the fuel's methane content is 75.41 volume percent, its ethane content is 2.33 volume percent, its ethylene content is 5.08 volume percent, its propane content is 1.54 volume percent, its propylene content is 1.86 volume percent, its nitrogen content is 9.96 volume

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percent, and its hydrogen content is 3.82 volume percent. Appendix G.D contains the Combustion Work Sheet, Excess Air and Relative Humidity Work Sheet, and Stack Loss Work Sheet from Appendix G.B filled out for this example.

The fuel's *LHV* is determined by entering the fuel analysis in Column 1 of the Combustion Work Sheet (see Appendix G.D) and dividing the total heats of combustion (Column 5) by the total fuel weight (Column 3). Therefore, *LHV* = 335,629/18.523 = 18,120 British thermal units per pound of fuel (42,147 kilojoules per kilogram of fuel).

The radiation heat loss, Q_r, is determined by multiplying *LHV* by the radiation loss expressed as a percentage. Therefore, $Q_r = 0.025 \times 18{,}120 = 453.0$ British thermal units per pound of fuel (1053.7 kilojoules per kilogram of fuel).

The stack heat loss, Q_s, is determined from a summation of the heat content of the flue gas components at the exit flue gas temperature, T_e (see Stack Loss Work Sheet, Appendix G.D). Therefore, Q_s = 1181.2 British thermal units per pound of fuel at 300°F (2747.5 kilojoules per kilogram of fuel at 148.89°C).

The sensible heat corrections—H_a for combustion air and H_f for fuel—are determined as follows:

$$\begin{aligned} H_a &= Cp_{air} \times (T_t - T_d) \text{ (Pounds of air per pound of fuel)} \\ &= 0.24\,(28 - 60)\,(14.344 + 3.201) \\ &= -134.7 \text{ British thermal units per pound of fuel} \\ &= -313.3 \text{ kilojoules per kilogram of fuel} \\ H_f &= Cp_{fuel} \times (T_f - T_d) \\ &= 0.525\,(100 - 60) \\ &= 21.0 \text{ British thermal units per pound of fuel} \\ &= 48{:}8 \text{ kilojoules per kilogram of fuel} \end{aligned}$$

The net thermal efficiency can then be calculated as follows [see equation (G-1)]:

$$e = \frac{(18{,}120 - 134.7 + 21) - (453.0 + 1181.2)}{(18{,}120 - 134.7 + 21)} \times 100$$

$$= 90.9 \text{ percent}$$

To determine the gross thermal efficiency, follow the procedure in G.3.1.2 (see also G.3.2.1).

G.3.2.3 Gas-Fired Heater With Preheated Combustion Air From an External Heat Source

This example (see Figure G-5) uses the same data that were used in G.3.2.2 except for the following changes: The air temperature (T_t) is 300°F (148.89°C), the flue gas temperature to the stack (T_e) is 500°F (260°C), and the flue gas analysis indicates that the oxygen content (on a dry basis) is 3.5 volume percent. Appendix G.E contains the Excess Air and Relative Humidity Work Sheet and Stack Loss Work Sheet from Appendix G.B filled out for this example.

LHV and Q are determined exactly as they were in G.3.2.2. Therefore, *LHV* = 18,120 British thermal units per pound of fuel (42,147 kilojoules per kilogram of fuel), and Q_f = 453.0 British thermal units per pound of fuel (1053.7 kilojoules per kilogram of fuel).

In this example the oxygen reading was taken on a dry basis, so the values for pounds of water per pound of fuel should be entered as zero when correcting for excess air. (see Excess Air and Relative Humidity Work Sheet, Appendix G.E). The calculation for total pounds of H_2O per pound of fuel (corrected for excess air) is again performed using values for water and moisture (see Excess Air and Relative Humidity Work Sheet).

The stack loss, Q_s, is determined from a summation of the heat content of the flue gas components at the stack temperature, T_e (see Stack Loss Work Sheet, Appendix G.E). Therefore, Q_s = 2099.9 British thermal units per pound of fuel at 500°F (4884.4 kilojoules per kilogram of fuel at 260°C).

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The sensible heat corrections, H_a and H_f, are determined as they were in G.3.2.2, but H_a changes because of the different temperatures and quantities:

$H_a = Cp_{air} \times (T_t - T_d)$(Pounds of air per pound of fuel)
$= 0.24\ (300 - 60)\ (14.344 + 2.619)$
$= 977.1$ British thermal units per pound of fuel
$= 2272.7$ kilojoules per kilogram of fuel

$H_f = 21.0$ British thermal units per pound of fuel
$= 48.8$ kilojoules per kilogram of fuel

The net thermal efficiency can then be calculated as follows [see equation (G-1)]:

$$e = \frac{(18{,}120 + 977.1 + 21) - (453.0 + 2099.9)}{(18{,}120 - 977.1 + 21)} \times 100$$

$= 86.6$ percent

To determine the gross thermal efficiency, follow the procedure given in G.3.1.2 (see also G.3.2.1).

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APPENDIX G.A—MODEL FORMAT FOR DATA SHEETS

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LABORATORY DATA SHEET

Job No. _____
Date of Report _____
Page 1 of 2

I. GENERAL INFORMATION

Owner: _____ Plant location: _____
Unit: _____ Site elevation: _____
Heater no.: _____ Service: _____

Test run date:						
Test run time:						
Run no.:						

II. FUEL GAS SAMPLE

Sample taken by:						
Sample no.:						
Sampling location:						
Date taken:						
Time taken:						

Fuel gas analysis, volume percent						
Hydrogen:						
Methane:						
Ethane:						
Other C_2:						
Propane:						
Other C_3:						
Butane:						
Other C_4:						
Pentane plus:						
Carbon monoxide:						
Hydrogen sulfide:						
Carbon dioxide:						
Nitrogen:						
Oxygen:						
Other inerts:						
Total:						

Remarks: _____

III. FUEL OIL SAMPLE

Sample taken by:						
Sample no.:						
Sampling location:						
Date taken:						
Time taken:						
Sample temperature, °F:						

Analysis, weight percent						
Carbon:						
Hydrogen:						

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LABORATORY DATA SHEET

Job No. _____
Date of Report _____
Page 2 of 2

Carbon-hydrogen ratio:*						
Sulfur:						
Ash:						
Nitrogen:						
Oxygen:						
Water:						
Other:						
Total:						
Calorimeter heating value:						
Vanadium, parts per million:						
Sodium, parts per million:						
Gravity, °API:						
Additive used:						

IV. PROCESS STREAM SAMPLE

Sample taken by:						
Sample no.:						
Sampling location:						
Date taken:						
Time taken:						

Sample test conditions

Temperature, °F:						
Pressure, psig:						
Name of fluid:						
Liquid gravity, °API:						
Vapor molecular weight:						

ASTM liquid distillation

Initial boiling point:						
10 percent vaporized:						
20 percent vaporized:						
30 percent vaporized:						
40 percent vaporized:						
50 percent vaporized:						
60 percent vaporized:						
70 percent vaporized:						
80 percent vaporized:						
90 percent vaporized:						
Endpoint:						

V. GENERAL CONDITIONS

Remarks: _____

*May be entered instead of carbon and hydrogen contents.

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RAW TEST DATA SHEET

Job No. _____
Date of Report _____
Page 1 of 3

I. GENERAL INFORMATION

Owner: _____
Unit: _____
Heater no.: _____
Manufacturer: _____

Plant location: _____
Site elevation: _____
Service: _____

Test run date:						
Test run time:						
Run no.:						
Recorded by:						

II. GENERAL CONDITIONS

Ambient air temperature, °F:						
Wind direction:						
Wind velocity, miles per hour:						
Plant barometric pressure, inches Hg:						
Radiation loss. percent						
Relative humidity, percent:						

III. COMBUSTION DATA

Fuel gas

Flow meter reading:						
Flow meter factor and data base:						
Pressure at flow meter, psig:						
Temperture at flow meter, °F:						
Pressure at burners, psig:						

Fuel oil (supply)

Flow meter reading:						
Flow meter factor and data base:						
Pressure at flow meter, psig:						
Temperature at flow meter, °F						
Pressure at burners, psig:						

Fuel oil (return)

Flow meter reading:						
Flow meter factor and data base:						
Pressure at flow meter, psig:						
Temperature at flow meter, °F:						

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RAW TEST DATA SHEET

Job No. _____
Date of Report _____
Page 2 of 3

Atomizing medium

Flow meter reading:						
Flow meter factor and data base:						
Pressure at flow meter, psig:						
Temperature at flow meter, °F:						
Pressure at burners, psig:						

IV. PROCESS STREAM DATA*

Flow

Flow meter reading:						
Flow meter factor:						
Flow pressure in, psig:						
Flow temperature in, °F:						
Flow pressure out, psig:						
Combined temperature out, °F:						

Steam injection

Location:						
Total consumption, pounds per hour:						

V. AIR AND FLUE GAS DATA

Pressure, inches H_2O

Draft at burners:						
Draft at firebox roof:						

*Similar data should be recorded for secondary streams such as boiler feed water, steam generation, and steam superheat.

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RAW TEST DATA SHEET

Job No. _____
Date of Report _____
Page 3 of 3

	Run no.					Run no.					Run no.				
	Traverse Readings				Average	Traverse Readings				Average	Traverse Readings				Average
Temperature, °F															
Air into preheater:															
Air out of preheater:															
Flue gas out of preheater:†															
Flue gas in stack:†															
Flue gas analysis, volume percent															
Oxygen content:†															
Combustibles and carbon monoxide:															

VI. ASSOCIATED EQUIPMENT

Air heater						
Nameplate size:						
Type:						
Bypass (open/closed):						
External preheat (on/off):						
Burners						
No. in operation:						
Type of fuel:						
Burner type:‡						

Remarks: _____

† Readings are to be taken after the last heat-absorbing surface.
‡The burner type should be designated as ND (natural draft), FD (forced draft). or FD/PA (forced draft preheated air).

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APPENDIX G.B—MODEL FORMAT FOR WORK SHEETS

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LOWER HEATING VALUE (LIQUID FUELS) WORK SHEET

Job No. _____
Date of Report _____
Page 1 of 1

Higher heating value (*HHV*), from calorimeter test, in British thermal units per pound of fuel: _____.
Carbon-hydrogen ration, from analysis: _____.
Impurities, from analysis, weight percent
Water vapor: _____
Ash: _____
Sulfur: _____
Sodium: _____
Other: _____
Total: _____.

Percent hydrogen = $\frac{100 - \text{Weight percent impurities}}{\text{Carbon - hydrogen ratio} + 1.0}$: _____.

$LHV = HHV - (9 \times 1059.7)\frac{\text{Percent hydrogen}}{100}$, in British thermal units per pound of fuel: _____.

Percent carbon = 100 – (Percent hydrogen + Percent impurities): _____.

INSTRUCTIONS

Calculate the values for percent hydrogen, lower heating value (*LHV*), and percent carbon. Enter these values in the appropriate columns of the Combustion Work Sheet.

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COMBUSTION WORK SHEET

Fuel Component	Column 1 Volume Fraction	Column 2 Molecular Weight	Column 3 (1 × 2) Total Weight (pounds)	Column 4 Net Heating Value (British thermal units per pound)	Column 5 (3 × 4) Heating Value (British thermal units)
Carbon		12.0		—	
Hydrogen		2.016		51,600	
Oxygen		32.0		—	
Nitrogen		28.0		—	
Carbon Monoxide		28.0		4,345	
Carbon Dioxide		44.0		—	
Methane		16.0		21,500	
Ethane		30.1		20,420	
Ethylene		28.1		20,290	
Propane		44.1		19,930	
Propylene		42.1		19,690	
Butane		58.1		19,670	
Butylene		56.1		19,420	
Pentane		72.1		19,500	
Hexane		86.2		19,390	
Sulfur*		32.1		—	
Hydrogen sulfide*		34.1		6,550	
Total					
Total per pound of fuel					

INSTRUCTIONS

If composition is expressed as volume percent, insert in Column 1; if composition is expressed as weight percent, insert in Column 3. Total all of the columns on the "Total" line and divide all of the column totals by the Column 3 total to obtain the values for the "Total per pound of fuel" line. The Excess Air and Relative Humidity Work Sheet and the Stack Loss Work Sheet use the totals per pound of fuel to calculate stack loss; for example, if one of the work sheets asked for "pounds of CO_2," the value would be taken from the "Total per pound of fuel" line in Column 9.

*SO_2 is to be included in the CO_2 column. Although this is inaccurate, the usual small quantities will not affect any of the final results.

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Job No. _____
Date of Report _____
Page 2 of 2

Column 6	Column 7 (3x6)	Column 8	Column 9 (3x8)	Column 10	Column 11 (3x10)	Column 12	Column 13 (3x12)
Air Required (pounds of air per pound)	Air Required (pounds)	CO_2 Formed (pounds of CO_2 per pound)	CO_2 Formed (pounds)	H_20 Formed (pounds of H_2O per pound)	H_20 Formed (pounds)	N_2 Formed (pounds of N_2 per pound)	N_2 Formed (pounds)
11.51		3.66		—		8.85	
34.29		—		8.94		26.36	
–4.32		—		—		–3.32	
—		—		—		1.00	
2.47		1.57		—		1.90	
—		1.00		—		—	
17.24		2.74		2.25		13.25	
16.09		2.93		1.80		12.37	
14.79		3.14		1.28		11.36	
15.68		2.99		1.63		12.05	
14.79		3.14		1.28		11.36	
15.46		3.03		1.55		11.88	
14.79		3.14		1.28		11.36	
15.33		3.05		1.50		11.78	
15.24		3.06		1.46		11.71	
4.31		2.00		—		3.31	
6.08		1.88		0.53		4.68	

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EXCESS AIR AND RELATIVE HUMIDITY WORK SHEET (See Note)

Job No. _____
Date of Report _____
Page 1 of 2

Atomizing steam: _____ pounds per pound of fuel (assumed or measured)

CORRECTION FOR RELATIVE HUMIDITY

$$\text{Moisture in air} = \frac{\rho_{vapor}}{14.696} \times \frac{\text{Relative humidity}}{100} \times \frac{18}{28.85}$$

$$= \frac{\quad}{14.696} \times \frac{\quad}{100} \times \frac{18}{28.85}$$

= _____ pounds of moisture per pound of air (a)

Where:

ρ_{vapor} = vapor pressure of water at the ambient temperature, in pounds per square inch absolute (from steam tables).

$$\text{Pounds of wet air per pound of fuel required} = \frac{\text{Air required}}{1 - \text{Moisture in air}}$$

$$= \frac{_____ (7)}{1 - _____ (a)}$$

= _____ (b)

Pounds of moisture per pound of fuel = Pounds of wet air per pound of fuel required – Air required

= _____ (b) – _____ (7)

= _____ (c)

Pounds of H_2O per pound of fuel = H_2O formed + Pounds of moisture per pound of fuel + Atomizing steam.

= _____ (11) + _____ (c) + _____

= _____ (d)

CORRECTION FOR EXCESS AIR*

Pounds of excess air per pound of fuel

$$= \frac{(28.85 \times \text{Percent } O_2)\left(\frac{N_2 \text{ formed}}{28} + \frac{CO_2 \text{formed}}{44} + \frac{H_2O \text{ formed}}{18}\right)}{20.95 - \text{Percent } O_2\left[\left(1.6028 \times \frac{\text{Pounds of } H_2O}{\text{Pounds of air required}}\right) + 1\right]}$$

$$= \frac{(28.85 \times _____)\left(\frac{_____ (13)}{28} + \frac{_____ (9)}{44} + \frac{_____ (d)}{18}\right)}{20.95 - _____\left[\left(1.6028 \times \frac{_____ (c)}{_____ (7)}\right) + 1\right]}$$

= _____ (e)

$$\text{Percent excess air} = \frac{\text{Pounds of excess air per pound of fuel}}{\text{Air required}} \times 100$$

$$= \frac{_____ (e)}{_____ (7)} \times 100$$

= _____ (f)

*If oxygen samples are extracted on a dry basis, a value of zero shall be inserted in equation (e) where a value is required from equations (c) and (d). If oxygen samples are extracted on a wet basis, the appropriate calculated value shall be inserted.

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EXCESS AIR AND RELATIVE HUMIDITY WORK SHEET

Job No. _____
Date of Report _____
Page 2 of 2

Total pounds of H_2O per pound of fuel (corrected for excess air)

$$= \left[\frac{\text{Percent excess air}}{100} \times \text{Pounds of moisture per pound of fuel}\right] + \text{Pounds of } H_2O \text{ per pound of fuel}$$

$$= \left[\frac{_____\ (f)}{100} \times _____\ (c)\right] + _____\ (d)$$

$$= _____ \quad (g)$$

Note: All values used in the calculations above shall be on a per-pound-of-fuel basis. Numbers in parentheses indicate values to be taken from the "Total per pound of fuel" line of the Combustion Work Sheet, and letters in parentheses indicate values to be taken from the corresponding lines of this work sheet.

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STACK LOSS WORK SHEET

Job No. _____
Date of Report _____
Page 1 of 1

Exit flue gas temperature, T_e: _____ °F

Component	Column 1 Pounds of Component Formed per Pound of Fuel	Column 2 Enthalpy at T (British thermal uniits per pound formed)	Column 3 Heat Content (British thermal units per pound of fuel)
Carbon dioxide			
Water vapor			
Nitrogen			
Air			
Total			

INSTRUCTIONS

In Column 1 above, insert the values from the "Total per pound of fuel" line of the Combustion Work Sheet for carbon dioxide (Column 9) and nitrogen (Column 13). Insert the value from equation (e) of the Excess Air and Relative Humidity Work Sheet for air, and insert the value from equation (g) of the Excess Air and Relative Humidity Work Sheet for water vapor.

In Column 2 above, insert the enthalpy values from Figures G.B-1 and G.B-2 for each flue gas component.

In Column 3 above, for each component insert the product of the value from Column 1 and the value from Column 2. This is the heat content at the exit gas temperature.

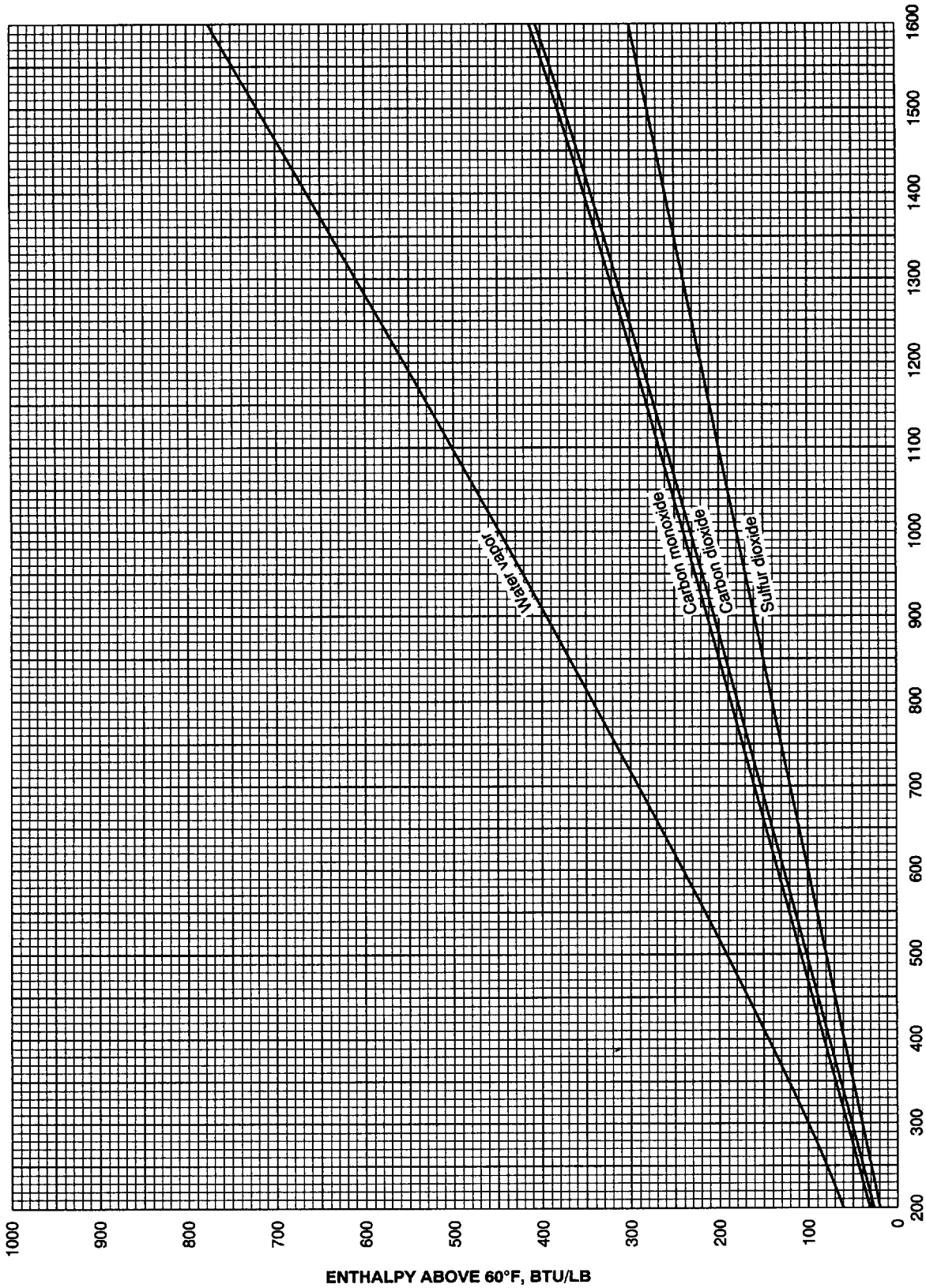
Total the values in Column 3 to obtain the heat loss to the stack, Q_s.

Therefore,

$Q_s = \Sigma$ Heat content at T_e = _____ British thermal units per pound of fuel.

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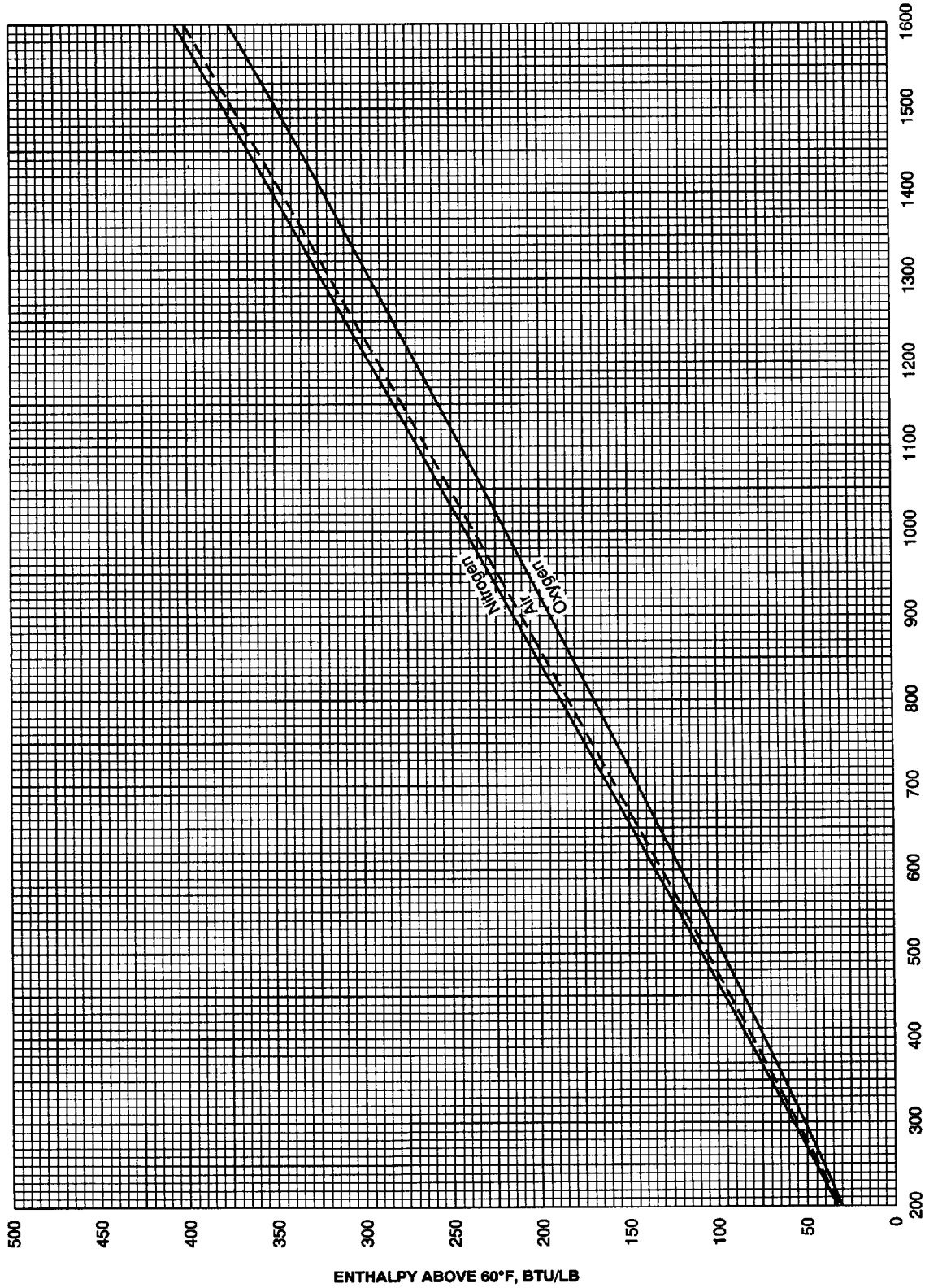
API STD*560 95 ■ 0732290 0550297 492 ■



Note: This figure is taken from the *Technical Data Book—Petroleum Refining* (English edition), Chapter 14, "Combustion," p. 14–25, American Petroleum Institute, Washington, D.C., 1966.

Figure G.B-1—Enthalpy of H_2O, CO, CO_2, and SO_2

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Note: This figure is taken from the *Technical Data Book—Petroleum Refining* (English edition), Chapter 14, "Combustion," p. 14–25, American Petroleum Institute, Washington, D.C., 1966.

Figure G.B-2—Enthalpy of Air, O_2, and N_2

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APPENDIX G.C—SAMPLE WORK SHEETS FOR AN OIL-FIRED HEATER WITH NATURAL DRAFT

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LOWER HEATING VALUE (LIQUID FUELS) WORK SHEET

Job No. *Sample Work Sheet for G.3.2.1*
Date of Report
Page *1* of *1*

Higher heating value (*HHV*), from calorimeter test, in British thermal units per pound of fuel:	*18,300* .
Carbon-hydrogen ration, from analysis:	*8.065*.
Impurities, from analysis, weight percent	
Water vapor:	
Ash:	
Sulfur:	*1.80*
Sodium:	
Other:	*0.95*
Total:	*2.75* .
Percent hydrogen = $\frac{100 - \text{Weight percent impurities}}{\text{Carbon - hydrogen ratio} + 1.0}$:	*10.73* .
$LHV = HHV - (9 \times 1059.7)\frac{\text{Percent hydrogen}}{100}$, in British thermal units per pound of fuel:	*17,277* .
Percent carbon = 100 – (Percent hydrogen + Percent impurities):	*86.52* .

INSTRUCTIONS

Calculate the values for percent hydrogen, lower heating value (*LHV*), and percent carbon. Enter these values in the appropriate columns of the Combustion Work Sheet.

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COMBUSTION WORK SHEET

Fuel Component	Column 1 Volume Fraction	Column 2 Molecular Weight	Column 3 (1 × 2) Total Weight (pounds)	Column 4 Net Heating Value (British thermal units per pound)	Column 5 (3 × 4) Heating Value (British thermal units)
Carbon		12.0	*0.8652*	—	
Hydrogen		2.016	*0.1073*	51,600	
Oxygen		32.0		—	
Nitrogen		28.0		—	
Carbon Monoxide		28.0		4,345	
Carbon Dioxide		44.0		—	
Methane		16.0		21,500	
Ethane		30.1		20,420	
Ethylene		28.1		20,290	
Propane		44.1		19,930	
Propylene		42.1		19,690	
Butane		58.1		19,670	
Butylene		56.1		19,420	
Pentane		72.1		19,500	
Hexane		86.2		19,390	
Sulfur*		32.1	*0.0180*	—	
Hydrogen sulfide*		34.1		6,550	
Inerts			*0.0095*		
Total			*1.0000*		
Total per pound of fuel			*1.0000*		

INSTRUCTIONS

If composition is expressed as volume percent, insert in Column 1; if composition is expressed as weight percent, insert in Column 3. Total all of the columns on the "Total" line and divide all of the column totals by the Column 3 total to obtain the values for the "Total per pound of fuel" line. The Excess Air and Relative Humidity Work Sheet and the Stack Loss Work Sheet use the totals per pound of fuel to calculate stack loss; for example, if one of the work sheets asked for "pounds of CO_2," the value would be taken from the "Total per pound of fuel" line in Column 9.

*SO_2 is to be included in the CO_2 column. Although this is inaccurate, the usual small quantities will not affect any of the final results.

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Job No. *Sample Work Sheet for G.3.2.1*
Date of Report
Page *2* of *2*

Column 6	Column 7 (3x6)	Column 8	Column 9 (3x8)	Column 10	Column 11 (3x10)	Column 12	Column 13 (3x12)
Air Required (pounds of air per pound)	Air Required (pounds)	CO_2 Formed (pounds of CO_2 per pound)	CO_2 Formed (pounds)	H_20 Formed (pounds of H_2O per pound)	H_20 Formed (pounds)	N_2 Formed (pounds of N_2 per pound)	N_2 Formed (pounds)
11.51		3.66	*3.167*	—		8.85	*7.657*
34.29		—	—	8.94		26.36	*2.828*
–4.32		—		—		–3.32	
—		—		—		1.00	
2.47		1.57		—		1.90	
—		1.00		—		—	
17.24		2.74		2.25		13.25	
16.09		2.93		1.80		12.37	
14.79		3.14		1.28		11.36	
15.68		2.99		1.63		12.05	
14.79		3.14		1.28		11.36	
15.46		3.03		1.55		11.88	
14.79		3.14		1.28		11.36	
15.33		3.05		1.50		11.78	
15.24		3.06		1.46		11.71	
4.31	*0.078*	2.00	*0.036*	—		3.31	*0.060*
6.08		1.88		0.53		4.68	
	13.715		*3.203*		*0.959*		*10.545*
	13.715		*3.203*		*0.959*		*10.545*

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EXCESS AIR AND RELATIVE HUMIDITY WORK SHEET (See Note)

Job No. *Sample Work Sheet for G.3.2.1*
Date of Report
Page *1* of *2*

Atomizing steam: *0.50* pounds per pound of fuel (assumed or measured)

CORRECTION FOR RELATIVE HUMIDITY

$$\text{Moisture in air} = \frac{\rho_{vapor}}{14.696} \times \frac{\text{Relative humidity}}{100} \times \frac{18}{28.85}$$

$$= \frac{0.5068}{14.696} \times \frac{50}{100} \times \frac{18}{28.85}$$

$$= 0.0107 \text{ pounds of moisture per pound of air} \quad \text{(a)}$$

Where:

ρ_{vapor} = vapor pressure of water at the ambient temperature, in pounds per square inch absolute (from steam tables).

$$\text{Pounds of wet air per pound of fuel required} = \frac{\text{Air required}}{1 - \text{Moisture in air}}$$

$$= \frac{13.715\ (7)}{1 - 0.0107\ (a)}$$

$$= 13.86 \quad \text{(b)}$$

Pounds of moisture per pound of fuel = Pounds of wet air per pound of fuel required – Air required

$$= 13.86\ (b) - 13.715\ (7)$$

$$= 0.145 \quad \text{(c)}$$

Pounds of H_2O per pound of fuel = H_2O formed + Pounds of moisture per pound of fuel + Atomizing steam.

$$= 0.959\ (11) + 0.145\ (c) + 0.50$$

$$= 1.604 \quad \text{(d)}$$

CORRECTION FOR EXCESS AIR*

Pounds of excess air per pound of fuel

$$= \frac{(28.85 \times \text{Percent } O_2)\left(\frac{N_2 \text{ formed}}{28} + \frac{CO_2 \text{formed}}{44} + \frac{H_2O \text{ formed}}{18}\right)}{20.95 - \text{Percent } O_2\left[\left(1.6028 \times \frac{\text{Pounds of } H_2O}{\text{Pounds of air required}}\right) + 1\right]}$$

$$= \frac{(28.85 \times 5.0)\left(\frac{10.545\ (13)}{28} + \frac{3.203\ (9)}{44} + \frac{1.604\ (d)}{18}\right)}{20.95 - 5.0\left[\left(1.6028 \times \frac{0.145\ (c)}{13.715\ (7)}\right) + 1\right]}$$

$$= 4.896 \quad \text{(e)}$$

$$\text{Percent excess air} = \frac{\text{Pounds of excess air per pound of fuel}}{\text{Air required}} \times 100$$

$$= \frac{4.896\ (e)}{13.715\ (7)} \times 100$$

$$= 35.7 \quad \text{(f)}$$

*If oxygen samples are extracted on a dry basis, a value of zero shall be inserted in equation (e) where a value is required from equations (c) and (d). If oxygen samples are extracted on a wet basis, the appropriate calculated value shall be inserted.

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EXCESS AIR AND RELATIVE HUMIDITY WORK SHEET

Job No. *Sample Work Sheet for G.3.2.1*
Date of Report
Page *2* of *2*

Total pounds of H_2O per pound of fuel (corrected for excess air)

$$= \left[\frac{\text{Percent excess air}}{100} \times \text{Pounds of moisture per pound of fuel}\right] + \text{Pounds of } H_2O \text{ per pound of fuel}$$

$$= \left[\frac{35.7\ \text{(f)}}{100} \times 0.145\ \text{(c)}\right] + 1.604\ \text{(d)}$$

$$= 1.656 \qquad \text{(g)}$$

Note: All values used in the calculations above shall be on a per-pound-of-fuel basis. Numbers in parentheses indicate values to be taken from the "Total per pound of fuel" line of the Combustion Work Sheet, and letters in parentheses indicate values to be taken from the corresponding lines of this work sheet.

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STACK LOSS WORK SHEET

Job No. *Sample Work Sheet for G.3.2.1*
Date of Report
Page *1* of *1*

Exit flue gas temperature, T_e: *450* °F

Component	Column 1 Pounds of Component Formed per Pound of Fuel	Column 2 Enthalpy at T (British thermal uniits per pound formed)	Column 3 Heat Content (British thermal units per pound of fuel)
Carbon dioxide	*3.203*	*86*	*275.46*
Water vapor	*1.656*	*175*	*289.80*
Nitrogen	*10.545*	*97.5*	*1028.14*
Air	*4.896*	*95*	*465.12*
Total	*20.300*	—	*2058.52*

INSTRUCTIONS

In Column 1 above, insert the values from the "Total per pound of fuel" line of the Combustion Work Sheet for carbon dioxide (Column 9) and nitrogen (Column 13). Insert the value from equation (e) of the Excess Air and Relative Humidity Work Sheet for air, and insert the value from equation (g) of the Excess Air and Relative Humidity Work Sheet for water vapor.

In Column 2 above, insert the enthalpy values from Figures G.B-1 and G.B-2 for each flue gas component.

In Column 3 above, for each component insert the product of the value from Column 1 and the value from Column 2. This is the heat content at the exit gas temperature.

Total the values in Column 3 to obtain the heat loss to the stack, Q_s.

Therefore,

$Q_s = \Sigma$ Heat content at T_e = *2058.5* British thermal units per pound of fuel.

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APPENDIX G.D—SAMPLE WORK SHEETS FOR A GAS-FIRED HEATER WITH PREHEATED COMBUSTION AIR FROM AN INTERNAL HEAT SOURCE (G.3.2.2)

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COMBUSTION WORK SHEET

Fuel Component	Column 1 Volume Fraction	Column 2 Molecular Weight	Column 3 (1 × 2) Total Weight (pounds)	Column 4 Net Heating Value (British thermal units per pound)	Column 5 (3 × 4) Heating Value (British thermal units)
Carbon		12.0		—	
Hydrogen	*0.0382*	2.016	*0.0770*	51,600	*3,973*
Oxygen		32.0		—	
Nitrogen	*0.0996*	28.0	*2.789*	—	—
Carbon Monoxide		28.0		4,345	
Carbon Dioxide		44.0		—	
Methane	*0.7541*	16.0	*12.066*	21,500	*259,419*
Ethane	*0.0233*	30.1	*0.701*	20,420	*14,314*
Ethylene	*0.0508*	28.1	*1.428*	20,290	*28,974*
Propane	*0.0154*	44.1	*0.679*	19,930	*13,532*
Propylene	*0.0186*	42.1	*0.783*	19,690	*15,417*
Butane		58.1		19,670	
Butylene		56.1		19,420	
Pentane		72.1		19,500	
Hexane		86.2		19,390	
Sulfur*		32.1		—	
Hydrogen sulfide*		34.1		6,550	
Total	*1.0000*		*18.523*		*335,629*
Total per pound of fuel	*1.0000*		*1.0000*		*18,120*

INSTRUCTIONS

If composition is expressed as volume percent, insert in Column 1; if composition is expressed as weight percent, insert in Column 3. Total all of the columns on the "Total" line and divide all of the column totals by the Column 3 total to obtain the values for the "Total per pound of fuel" line. The Excess Air and Relative Humidity Work Sheet and the Stack Loss Work Sheet use the totals per pound of fuel to calculate stack loss; for example, if one of the work sheets asked for "pounds of CO_2," the value would be taken from the "Total per pound of fuel" line in Column 9.

*SO_2 is to be included in the CO_2 column. Although this is inaccurate, the usual small quantities will not affect any of the final results.

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Job No. *Sample Work Sheet for G.3.2.2*
Date of Report
Page 2 of 2

Column 6	Column 7 (3x6)	Column 8	Column 9 (3x8)	Column 10	Column 11 (3x10)	Column 12	Column 13 (3x12)
Air Required (pounds of air per pound)	Air Required (pounds)	CO_2 Formed (pounds of CO_2 per pound)	CO_2 Formed (pounds)	H_20 Formed (pounds of H_2O per pound)	H_20 Formed (pounds)	N_2 Formed (pounds of N_2 per pound)	N_2 Formed (pounds)
11.51		3.66		—		8.85	
34.29	*2.640*	—	—	8.94	*0.688*	26.36	*2.030*
–4.32		—		—		–3.32	
—	—	—	—	—	—	1.00	*2.789*
2.47		1.57		—		1.90	
—		1.00		—		—	
17.24	*208.018*	2.74	*33.061*	2.25	*27.149*	13.25	*159.875*
16.09	*11.279*	2.93	*2.054*	1.80	*1.262*	12.37	*8.671*
14.79	*21.120*	3.14	*4.484*	1.28	*1.828*	11.36	*10.222*
15.68	*10.647*	2.99	*2.030*	1.63	*1.107*	12.05	*8.182*
14.79	*11.581*	3.14	*2.459*	1.28	*1.002*	11.36	*8.895*
15.46		3.03		1.55		11.88	
14.79		3.14		1.28		11.36	
15.33		3.05		1.50		11.78	
15.24		3.06		1.46		11.71	
4.31		2.00		—		3.31	
6.08		1.88		0.53		4.68	
	265.285		*44.088*		*33.036*		*206.664*
	14.322		*2.380*		*1.784*		*11.157*

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EXCESS AIR AND RELATIVE HUMIDITY WORK SHEET*

Job No. *Sample Work Sheet for G.3.2.2*
Date of Report _____
Page *1* of *2*

Atomizing steam: *0* pounds per pound of fuel (assumed or measured)

CORRECTION FOR RELATIVE HUMIDITY

$$\text{Moisture in air} = \frac{\rho_{vapor}}{14.696} \times \frac{\text{Relative humidity}}{100} \times \frac{18}{28.85}$$

$$= \frac{0.0707}{14.696} \times \frac{50}{100} \times \frac{18}{28.85}$$

$$= 0.0015 \text{ pounds of moisture per pound of air} \quad \text{(a)}$$

Where:

ρ_{vapor} = vapor pressure of water at the ambient temperature, in pounds per square inch absolute (from steam tables).

$$\text{Pounds of wet air per pound of fuel required} = \frac{\text{Air required}}{1 - \text{Moisture in air}}$$

$$= \frac{14.322\ (7)}{1 - 0.0015\ (a)}$$

$$= 14.344 \quad \text{(b)}$$

Pounds of moisture per pound of fuel = Pounds of wet air per pound of fuel required – Air required

$$= 14.344\ (b) - 14.322\ (7)$$

$$= 0.022 \quad \text{(c)}$$

Pounds of H_2O per pound of fuel = H_2O formed + Pounds of moisture per pound of fuel + Atomizing steam.

$$= 1.784\ (11) + 0.022\ (c) + 0$$

$$= 1.806 \quad \text{(d)}$$

CORRECTION FOR EXCESS AIR*

Pounds of excess air per pound of fuel

$$= \frac{(28.85 \times \text{Percent } O_2)\left(\frac{N_2 \text{ formed}}{28} + \frac{CO_2 \text{formed}}{44} + \frac{H_2O \text{ formed}}{18}\right)}{20.95 - \text{Percent } O_2 \left[\left(1.6028 \times \frac{\text{Pounds of } H_2O}{\text{Pounds of air required}}\right) + 1\right]}$$

$$= \frac{(28.85 \times 3.5)\left(\frac{11.157\ (13)}{28} + \frac{2.380\ (9)}{44} + \frac{1.806\ (d)}{18}\right)}{20.95 - 3.5 \left[\left(1.6028 \times \frac{0.022\ (c)}{14.322\ (7)}\right) + 1\right]}$$

$$= 3.201 \quad \text{(e)}$$

$$\text{Percent excess air} = \frac{\text{Pounds of excess air per pound of fuel}}{\text{Air required}} \times 100$$

$$= \frac{3.201\ (e)}{14.322\ (7)} \times 100$$

$$= 22.35 \quad \text{(f)}$$

*If oxygen samples are extracted on a dry basis, a value of zero shall be inserted in equation (e) where a value is required from equations (c) and (d). If oxygen samples are extracted on a wet basis, the appropriate calculated value shall be inserted.

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EXCESS AIR AND RELATIVE HUMIDITY WORK SHEET

Job No. *Sample Work Sheet for G.3.2.2*
Date of Report
Page *2* of *2*

Total pounds of H_2O per pound of fuel (corrected for excess air)

$$= \left[\frac{\text{Percent excess air}}{100} \times \text{Pounds of moisture per pound of fuel}\right] + \text{Pounds of } H_2O \text{ per pound of fuel}$$

$$= \left[\frac{22.35\ \text{(f)}}{100} \times 0.022\ \text{(c)}\right] + 1.806\ \text{(d)}$$

$$= 1.811 \qquad \text{(g)}$$

Note: All values used in the calculations above shall be on a per-pound-of-fuel basis. Numbers in parentheses indicate values to be taken from the "Total per pound of fuel" line of the Combustion Work Sheet, and letters in parentheses indicate values to be taken from the corresponding lines of this work sheet.

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STACK LOSS WORK SHEET

Job No. *Sample Work Sheet for G.3.2.2*
Date of Report
Page *1* of *1*

Exit flue gas temperature, T_e: *300* °F

Component	Column 1 Pounds of Component Formed per Pound of Fuel	Column 2 Enthalpy at T (British thermal uniits per pound formed)	Column 3 Heat Content (British thermal units per pound of fuel)
Carbon dioxide	*2.380*	*50*	*119.00*
Water vapor	*1.811*	*105*	*190.16*
Nitrogen	*11.157*	*60*	*669.42*
Air	*3.201*	*57.5*	*202.61*
Total	*18.549*	—	*1181.19*

INSTRUCTIONS

In Column 1 above, insert the values from the "Total per pound of fuel" line of the Combustion Work Sheet for carbon dioxide (Column 9) and nitrogen (Column 13). Insert the value from equation (e) of the Excess Air and Relative Humidity Work Sheet for air, and insert the value from equation (g) of the Excess Air and Relative Humidity Work Sheet for water vapor.

In Column 2 above, insert the enthalpy values from Figures G.B-1 and G.B-2 for each flue gas component.

In Column 3 above, for each component insert the product of the value from Column 1 and the value from Column 2. This is the heat content at the exit gas temperature.

Total the values in Column 3 to obtain the heat loss to the stack, Q_s.

Therefore,

$Q_s = \Sigma$ Heat content at $T_e =$ *1181.2* British thermal units per pound of fuel.

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APPENDIX G.E—SAMPLE WORK SHEETS FOR A GAS-FIRED HEATER WITH PREHEATED COMBUSTION AIR FROM AN EXTERNAL HEAT SOURCE (G.3.2.3)

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EXCESS AIR AND RELATIVE HUMIDITY WORK SHEET (See Note)

Job No. *Sample Work Sheet for G.3.2.3*
Date of Report _____
Page *1* of *2*

Atomizing steam: *0* pounds per pound of fuel (assumed or measured)

CORRECTION FOR RELATIVE HUMIDITY

$$\text{Moisture in air} = \frac{\rho_{vapor}}{14.696} \times \frac{\text{Relative humidity}}{100} \times \frac{18}{28.85}$$

$$= \frac{0.0707}{14.696} \times \frac{50}{100} \times \frac{18}{28.85}$$

$$= 0.0015 \text{ pounds of moisture per pound of air} \quad \text{(a)}$$

Where:

ρ_{vapor} = vapor pressure of water at the ambient temperature, in pounds per square inch absolute (from steam tables).

$$\text{Pounds of wet air per pound of fuel required} = \frac{\text{Air required}}{1 - \text{Moisture in air}}$$

$$= \frac{14.322\,(7)}{1 - 0.0015\,(a)}$$

$$= 14.344 \quad \text{(b)}$$

Pounds of moisture per pound of fuel = Pounds of wet air per pound of fuel required – Air required

$$= 14.344\,(b) - 14.322\,(7)$$

$$= 0.022 \quad \text{(c)}$$

Pounds of H_2O per pound of fuel = H_2O formed + Pounds of moisture per pound of fuel + Atomizing steam.

$$= 1.784\,(11) + 0.022\,(c) + 0$$

$$= 1.806\,(d)$$

CORRECTION FOR EXCESS AIR*

Pounds of excess air per pound of fuel

$$= \frac{(28.85 \times \text{Percent } O_2)\left(\frac{N_2 \text{ formed}}{28} + \frac{CO_2 \text{formed}}{44} + \frac{H_2O \text{ formed}}{18}\right)}{20.95 - \text{Percent } O_2\left[\left(1.6028 \times \frac{\text{Pounds of } H_2O}{\text{Pounds of air required}}\right) + 1\right]}$$

$$= \frac{(28.85 \times 3.5)\left(\frac{11.157\,(13)}{28} + \frac{2.380\,(9)}{44} + \frac{0\,(d)}{18}\right)}{20.95 - 3.5\left[\left(1.6028 \times \frac{0\,(c)}{14.322\,(7)}\right) + 1\right]}$$

$$= 2.619 \quad \text{(e)}$$

$$\text{Percent excess air} = \frac{\text{Pounds of excess air per pound of fuel}}{\text{Air required}} \times 100$$

$$= \frac{2.619\,(e)}{14.322\,(7)} \times 100$$

$$= 18.3 \quad \text{(f)}$$

*If oxygen samples are extracted on a dry basis, a value of zero shall be inserted in equation (e) where a value is required from equations (c) and (d). If oxygen samples are extracted on a wet basis, the appropriate calculated value shall be inserted.

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EXCESS AIR AND RELATIVE HUMIDITY WORK SHEET

Job No. *Sample Work Sheet for G.3.2.3*
Date of Report
Page *2* of *2*

Total pounds of H_2O per pound of fuel (corrected for excess air)

$$= \left[\frac{\text{Percent excess air}}{100} \times \text{Pounds of moisture per pound of fuel}\right] + \text{Pounds of } H_2O \text{ per pound of fuel}$$

$$= \left[\frac{18.3\ \text{(f)}}{100} \times 0.022\ \text{(c)}\right] + 1.768\ \text{(d)}$$

$$= 1.772 \qquad \text{(g)}$$

Note: All values used in the calculations above shall be on a per-pound-of-fuel basis. Numbers in parentheses indicate values to be taken from the "Total per pound of fuel" line of the Combustion Work Sheet, and letters in parentheses indicate values to be taken from the corresponding lines of this work sheet.

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STACK LOSS WORK SHEET

Job No. *Sample Work Sheet for G.3.2.3*
Date of Report
Page *1* of *1*

Exit flue gas temperature, T_e: *500* °F

Component	Column 1 Pounds of Component Formed per Pound of Fuel	Column 2 Enthalpy at T (British thermal uniits per pound formed)	Column 3 Heat Content (British thermal units per pound of fuel)
Carbon dioxide	*2.380*	*100*	*238.0*
Water vapor	1.772	*200*	*354.4*
Nitrogen	*11.157*	*110*	*1227.3*
Air	*2.619*	*107*	*280.2*
Total	*17.928*	—	*2099.9*

INSTRUCTIONS

In Column 1 above, insert the values from the "Total per pound of fuel" line of the Combustion Work Sheet for carbon dioxide (Column 9) and nitrogen (Column 13). Insert the value from equation (e) of the Excess Air and Relative Humidity Work Sheet for air, and insert the value from equation (g) of the Excess Air and Relative Humidity Work Sheet for water vapor.

In Column 2 above, insert the enthalpy values from Figures G.B-1 and G.B-2 for each flue gas component.

In Column 3 above, for each component insert the product of the value from Column 1 and the value from Column 2. This is the heat content at the exit gas temperature.

Total the values in Column 3 to obtain the heat loss to the stack, Q_s.

Therefore,

$Q_s = \Sigma$ Heat content at $T_e =$ *2099.9* British thermal units per pound of fuel.

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APPENDIX G.F—ESTIMATING THERMAL EFFICIENCY FOR OFF-DESIGN OPERATING CONDITIONS

G.F.1 Introduction

This appendix provides a method for estimating thermal efficiency of fired process heaters at operating conditions other than the design or known operating conditions. This method is intended to be used as a short-cut procedure when it is impractical or unjustified to make detailed calculations.

G.F.2 Purpose and Scope

This method uses a series of empirical relationships to estimate the exit flue gas temperature at the off-design condions. This temperature, in turn, can be used to estimate the corresponding thermal efficiency. This method is intended for use with single-service heaters without air preheaters.]

These correlations have interent inaccuracies associated with all simplified correlations used to describe complex relationships. The method should be limited to estimating efficiencies for heater operations between 60 percent to 140 percent of design or known duty and with an inlet fluid temperature in the range of plus or minus 200°F of the design or known inlet temperature.

G.F.3 Estimation of Exit Flue Gas Temperature

The following equation can be used to estimate the exit flue gas temperature from the convection section of a fired process heater at alternative operating conditions, based on the heater's design or known operating conditions.

$$T_{e_2} = T_{i_2} + f_1 f_2 f_3 f_4 \left(T_{e_1} - T_{i_1}\right) \qquad \text{(G-3)}$$

Where:

f_1 = heat duty factor

$$= \left[\frac{Q_{a_2}}{Q_{a_1}}\right]^m \qquad \text{(G-4)}$$

$$m = \frac{1}{0.5 + 0.00125\,(T_{e_1} - T_{i_1})}$$

f_2 = coil inlet temperature factor

$$= \left[\frac{T_{i_2} + 460}{T_{i_1} + 460}\right]^{-0.4} \qquad \text{(G-5)}$$

f_3 = coil temperature rise factor

$$= 0.8 + 0.2\left[\frac{T_{o_2} - T_{i_2}}{T_{o_1} - T_{i_1}}\right] \qquad \text{(G-6)}$$

f_4 = excess air factor

$$= \left[\frac{AIR_2}{AIR_1}\right]^n \qquad \text{(G-7)}$$

$$n = \left[\frac{180}{T_{e_1} - T_{i_1}}\right]^{0.35}$$

AIR = total air flow relative to stoichiometric air required (e.g., 30 percent excess air = 1.30)
Q_a = heat absorbed, in million Btu per hour.
T_e = exit flue gas temperature, in degrees °F.
T_i = coil inlet temperature, in degrees °F.
T_o = coil outlet temperature, in degrees °F.
Subscript 1 = design or known condition.
Subscript 2 = off-design or unknown condition.

G.F.4 Sample Calculation

Use of the equations in G.F.3 can be shown with a sample calculation. For a heater with fuel and air conditions equal to those of Sample Calculations G.3.2.1 (oil-fired heater) and the following design conditions, estimate the exit flue gas temperature and efficiency at a 60 percent alternative operation.

	Design Conditions	80% Operation
Q_a, MM Btu/h	20.0	12.0
Flow rate, lb/h	93,600	68,100
T_i, °F	300	330
T_o, °F	700	680
Excess air, %	20	30
Radiation heat loss, %	1.5	2.0*
Exit flue gas temperature, °F	450	(to be determined)
Net efficiency, %	88.0	(to be determined)

*Estimated heat loss at reduced load

Using Equation (G-4):

$$f_1 = \left(\frac{12.0}{20.0}\right)^m$$

$$m = \frac{1}{0.5 + 0.00125\,(450 - 300)} = 1.455$$

$f_1 = (p.6)^{1.455}$

= 0.476 (heat duty factor)

Using Equation (G-5):

$$f_2 = \left(\frac{330 + 460}{300 + 460}\right)^{-0.4}$$

= 0.985 (coil inlet temperature factor)

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Using Equation (G-6):

$$f_2 = 0.8 + 0.2\left(\frac{680-330}{700-300}\right)$$

$$= 0.975 \text{ (coil temperature rise factor)}$$

Using Equation (G-7):

$$f_2 = \left(\frac{1.30}{1.20}\right)^n$$

$$n = \left(\frac{180}{450-300}\right)^{0.35} = 1.066$$

$$f_4 = (1.083)^{1.066}$$

$$= 1.089 \text{ (excess air factor)}$$

Using Equation (G-3) to find the estimated flue gas exit temperature:

$$T_{e2} = 330 + (450-300)(0.476)(0.985)(0.975)(1.089)$$

$$= 330 + (150)(0.498)$$

$$= 405° \text{ F}$$

Using the Stack Loss Work Sheet from Appendix G.C, at 405° F flue gas temperature and 30 percent excess air:

Q_s = 1749.7 British thermal units per pound of fuel

Using Sample calculations G.3.2.1, the net efficiency is:

$$e = \frac{(17,277 + 89.5 + 139.2 + 53.9) - (354.5 + 1749.7)}{(17,277 + 89.5 + 139.2 + 53.9)} \times 100$$

= 88.1 percent

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